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# GUIDELINES ON PROGRAMMATIC MANAGEMENT OF DRUG-RESISTANT TUBERCULOSIS IN INDIA

2017

REVISED NATIONAL TUBERCULOSIS CONTROL PROGRAMME
Central TB Division, Directorate General of Health Services,
Ministry of Health & Family Welfare, Nirman Bhavan, New Delhi

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**Government of India**
**Department of Health and Family Welfare**
**Ministry of Health & Family Welfare**

Dated : 28th December, 2017

## Message

RNTCP has recently developed National Strategic Plan (NSP) 2017-25 to work towards achieving the goals of eliminating TB by 2025. This necessitates a paradigm shift in approach and strategy. We have seen excellent commitment and the progress achieved through the previous NSP period like achieving the MDG goals and scaling up services for drug resistant TB as well as TB HIV across India, yet much more is required to be done to accelerate the march towards a TB free India.

Drug Resistant TB (DR-TB) is one of the major impediment to achieve NSP goal of ending TB in India. RNTCP has updated the Guidelines for programmatic management of drug resistant TB – 2017. This now aligns with the WHO End TB Strategy, Sustainable Development Goals (SDG) and WHO PMDT Guidelines - 2016 in order to advance the country to scaling-up universal access to drug susceptibility testing for all diagnosed and notified TB patients and decentralized patient centric care of the drug resistant TB.

Patient support in terms of counseling, linking eligible patients with available services is essential for timely diagnosis and successful treatment of DR-TB patient. Additionally, all the DR-TB patients should also be linked to other supportive systems for drug safety monitoring and management, nutritional support, vocational rehabilitation and financial protection.

These guidelines will be used for training of all health care workers at various levels for its smooth and seamless implementation. It also promotes digital data management and monitoring systems through Nikshay/e-Nikshay.

Implementing these guidelines will call for greater investments in improving access through decentralization of specialized service delivery facilities and efficient ICT enabled systems for programmatic management of drug resistant TB in India. The National Strategic Plan 2017-25 has been simultaneously developed in alignment with these investment requirements as well as the results framework for monitoring the progress. I would like to encourage all the states to avail the funding through the annual planning processes under the national health mission.

I would encourage all programme managers, providers, stakeholders, civil society and community members to work together for comprehensive implementation of these guidelines to combat and win over the menace of DR-TB in India.

(Preeti Sudan)

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## MESSAGE

Programmatic management of drug resistant TB (PMDT) was introduced under RNTCP in 2007 with the first version of the erstwhile DOTS Plus Guidelines. This was aligned to recommendations of World Health Organization (WHO). RNTCP revised the PMDT Guidelines in 2012 to align it with available evidence based recommendation in the updated WHO guidelines (2011). A recently conducted nationwide drug resistance survey also suggested that the drug resistance to first line anti-TB drugs reveals that the proportion of multidrug-resistant (MDR-TB) among new and previously treated TB patients are within the range of previous estimates.

With India's commitment to end TB by 2025, combating drug resistance TB will be essential to meet this ambitious target. It is now time to decentralize DR-TB services to improve access, minimize patient travel and maximize patient satisfaction. Early diagnosis of drug resistance among all diagnosed TB patients need to be ensured by providing rapid molecular technology (CBNAAT and LPA) at more decentralized level with the help of strong specimen collection and transportation system from the first point of contact where the patients choose to seek care. Similarly treatment of DR-TB services needs to be available at least up to the district level through a team of specialists at district hospitals, medical colleges or through available partnership options substantiated with patient support systems.

In 2016, WHO recommended use of shorter MDR-TB regimen for treatment of MDR/RR-TB patient who are susceptible to any fluoroquinolone and/or second line injectable drugs using second line – line probe assay as well as use of newer drugs i.e. Bedaquiline with a suitable background regimen for the management MDR-TB patients with advanced diseases or additional resistance to second line drugs. The guidelines are aligned to these recommendations. Further the regimen for H mono/poly drugs are already incorporated in these guidelines. Apart from this, the approaches of DST guided individualized regimen, active drug safety monitoring, palliative care, management of NTM and airborne infection control are also included in the guidelines. Use of information communication technology for adherence monitoring, drug and logistics management (Nikshay-aushadhi) and also for data management (Nikshay/eNikshay) is integral part of these guidelines.

I am very hopeful that these guidelines will play a pivotal role in enhancement of services provided for DR-TB patients in the country, an essential intervention to ensure that we witness the end of TB from India.

Jr Prasad

(Dr. Jagdish Prasad)

World Health Organization
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# Message from the WHO Representative to India

Globally, countries have made substantial progress in the fight against tuberculosis (TB) in the last decade and a half. However, an estimated 10.4 million people developed TB in 2016, and 1.3 million died of the disease in the world.

Drug-resistant TB is a continuing threat. In 2016, an estimated 601 000 new cases with resistance to rifampicin (RR-TB), the most effective first-line drug, of which 490 000 had multidrug-resistant TB (MDR-TB) emerged globally. Almost half (47%) of these cases were in India, China and the Russian Federation. In 2016, India witnessed a coverage of testing for rifampicin resistance of 20% for new TB patients, 67% for previously treated TB patients, and 41% overall among notified TB patients leaving a large unmet need.

WHO is proud to be associated with the Central TB Division, Directorate General of Health Services and Ministry of Health & Family Welfare, Government of India to reach the unreached with early diagnosis and appropriate treatment of patients suffering from various forms of TB and drug resistant TB.

The first ever National anti-TB Drug Resistance Survey conducted by RNTCP in 2014-16 with technical assistance from WHO revealed the current epidemiology of various forms of DR-TB in India. The proportion of MDR-TB is 2.84% and 11.60% in new and previously treated TB patients respectively, with geographic variations. Development of the **Guidelines for Programmatic Management of Drug-resistant Tuberculosis in India – 2017** is a major step forward in addressing this epidemiological reality and provide evidence-based guidance for meeting this unmet need. I am glad to state that our TB team at the WHO Country Office for India and the network of proficient technical consultants has worked closely with the Central TB Division in the development of these guidelines.

These guidelines offer technical and operational guidance to treating doctors, care providers and programme managers for early and quality diagnosis and management of DR-TB.

The government of India must be applauded for developing the ambitious National Strategic Plan for TB elimination (2017-25) that embraces the investment required for the paradigm shift in policy and strategies enshrined in these guidelines. India stood out as a country in which, the budget envelope for TB was substantially increased in the NSP (2017-25) following political commitment from the Prime Minister to the goal of ending TB by 2025.

The Guidelines for PMDT in India adds to the wide range of important TB documents from India and will be of immense help for the Global DR-TB control. Wider dissemination of these guidelines through various platforms and trainings of public and private doctors is vital.

WHO is committed to providing necessary technical guidance and assistance to the central and state governments in translating these guidelines into quality services for the patients and also continue evidence generation through surveillance and research for future refinement of the policies and strategies to align with the growing global evidences. We will also continue to work with all stakeholders and partners on our common resolve of fighting TB.

Actions taken today must build on the principles of government stewardship, engagement of civil society, empowering patient, reaching out to private sector, human rights and equity, and adaptation to the unique context of diverse epidemics and settings.

**Henk Bekedam**
WHO Representative to India

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Dated : 04/01/2018
Dated.....

# FOREWORD

The emergence of resistance to drugs used to treat tuberculosis (TB), particularly multidrug-resistant TB (MDR-TB), has become a significant public health problem in a number of countries and an obstacle to end TB. In India, available information from several drug resistance surveillance studies conducted in the past and the recently concluded national drug resistance survey suggest that the proportion of MDR/RR-TB among TB patients is relatively low. However, this translates into a large absolute number of patients that need to be treated with second line anti-TB drugs. Specific measures are being taken within the RNTCP to address the DR-TB challenge through appropriate strengthening of the health system for an effective management of the disease and prevention of transmission of DR-TB.

The term "Programmatic Management of Drug-Resistant TB" (PMDT) refers to programme based DR-TB diagnosis, treatment and cure. These guidelines promote full integration of basic TB management and PMDT activities under the RNTCP, so that patients with TB are evaluated for drug-resistance and placed on an appropriate treatment regimen and properly managed from the outset of treatment or as early as possible.

These Guidelines on Programmatic Management of Drug Resistant TB (PMDT) in India (2017) are an advancement over the RNTCP Technical & Operational Guidelines, 2016 and the Guidelines for use of Bedaquiline under conditional access through RNTCP PMDT in India, 2016 and would now supersede those. They have been duly updated and aligned with the WHO End TB Strategy, Sustainable Development Goals (SDG) and WHO PMDT Guidelines, 2016 in order to cover advancing the country to scaling-up universal access to drug susceptibility testing for all diagnosed and notified TB patients. As part of the national strategic plan (NSP) 2017-25, decentralization of diagnosis and treatment of MDR-TB to district level, the recently endorsed WHO recommended second-line probe assay (SL-LPA), rapid molecular drug susceptibility test (DST) for second-line drugs, shorter MDR-TB regimen, DST guided regimen to cover all variety of DR-TB including Isoniazid (H) mono-poly DR-TB, use of newer drugs like Bedaquiline, revised recording reporting systems, e-NIKSHAY and pharmacovigilance systems for active drug safety monitoring and management (aDSM) have been outlined. As global evidence evolves, in future, these guidelines will be updated based on the revision of WHO DR-TB guidelines.

(Dr. Sunil D Khaparde)

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The deliberations and workshop involved discussions with several stakeholders, extensive review of literature and aligning with current and emerging evidence to address the prevailing epidemiology of drug-resistant TB (DR-TB) in India from initiatives such as:

- the recently concluded first national drug resistance survey (2014-16);
- lessons learnt from the roll-out of Bedaquiline containing regimen in 6 sites in 2016;
- revised technical and operational guidelines for TB;
- WHO PMDT Guidelines (2016) updates that recommend shorter multidrug resistant TB (MDR-TB) regimen with second line – line probe assay and other evidence based updates on management of DR-TB; and
- National Strategic Plan for TB elimination in India (2017-25) which in turn is aligned with the WHO End TB Strategy.

In October 2016, the National Experts Committee on Diagnosis and Treatment of TB recommended adoption of the updated WHO PMDT Guidelines (2016) along with geographical expansion of access to Bedaquiline with active drug safety monitoring in India.

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These guidelines are consistent with the WHO PMDT guidelines (2016), Companion Handbook to WHO Guidelines on PMDT (2014). The technical and operational aspects listed herein are intended to complement the existing RNTCP technical and operational guidelines and the National Strategic Plan for TB Elimination (2017-25) in India. These guidelines will now be implemented across India to gain and document experiences on feasibility, safety monitoring and enhancement in treatment outcomes of DR-TB patients to further guide the country in its refinement as national and global evidence emerges in future.

## Abbreviations

| | |
|---|---|
| aDSM | Active drug safety monitoring and management |
| ADR | Adverse drug reaction |
| AE | Adverse event |
| AFB | Acid Fast bacilli |
| AIC | Airborne infection control |
| AIDS | Acquired Immune Deficiency Syndrome |
| ALT | Alanine aminotransferase |
| Am | Amikacin |
| AMC | ADR monitoring center |
| Amx/Clv | Amoxicillin/clavulanate |
| ART | Anti-retroviral therapy |
| ARV | Anti-retroviral |
| AST | Aspartate aminotransferase |
| ATS | American Thoracic Society |
| Bdq | Bedaquiline |
| BPL | Below poverty line |
| CAP | Conditional Access Programme |
| CBNAAT | Cartridge Based Nucleic Acid Amplification Test |
| CEM | Cohort event monitoring |
| Cfz | Clofazimine |
| Clr | Clarithromycin |
| Cm | Capreomycin |
| CMO | Chief Medical Officer |
| CP | Continuation phase |
| CPT | Co-trimoxazole preventive therapy |
| Cs | Cycloserine |
| CTD | Central TB Division |
| CUP | Compassionate Use Programme |
| C-DAC | Centre for Development of Advanced Computing |
| C-DST | Culture and Drug Susceptibility Test |
| CL-HIV | Children living with HIV |
| DAIDS | Division of AIDS |
| DBT | Direct Beneficiary Transfer |
| DCGI | Drugs Controller General of India |
| DDG | Deputy Director General |
| DDS | District drug store |
| DDR-TBC | District DR-TB Centre |
| DG | Director General |

| | |
|---|---|
| DGHS | Directorate General of Health Services |
| Dlm | Delamanid |
| DMC | Designated microscopy centre |
| DOT | Directly observed treatment |
| DOTS | Core approach underpinning the Stop TB strategy for TB control |
| DRT | Drug resistance testing |
| DR-TB | Drug-resistant tuberculosis |
| DR-TBC | Drug-Resistant Tuberculosis Centre |
| DSMC | Drug Safety Monitoring Committee |
| DST | Drug susceptibility testing |
| DTO | District TB officer |
| DVDMS | Drug vaccine distribution management system |
| E | Ethambutol |
| ECG | Electrocardiogram |
| ECHO | Extension of Community Health Care Outcomes |
| EP-TB | Extra-pulmonary tuberculosis |
| EQA | External quality assurance |
| Eto | Ethionamide |
| EU | European Union |
| FDA | Food and Drug Administration |
| FEFO | First expiry first out |
| FL-LPA | First line-line probe assay |
| FQ | Fluoroquinolone |
| GLC | Green Light Committee |
| GFATM | Global Fund for AIDS, Tuberculosis & Malaria |
| Gfx | Gatifloxacin |
| GMSD | General Medical Stores Depot |
| GoI | Government of India |
| H | Isoniazid |
| Hh | High dose isoniazid |
| HRCT | High resolution CT scan |
| ICH | International Conference on Harmonization |
| ICT | Information communication technology |
| ICMR | Indian Council for Medical Research |
| IP | Intensive phase |
| Ipm | Imipenem |
| IPAQT | Initiative for promoting affordable & quality TB test |
| IQC | Internal Quality Control |
| IRL | Intermediate reference laboratory |
| ISO | International Standard Organization |

| | |
|---|---|
| Km | Kanamycin |
| LC | Liquid culture |
| LFT | Liver function test |
| Lfx | Levofloxacin |
| L J | Lowenstein Jensen |
| LPA | Line probe assay |
| LT | Laboratory technician |
| LTFU | Lost to follow up |
| Lzd | Linezolid |
| MAC | Mycobacterium avium complex |
| MDR-TB | Multidrug-resistant TB |
| Mfx | Moxifloxacin |
| Mfxh | High dose Moxifloxacin |
| MGIT | Mycobacteria growth indicator tube |
| MIS | Management information system |
| MO | Medical Officer |
| MoHFW | Ministry of Health and Family Welfare |
| MO-DMC | Medical Officer-designated microscopy centre |
| MO-PHI | Medical Officer- peripheral health institute |
| MO-TC | Medical Officer TB control |
| MOTT | Mycobacterium other than tubercle bacilli |
| MoU | Memorandum of understanding |
| Mpm | Meropenem |
| MPR | Mixed pattern resistance |
| MR | Mono resistance |
| MSS | Monthly stock statement |
| NAAT | Nucleic Acid Amplification Test |
| NABL | National accreditation board for laboratories |
| NDRS | National Drug Resistance Survey |
| NDR-TBC | Nodal DR-TB Centre |
| NGO | Non-Government Organization |
| NGS | Next-Generation Sequencing |
| NHPS | National health protection scheme |
| NHM | National Health Mission |
| NIRT | National Institute for Research in Tuberculosis |
| NITRD | National Institute for Tuberculosis and Respiratory Diseases |
| NRL | National reference laboratory |
| NSP | National strategic plan |
| NTI | National TB institute |

| | |
|---|---|
| NTM | Non-Tuberculous Mycobacterium |
| OBR | Optimized background regimen |
| Ofx | Ofloxacin |
| OPD | Out Patient Department |
| PAS | p-aminosalicylic acid |
| Pdx | Pyridoxine |
| PDR | Poly drug resistance |
| PHI | Peripheral health institute |
| PK/PD | Pharmacokinetic/pharmacodynamics |
| PL-HIV | People living with HIV |
| PMDT | Programmatic management of drug-resistant tuberculosis |
| PP | Private Provider |
| PQC | Product quality compliance |
| PSM | Procurement and supply management |
| PT | Previously treated |
| PTE | Pre-treatment evaluation |
| Pto | Protionamide |
| PvPI | Pharmaco-vigilance programme of India |
| QA | Quality assurance |
| QSE | Quality System Elements |
| R | Rifampicin |
| RNTCP | Revised National Tuberculosis Control Programme |
| RR-TB | Rifampicin-resistant tuberculosis |
| R&R | Recording & reporting |
| RT-MERM | Real time medication event reminder monitor device |
| S | Streptomycin |
| SA | Statistical Assistant |
| SAE | Serious adverse event |
| SDG | Sustainable Development Goals |
| SDS | State drug store |
| SLD | Second line anti-TB drugs |
| SLDST | Second line drug susceptibility testing |
| SLI | Second line injectable |
| SL-LPA | Second line-line probe assay |
| SME | Supervision, Monitoring & Evaluation |
| SoP | Standard operating procedures |
| SPC | Specimen Processing Control |
| STLS | Senior TB Laboratory Supervisor |
| STO | State TB Officer |

| | |
|---|---|
| STR | Standardized treatment regimen |
| STS | Senior treatment supervisor |
| TALFU | Treatment after lost to follow up |
| TAT | Turn-around time |
| TB | Tuberculosis |
| TBHV | TB Health Visitor |
| Thz | Thioacetazone |
| ToR | Terms of reference |
| Trd | Terizidone |
| TU | TB Unit |
| UDST | Universal Drug Susceptibility Testing |
| ULN | Upper limit of normal |
| UPT | Urine pregnancy test |
| USAID | United States Agency for International Development |
| USFDA | United States Food & Drug Administration |
| WCO India | World Health Organization Country Office for India |
| WHO | World Health Organization |
| XDR-TB | Extensively-drug resistant TB |
| Z | Pyrazinamide |

# Chapter 1: Background and framework for effective control of drug-resistant tuberculosis

Globally, the first WHO endorsed PMDT services began in the year 2000. At that time, the Green Light Committee (GLC) was established to promote access to high quality second-line drugs for appropriate use in TB programmes. In 2002, the Global Fund to Fight AIDS, TB and Malaria (GFATM) started financing TB programmes, including DR-TB, thus greatly reducing the economic barrier to countries for DR-TB services. Since then, PMDT services have expanded rapidly. Based on data and experience from these projects and practices, further scientific evidence continues to evolve with respect to services for DR-TB.

## 1.1 History of PMDT in India

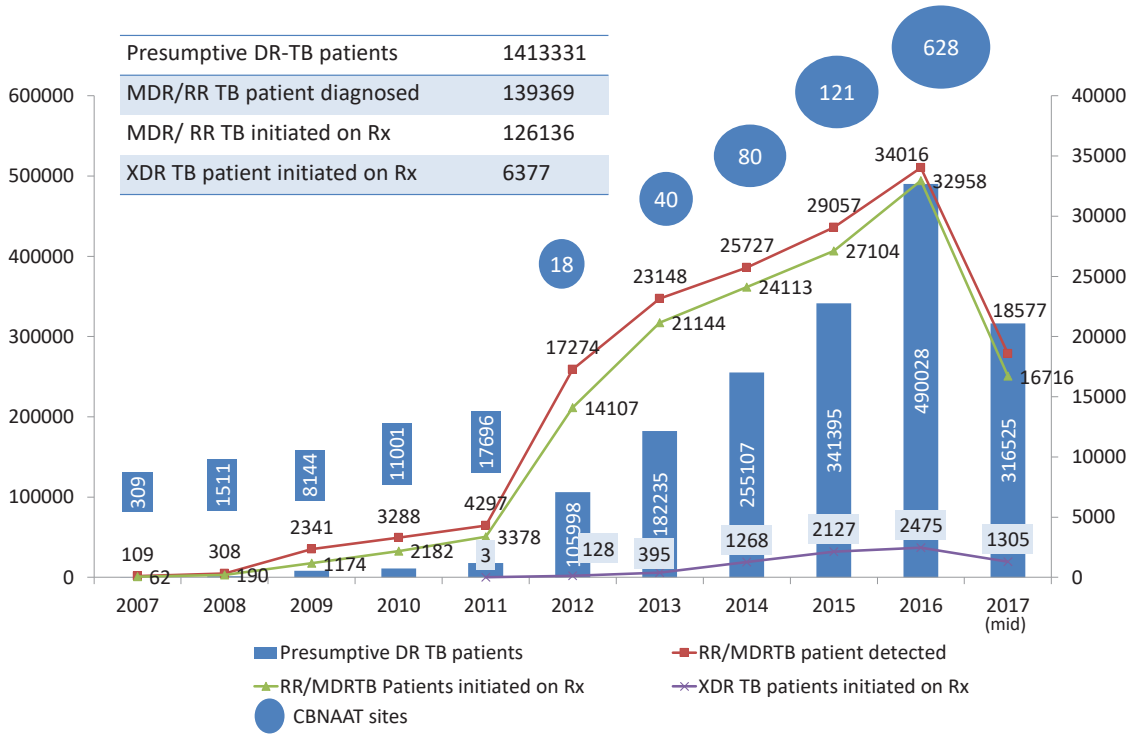
After successfully establishing RNTCP services across the country in 2006, the PMDT services were introduced in 2007 and complete geographic coverage was achieved by 2013. [1] To begin with, DR-TB services were offered to the subset of TB patients with the highest risk to develop drug resistance *i.e.,* treatment failures. This was followed by a horizontal and vertical scale-up. Definite criteria were set to assess the risk and eligibility for the drug susceptibility test (DST). The DST was thus offered to TB patients who remained smear positive during follow-up; to previously treated patients; those who were HIV positive and people who had contact with a known DR-TB patient. [2] This would then lead to universal DST, *i.e.,* DST to all diagnosed and notified TB patients. To conduct this, huge laboratory capacity in terms of geographic coverage, DST technology, trained laboratory personnel, quality assurance and certification were required. From a few national reference laboratories (NRL), and state level Intermediate reference laboratories (IRL) with solid or liquid culture and DST facilities, the country expanded its capacity to a wide network of state and regional level intermediate reference laboratories with solid and liquid culture DST and Line Probe Assay (LPA) and district level network of Cartridge Based Nucleic Acid Tests (CBNAAT). [1] [3]

Providing treatment to diagnosed DR-TB patients is extremely important. To begin with, only MDR-TB patients were offered treatment with a standard second-line regimen. Later, treatment with standard regimen was offered to extensively drug resistant (XDR) TB patients and MDR-TB with additional resistance to quinolone or second-line injectable. Procurement and supply chain management of second-line drugs is complex, since no standardized patient-wise boxes are manufactured, shorter shelf-life of some drugs and drugs do need temperature regulated storage and repacking. [2]

During 2011-12, there was a massive scale-up of all these facilities with concerted efforts of multiple stakeholders, resulting in countrywide coverage by 2013. Later in 2014, baseline

second-line DST facilities were established in a few intermediate reference laboratories, which also got scaled-up to the entire country in 2015. The progress of DR-TB treatment coverage is shown in Figure 1.1below [1]:

**Figure 1.1 DR TB finding and treatment initiation effort, 2007-17**




Detection of DR-TB through RNTCP has been progressively rising with increased access to various forms of DST. In 2016, RNTCP was detected and treatment initiated in about 34016 patients of MDR-TB and 2476 patients of XDR-TB. [1]

In 2016, new drugs Bedaquiline (Bdq) was made accessible to DR-TB patients through a conditional access programme (CAP) under RNTCP. [4] In 2016, with the release of the revised Technical and Operational Guidelines, regimens to treat other forms of drug resistance, such as mono and poly resistance to first and second-line drugs were also included. [5]

Offering treatment to DR-TB patients was far more complex than today due to limited capacity of diagnosis and treatment. However, it still requires multidisciplinary drug resistant TB centres (DR-TBC) for pretreatment evaluation, observation and management of adverse drug reactions (ADR), where trained specialists, supportive human resources, efficient management information systems and AIC facilities are assured.

## 1.2 Magnitude of the DR-TB problem in India

Drug-resistant TB has been known from the time anti-TB drugs were first introduced for the treatment of TB. Currently, the WHO estimated incidence of Rifampicin (R) and MDR-TB in India is estimated to be around 147000. This translates to around 11 patients per 100 000 population annually as per the Global TB Report, 2017 [6]. Newer and higher quality information from various sources helped update the estimates of DR-TB. This revision in estimates has resulted in a rise of the perceived burden of DR-TB. However, this is not indicative of the problem of drug resistance being on the rise. The first national anti-TB drug resistance survey for 2014-16 (NDRS) concluded recently.[7] The following are the key findings of the NDRS report:

- under the survey 526 (92.5%) designated microscopy centres (DMC) participated and the quality of specimens collected was good, with >97% specimens accepted;
- 95% patient enrolments were completed within the planned period; and
- recovery rate for cultures was 94.6% in new and 92.8% in previously treated patients.

Table 1.1 Summary of DST patterns from the National Drug Resistance Survey

| DST pattern | New | Previously treated | All |
|---|---|---|---|
| Total DST results available | 3065 | 1893 | 4958 |
| Sensitive to all drugs | 2374 (77.46 %) (75.93-78.92%) | 1196 (63.18 %) (60.96-65.36%) | 3570 (72.01 %) (70.73-73.25%) |
| Any drug resistance | 691 (22.54 %) (21.10-24.10%) | 697 (36.82 %) (34.64-39.04%) | 1388 (28.00 %) (26.77-29.29%) |
| MDR | 87 (2.84%) (2.28-3.49%) | 220 (11.62 %) (10.21-13.15%) | 307 (6.19%) (5.54-6.90%) |
| MDR with second-line anti-TB (SLI) resistance | 6 (6.90 %) (2.57-14.41%) | 5 (2.27 %) (0.74-5.22%) | 11 (3.58 %) (1.80-6.32%) |
| MDR with fluoroquinolone (FQ) resistance | 21 (24.14 %) (15.60-34.50%) | 46 (20.91 %) (15.73-26.89%) | 67 (21.82 %) (17.33-26.87%) |
| XDR | 2 (2.30 %) (0.28-8.06%) | 2 (0.91 %) (0.11-3.25%) | 4 (1.30 %) (0.36-3.30%) |

- any drug resistance among new patients is 22.54%, among previously treated (PT) patients is 36.82% and among all patients 28.02%;
- further, any H resistance (16% in all with 11.6% in new and 25.09% in previously treated patients) being the driver for R resistance;
- almost all RR-TB patients are resistant to H with/without other first or second-line drugs;
- MDR-TB is 6.19% in all patients (2.84% among new and 11.6% among previously treated patients);
- among MDR-TB patients, additional resistance to other first-line drugs is high at 74.3%. The drug specific resistance in combination with MDR-TB includes:
    - any Streptomycin resistance – 70% in new and 59.09% in PT;
    - any E resistance - 45.98% in new and 46.36% in PT; and
    - any Z resistance – 31.03% in new and 20.45% in PT.
- among MDR-TB patients, additional resistance to other second-line drugs includes:
    - any FQ (Ofx, Lfx, Mfx) is 21.82% (24.14% in new and 20.91% in previously treated patients);
    - any SLI drugs (Km, Am, Cm) is 3.58% (6.9% in new and 2.27% in previously treated patients); and
    - XDR-TB is only 1.3% in all MDR-TB patients (2.3% in new and 0.91% in previously treated patients).
- state level and cluster-wise variations clearly indicate that although the national DR TB situation is well within the range of previous state level surveys, there exist focal epidemics of DR TB in some states.

Programmatically, MDR-TB (at-least to H & R), Rifampicin resistance (RR-TB) and XDR-TB (at-least H, R, second-line injectable [SLI] and fluoroquinolones [FQ]) received priority. As facilities for detecting other varieties of resistance became increasingly available, making available regimens for their treatment also became a programmatic priority.

While prevention of development of drug resistance is of paramount importance for ending TB, early detection and immediate enrolment as well as completion of an effective treatment regimen are keys to interrupt on-going transmission, to prevent death and reduce chances of sequelae post-treatment. The programme has so far been able to successfully treat 46% of the cohorts of patients initiated on treatment 30-33 months ago. However, treatment outcomes vary from state-to-state. While few states were able to achieve more than 70% treatment success rate among the diagnosed MDR/RR-TB patients, others could achieve less than 40%, suggesting the variability of the health system to deliver treatment, care and cure. Main reasons attributed for the attrition were death and lost to follow-up (LTFU) during treatment. [1]

## 1.3 Causes of drug-resistant tuberculosis

Drug-resistant TB, like TB is a disease of poverty, expressed through microbial, clinical and programmatic channels. From a microbiological perspective, the resistance is caused by a genetic mutation that makes a drug ineffective against the mutant bacilli. In clinical settings, an inadequate or poorly administered treatment regimen allows drug-resistant mutants to become the dominant strain in a patient infected with TB. Clinical characteristics of patients have also been recognized where appropriately administered drugs have not achieved necessary drug levels to deal with all populations of mycobacteria. From a programmatic perspective, weak TB services lead to delay in detection and effective treatment of drug resistance and are unequipped to support patients to keep adherence to treatment and prevent ongoing transmission.

## 1.4 Prevention of drug-resistance

The problem of DR-TB cannot be addressed completely by standalone systems for detection and treatment of drug-resistance. Strong systems to detect, successfully treat and ensure long term disease-free status of TB patients, are required to prevent emergence of resistance. Thus, basic TB diagnostic and treatment services should receive priority and systems for early detection and treatment of drug-resistant forms of TB which should be integrated into existing TB services.

Improperly treated patients with resistant strains of TB will constitute a source of ongoing transmission of resistant strains. Health care facilities and congregate settings lacking of proper infection control measures as recommended by WHO contribute to maintain transmission. The interruption of transmission and not only rapid detection and immediate enrolment on effective regimens are therefore necessary to prevent the emergence of new DR-TB patients. Measures to prevent incidence and transmission of TB are also effective in prevention of drug-resistance. The framework for PMDT presented in this document is to be considered along with management of drug susceptible TB patients as a continuum of care.

## 1.5 Move to patient-centric care

Successful treatment and care can only result when patient preferences, values and needs are satisfactorily addressed along with PMDT services. This includes ensuring that the diagnosis of DR-TB is early, accurate and affordable; and the most effective treatment is delivered early and provided in a manner that is easily accessible to the patient, affordable and socially acceptable. At the same time it must ensure that the confidentiality and dignity of the patient is protected. It is the responsibility of the health system to make sure that the patient is treated successfully within the society s/he belongs to, enjoying all support which the community would otherwise provide to its members so that the new chain of infection is arrested at source and the cured member enriches his/her material, social and cultural

assets. [3] Prevention, management and mitigation of stigma and discrimination are essential elements of a patient-centered care approach to TB management.

## 1.6 Universal drug susceptibility testing

The programme is committed to providing Universal Drug Susceptibility Testing (UDST) for all diagnosed and notified TB patients.[3] Depending upon resource availability, this service is progressively being made available throughout the country. A range of rapid molecular tests are available for UDST, such as Cartridge Based Nucleic Acid Amplification Test (CBNAAT) and Line Probe Assay (LPA).

In the past, patients were offered DST based on their risk of developing drug resistance. Over time, DST has been increasingly made available to the patient to lower levels of risk. Currently, all patients of TB apart from those not previously exposed to anti-TB treatment (ATT) are routinely offered atleast the R resistance test. The revised algorithm for presumptive TB in the RNTCP Technical and Operational Guidelines, 2016 gives provision for testing presumptive patients for TB on CBNAAT for reliable and early microbiological confirmation and in these patients, R resistance information is available as a byproduct. [5]

There are several additional DST technologies in the development pipeline. These tests will be made available progressively as they are endorsed for use by WHO. Recently, WHO endorsed LPA for use for DST of FQ class and SLI class [8]. Currently, LPA labs are in preparation for rolling out this facility while DST on CBNAAT is offering more accurate and wider testing of resistance to R which is also in various phases of development.

## 1.7 DST guided treatment

The 2012 PMDT Guidelines were largely prioritized for dealing with RR, MDR and XDR types of drug resistant TB. [2] This version will address management of all variants of DR-TB, including H mono/poly resistance (R sensitive) and various types of poly-drug resistance (mixed patterns beyond MDR/RR and XDR). [9][4]

## 1.8 Decentralized DR-TB management

Previously, guidance was largely provided for centralized PMDT treatment initiation in inpatient settings. This version provides guidance on both inpatient and outpatient based treatment initiation for providing increased access to PMDT services. Inpatient-care for those who require it on medical grounds and drug supply chain management have been decentralized down to the district level.

## 1.9 Customized treatment supervision and patient support

Traditionally, treatment supervision methods were limited to direct observation of therapy (DOT) by a trained person other than family members. Move to a patient-centric approach would mean giving more priority for patient's needs and preferences in view of better adherence in such a way that ethics and standards for DOT are met. In some patients, a family member might be able to ensure better treatment supervision and adherence as compared to an external individual visiting the home. With the development of information communication technology (ICT), there are options by which patients can reliably self-report drug consumption, be monitored and supported by various levels simultaneously. [3] The newer guideline favours the principle of adherence monitoring which has to be applied logically and judicially. These also provide options, whereby, the most appropriate modality of adherence monitoring may be used as a collective decision for the patient, treatment supporter and the Medical Officer (MO).

The necessity for supporting patient needs related to TB care in addition to treatment is increasingly being recognized. This version of the PMDT Guidelines includes a section on treatment support.

## 1.10 Newer drugs and regimen

There are newer drugs like Bedaquiline (Bdq) and Delamanid (Dlm) currently approved for conditional use by stringent drug regulatory authorities and for which WHO has provided interim guidance for their use under programmatic settings. These drugs, still under development, are expected to improve treatment outcomes among selected populations. [10] [11] Similarly, some of the earlier drugs like clofazimine and linezolid are being repurposed and recommended for use in the management of DR-TB. A few shorter MDR-TB regimens are being studied worldwide using combination therapy with newer drugs. However, till further evidence is available, WHO has recommended a shorter MDR-TB regimen for use for MDR/RR-TB patients without any additional resistance to FQ/SLI class based on available evidence. [12]

This version of the guidelines on the Programmatic Management of Drug-Resistant TB (PMDT) in India, 2017, integrates use of the shorter MDR-TB regimen and newer drug containing regimen under RNTCP with a DST guided regimen design.

# Chapter 2: Structure and responsibilities

This chapter describes the structure and responsibilities for PMDT at various levels and the service cascade for DR-TB.

Diagnosis of DR-TB starts with identification of patients presumed to have DR-TB. This will keep on changing with availability of new technologies to detect resistance and the availability of resources to detect resistance and the prevalence of various types of resistance in the background population. Thus, every newly diagnosed TB patient could be a presumptive DR-TB in a population with a high background prevalence of resistance to the most potent anti-TB drugs. [13] Similarly, even the treatment failures may be presumed to fail due to reasons other than drug resistance in a population with very low background prevalence of resistance.

When PMDT (erstwhile DOTS Plus) was rolled-out in India in 2007, failures after treatment with Category 1 (2[HREZ]3 + 4[HR]3 for new TB patients or Category 2 (2[HREZS]3 + 1[HREZ] + 5[HRE]3 for previously treated TB patients only were considered as presumptive MDR-TB patients. Later, patients remaining smear positive during any follow-up, patients with history of previous TB treatment, people living with HIV (PLHIV) with TB and TB patients who are close contacts of DR-TB were added to the presumptive list (erstwhile Criteria C). [2] These vertical expansions were aligned with the horizontal geographic expansion since 2010. By 2013, the entire population had access to rapid molecular DRT (CBNAAT/LPA) at least for R and by 2014 all districts initiated DST as per Criteria C.

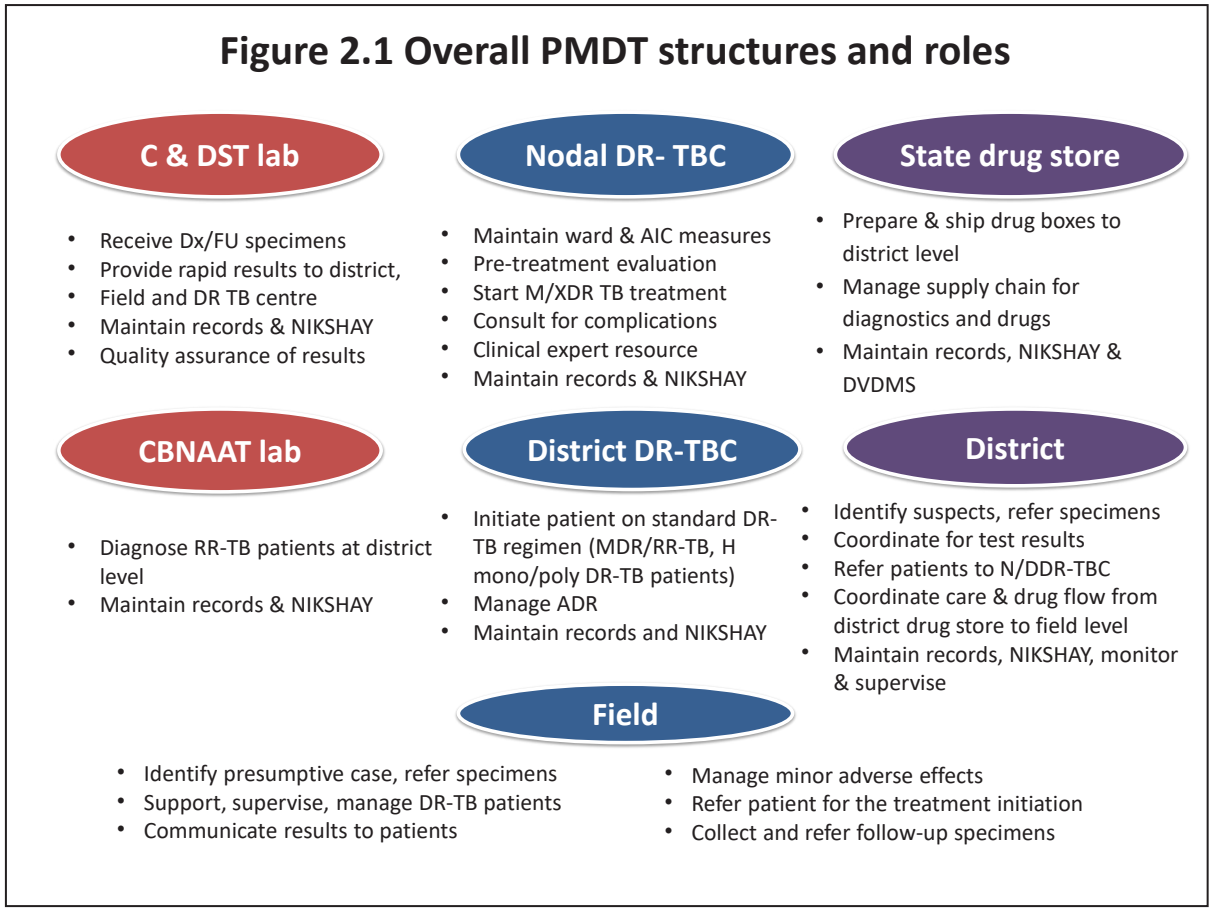
However, all eligible presumptive DR-TB patients may not be subjected to DST if the patients are not closely monitored and health service staff and Medical Officers are not keen. The key adjectives to 'identification' of presumptive DR-TB patients are 'early' and 'complete'. From identification of presumptive DR-TB patients to initiating treatment of confirmed DR-TB patients, there are a chain of events that are prone for loss of patients. The first event is the patient becoming presumptive DR-TB as per the prevailing criteria. This is followed by identification, referral, passive reporting/active tracking of referred presumptive patients, collection and transportation of biological specimen, rejection and retesting of specimens if it occurs, reporting of results, referral of confirmed patients for treatment and patients reaching DR-TB appropriate service delivery sites for initiating treatment. At each point, a proportion of patients are lost.

The “WHO End TB Strategy” recommends every country to advance towards knowing at-least the RR-TB status of all TB patients at the time of diagnosis to guide appropriate treatment decisions.

## 2.1 State-level structure and responsibilities

While a national expert technical working group has developed national policies, technical and operational guidelines, the state-level is where the majority of planning activities, implementation and monitoring occur. The overall structures and roles are summarized in Figure 2.1 below. The State PMDT Committee is responsible for developing the plan of action for implementation, expansion, maintenance, supervision, monitoring and quality enhancement of PMDT services in the respective state. The composition and terms of reference of the State PMDT Committee are detailed in Annexure 1.

**Figure 2.1 Overall PMDT structures and roles**




## 2.2 Drug resistant-tuberculosis centre

Programmatic and clinical management of DR-TB is complex but feasible when the health system is strengthened to effectively integrate what is necessary. Treatment of drug-resistant TB is not completely based on centralized and institutionalized care for the entire duration. In fact, clinical care needs the presence of a clinical and patient support expert resource centre. This is the DR-TB Centre which is a 20-30 bedded tertiary care facility established to serve a population of approximately 10 million, with an airborne infection

control compliant ward, facilities for pretreatment evaluations, treatment initiations, follow-up monitoring and management of adverse drug reactions, prevention and relief of physical and social suffering caused by the disease and its treatment, complications and comorbidities. [2] All these activities are supported by the programme staff in addition to having counselling for patients and undertaking data management.

Currently, 147 DR-TBCs are established across India (2017), one for approximately every 10 million population, including some in private institutes partnering with RNTCP. [1] About 5 to 10 districts are attached to each centre. DR-TB patients are admitted for a short period and once stabilized on treatment, discharged with advance intimation to the districts and referred back to their districts for continuation and completion of treatment. During treatment, they are referred back to DR-TBCs for change of regimens and management of ADRs. [2]

However, periodic monitoring and review by the programme revealed that many challenges are currently being faced by the programme. These relate to delayed initiation of treatment, inadequate bed capacity, compromised follow-up, insufficient and untimely relief of suffering and poor accessibility. In most situations, patients have to travel long distances, including loss of work hours and family income, resulting in catastrophic expenditures. Further, the travel poses additional risk of transmitting infection in transit.

Since March 2016, 500 CBNAAT machines have been made functional in addition to 128 existing machines to cover access to most of the districts in India. [1] These machines are currently utilized at the district level for testing presumptive DR-TB patients and presumptive TB patients among key populations to detect presence of M. Tb in the biological specimen with concomitant detection of RR-TB if present. A 35% rise in MDR/ RR-TB patients notified was observed in Q2 2016 against Q1 2016 which is expected to further increase in future. [1] This is likely to increase the demand for treatment significantly. Treatment and care capacity needs to be enhanced to meet with this diagnostic capacity in every district in India. This will empower districts to enable the “test and treat approach” to eliminate delays in diagnostic and treatment initiation pathways for all MDR/RR-TB patients within the district. However, until initiation of treatment, infection control services and advice need to be available for all diagnosed DR-TB patients.

Hence, to decentralize the pretreatment evaluation, treatment initiation of RR-TB or H mono/poly DR-TB and follow-up processes, two distinct types of DR-TBCs will be accordingly established. The existing nodal DR-TB centre (NDR-TBC) will continue for approximately 10 million populations. District DR-TB centre (DDR-TBC) should be established for at least every district. Some of the states have already established these centres. Additionally, DDR-TBC that could function on an outpatient department (OPD) basis are considered to manage DR-TB patients on OPD basis. Structure and function of these centres and provisions under

RNTCP to upgrade them to function as RNTCP designated DDR-TBC and NDR-TBC are detailed below.

Decentralization is necessary to ensure prompt treatment initiation with scale-up to universal DST, shorter MDR-TB regimen and DST guided regimen with/without newer drugs, and delivery of patient care including nutritional assessment and support, palliative and end-of-life care and management of co-morbidities.

### 2.2.1 District DR-TB centre

The DDR-TBC is responsible for the initiation and management of uncomplicated DR-TB patients like RR-TB or H mono/poly DR-TB in a district, not only on inpatient basis, but also on outpatient basis, wherever advisable and possible.

**Rationale for setting up District DR-TB centres**

The advantages of decentralized "test and treat approach" are:

- early and faster initiation of treatment of all diagnosed DR-TB patients;
- bringing care closer to the residence of majority of the DR-TB patients;
- significant reduction in catastrophic expenditure including loss of work hours and family income;
- rationally minimizing the need and duration for hospitalization;
- minimizing travel of patients, thereby transmission risks during travels;
- accountability of the district programme management units; and
- rationalizing utilization of existing DR-TBCs to enable them to concentrate in more complex clinical decisions and ensuring quality assurance of treatment and research.

The 147 existing DR-TBCs will be designated as NDR-TBCs, serving as referral centres for DDR-TBCs. The NDR-TBC located at a district may also serve as DDR-TBC for the same district based on the work load as well as a referral centre for other districts.

**Process of establishing DDR-TB centres**

The DDR-TBC can be established at institutes in a certain order of preference, namely, medical colleges, district hospitals, TB hospitals and NGO/private/corporate institutes/other sector hospitals with the availability of required clinical expertise.

It must be ensured that the DDR-TBC is in close proximity of the CBNAAT site/ LPA lab in the district. There should be atleast one DDR-TBC available in the district which should be identified by the State TB officer (STO) in consultation with the District TB Officer (DTO) and Chief Medical Officer (CMO) of the district. However, more than one DDR-TBC can be

established wherever needed to improve access. The district authorities may take a decision to establish more than one centre based on the workload of the first centre and the distance taken by patients to travel. This should be in consultation with CTD. The composition and terms of reference of the DDR-TBC Committee is detailed in Annexure 1 and a checklist to guide preparation of DDR-TBC is enclosed at Annexure 2.

Each DDR-TBC should preferably have an airborne infection control (AIC) compliant indoor facility to accommodate at least 4 DR-TB patients with male and female separation (wherever indoor DDR-TBC is proposed) outdoor facilities and specialists trained in clinical management of DR-TB patients. Patients who are ambulatory can be managed on an OPD basis. DDR-TBCs could operate in public or private sector institutions, linked with appropriate schemes for private sector engagement under National Partnership guidelines of RNTCP.

The DDR-TBC should be established preferably under auspices of the Department of Respiratory Medicine/General Medicine. The committee may be formed with members of various clinical specialties in the same institution if available or through other public/private sector institutions. This committee should meet every month or whenever required. The DDR-TBCs will continue to be attached to concerned NDR-TBCs for referral and expert consultations.

The DDR-TBC would leverage upon as well as strengthen the capacity of the health system to respond not only to DR-TB but also to its most common comorbidities, which have a significant country burden as well (e.g. HIV, diabetes, hepatitis C, mental health) and other clinical and public health services such as nutritional assessment and management of undernutrition, respiratory infection control, patient-centered care, pharmacovigilance, digital health, palliative/end-of-life care services.

**Requirements from institute and provision from RNTCP for establishing DDR-TBCs**

Requirements from the institute:

- should preferably be a tertiary/secondary care institute;
- wherever indoor DDR-TBC is proposed, at least 2 beds each, for male and female separated structurally or temporally within existing wards/ separate isolation areas if available or necessary;
- for indoor and outdoor DDR-TBC, an outpatient clinic and a separated well ventilated waiting area in an open-air and shaded area to be made available;
- administrative, environmental and personal protective measures for airborne infection control to be in place in all indoor and outdoor facilities; [14]
- all PMDT services to be provided free of cost to the patient;
- DDR-TBC Committee to be formally established with minimum set of experts available at the district level as per Annexure 1;

- relevant specialties like physician (Nodal Officer), psychiatrist, dermatologist, ENT specialist, nutritionist, anesthesiologist or palliative/end of life care specialist, cardiologist and gynaecologist *etc.,* to be made available. This can be direct or through honorary visits mainly to provide specialist consultations;
- State-level training of DDR-TBC committee doctors (including Chairperson), where trainers including from NDR-TBC committee members can serve as facilitators;
- routine clinical, radiological, audiological, biochemical investigation and ECG facility to be made available for pretreatment evaluation and follow-up monitoring at no cost to patient;
- ancillary drugs to be provided as per DDR-TBC Committee`s advice at no cost to patients;
- doctors, nursing and support staff should be available from the institute;
- records and reports to be maintained for PMDT;
- NIKSHAY entries to be done on real time basis with regular electronical updates; and
- financial requirements to be availed through institute/state budgets or National Health Mission (NHM).

Provision under RNTCP:

- Existing MO DTC/ 2$^{nd}$MO of DTC/ MO Medical College will provide support to the physician in-charge serving as nodal officer of the DDR-TBC;
- DR-TB Counsellor for counselling patients as per RNTCP NSP 2017-25;
- District DR-TB TB-HIV Supervisor to maintain all records and reports including NIKSHAY entry and coordination with all PHI and TB Unit staff;
- training concerned staff and availability of formats and registers for PMDT;
- Ensuring availability of second-line anti-TB drugs; and
- Computer, internet facility and NIKSHAY login ID for the institute.

**Functions of the DDR-TB centres**

The package of services at these DDR-TBCs would include:

***Pre-treatment evaluation (PTE)***: All investigations (detailed in the relevant section) would be done within the district, at no cost to the patients. If there are some investigations which are not available within the district, then adequate linkages will be established with the private laboratories with proper memorandum of understanding (MoU) mechanism as per schemes available under the National Partnership Guidelines. Since the drugs used for the treatment of DR-TB patients are known to produce adverse effects, a proper pretreatment evaluation is essential to identify patients who are at increased risk of developing such adverse effects.

***Providing counselling to patient and family members:*** Counsellor at the DDR-TBC will provide counselling and health education, an essential part of the social support to the DR-TB patient and his/her family members about the disease, necessity of taking regular and adequate treatment, possible adverse events and memorandum of understanding (steps to be taken, mechanisms of TB transmission, prevention and mitigation of stigma and discrimination, nutritional counselling and assistance to avail social support and social protection schemes. Health education and counselling is provided to all patients and family members at different levels of health care, right from one at the periphery to those at the DR-TBC facility. It is started at the initial point of contact and carried out on a continuous basis during all visits by the patient at the health facility. The continuous counselling and motivation is essential, not only for the patient but also family members and caregivers to ensure treatment adherence.

***Treatment initiation of RR-TB or H mono/poly DR-TB patients***: Treatment initiation based on results of first-line DST with CBNAAT or LPA will be done by the DDR-TBC committee. Treatment of the patient can be initiated by the nodal person of DDR-TBC with post facto approval of other committee members after pretreatment evaluation of the patient is completed. PMDT treatment card and patient treatment booklet are opened; first dose is given under supervision and registration of patients done at the DDR-TBC using the revised PMDT treatment register. Before starting treatment, the patient should be instructed in detail about potential adverse effects that could be produced by the prescribed drug regimen and if and when they occur, to notify the treatment supporter or healthcare provider.

***Referral of patients to NDR-TBC for regimen change:*** The regimen may have to be changed in any of the following circumstances:

- additional laboratory confirmed resistance to second-line drugs;
- severe adverse events known before treatment or during follow-up treatment;
- seriously ill patient with very low general condition or as evaluated by DDR-TBC any time before or during treatment; and
- patient with failing regimen or returning after treatment interruption of >1 month or emergence of any exclusion criteria for standard regimen.

***Providing travel enablers***: This will be provided to the patient and one attendant during PTE, initiation of treatment and follow-up visits as per RNTCP guidelines.

***Follow-up monitoring***: DDR-TBC committee would ensure timely follow-up for all DR-TB patients initiated on treatment as per the follow-up schedule for clinical, biochemical and

culture including audiometry, mental health monitoring and ECG monitoring for QTC interval as applicable.

***Management and monitoring of adverse drug reactions***: The DDR-TBC committee doctors, nurses, treatment supporters and supervisory staff, will prevent, monitor, manage, document all the adverse events routinely and the doctor will report them as per the aDSM framework. The PTE serves to establish a baseline and may identify patients who are at increased risk for adverse effects or poor outcomes. Laboratory investigation including audiometry and periodic ECGs will be done as per the follow-up schedule. These are invaluable for detecting certain adverse effects that are more occult and before serious harm is done. Most of the ADRs can be prevented and managed by the DDR-TBC. If required, the patients may be referred to the NDR-TBC for the management of the serious ADRs after providing initial management.

***Recording and reporting***: All records and reports for DR-TB patients managed at the DDR-TBC will be maintained by the respective DDR-TBC staff at the district level.

***NIKSHAY:*** Real time entries, periodic updating of patient information and analysis for corrective actions need to be ensured from each DDR-TBC.

***Airborne infection control measures:*** These would be implemented as per the National infection control guidelines. [14] During routine household visits, patient and family members will be advised on measures needed for infection control by the concerned health staff.

***Nutritional assessment:*** Nutritional assessment will be undertaken for all patients and the corresponding nutritional advice and support will be provided, according to guidelines for nutritional assessment and support for TB patients in India.

***Mental health:*** Assessment of mental health is essential in all patients at the time of diagnosis and regularly monitored in all patients especially those receiving cycloserine.

***Palliative care:*** Palliative/end-of-life care with priority for patients with XDR-TB and other MDR-TB patients with poor prognosis will be provided in consultation with the NDR-TBC.

### 2.2.2 Nodal DR-TB centre

Patients with additional resistance to second-line drugs, drug intolerance, contraindications, failing regimen, patients returning after treatment interruption of >1 month, emergence of any exclusion criteria for standard regimen for RR-TB or H mono-poly DR-TB regimen, non-TB *mycobacterium* (NTMs) and those needing palliative care would be managed at NDR-TBC.

**Process of establishing the NDR-TB centre**

The existing DR-TBC must be upgraded to function as NDR-TBC. Additional NDR-TBC as required must be established under the auspices of the Department of Respiratory Medicine or Department of Medicine (if the former department does not exist). NDR-TBC Committee is a clinical committee where the Head of Department (HoD) or a senior faculty member of the department of Respiratory Medicine (General Medicine) is the Chairperson and HoDs or senior faculty members of other specialties are members. Clinical function of these committees should be adequately supported by the administrative or management committees of the institution in which the STO must be ex-officio member. (composition and terms of reference of NDR-TBC Committee are detailed in Annexure 1).

The requirements from the institute (listed below) must be provided by the Government medical college/ institutes, including free laboratory investigations and ancillary drug supply as part of their commitment for which no reimbursement will be available from the programme. To manage paediatric DR-TB, a specific AIC compliant isolation area in the paediatric ward in the institute of NDR-TBC must be designated. However, government medical colleges/ institutes will be eligible for all provisions from RNTCP listed above along with one-time provision for upgradation of the ward to incorporate AIC measures as per norms of costing.

Private and NGO hospitals are considered to serve as NDR-TBC at places where a government medical college is not available. A scheme is available for such engagement under National partnership guideline of RNTCP.

**Requirements from the institute and provision from RNTCP to establish Nodal DR-TB centres**

Requirements from the institute:

- should preferably be a tertiary care institute;
- separate ward for male and female patients should be available with at least 10 beds in each;
- all PMDT services (beds, investigations, ECG and ancillary drugs for management of adverse drug reactions) to be provided free of cost to the patient;
- relevant specialties including respiratory medicine, general medicine, psychiatry, dermatology, ENT, ophthalmology, gynaecology, paediatrician, anaesthesiologist and cardiologist should be available directly or through linkages;
- NDR-TBC committee to be formed;
- national training of NDR-TBC committee members (including Chairperson);
- National AIC Guidelines to be implemented in DR-TB wards and outpatients setting. (Annexure 3); and

- routine clinical laboratory investigation facility to be made available for pretreatment evaluation and monitoring;
- ancillary drugs should be available;
- management of adverse drug reactions (ADR) as per PMDT guidelines;
- doctors, nursing and support staff should be available from the institute;
- records and reports to be maintained for PMDT; and
- quarterly reports to be submitted electronically.

Provision under RNTCP

- remuneration of Senior Medical Officer & Statistical Assistant and Counsellor;
- training, formats and registers for PMDT;
- second-line anti TB drugs; and
- computer, internet and NIKSHAY ID for the facility.

**Functions and responsibilities of Nodal DR-TB Centres**

The package of services at these NDR-TBCs would include:

***Pretreatment evaluation***: Patients should be examined by specialist members. The pretreatment investigations should be done free of cost. If there are some investigations which are not available within the public sector, then adequate linkages need to be established with the private sector with proper MoU mechanism as per schemes available under the National Partnership Guidelines. Pretreatment evaluation should include a thorough clinical evaluation by a physician, chest radiograph, haematological and biochemical tests detailed below. Since the drugs used for treatment of DR-TB are known to produce adverse effects, a proper pretreatment evaluation is essential to identify patients who are at increased risk of developing such adverse effects. A thorough clinical examination should be done during pretreatment evaluation (details of pretreatment evaluation are given in relevant sections);

***Counselling to patient and family members:*** As detailed in DDR-TBC section;

***NIKSHAY***: As detailed in DDR-TBC section;

***Travel enablers:*** These will be provided to the patient and one attendant during PTE, initiation of treatment and follow-up visits as per RNTCP guidelines;

***Follow-up monitoring and management of adverse drug reactions:*** As detailed in DDR-TBC section, patients referred from DDR-TBC will be appropriately managed;

***Providing clinical decision, referral, management support and training to districts;***

***Airborne infection control measures:*** As detailed in DDR-TBC section;

***Nutritional assessment:*** As detailed in DDR-TBC section;

***Mental health:*** As detailed in DDR-TBC section; and

***Palliative care:*** Palliative/end-of-life care with priority for patients with XDR-TB and other MDR-TB patients with poor prognosis will be provided. In addition, guidance will be provided by NDR-TBC to the DDR-TBC and concerned health staff for palliative care at community level.

## 2.3 DR-TB services for patients seeking care in the private sector

The programme recognizes the fact that although free quality diagnostic and treatment services are available in the public sector, a significant number of patients are seeking health services from the large unorganized private/other sector. Reaching out to these patients is equally important, especially to deliver essential public health services to prevent the spread of disease, emergence of drug-resistance, to support TB patients on treatment and address comorbidities which adversely affect treatment outcomes. Patients seeking care in private/other sectors are equally eligible for diagnostics, drugs and all supportive services offered to patients seeking care in the public sector.

Screening for presence of drug resistance, access to quality diagnosis of DR-TB, access to second-line anti-TB drugs including newer drugs and shorter regimen for the patient diagnosed with drug-resistance TB available under RNTCP can be systematically offered to patients seeking care in the private/other sector. This includes all treatment support services like pretreatment investigations, counselling, treatment adherence support, management of adverse drug reaction *etc.,* as part of public health actions to be taken by the public sector after any TB patient is notified.

A strong and sustainable partnership between the programme and providers is necessary to establish linkages to ensure availability of all services for any DR-TB patient irrespective of where the patient chooses to seek care. One of the options for such linkages available within the programme currently is the National Guidelines for Partnership (2014). [15] The Guideline provides options for purchasing services for diagnosis, specimen transportation, DR-TB centres and treatment support to all type of TB patients. It also provides some programme management options like packaging of anti-TB drugs which may be used based on the requirement for repackaging second-line anti-TB drugs at the State drug store.

States can innovate and use flexibility of budget options provided under the NGO-PP head (H.9) under RNTCP to extend services in private sector. As per norms of existing budget, the

State can use 10% of the NGO-PP head budget to involve NGOs or private providers and use 30% of the NGO-PP head budget for innovative interventions for patients who seek care in private/other sectors. [16] PMDT services for patients seeking care in the private/other sectors can be accessed from RNTCP at all levels. However, this would be guided by the willingness on part of the patient as well as provider and hence there needs to be flexibility and choices for the provider and patients to avail these services in various situations. Access to DST must be ensured in consultation with providers. With expansion of CBNAAT services up to district level, states need to arrange for DST of notified TB patients as per eligibility criteria followed under the programme.

As states advance DST coverage to universal DST, they need to ensure that all notified TB patients undergo DST at least for R. To avail services available under RNTCP, efficient specimen collection and transport system should be established from the private/other providers to the nearby CBNAAT laboratory. Cost of such specimen transportation system should be borne from the relevant budget head of the patient support system. As an option, the NGO/any other agency may be engaged for specimen collection and transport using partnership options under the National Partnership Guidelines. Laboratory services for C-DST of first-second-line drugs as well as DR-TB treatment services can also be purchased from the private/other sector as per the National Partnership Guidelines, if required. [15]

For patients diagnosed as DR-TB patients in the private sector, efforts should be made to put in place patient-friendly and provider-convenient system of referral to DR-TBCs. For such patients, the DR-TBC should provide public health action which includes contact investigation, family counselling, treatment adherence support, comorbidity testing, follow-up investigation, reporting of treatment outcomes, pharmacovigilance and social protection linkages.

For increasing capacity of DR-TB treatment services and utilizing experts/health institutes in expanding access, the State should consider engaging private health institutes or provider for DR-TB Centres. There are three options available under National Partnership Guidelines. [15] namely, DR-TB centres with wards/ on outpatient basis/ private specialist to be provided honorarium for supporting public sector DR-TB patients in clinical management.

If the provider has the capacity to initiate an appropriate DR-TB treatment regimen as per WHO recommended principles of designing a regimen [12] and is willing to avail treatment support services including second-line drugs from RNTCP, efforts should be made to partner with such institutes to serve as a DR-TB Centre as per National Partnership Guidelines.

A strong and sustainable partnership between the programme and providers is necessary to establish linkages to ensure availability of all services for any DR-TB patient irrespective of where the patient chooses to seek care.

**Access to PMDT services including newer drugs to patients seeking care in private/other sector**

PMDT services including newer drugs like Bedaquiline would be available through regulated access from RNTCP and can be provided to the patient seeking services in private/other sector. Again, priority providers should be mapped by every state for partnership. The mechanism for access to PMDT services including newer drugs for patient seeking care in the private sector are detailed in Chapter 7.

## 2.4 Preparation for introduction of newer regimen and drugs

In India, RNTCP has introduced a new drug Bedaquiline (Bdq) as per WHO interim recommendations in 2016 through conditional approval of DCGI. [4] Bdq is currently available through RNTCP under the conditional access programme and is not available to the open market in India. There are few prerequisites proposed for states to expand access under the RNTCP programme:

- availability or linkage with a laboratory with expanded second-line DST;
- establishment of district DR-TB centre in every district as per guidelines;
- availability of a nodal DR-TB centre with all requisite specialists as per guidelines;
- availability of human resources trained for introduction of newer drugs and regimen including staff at DR-TB centres, labs, drug stores, state, districts and field level;
- availability or access to all requisite laboratory investigations for pre-treatment evaluation and follow-up monitoring including electrocardiography (ECG) as per technical specifications (Annexure 4);
- mechanism for active drug safety monitoring and management (aDSM) [17]; and
- updated recording and reporting systems with NIKSHAY readiness.

## 2.5 Coordination

As RNTCP embarks on PMDT activities for the management of DR-TB, coordination of activities at all levels is critical. Coordination needs to include the contribution of all key stakeholders, organizations and external partners, as considered below:

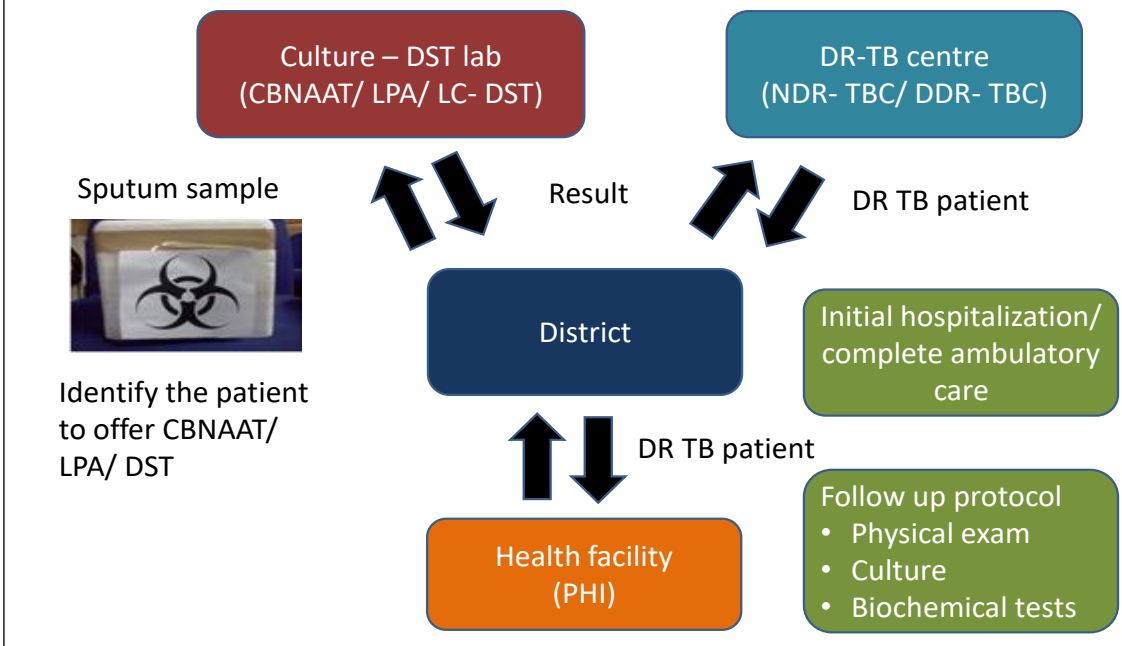
***Central TB Division (CTD), MoHFW, GoI:*** The CTD is the central coordinating body for activities described in the framework. Commitment of necessary resources, particularly towards a strong central management team, ensures all aspects are in place from the procurement of second-line anti-TB drugs to appropriate implementation and monitoring of PMDT service. As needed, partnerships with all relevant health care providers can be built. The CTD is supported by a National PMDT Committee, comprising members from CTD, the three central TB institutes (NTI, NIRT and NITRD), medical colleges and WHO.

***Local health system***: RNTCP PMDT activities will be tailored to fit into the respective state and district levels' infrastructure. The exact organizational structure of RNTCP PMDT services may vary between different settings depending on how local health care is provided. Transfer between hospitals to outpatient settings or between Treatment centres requires great care, advance planning and good communication. Given the type of care required in the treatment of DR-TB, a team of health workers including physicians, nurses and social workers (wherever available) should be used.

***Community level***: Community involvement and communication with community leaders can greatly facilitate implementation of PMDT and may respond to needs that cannot be met by medical services alone. Community education, involvement and organization around TB issues can encourage a feeling of community ownership of TB programmes and reduce stigma. In some circumstances, communities can help address the patient's interim needs including provision of treatment support, food and/or housing, vocational support *etc.,*

## 2.6 Overview of model of care

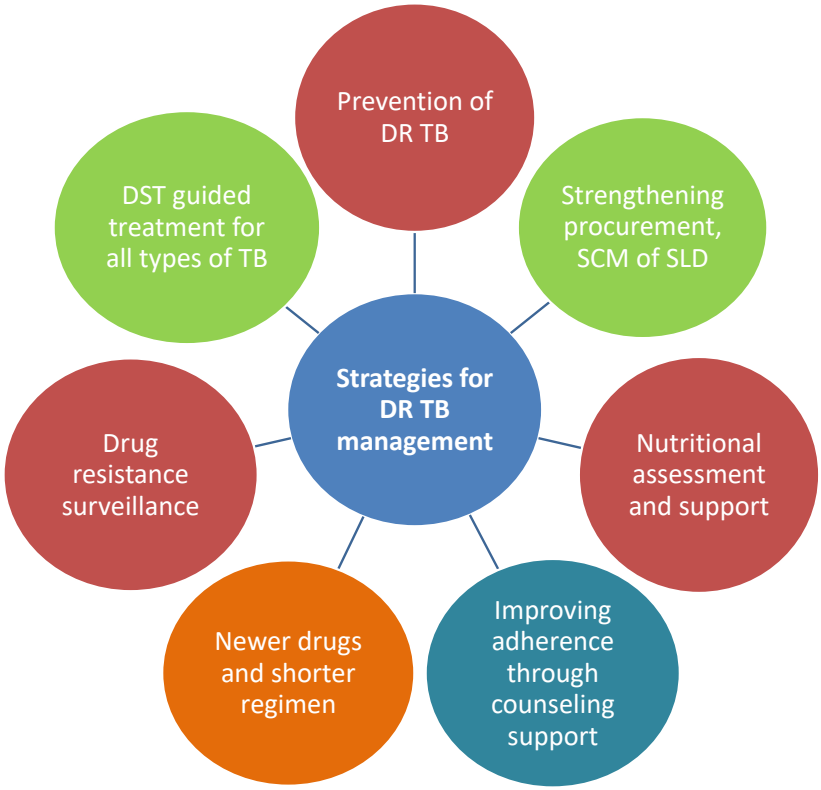
**Figure 2.2: Overview of model of care in RNTCP PMDT**




Integration of PMDT services will require multiple care levels to work in coordination. No longer can the field level unit be totally self-sufficient. The care at the field level is supported by the laboratory and the DR-TBC, coordinated by the district and supported by the State. This is depicted in the Figure 2.2.

Supportive services for DR-TB patients are an essential component of comprehensive DR-TB care. Core strategies like screening for drug resistance, DST guided treatment, treatment regimen containing newer drugs, ensuring uninterrupted supply of diagnostics and second line drugs, ancillary drugs for management of adverse drug reactions, treatment adherence support, patient education and counseling are essential, however, these need to be supplemented by appropriate nutritional support, linkage to various social protection measures like direct benefit transfers to make it more comprehensive. Prevention of DR-TB transmission through adequate infection control as well as averting amplification by appropriate treatment of drug sensitive TB, remains the top priority to fight with DR-TB problem. Strategies for DR-TB management are depicted in figure 2.3 and included in the NSP (2017-25). [3]

**Figure 2.3 Strategies for DR-TB management**



# Chapter 3: National strategic plan for universal access to quality diagnosis and treatment of DR-TB

This chapter provides a brief overview of the RNTCP PMDT vision under the National Strategic Plan (2017-25) for Tuberculosis elimination by 2025. The National Strategic Plan (NSP) sets out the strategic direction and key initiatives that MoHFW, GoI will undertake in the period, 2017 to 2025 for working towards achieving the goals of eliminating TB by 2025. This includes the strategy for prevention and management of DR-TB; strategy to strengthen laboratory capacity and treatment services; and nationwide scale-up of new interventions under PMDT. [3]

## 3.1 PMDT expansion plans and progress

RNTCP expanded PMDT services to all districts in the country by March 2013. The systematic participatory planning undertaken by CTD with all states for phased scale-up of PMDT services in 2010 keeping available/planned resources with secured funding in mind and intensive regional review for course correction over the last five years have enabled the programme to successfully scale-up PMDT services as per the national PMDT and lab scale-up plans (2009-2014). The enormous efforts put in and cooperation extended to CTD by every state is well appreciated. However, states since 2011 were expected to strive to consolidate the current policy of offering DST to presumptive DR-TB and CBNAAT to presumptive TB among PLHIV, children and EP-TB with baseline second line DST to RR-TB patients.

The National Strategic Plan (2012-17) for RNTCP set the goal of universal access to quality diagnosis and treatment for all TB patients (including DR-TB and TB HIV). [18] The year 2014 saw large strides being undertaken with many new initiatives and policy changes in RNTCP-PMDT. Some examples include the launch of the first nationwide anti-TB drug-resistance survey of India and formulation of guidelines on DST guided treatment for drug-resistant TB patients.

The national PMDT and lab scale-up plans were accordingly updated for the next five years (2015-20) to align them with this goal with part-funding secured through the Global Fund NFM. The revised RNTCP Technical and Operational Guidelines 2016 [5] (updated in accordance to the WHO End TB Strategy [13]), guidelines for use of Bedaquiline under conditional access in RNTCP PMDT [4] and expansion of CBNAAT services were launched by the Hon'ble Health Minister of India in March 2016. [1]

## 3.2 National Strategic Plan (2017-25) and PMDT

PMDT is one of the thrust areas of the RNTCP National Strategic Plan. India's achievements in the TB programme over the past decade have been remarkable. India achieved complete geographical coverage for diagnostic and treatment services for MDR-TB in 2013, with a remarkable 139369 persons with MDR-TB diagnosed and 126136 put on treatment till 2016.

The National PMDT scale-up plan for 2017- 2020, an operational plan, was developed by consolidating state-wise PMDT micro-plans developed during the series of regional PMDT review meetings with 35 states organized by CTD at north, south, west, east and northeast zone in the year 2015-2016. Outputs included clarity and transparency on national training and district appraisal needs, laboratory scale-up requirements, national/state/district responsibilities understood by all and scale-up plan of Bedaquiline, shorter MDR-TB regimen and DST guided treatment.

Specific objectives for PMDT under NSP are:

- by end 2017, complete nationwide geographical coverage of access to baseline second-line DST using SL-LPA, access to shorter MDR-TB regimen and newer drugs like Bedaquiline;
- by 2025 ensure Universal access to rapid molecular DRT for all diagnosed TB patients; Universal access to DST guided treatment and expands access to newer drug; and management of NTM.

## 3.3 RNTCP strategy & intervention for prevention and management of DR-TB under NSP (2017-25)

The RNTCP response to DR-TB revolves around strategy to prevent emergence and stop transmission of DR-TB. These are enumerated below:

**Prevention of DR-TB**

- sustain the highest quality care for drug sensitive TB patients;
- promote rational use of anti-TB drugs; and
- implement infection control measures.

**Stopping transmission of DR-TB**

- ensure early diagnosis, respiratory isolation (segregation of active TB patients in waiting rooms, wards, fast tracking) and prompt effective treatment initiation of all forms of DR-TB;
- improve laboratory capacity for rapid diagnosis of DR-TB (expanded below);

- initiate and rapidly scale-up services to all types of DR-TB patients (expanded below);
- ensure effective DST guided treatment of DR-TB patients; and
- evaluate the extent of second-line anti-TB drug resistance and management strategies.
- ensure full implementation of infection control guidelines

The specific strategies laid down in the NSP for early diagnosis, prompt and appropriate treatment as well quality care, social support and protection for DR-TB patients are enlisted below:

**Laboratory systems for drug-resistant TB:**

- universal DST to at least R for all diagnosed TB patients through offer of rapid molecular tests will be rolled out in a phased manner starting 2017;
- introduce and scale-up diagnostics for NTM detection and DST;
- strengthen surveillance systems including introduction and scale- up of next generation sequencing (NGS) platforms;
- scale-up effective mechanisms of affordable diagnostics for TB in private sector will be done including provision of services by the programme, giving diagnostics to the private sector or reimbursement of the cost;
- have the programme empanel accredited laboratories for diagnostics; get NABL accreditation for public sector laboratories as per the lab scale up plan; maintain linkages with NABL for providing proficiency panels to private and corporate sector laboratories for quality assured diagnostics; and have DST expand capacity and accessibility;
- implement a laboratory information management system and link it to e-NIKSHAY during this plan period; and
- purchase and ensure notification through laboratories from the private sector and link them to laboratory surveillance.

**Treatment of DR-TB**

- Decentralize DR-TB treatment: As RNTCP moves to universal DST with rapid molecular tests available at every district, second-line DST testing, shorter MDR-TB treatment and DST guided treatment with or without newer drugs, volume of patients to be managed at centralized DR-TB centres will delay treatment initiation and increase numbers lost to follow-up. To mitigate this anticipated loss and promote DR-TB treatment initiation within 24-48 hours, RNTCP will decentralize DDR-TBC to initiate standard regimen for H mono-poly DR-TB and MDR/RR-TB regimen (shorter/ conventional longer) at every district level established within close proximity of CB-NAAT site. Patients with additional resistance to second-line drugs, drug intolerance or those seriously ill will need regimen modification and will be managed at the NDR-TBC;

- Manage the H mono/poly DR-TB patients: H mono/poly resistance is known to be around thrice as prevalent as RR-TB. A specific 9-12 month treatment regimen has been initiated to manage H mono-poly resistance with available first-line drugs strengthened with a fluoroquinolone and a second-line injectable. This regimen will be scaled-up across the country as per expansion of diagnostic and treatment capacity;
- scale-up shorter MDR-TB regimen across the country by the end of 2017 for all RR-TB patients that meet criteria for this regimen;
- Finalize the regimens containing newer drugs: Access to Bedaquiline will be expanded across India by the end of 2017. RNTCP will also introduce another new drug Delamanid by the end of 2017, after conditional approval from DCGI and undertake research with support of ICMR to evaluate the use of these newer drugs in combination therapy to reduce the duration of DR TB regimens to 4-6 months;
- Ensure DST guided DR-TB regimen and manage XDR TB patients with or without resistance to any other first or second-line drugs, who do not consent or are not eligible for newer drugs. Do this with an appropriate regimen design based on their DST results. This approach will be scaled-up along with Bedaquiline expansion across India by the end of 2017; and
- RNTCP to initiate addressing non TB *Mycobacteria* and scale-up along with NTM diagnosis and treatment across India by the end of 2018.

**Social protection & supportive systems**

- patients with drug resistance in whom an appropriate regimen cannot be formed, even with addition of newer drugs, as per the WHO recommended regimen, will be offered palliative care through the nodal DR-TB centres or at the community level under guidance of the nodal DR-TB Centre. Necessary palliative services including pain relief, surgery, prosthesis, psychosocial support and respiratory physiotherapy will be provided;
- any additional cost incurred by patients for diagnosis, treatment, other social determinants, palliative care and rehabilitation will be covered under existing insurance schemes like National Health Protection Scheme (NHPS) that provides health insurance cover of up to Rs. 1 lakh to the poor. The scheme is projected to benefit about 10 crore families in the first phase. The main beneficiaries of the scheme would be families belonging to the below poverty line (BPL) category or those in the list of deprivations as per the socioeconomic caste census data; and
- the guidance document for nutritional assessment and supplementation for TB patients in India (2016) provides technical and operational guidance to states to establish systems of nutritional assessment and linkages with various social welfare schemes and public distribution systems to provide nutritional supplementation to the family of patients suffering from TB and DR-TB in India with successful models being implemented by some states like Kerala, Chhattisgarh *etc.,*

## Table 3.1 Result framework for PMDT indicators in NSP (2017-25)

| Indicators | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Proportion of notified TB patients offered DST | 25% | 30% | 50% | 60% | 70% | 80% | 90% | 95% | 98% | 99% | 100% |
| No. of presumptive DRTB patients to be examined | 341395 | 444933 | 600000 | 7000000 | 9000000 | 1100000 | 1200000 | 1300000 | 1400000 | 1500000 | 1500000 |
| No. of MDR/RR TB patients notified | 29057 | 36000 | 53460 | 66000 | 78975 | 92000 | 92000 | 78250 | 69000 | 61500 | 55000 |
| No of MDR/RR TB patients initiated on treatment | 27104 | 33000 | 48114 | 59400 | 71078 | 82800 | 82800 | 70425 | 62100 | 55350 | 49500 |
| No. of H mono-poly resistant TB patients notified | 11802 | 13144 | 16038 | 79200 | 189540 | 276000 | 276000 | 234750 | 207000 | 184500 | 165000 |
| No. of H mono-poly resistant TB patients initiated on treatment | 100 | 200 | 14434 | 71280 | 170586 | 248400 | 248400 | 211275 | 186300 | 166050 | 148500 |
| No. of pre XDR TB patients notified | 1837 | 1785 | 13365 | 22500 | 25125 | 27000 | 27000 | 23250 | 20250 | 17250 | 15000 |
| No. of pre XDR TB patients initiated on treatment | 1670 | 1550 | 12029 | 20250 | 22613 | 24300 | 24300 | 20925 | 18225 | 15525 | 13500 |
| No. of NTM patients notified | 580 | 720 | 1974 | 2520 | 2814 | 3024 | 3024 | 2604 | 2268 | 1932 | 1680 |
| No. of NTM patients initiated on treatment | 0 | 0 | 200 | 500 | 1500 | 2419 | 2419 | 2344 | 2041 | 1739 | 1512 |
| No. of XDR TB patients notified | 2340 | 1340 | 2406 | 2970 | 3554 | 4140 | 4140 | 3521 | 3105 | 2768 | 2475 |
| No. of XDR TB patients initiated on treatment | 2127 | 1197 | 2165 | 2673 | 3198 | 3726 | 3726 | 3169 | 2795 | 2491 | 2228 |
| Treatment success rate for MDR/RR TB | 46% | 46% | 48% | 48% | 56% | 65% | 70% | 72% | 73% | 74% | 75% |
| Proportion of notified TB patients (including DR-TB) using ICT supported adherence systems | 1% | 2% | 10% | 30% | 50% | 80% | 90% | 90% | 90% | 90% | 90% |
| Proportion of notified TB patients (including DR-TB) receiving financial support through DBT | 0% | 0% | 2% | 50% | 70% | 80% | 90% | 90% | 90% | 90% | 90% |
| Proportion of tertiary and secondary facilities with budgeted action plan for AIC in TB facilities | 0 | 0 | 0 | 40% | 60% | 80% | 100% | 100% | 100% | 100% | 100% |
| Number of District DR-TB Centre established | 0 | 143 | 250 | 700 | 750 | 790 | 810 | 820 | 830 | 840 | 850 |
| No. of rapid molecular laboratories established | 123 | 628 | 835 | 1335 | 1835 | 2335 | 3335 | 4335 | 5335 | 6335 | 7335 |
| No. of first-line DST (Phenotypic) laboratories established | 62 | 66 | 80 | 95 | 110 | 125 | 125 | 125 | 125 | 125 | 125 |
| No. of first-line DST (Genotypic) laboratories established | 44 | 46 | 54 | 54 | 54 | 54 | 54 | 54 | 54 | 54 | 54 |
| No. of second-line DST (Genotypic) laboratories established | 0 | 0 | 54 | 54 | 54 | 54 | 54 | 54 | 54 | 54 | 54 |
| No. of second-line DST (Phenotypic) laboratories established | 18 | 26 | 40 | 55 | 70 | 85 | 100 | 115 | 130 | 145 | 160 |
| Percentage of districts covered for call center support for treatment adherence (including ICT intervention districts) | 4% | 4% | 18% | 50% | 80% | 100% | 100% | 100% | 100% | 100% | 100% |

# Chapter 4: Laboratory services for programmatic management of drug-resistant TB

This chapter describes the concept of certification of laboratories for quality-assured results, case definitions for different drug-resistant TB (mono-resistant, poly-resistant, MDR-TB, XDR-TB), laboratory services needed to diagnose and treat DR-TB patients. In addition, the section elaborates the process of liaising with the NRL for setting up and maintaining the entire gamut of laboratory services across the health sector

## 4.1 Vision

The vision of the programme is to provide Universal DST to all diagnosed TB patients, while offering an upfront CBNAAT to all presumptive TB patients among key populations. [3] [13] Revised diagnostic algorithm for DR-TB will ensure comprehensive diagnostic and drug susceptibility testing of patients to enable timely treatment decision with appropriate regimen. Towards this end, a staggered testing algorithm utilizing newer rapid diagnostic technologies such as CBNAAT, first and second-line LPA in line with WHO guidelines and expert committee meetings is detailed.

The programme must strive to provide every TB patient with access to at least R susceptibility testing *i.e.,* UDST.

## 4.2 Laboratory services required for PMDT

Optimal management of DR-TB requires both *Mycobacterial* and clinical laboratory services. At a minimum, the state level Intermediate Reference Laboratory (IRL) or any other RNTCP-certified Culture & DST laboratory should provide diagnostic culture on liquid and/or solid media; testing for resistance/susceptibility to at least R by RNTCP approved genotypic or phenotypic methods; and confirmation of the species as *M.tuberculosis* by Immuno-chromatographic assay.

Clinical laboratory services are required for proper evaluation and monitoring of patients, including basic haematology, biochemistry, serology, and urine analysis. These would also be available at the DR-TBCs identified by the state through public or private facility. Refer to pretreatment evaluation for details in Chapter 6.

## 4.3 Definition of accreditation and certification

Laboratory accreditation means third-party certification by an authorized agency using internationally approved standards for evaluating competence of laboratories to perform

specific type(s) of testing and is a formal recognition of competent laboratories. It includes all aspects of the laboratory such as physical infrastructure, biosafety, competencies of staff, processes, procedures and quality system elements (QSE) as per National Accreditation Board for Testing and Calibration Laboratories (NABL) standards (ISO 15189)

Certification is a process by which a specific procedure being performed in the laboratory *i.e.,* DST in TB labs is being quality assured by means such as standard EQA system (retesting and panel testing) by a higher level laboratory to ensure quality of that service.

RNTCP previously used the terminology of accreditation for quality assurance of its laboratory network. However though most aspects of the QSE are being used in this exercise, it does not fulfill the criteria of RNTCP being an authorized agency like ISO, NABL *etc.,* for providing accreditation. Henceforth, the term certification will be used to describe the quality assurance process for maintenance of TB DST EQA in its network of laboratories.

## 4.4 Definitions and classification of DR-TB patients

The following are the definitions and classification for DR-TB patients. [5][19]

***Presumptive DR-TB*:** It refers to the following patients in order of their risk:

- TB patients found positive on any follow-up sputum smear examination during treatment with first line drugs including treatment failures;
- paediatric TB non-responders;
- TB patients who are contacts of DR-TB;
- previously treated TB patients;
- new TB patients with HIV co-infection;
- all notified new TB patients.[1]

A patient is confirmed to have drug resistant TB, only when the results are from a RNTCP quality-assured Culture & DST Laboratory and by a RNTCP-endorsed testing method. Such patients are classified according to the following definition:

***Mono-resistance TB (MR):*** A TB patient, whose biological specimen is resistant to one first-line anti-TB drug only.

***Poly-drug resistance TB (PDR):*** A TB patient, whose biological specimen is resistant to more than one first-line anti-TB drug, other than both H and R.

[1]States to advance in phased manner as per PMDT scale-up plan for universal DST based on lab capacity and policy on use of diagnostics. Although new TB patients from above mentioned categories are at the lower risk of drug resistance, testing them further for presence of drug resistance at the time of diagnosis is a necessary standard of care.

***Rifampicin resistance (RR)*:** A TB patient, whose biological specimen is resistant to R, detected using phenotypic or genotypic methods, with or without resistance to other anti-TB drugs. It includes any resistance to R, in the form of mono-resistance, poly-resistance, MDR or XDR.

***Multidrug resistance TB (MDR):*** A TB patient, whose biological specimen is resistant to both H and R with or without resistance to other first-line anti-TB drugs. MDR-TB patients may also have additional resistance to any/all FQ **OR** any/all SLI anti-TB drug.

***Extensive drug resistance (XDR):*** A MDR-TB patient whose biological specimen is additionally resistant to at least a FQ (Ofx, Lfx Mfx) and a SLI anti-TB drug (Km, Am, Cm).

It is to be noted that R resistance is quite rare without H resistance. Majority of DST results with R resistance will also be H resistant, *i.e.,* MDR-TB. This has been substantiated in the National Drug Resistance Survey (2014-16). [7] Therefore, RNTCP has taken the programmatic decision that patients, who have any R resistance, should be managed as if they are an MDR-TB patient and this is in line with WHO global guidelines for PMDT.

## 4.5 Methods for drug susceptibility testing

Presently drug resistance detection using the following technologies is available for diagnosis of Drug Resistant TB through rapid molecular diagnostic testing:

- Line Probe Assay (LPA) for detection of MTB complex and resistance to first line drugs R, H and second-line drugs class FQ and class SLID [20] [8]; and
- Cartridge Based - Nucleic Acid Amplification Test (CBNAAT) Xpert MTB/Rif testing using the Gene-Xpert platform for now [20].

***Growth-based phenotypic drug susceptibility testing***: Culture though a highly sensitive and specific method for TB diagnosis, requires 2-8 weeks to yield results and hence does not help in early detection. However culture will be used for long-term follow-up of patients on Drug Resistant TB treatment and help detect early recurrence in both drug sensitive and drug-resistant TB. The growth-based phenotypic culture methods include automated Liquid culture systems *e.g.,* BACTEC MGIT 960, BacTAlert or Versatrek *etc.,* and solid (Löwenstein Jensen) media.

***Rapid molecular Drug Resistance Testing***: Line Probe Assay (LPA) provides rapid diagnosis of R and H resistance as well as resistance to class FQ and class SLID. LPA can yield results in 72 hours. Nucleic Acid Amplification Test (NAAT) provides accurate and rapid diagnosis of TB by detecting *M.tb* and R resistance conferring mutations. [20] The test can be performed on both respiratory and non-respiratory specimens and yields results in 2 hours. Presently, under RNTCP, its use is recommended for diagnosis of DR-TB in presumptive DR-TB patients

**and** TB in children, EP TB and in Key Population such as PLHIV, socially and clinically vulnerable groups and for active case finding efforts.

Drug resistance status is determined by either of the following methods:

***Drug Resistance Tests (DRT) using molecular methods:*** This can be performed on sputum specimen (direct) or on culture isolates (indirect) for diagnostic purpose. The methods are PCR-based and cannot be used for determining response to treatment. The tests that belong to this group include:

- CBNAAT: can be performed on smear positive, smear negative and extra pulmonary specimen. The test detects TB and resistance to R; and
- Line Probe Assay (LPA): is performed on smear positive specimen. The test detects TB and resistance to R and H (FL- LPA) as well as class FQ and class SLI (SL-LPA)

***Drug Susceptibility Test (DST)***: These are growth based tests and can be performed on L J culture isolates or in Liquid culture system Mycobacteria growth indicator tube (MGIT) for both pulmonary and EP specimen. Most commonly used method for testing is the economic variant of the proportion sensitivity. MGIT is preferred method for DST **and** both first and second-line anti-TB drugs can be tested. Following drugs can be tested for susceptibility by liquid culture:

- first-line drugs: R, H, Z*
- second-line drugs: Lfx, Mfx, Km, Cm, Am
- other drugs: Lzd, Cfz*, Bdq*, Dlm* *etc.,*

**when standardized, WHO endorsed and approved for use in programme*

## 4.6 Organization and development of the laboratory network

RNTCP has a three-tier laboratory network. The first is the designated microscopy centres (DMCs) providing sputum smear microscopy services. The second tier includes the Intermediate Reference Laboratories (IRL) and other C-DST laboratories. IRL has 3 main functions namely imparting trainings, ensuring quality assurance [QA] and providing TB Culture and DST services. The third-tier is the National Reference Laboratories (NRL). The overall quality assurance of IRL and other C-DST laboratories (medical colleges, NGO, private *etc.,*) is the responsibility of the designated NRL to which the State is linked. NRLs are also responsible for all training activities and EQA for C-DST as well as molecular diagnostics (LPA & CB-NAAT) for state level staff. [21]

## 4.7 Choice of diagnostic technology

The programme has substantially scaled-up the laboratory capacity of various C-DST laboratories. The choice of technology to be used for diagnosis of DR-TB has been determined as per recommendations of the National Laboratory Committee. The choice of technologies is given below.

| DR diagnostic technology | Choice |
|---|---|
| CBNAAT/LPA | First |
| Liquid culture isolation and LPA DST | Second |
| Liquid culture isolation and liquid DST | Third |

All the follow-up cultures will be done through the liquid culture. If the state is facing challenges, with respect to follow up culture by LC, this may be communicated to the Central TB Division for facilitation.

## 4.8 Specimen collection and transportation to C-DST laboratories

Obtaining good quality specimens of adequate volume is critical to ensuring correct diagnosis. The Laboratory technician needs to explain the process of collecting "a good quality sputum specimen" while adhering to AIC measures. Programme recommends collection of sputa one spot and one morning, **OR** 2 spot specimens collected within a gap of at least one hour (if the patient is coming from a long distance **OR** s/he is unlikely to return to give the second specimen). Ideally, a sputum specimen should have a volume of 2-5 ml and preferably be mucopurulent. Care should be taken to ensure that specimens sent for molecular testing are not heavily blood stained or contaminated. It is advisable to let the haemoptysis subside before collecting the specimen. The patient must be advised to collect the specimen in a sterile container (50 ml conical tube) after thorough rinsing of the mouth with clean water. Specimens should be transported to the laboratory as soon as possible after collection. In case of leakage or spillage of specimen during transportation leading to non-testing of specimen, special care should be taken to ensure recollection of specimen.

## Figure 4.1 Technical specifications of transport box for sputum specimen transportation in cool chain

Thermocol transport box
Size:
Thickness of box- 2.5 cm
Outer dimensions:
Length- 18.5 cm, Breadth-13 cm, Height- 12 cm (without lid), Height- 14 cm (with lid)
Inner dimensions:
Length- 14.5 cm, Breadth- 8 cm, Height- 12 cm (without lid), Height- 13 cm (with inner part of lid)
No. of gel pack required: 2 , Weight of fully packed consignment box: 400 grams, Approximate cost of courier charges: 60-70 Rupee per box
Gel packs maintain a temperature between 12-20 degree celsius for upto approximately 48 hours in tightly packed thermocol boxes (average outside temperature 35˚C). If conditioned in the deep freezer (temperature between -20 to -15˚C) for a minimum of 48 hours to a maximum of 72 hours before use. (This is onetime use box. Thermocol boxes and gel packs are not reused)

As per the diagnostic algorithm, two fresh specimen need to be collected at designated collection centres (DMC or PHI level) by trained LT and transported in a cool chain on the same day to the nearest CBNAAT lab for all eligible patients. At the CBNAAT site, the test is performed. For all results reported as TB (RS and RR), the second specimen need to be repacked by the CBNAAT LT and transported to the RNTCP C-DST laboratory on the same day in cool chain. The request form (RNTCP request form for examination of biological specimen for TB) updated with the CBNAAT result should be sent along with the transported specimen. Simultaneously, the result of CBNAAT test is to be communicated to the patient and provider. At the C-DST laboratory, SL- LPA is performed for RR-TB and FL-LPA for RS-TB. All specimens need to be delivered to the RNTCP C-DST laboratory within 48-72 hours of collection. Ideally, an agency (courier/speed post) should be identified for this purpose by the concerned DTO. NGOs may be engaged as per partnership guidelines for specimen transportation in cool chain. If none is available, transporter need to be identified from the health system/ community to transport the specimen in bio safe conditions with appropriate enablers. Models for packaging specimens are given in Figure 4.1 above. The following points are critical for the collection of fresh sputum specimens at designated collection points:

- 50-ml conical bottom tubes (made of polypropylene material) and the 3 layer packing materials like thermocol box, ice gel pack (pre-freezed at -20 degree for 48 hours), request form for examination of biological specimen for TB, polythene bags, tissue paper

roll as absorbent, parafilm tapes, brown tape for packaging box, permanent marker pen, labels, BIO-HAZARD sticker, scissors, spirit swab etc. should be supplied to the DMCs for collection of sputum through the DTO;

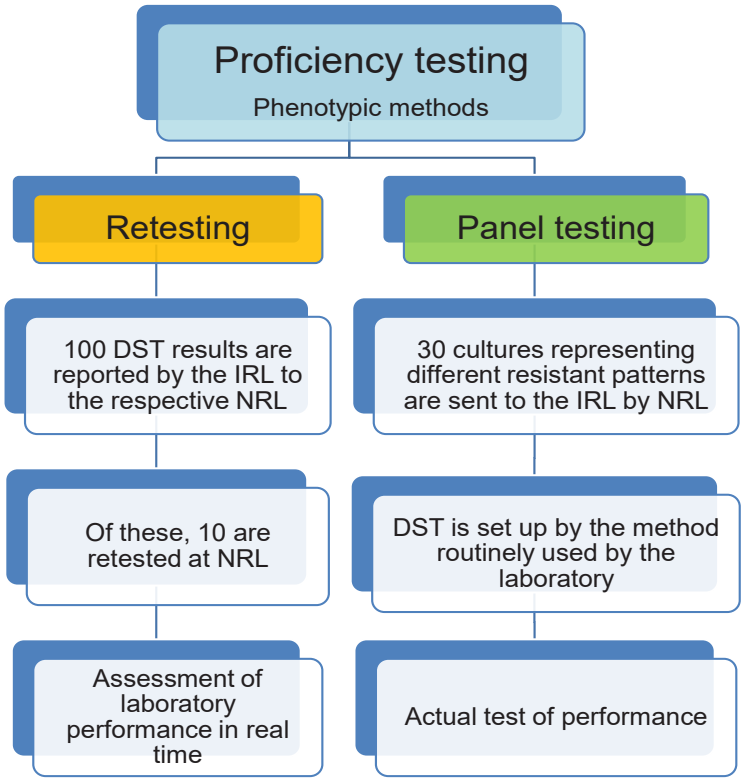
- 50-ml conical tubes should carry a label indicating the date of collection of the specimens along with patient's details like name, date of specimen collection, name of DMC/DTC, Lab. No:- XYZ, specimen A or B;
- LT at DMCs should be trained to carefully to pack the sputum specimens in the cool box to avoid spillage of the specimens;
- LT of DMC issuing the conical tubes to the patients should give clear instructions to the patients on correct technique of collection of the sputum. Also the date of issue of the conical tubes to the patient should be recorded;
- LT of DMC should ensure that the request form for examination of biological specimen for TB is packed in a separate plastic zip pouch and placed in the cool box before sealing the lid of the box. Also, the BIO-HAZARD symbol should be pasted on the external side of the cool box along with the label indicating the postal address of the C-DST Lab assigned;
- LT of DMC should promptly inform the specimen transport agency like a courier/ speed post service to collect and transport the specimens;
- as per national guidelines on biomedical waste management (Annexure 5) the containers used for transporting sputum specimens to the C-DST laboratory should be labelled with a "BIO-HAZARD" sticker;
- for every presumptive DR-TB patient referred by MO-DMC, date of referral and transport of sputa specimens to C-DST laboratory should be entered in "Remarks" column of the respective DMC lab register and TB notification register and in the referral for C-DST register held at the DTC. Alternatively, presumptive DR-TB patient referred to nearby DMC selected for specimen collection and transport for C-DST maybe provided two conical tubes by the concerned DMC LT/MO and instructed on collecting two specimens (one early morning and one supervised spot);
- these specimens will be taken by the patient/relative to the DMC selected for specimen collection from where these will be packed in cool boxes and transported to the C-DST laboratory; and
- once the sputum has been transported to the C-DST laboratory, the presumptive DR-TB patient should return to continue their RNTCP first-line treatment.

The specimen collection and transportation from EP-TB sites and children need to be done as per the SOP under RNTCP (Annexure 6) [22]

## 4.9 Certification and quality assurance for C-DST laboratories

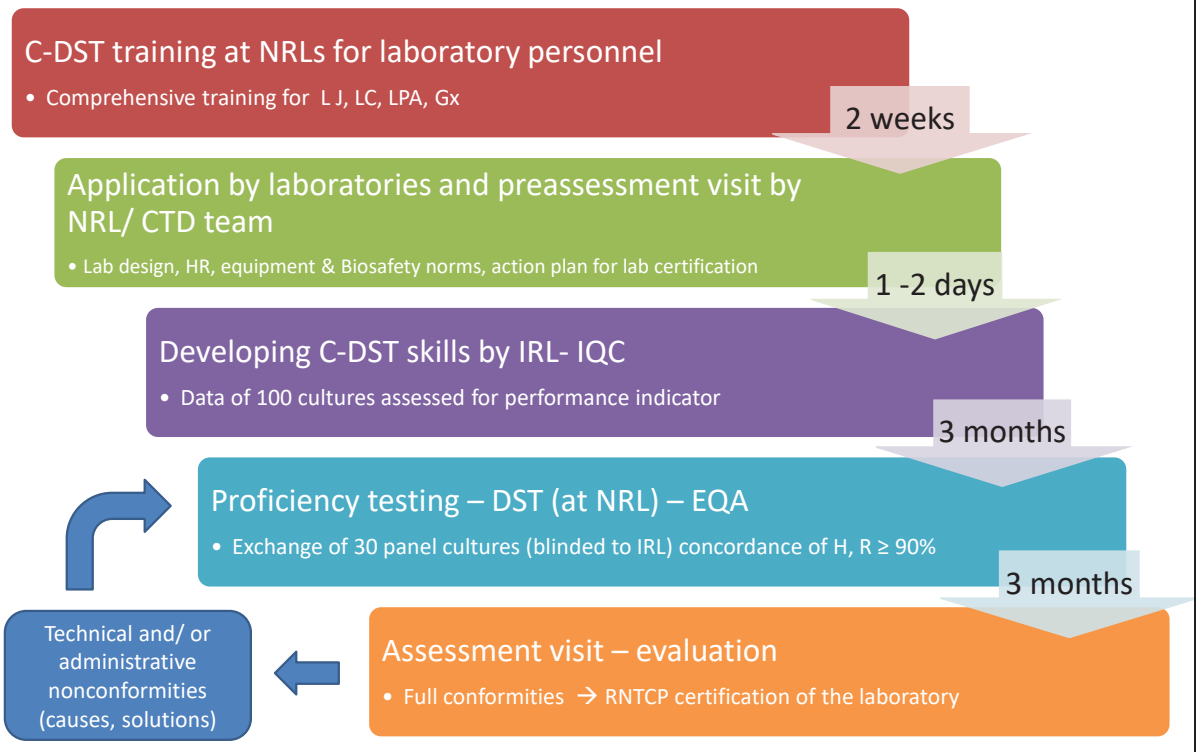
The components of Quality Assurance for C-DST include Internal Quality Control (IQC) and External Quality Assessment (EQA) mechanisms. IQC of L J media is performed as a routine laboratory protocol and involves testing each batch of media for contamination as well as the use of control strain (H37RV) for growth parameters. IQC for MGIT is instrument guided. EQA is not performed for culture. IQC of DST involves use of control strain (H37RV) as well as mono resistant strains (R mono and H mono) with every batch of DST performed.

**Figure 4.2: External Quality Control for TB Culture and DST laboratories**




EQA for both L J as well as MGIT is performed in two stages, initial retesting as one time activity where the NRL retests 10 strains out of 100 performed by the participating laboratory. This is assessment of the laboratory in real time. As a second stage, the participating laboratory is required to perform DST for 30 panel strains received annually from the NRL. This is the actual test of performance. For further details refer to Guidance for accreditation of laboratories under RNTCP for *Mycobacterial* culture & DST

**Figure 4.3: RNTCP certification process for TB culture and DST laboratories**




***Quality assurance for LPA*:** Initially, the NRL retests DNA extracts of 20 strains out of 50 performed in duplicates at the participating laboratory. This is followed by annual proficiency with panel strains. Panel Testing benchmark includes:

- invalid LPA results – less than 10%;
- contamination of negative control – clean in all runs;
- internal concordance – greater than 95%; and
- external concordance – greater than 95%.

***Quality assurance for CBNAAT*:** Each CBNAAT cartridge contains internal controls, Specimen Processing Control (SPC) and Probe Check Control. If the probe check fails, then the test is stopped and an error result is obtained. Troubleshooting is required based on the error code generated. Error rates higher than 5% should be investigated. Meanwhile, SPC must be positive when the result is MTB not detected. It can be negative/positive when the result is MTB detected. The test result is considered invalid if the SPC is negative when the test result is found negative.

Visits to CBNAAT sites should be planned at regular intervals to assess laboratory performance by district, state, IRL, NRL, CTD using the available standardized supervisory checklist for CBNAAT. CBNAAT sites in the districts should also be visited by IRL/NRL during

their visits for EQA of sputum smear microscopy. Poorly performing sites should be prioritized for onsite visits. All newly inducted laboratory staff members must undergo an induction training and periodic refresher training, as prescribed under RNTCP.

## 4.10 Initiative for Promoting Affordable and Quality TB Test (IPAQT) for private sector

TB diagnosis in the private sector was driven by extensive use of unreliable serological test and high cost of WHO endorsed diagnostics tests. IPAQT (Initiative for Promoting Affordable and Quality TB Tests) is an initiative which started in 2013 to bring RNTCP or WHO-approved TB tests at affordable prices to patients in the private sector. IPAQT brought together various private labs with the support of test manufacturers and other major stakeholders to cut down the cost for quality TB tests by up to 50% in the private sector. IPAQT is an initiative of nonprofit stakeholders and over 100 private sector labs/hospitals (more than 5000 collection centres) with a pan-India presence that have come together to provide WHO approved tests for TB. Each patient diagnosed by these IPAQT labs must notify to the programme surveillance system (further details available at http://www.ipaqt.org/) and appropriate public health action must be offered for all noticed TB & DR-TB patients. [23]

## 4.11 Non Tuberculous *Mycobacteria* (NTM)

A large number of M*ycobacteria* other than *Mycobacterium tuberculosis* are being increasingly recognized as a cause of human disease. Commonly referred to as non-TB *Mycobacteria* (NTM), they are also known as atypical *mycobacteria*, anonymous *mycobacteria* or *mycobacteria* other than *tubercle bacilli* (MOTT). NTM are ubiquitously distributed in the environment and hence also known as environmental mycobacteria. They are distinct from *M.tb* in their characteristics that they can survive outside the human or animal host. They are generally nonpathogenic or opportunistic pathogens and most commonly causes disease when there is immunosuppression or injury, except for few species which infect immune-competent humans.

Often these bacteria inhabit the respiratory passages in the form of commensal organisms. Pulmonary infection from NTM though rare, can cause disease similar to TB. They more commonly infect the skin, soft tissue, lymph nodes, implant devices, wounds, bones and joints. Disseminated NTM disease is mostly seen in patients who are immunosuppressed or who have Acquired Immunodeficiency Syndrome (AIDS).

Though NTM are widely distributed in the environment, the clinical infection is rare. They may be falsely recovered from clinical specimens due to laboratory contamination or contamination of medical instruments. Chronic pulmonary infection due to *M. avium* complex and *M. Kansasii* generally occurs in elderly persons especially males who are smokers or who have preexisting lung lesions. Cervical lymphadenopathy occurs in children

due to *M. Scrofulaceum*, while skin and soft tissue infections may develop from *M. Fortuitum*, *M. Chelonei, M. Xenopi* and *M. Ulcerans*. Exposure of humans to NTM may occur while bathing, swimming and drinking and the organism can also gain entry through cuts and abrasions. However, the risk of infection is generally less. Disseminated lesions are found in immunocompromised patients due to infection from *M. Avium* complex. Sometimes, *M. Chelonei* may cause very indolent pulmonary infection.

**Diagnosis of NTM**

Because of their omnipresence in our environment, isolation of NTM from non-sterile body sites does not imply true infection or disease, per se. Repetitive isolation, signs of clinical disease, radiological abnormalities, the exact species isolated and predisposing conditions of the patient involved, are all helpful in determining whether the isolated *mycobacteria* are to be considered causative agents of the patient's disease. In normally sterile sites, isolation of NTM, preferably backed up by histological evidence of granulomatous inflammation, suffices for the diagnosis of NTM disease. [24] [25]

Table 4.1: Most frequently isolated Non TB *Mycobacteria* and their sites of infection

| Species | Main site of infection | Growth rate |
|---|---|---|
| **M. Avium complex (M. Aviaum, M. Intracellulare, minor species)** | Pulmonary, lymph nodes, disseminated disease | Slow |
| **M. Kansasii** | Pulmonary, disseminated disease | Slow |
| **M. Xenopi** | Pulmonary | Slow |
| **M. Malmoense (NW Europe)** | Pulmonary | Slow |
| **M. Ulcerans** | Skin | Slow |
| **M. Marinum** | Skin | Intermediate |
| **M. Abscessus** | Pulmonary, skin | Rapid |
| **M. Chelonae** | Skin, soft tissues, disseminated disease | Rapid |
| **M. Fortuitum** | Skin, soft tissues, pulmonary | Rapid |
| **M. Scrofulaceum** | Lymph nodes | |
| **M. Haemophilum** | Disseminated disease | |

The minimum evaluation of a patient presenting with features suggestive of non-TB *Mycobacterial* (NTM) lung disease should include the following:

- chest radiograph or chest high-resolution computed tomography (HRCT) scan. HRCT may be done in settings where access to this technology is available. However, it is not mandatory for evaluation and decision to treat the patient;
- three or more sputum specimens for acid-fast bacilli (AFB) analysis;
- exclusion of other disorders, such as TB;

- expert consultation should be obtained when NTM are recovered that are either infrequently encountered or that usually represent environmental contamination. Patients suspected of having NTM lung disease but who do not meet the diagnostic criteria should be followed until the diagnosis is firmly established or excluded; and
- making the diagnosis of NTM lung disease does not, per se, necessitate the institution of therapy, which is a decision based on potential risks and benefits of therapy for individual patients.

Clinical, radiological and microbiological criteria are equally important and all must be met to make a diagnosis of NTM lung disease. The following criteria apply to symptomatic patients with radiographic opacities, nodular or cavitary or an HRCT scan that shows multifocal bronchiectasis with multiple small nodules. These criteria best fit with *Mycobacterium avium complex* (MAC), *M. Kansasii* and *M. Abscessus*.

**Clinical**

- pulmonary symptoms include cough, hemoptysis, fever, weight loss or organ specific signs and symptoms *etc.;* and
- exclusion of any other etiologies.

**Radiological**

- radiological findings pertain to nodular or cavitary opacities on chest radiograph; and/or
- HRCT scan that shows multifocal bronchiectasis with multiple small nodules.

**Microbiological**

Table 4.2: Microbiologic criteria for diagnosis of NTM lung disease

| Specimen | Result |
|---|---|
| At least three results available with; OR | Two positive cultures regardless of the results of AFB smear |
| Single available bronchial wash or lavage with; OR | One positive culture regardless of the results of AFB smear |
| Tissue biopsy with: | Compatible histopathology- (granulomatous inflammation) and a positive biopsy culture for NTM<br>Compatible histopathology- (granulomatous inflammation) and a positive sputum or bronchial wash culture for NTM |

**Guidelines for making the diagnosis of NTM-pulmonary disease**

- clinical features of an indolent, respiratory disease include cough, expectoration, fever and other constitutional symptoms;

- positive smear for AFB and/or heavy growth of NTM (at least 1+ on solid media) on culture in respiratory specimens with the same species being identified repeatedly;
- histopathological features of *mycobacterial*/granulomatous disease or culture of NTM from biopsy specimens;
- radiological features of nodular infiltrates with or without cavitation and/or bronchiectasis lesions;
- underlying host conditions include immunosuppression, AIDS, alcoholism, COPD, cystic fibrosis, diabetes, malignancies, prior TB, esophageal motility disorders *etc.;*
- absence of other causes of pulmonary lesions, such as TB, aspergillosis, *etc.;*
- persistence of AFB after anti-TB treatment for two weeks or more with CBNAAT or LPA report not detecting *M.tb*; and
- smear positive, CBNAAT negative, LPA-TUB band absent with or without R resistance in patients of presumed DR at diagnosis also need to be evaluated.

The diagnostic processes for NTM to be followed at C-DST laboratories are detailed in Annexure 7. The laboratory staff would be separately trained in these standard operating procedures for laboratory confirmation of NTM. For more information on NTM including extra-pulmonary NTM, microbiologists are encouraged to refer to the latest American Thoracic Society (ATS) guideline.

# Chapter 5: Case-finding

This chapter describes the RNTCP strategy for timely case-finding and confirmation of diagnosis among presumptive DR-TB patients as well as vulnerable groups. Early identification and prompt initiation of treatment will prevent the patient from spreading the disease to others, developing a resistant strain to more drugs and progressing to a chronic state of permanent lung damage.

## 5.1 Case-finding strategy

Vulnerable groups to be offered upfront CBNAAT. A vulnerable group is any group of people in which the prevalence or incidence of TB is significantly higher than in the general population. The recommended vulnerable groups to be considered for intensified case finding may be classified as follows [5]:

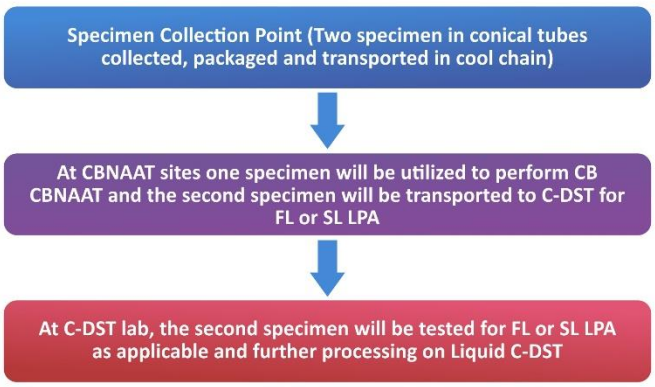
Table 5.1: Classification of vulnerable groups

| Clinical | Social | Geographical |
|---|---|---|
| Clients attending HIV care settings | Prisoners | Urban slums |
| Substance abuse including smokers | Occupations with risk of developing TB (enumerate from TOG) | Hard-to-reach areas |
| Comorbidities like diabetes mellitus, malignancies, patients on dialysis and on long term immunosuppressant therapy | People in congregated settings – night shelters, de-addiction centres, old age homes | Indigenous and tribal populations |
| Health care workers | | |
| Household & workplace contacts | | |
| Patients with past history of TB | | |
| Malnourished | | |
| Antenatal mothers attending antenatal /MCH clinics | | |

The Medical Officer, Peripheral Health Institute (MO PHI) along with DMC LT will be responsible for identification of all diagnosed and notified TB patients and other key and vulnerable patient. This will be in addition to presumptive DR-TB patients who have failed treatment with first-line drugs, paediatric TB non-responders, TB patients who are contacts of DR-TB, TB patients who are found positive on any follow-up sputum smear examination during treatment with first line drugs, previously treated TB patients and/or TB patients with HIV co-infection. It should be ensured that detailed history is elucidated for every patient. The testing of new TB patients from non-key/vulnerable groups would be initiated after ascertaining the available CBNAAT capacity in the districts/states.

A patient who is identified for testing with CBNAAT should be referred by the respective MO-PHI to the nearby sputum collection centre for diagnosis by the rapid molecular method. CBNAAT is the preferred rapid molecular diagnostic test under RNTCP. At the CBNAAT site, two specimens will be collected as detailed earlier. In certain patients, second specimens may need to be tested for confirming R Resistance (in case of invalid, indeterminate, error or no results reported by CBNAAT machine). With availability of X-pert ultra-cartridges in the near future, the reconfirmation of RR-TB among new TB patients will no longer be required. Patient's results will be available within a day and the decision of starting the patient on the appropriate regimen can be taken after results are available.

**Figure 5.1 Operational process of specimen referral**




***Turnaround time*:** Genotypic testing is much faster than phenotypic methods. As these are not growth based tests, timely diagnosis and prompt treatment initiation is possible. DST results on Solid L J media have a turnaround time (TAT) of up to 84 days, liquid culture (through MGIT) up to 42 days, LPA up to 72 hours and CBNAAT by 2 hours.

Once the MO PHI confirms that the patient belongs to presumptive DR-TB, s/he should arrange for sending two sputa specimens, one of which is an early morning specimen and the other a "supervised" spot specimen, from the patient to the assigned C-DST laboratory along with the RNTCP request form (Request for examination of biological specimen for TB). For specimens to be collected and transported from the DMC, all necessary materials for specimen collection and transport need to be made available at the DMC identified by the DTO as specimen collection centre for C-DST.

Empty 50-ml conical tubes can be provided by the LT/ MO to the patient with guidance to collect fresh sputum specimen. The patient is to be advised to go on the next day with the early morning specimen and relevant medical records to the nearest specimen collection center for C-DST. The spot specimen can be collected in such cases when the patient arrives to submit the early morning specimen. Alternately, specimens can be collected at such DMCs and PHIs and transported in cool boxes by the staff on the same day to the nearest DMC identified as specimen collection centre. The specimen is further packed and transported in cool boxes through courier or speed post to the C-DST. If there is likely to be a delay in transporting the specimens, these should be stored in a refrigerator at the peripheral DMC/PHI with biosafety precautions.

## 5.2 Integrated DR-TB diagnostic algorithm

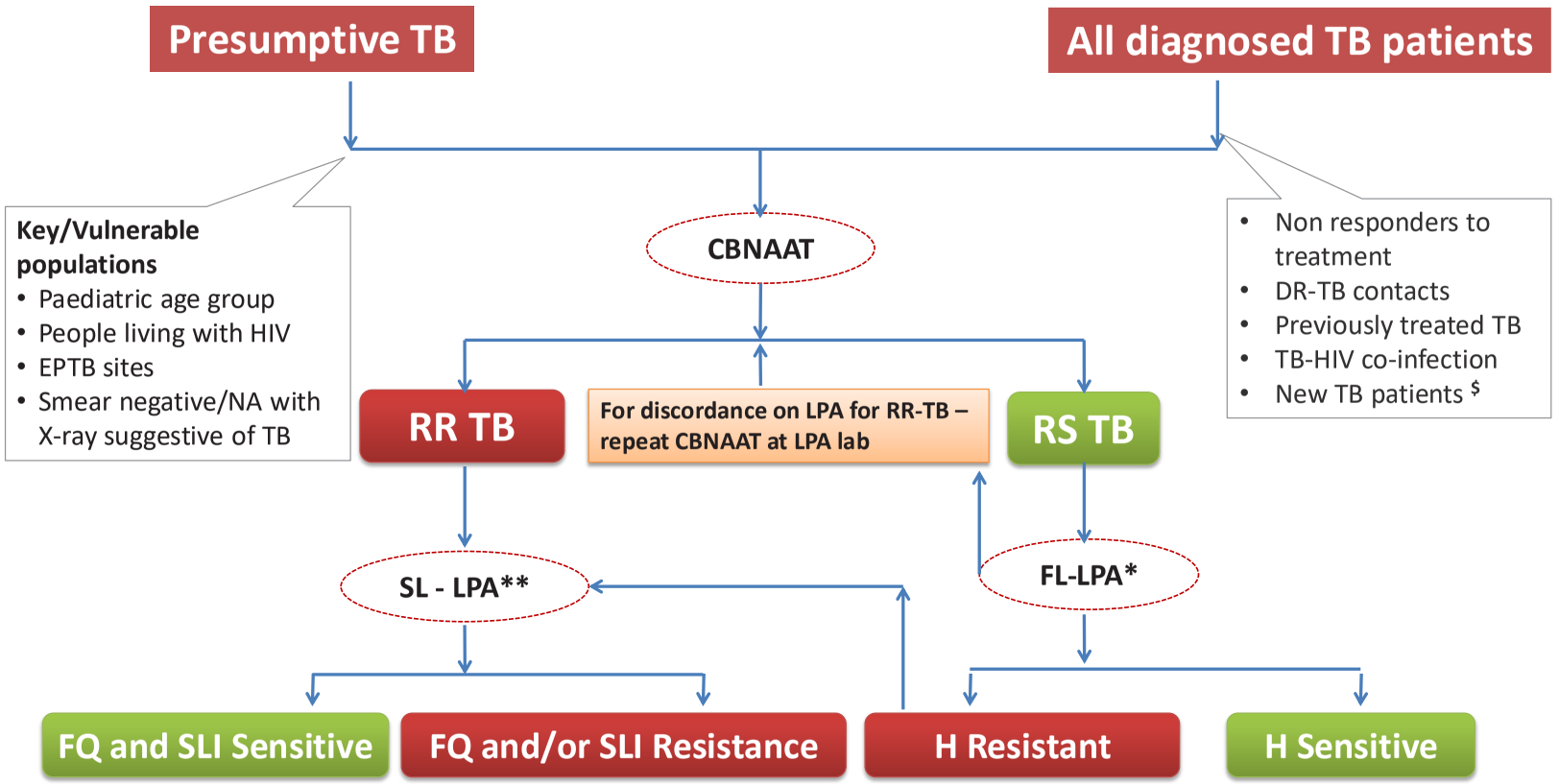
The vision of the programme is to offer DST to TB patients at the earliest time in their diagnostic process. The integrated diagnostic algorithm starts with two groups of patients who are either presumptive TB or diagnosed TB. The main objective of this algorithm is to segregate people based on risk assessment for DR-TB and offer DST guided treatment based on drug resistance status at least for R resistance at the time of diagnosis of TB *i.e.,* Universal DST. The subsequent time points when DST is offered if any of the following events occur during the course of a TB treatment schedule:

- bacteriologically positive after intensive phase of a course of TB/DR-TB treatment;
- failure to respond to treatment as per RNTCP definitions;
- recurrence of TB diagnosed after a course of TB treatment;
- for patients who are retrieved after lost to follow-up;
- for patients found to be HIV positive before or anytime during the course of TB treatment and
- any other reason as per treating physicians advice.

Two diagnostic specimens would be collected from the patients one early morning and one spot specimen wherever possible, but if there is a likelihood of the patient not returning for the second collection or travelling from long distance then 2 spot specimens may be collected with a gap of at least one hour. The DR-TB diagnostic algorithm is as given in Figure 5.2.

The left arm of the algorithm starts with persons presumed to have TB. From this group those belonging to the paediatric age group, PLHIV, EP group or with a smear negative chest X-ray suggestive of TB will be offered CBNAAT test. By virtue of using CBNAAT as the TB diagnostic test, the R status is also available simultaneously along with TB detection.

# Figure 5.2 DR-TB Diagnostic Algorithm




*Offer molecular testing for H mono/poly resistance to TB patients prioritized by risk as per the available lab capacity
**LC DST (Mfx 2.0, Km, Cm, Lzd) will be done only for patients with any resistance on baseline SL-LPA. DST to Z, Cfz, Bdq & Dlm would be considered for policy in future, whenever available, standardized & WHO endorsed.
$ States to advance in phased manner as per PMDT Scale up plan for universal DST based on lab capacity and policy on use of diagnostics

Patients other than key/vulnerable population would be evaluated for TB as per the RNTCP technical and operational guidelines. The right arm begins with offer of CBNAAT to all diagnosed or notified TB patients who are at risk of DR-TB as per risk group criteria and districts will transition to testing all patients diagnosed or notified as TB. Based on the result obtained on the CBNAAT, the patient would be classified as R resistant (RR TB) or R susceptible (RS TB) to guide decision of the appropriate treatment. As soon as the CBNAAT results are available, the reports must be immediately updated in Nikshay by the CBNAAT lab and communicated to the DDR-TBC, DTO, MO PHI and the patient.

For patients with CBNAAT result as *M.tb* detected (RR-TB & RS-TB), the second specimen will be reflexly transported in cool chain from the CBNAAT lab to the C-DST lab. In rare circumstances when CBNAAT may reveal R indeterminate or no result on the first specimen, the second specimen will be used at CBNAAT lab itself to repeat the test. In such cases, a fresh specimen is to be collected from the patient and transported in cool chain to the concerned C-DST lab. However, this may not be always possible for EP specimen.

At the C-DST laboratory (for all smear positive specimen) FL- LPA test will be performed for RS-TB and SL- LPA will be performed for RR-TB (Base line SL DST). DST to Mfx (2.0), Km, Cm and Lzd will be set up on liquid culture using the decontaminated deposits only for patients who are found to be resistant to FQ and/or SLI class. The results of the LC-DST for individual FQ and second line SLI will be provided based on a single breakpoint concentration and decisions on modification of regimen will be made by the NDR-TBC committee based on the results of LC-DST for each individual patient as detailed in the guidelines later.

It is to be noted that for the individual FQs the cross resistance between Ofx (2.0), Lfx (1.5) and low level Mfx (0.5) is almost complete as observed in NDRS and thus not useful for clinical management except for higher concentration of Mfx (2.0) DST [7][8] which then guides to take a decision to use it at a higher dose if sensitive or to move to a FQ free regimen if resistant.

For patients with discordant results of R resistance between CBNAAT and LPA, another CBNAAT test is performed in the C-DST lab using the decontaminated deposit. The final result will be on consensus of the 3 tests (2 CBNAAT and 1 LPA). If 2 of 3 are R resistant then the final result will be R resistant; if 2 of 3 are sensitive, then the final result will be sensitive to R.

If the specimen is found to be smear negative at C-DST lab, a culture would be set up and LPA will be indirectly conducted on the culture isolate.

If the result is H resistant, the SL-LPA will be reflexly performed on available specimen (DNA) and if any class resistance (FQ/SLI) detected on SL-LPA, DST to Mfx (2.0), Km, Cm and Lzd will be done as per the algorithm.

DST to Z, Cfz, Bdq and Dlm would be considered for policy in future, whenever available, standardized and WHO endorsed. Z LC-DST though standardized is not being performed in line with the recent WHO recommendations (2016) and will be reviewed later based on in-country evidence after evaluating the Nipro LPA, sequencing and LC-DST concordance or agreement studies.[12] [26] [27] [28]

## 5.3 Laboratory recording and reporting

Results of the smear, culture and DST / LPA/ CBNAAT results are entered in the culture and DST register as annexed, held at the laboratory. All results must be communicated to the concerned DTO, DR-TBC/ private provider through Nikshay as soon as results are available so that patient treatment decisions can be smoothly managed. However, for providers without access to Nikshay, alternative means (email, SMS *etc.,*) for communicating the results must be utilized. If the culture result shows early contamination (within 4 days), the same is informed to the DTO within 24 hours and s/he is expected to arrange sending a repeat specimen (one early morning and one spot) to the laboratory within 3 days.

If LPA is found to be invalid or the sputum is smear negative, the sputum specimen is inoculated on solid or liquid culture immediately. If the culture result is found to be positive, the culture isolate is subjected to LPA test for confirming MDR-TB / RR-TB.

## 5.4 Management of patients while DST results are awaited

Any TB patient whose RR-TB result is awaited, would be initiated on first-line anti-TB treatment and continued on the same if found to be RS-TB. If RR-TB is detected, the patient is immediately referred for pretreatment evaluation and treatment initiation for DR-TB. However, the first-line treatment should be stopped as applicable. Moreover, all presumptive TB among key/vulnerable population would be initiated on RR-TB or RS-TB regimen based on the results of CBNAAT. Patients must be counselled to practice covering the cough and other infection control measures like provision of personal protective equipment and support to prevent the transmission of infection.

## 5.5 Diagnosis of DR-TB in children

Very limited data is available for DR-TB in children. It is mainly due to primary transmission of drug-resistant TB to the child and less likely to be acquired from exposure to TB treatment. Paediatric DR-TB is likely to reflect DR-TB in adults, so DR-TB is common in children in settings with higher prevalence, mortality and morbidity of DR-TB in adults compared to drug-sensitive TB.

Children with recurrent TB, treatment after lost to follow-up and treatment after failure are presumed patients of DR-TB. Children usually have pauci-bacillary disease and are sputum

negative. So, these definitions are to be used in conjunction with clinico-radiological picture. Additionally, other children who are at high risk of DR-TB could be contacts of DR patients, children living with HIV (CLHIV) and paediatric non-responders. With recent experience gained in India through a multi-city project involving paediatrician, there is now a policy to use CBNAAT for diagnosis of TB in children that gives the opportunity of simultaneous diagnosis of RR-TB using an appropriate biological specimen.

***Approach to diagnose DR-TB in children*:** Careful history of patients, in particular, history of contact with DR-TB patients is critical information, as also considering child failing first-line TB treatment despite adherence. Clinical examination and investigations that are relevant for presumptive TB or EP-TB patients need to be carried out. It is important to try and get appropriate specimens from children. Failure to respond to TB treatment must rule out HIV-related lung disease which may not be TB. Bacteriological confirmation and drug susceptibility testing should be carried out whenever possible. For this, sputum (or other relevant specimens e.g. lymph node aspiration) must be collected from children with presumed DR-TB for CBNAAT ( Xpert MTB/RIF) or LPA or culture and drug sensitivity testing.

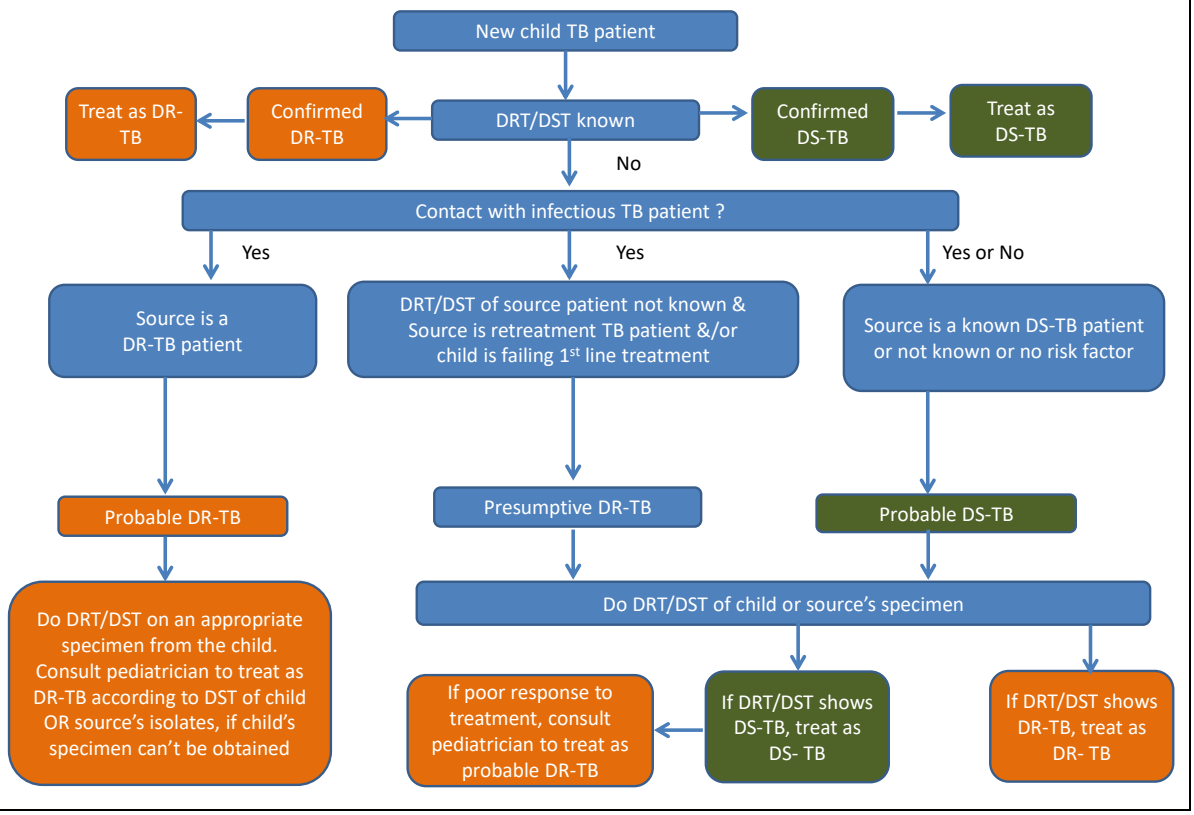
Microbiological confirmation should always be done to ensure correct diagnosis, all effort must be made to get appropriate clinical specimens from affected site. Sputum, gastric lavage, BAL, pleural tap, lymph-node aspiration, excision, cerebrospinal fluid (CSF) and laparoscopic tissue biopsy can be considered. Diagnosis of DR-TB in absence of microbiological confirmation must be thoroughly reviewed as it may be untenable. In presumptive DR-TB, if there is no microbiological confirmation, bacteriologically negative clinically diagnosed DR-TB can be considered after ruling out alternative diagnosis. [20] [29] [30]

***Probable MDR-TB among children*:** The case definitions of presumptive DR-TB and variety of DR-TB patients used for adults in section 4.4 of the previous chapter, would also apply to children. However, the term probable MDR-TB in children would be applied to children wherein bacteriologic confirmation is not available and the decision regarding diagnosis and initiation of treatment is taken by the NDR-TBC committee. Criteria for diagnosis of "probable MDR-TB" include children with signs and symptoms of active TB disease who in addition have the following risk factors. They should be considered as having "probable" MDR-TB and started on MDR-TB treatment, even in the absence of bacteriological confirmation. They include children who have:

- close contact with a known case of MDR-TB;
- close contact with a person who died whilst on TB treatment;
- close contact with a person who failed TB treatment;
- failure of a first-line regimen, recognizing that both bacteriological and clinical definitions of failure should be used; and
- previous treatment with second-line medications.

The algorithmic approach to diagnose DR-TB in children is shown in Figure 5.3. If an appropriate specimen is available from the child, the specimen processing will be in accordance to the integrated DR-TB diagnostic algorithm shown in Figure 5.2.

**Figure 5.3 Algorithmic approach to diagnosis of DR-TB in children**




All patients considered to have 'probable' MDR-TB should be presented to and discussed with a NDR-TBC Committee followed by a decision to treat which ought to be made in consultation with the paediatrician. This consideration of initiation of appropriate DR-TB regimen without bacteriological confirmation does not replace the need for a thorough and ongoing diagnostic evaluation, including consideration of non-TB causes, prior to initiation of DR-TB treatment. Children with a central nervous system disease and/or those with other life-threatening manifestations who meet the criteria for 'probable' MDR-TB should be initiated on therapy immediately, in consultation with the paediatrician in the NDR-TBC committee, given the high risk of mortality.

# Chapter 6: Pretreatment evaluation

The chapter provides the process of referral for pretreatment evaluation and pretreatment evaluation process.

## 6.1 Referral for pretreatment evaluation

It is crucial that patients with DR-TB be referred for treatment as soon as possible. If RR-TB/ H mono-poly DR-TB is confirmed, the DTO will trace the patient, with help of Medical Officers – TB Control (MO-TC) & PHI, Senior DR-TB TB-HIV Supervisor and Senior Treatment Supervisor (STS) and bring the patient to the DDR-TBC where s/he will be counselled by the counsellor. Counselling should include the following:

- information on lab results and reliability of lab results from CBNAAT/RNTCP certified C-DST laboratories;
- need for additional treatment;
- importance of rapid initiation of treatment and adherence to prescribed treatment;
- services RNTCP offers for PMDT;
- what the patient should do next;
- necessary infection control precautions;
- re-assurance to the family against panic or unnecessary stigmatization of the patient; and
- patient support system

After counselling, the patient is referred to the DDR-TBC with its DST result and PMDT referral for treatment form, for pretreatment evaluation and initiation of standard regimen for DR-TB as appropriate. In addition to those patients diagnosed as RR-TB or MDR-TB, patients with H mono/poly DR-TB, will also be referred to the DDR-TBC for pretreatment assessment. Those patients, who have RR-TB, will also be treated with a regimen for MDR-TB.

While the RR/MDR-TB/H mono/poly DR-TB patient is undergoing pretreatment evaluation, the Senior DR-TB TB-HIV Supervisor, STS and DTO should ensure an initial home visit to verify the address and meet the family members. A Treatment supporter (who can either be a health care worker, community worker/ volunteer or private practitioner) should be identified in consultation with the patient. The Treatment centre can either be at the sub centre of the health system or in the community. The family member, if identified as Treatment supporter should be trained to give medication under supervision at the residence, under close monitoring by TBHV/STS. The Treatment supporter should also be given training for drug administration, identification of adverse effects during treatment,

frequency of follow-up and record keeping. During IP, appropriate arrangement for injections should be done by the DTO.

## 6.2 Pretreatment evaluation for DR-TB patients

In majority of MDR/RR TB and H mono/poly DR-TB patients, pretreatment evaluation can be done on an outpatient basis, under intensive supervision. The DTO can arrange pretreatment evaluation at DDR-TBC or at sub-district level health facility, wherever possible. The patient should be fast-tracked for pretreatment evaluation and for infection control purposes and a separate space for specimen and blood collection should be identified.

Patients will be referred to the DDR-TBC/NDR-TBC with pretreatment evaluation results for initiation of treatment. The physician may decide for admission to DDR-TBC/NDR-TBC for pretreatment evaluation and initiation of treatment or get it done on an outpatient basis. Pretreatment evaluation should include, a thorough clinical evaluation by a physician. A specialist consultation along with reports of pre-treatment evaluation tests can be arranged, if required. Since the drugs used for the treatment of DR-TB have significant adverse effects, a pretreatment evaluation is essential to identify patients at increased risk of developing such adverse effects. This evaluation will vary as per the regimen class and is detailed in Table 6.1.[31]

Each of the DR-TBCs (DDR-TBC/NDR-TBC) must ensure that laboratory capacity and consultancy services from various specialists are available, either in-house supported under institutional/state government mechanism or through an outsourced mechanism. Tie up with private facility under Partnership Guidelines (innovation option) should be undertaken for investigations that are not available.

Pretreatment evaluation and treatment initiation must be done at the DR-TBC (DDR-TBC/NDR-TBC). The concerned DR-TBC committee provides counselling, initiates activities related to active drug safety monitoring (aDSM) like, assessing the baseline history of known adverse/serious adverse events (AE/SAE), biochemical investigations, ECG *etc.,* and initiates him/her on an appropriate treatment regimen. Care must be taken to correct any electrolyte imbalance before treatment initiation.

MDR/RR-TB patients (without additional resistance) and H mono/poly DR-TB patients can be initiated on a standard treatment regimen at DDR-TBC. The DDR-TBC should refer those patients to NDR-TBC for management of patients with additional drug resistance, drug intolerance, contraindication, failing regimen, return after treatment interruption of >1 month, emergence of exclusion criteria for standard regimen, for expert opinion, management of any complications warranting regimen change for consideration of newer drug containing regimen or DST guided regimen based on a detailed assessment.

Table 6.1 Preteratment evaluation of DR-TB Patients by regimen class

| SN | Pretreatment evaluations | Regimen for H Mono / poly DR-TB | Conv. MDR-TB regimen | Shorter MDR-TB regimen | Regimen for RR-TB with FQ/SLI ± Lzd resistance (without newer drugs) | Newer drugs containing regimen |
|---|---|---|---|---|---|---|
| 1 | Detailed history (including screening for mental illness, seizure disorder, drug/alcohol abuse, etc.) | ✓ | ✓ | ✓ | ✓ | ✓ |
| 2 | Previous history of ATT taken especially SLI/FQ | ✓ | ✓ | ✓ | ✓ | ✓ |
| 3 | Weight & height | ✓ | ✓ | ✓ | ✓ | ✓ |
| 4 | Thorough clinical examination | ✓ | ✓ | ✓ | ✓ | ✓ |
| 5 | Complete blood count with hemoglobin & platelets count | ✓ | ✓ | ✓ | ✓ | ✓ |
| 6 | Blood sugar to screen for Diabetes Mellitus | ✓ | ✓ | ✓ | ✓ | ✓ |
| 7 | Blood urea and S. Creatinine to assess renal function | ✓ | ✓ | ✓ | ✓ | ✓ |
| 8 | Urine examination – routine and microscopic | ✓ | ✓ | ✓ | ✓ | ✓ |
| 9 | UPT (for all women in the child-bearing age) | ✓ | ✓ | ✓ | ✓ | ✓ |
| 10 | Chest X-ray | ✓ | ✓ | ✓ | ✓ | ✓ |
| 11 | HIV counselling and testing* | ✓ | ✓ | ✓ | ✓ | ✓ |
| 12 | Audiogram | ✓ | ✓ | ✓ | ✓ | ✓ |
| 13 | Liver function tests# | ✓ | ✓ | ✓ | ✓ | ✓ |
| 14 | TSH levels to assess the thyroid function | | ✓ | ✓ | ✓ | ✓ |
| 15 | Mental healthevaluation | | ✓ | ✓ | ✓ | ✓ |
| 16 | Surgical evaluation | | ✓ | ✓ | ✓ | ✓ |
| 17 | ECG (if Mfx[h], Dlm, Bdq, Cfzused) | | | ✓ | ✓ | ✓ |
| 18 | Serum electrolytes – potassium, magnesium, calcium | | | | | ✓ |
| 19 | Serum proteins, lipase, amylase | | | | | ✓ |
| 20 | Ophthalmologist opinion to rule out chorioretinitis / uveitis | | | | | ✓ |

**All DR-TB patients will be offered referral for HIV counselling and testing at the nearest centre if the HIV status is not known or HIV test result is negative with results more than 6 months. If patient is HIV positive, refer to ART centre (if not on ART)*
*# including HBsAg at baseline*

## 6.3 Providing health education/ counselling to patient and family members

Providing counselling and health education to the MDR/RR-TB patient and family members about the disease, the mechanism of transmission and necessity of taking regular and adequate treatment, is of utmost importance. Health education and counselling is provided to all patients and family members at different levels of health care, from the periphery to the DDR-TBC facility. It is started at the initial point of contact and continued during all visits by the patient to a health facility. Confidentiality and informed decision making process according to sound ethics standards is paramount when performing education and counselling to patients and their family members.

DDR-TBC/NDR-TBC counsellors to provide counselling for all DR-TB patients on the following:

- nature and duration of treatment;
- importance of adherence to treatment and need for complete and regular treatment;
- possible side effects of drugs;
- mechanism of transmission; and
- consequences of irregular treatment or premature cessation of treatment.

It is advisable to involve close family members during the counselling, since family support is an essential component in the management. Patients should be advised to report any side effects immediately**.** Female patients should receive special counselling on family planning. The treatment must be presented as an option and include information on any uncertainty about the adverse effects of drugs as detailed in the patient education booklet.

The counsellors would be trained exclusively with a counsellors training module on an e-learning platforms by RNTCP. This covers the various approaches, steps involved in counselling, tools, activities to be undertaken as well as the records and reports to be maintained by the counsellors. A patient-based counselling register must be maintained for all patients for recording information about patients' situation and counselling services provided from the time of diagnosis till post-treatment follow-up period.

Pretreatment counselling must serve as an informed decision-making process that enables patients to make a duly informed decision regarding the use of all anti-TB drugs including newer drugs like Bdq. This activity must be recorded in the counsellor's register, PMDT treatment card and treatment book of the patient before initiating treatment.

# Chapter 7: Treatment of drug-resistant TB

This chapter provides guidance on the treatment of all forms of DR-TB patients.

## 7.1 Classes of anti -TB drugs recommended for treatment of DR-TB patients

The anti-TB drugs recommended for treatment of MDR/RR TB patients are grouped based on efficacy, experience of use and drug class and aligned with revised classification as per WHO PMDT Guidelines 2016. The same is explained in the table below. [12]

Table 7.1: Grouping of anti-TB drugs

| New grouping of drugs | | | |
|---|---|---|---|
| A. Fluoroquinolones | | Levofloxacin<br>Moxifloxacin<br>Gatifloxacin | Lfx<br>Mfx<br>Gfx |
| B. Second-line injectable agents | | Amikacin<br>Capreomycin<br>Kanamycin<br>(Streptomycin) | Am<br>Cm<br>Km<br>(S) |
| C. Other second-line agents | | Ethionamide / Prothionamide<br>Cycloserine / Terizidone<br>Linezolid<br>Clofazimine | Eto/Pto<br>Cs/Trd<br>Lzd<br>Cfz |
| D. Add-on agents (not part of the core MDR-TB regimen) | D1 | Pyrazinamide<br>Ethambutol<br>High-dose isoniazid | Z<br>E<br>$H^h$ |
| | D2 | Bedaquiline<br>Delamanid | Bdq<br>Dlm |
| | D3 | p-aminosalicylic acid<br>Imipenem-cilastatin<br>Meropenem<br>Amoxicillin-clavulanate<br>(Thioacetazone) | PAS<br>Ipm / Cls<br>Mpm<br>Amx-Clv<br>(T) |

## 7.2 Newer anti-TB drugs

After almost five decades of discovery of Rifampicin, the first new drug named Bedaquiline with anti-TB effect was approved for treatment of multidrug resistant TB by US FDA in late 2012. [10]This was followed by the approval of another new drug Delamanid by the stringent regulatory authority of various countries. [11]The drug development pipeline of new and repurposed drug has gained momentum in the recent past and more new molecules are expected to be approved in the future.

**Bedaquiline** is a new class of drug, diarylquinoline that specifically targets mycobacterial ATP synthase, an enzyme essential for the supply of energy to Mycobacterium TB. Strong bactericidal and sterilizing activities against *M.tb* have been shown in pre-clinical, laboratory and animal experiments. The drug has a high volume of distribution, with extensive tissue distribution, highly bound to plasma proteins and is hepatically metabolized. The drug has an extended half-life, which means that it is still present in the plasma up to 5.5 months post stopping BDQ. The dosing schedule has been established after extensive pharmacokinetic/ pharmacodynamic (PK/PD) studies in animals and humans and hence need to be administered as per the manufacturer's advice. BDQ has shown significant benefits in improving the time to culture conversion in MDR-TB patients. [10]

**Delamanid** as per an interim policy guidance document released by WHO in 2014, may be added to a WHO-recommended regimen in adult patients with pulmonary MDR-TB. [11] The process for introduction of Dlm under RNTCP is ongoing and guidance for the same will be released subsequently as an addendum to these guidelines

In June 2013, WHO published an interim policy guideline for use of BDQ in conjunction with WHO recommended MDR TB treatment that recommends that BDQ may be added to a WHO recommended regimen in adult patients with pulmonary MDR TB when an effective treatment regimen containing at least four second-line drugs in addition to Z cannot be designed or when there is documented evidence of resistance to any FQ and/or SLI in addition to MDR TB. In 2015, WHO published an implementation plan for introduction of BDQ for treatment of MDR TB at country level to support country efforts towards implementation of recommended new drugs or treatment regimens with particular emphasis on aDSM in absence of Phase III results on safety and efficacy of the drug. [10]

### 7.2.1 Regulatory approvals for Bedaquiline in India

After a thorough review of available evidence from various phased trials up to phase IIb; evidence emerging from several of the early implementing countries; WHO interim guidelines for use of Bdq; regulatory processes adopted by United States Food Drugs Administration (US FDA), European Union (EU) and stringent regulatory authorities of various countries was undertaken by the national expert committee on regulation of newer

anti-TB drugs in India. On 24 December 2014, the Apex Committee under the MoHFW, GoI for supervising clinical trials on new chemical entities in the light of directions of the Supreme Court of India approved the use of BDQ (100 mg) in adults aged 18 and above as part of a combination therapy of pulmonary TB due to MDR-TB.

Considering MDR-TB as a serious condition with high mortality and a disease of special relevance in the Indian health scenario, the committee recommended waiver of local clinical trials at this stage and approved BDQ with restriction that it shall be used under the RNTCP framework for conditional access through the PMDT programme for treatment of MDR-TB patients only. Soon after this, on 04 January 2015, the Drug Controller General of India (DCGI) granted an import license to Janssen (M/s. Johnson & Johnson Limited, India) that is guided by apex committee approval. The drug was approved for conditional access. This means, it shall be used under RNTCP PMDT framework for treatment of MDR-TB patients only with aDSM systems. [4] In March 2016, RNTCP introduced BDQ through a conditional access programme at six DR-TBCs in the country. Based on the lessons learnt from these initial sites, preparation for expansion of access to BDQ was initiated in all states in 2017.

### 7.2.2 Criteria for patients to receive Bedaquiline

**Inclusion criteria**

The criterion for patients to receive BDQ as approved by the Apex Committee is: adults aged > 18 years having pulmonary MDR-TB.

**Additional requirements**

- non-pregnant females or females not on hormonal birth control methods are eligible. They should be willing to continue practicing birth control methods throughout the treatment period or have been post-menopausal for past 2 years; and
- patients with controlled stable arrhythmia can be considered after obtaining cardiac consultation.

**Exclusion criteria**

- currently having uncontrolled cardiac arrhythmia that requires medication;
- having any of the following QT/QTc interval (Annexure 8) characteristics at screening:
    - marked prolongation of QT/QTc interval, e.g. repeated demonstration of QTcF (Fredericia correction) interval > 450 ms; and
    - history of additional risk factors for Torsade de Pointes, e.g. heart failure, hypokalaemia, family history of long QT syndrome;

BDQ is provided along with a background regimen based on DST results. Certain conditions as listed below should be taken into consideration while choosing the drugs for the background regimen in patients who:

- have evidence of chorioretinitis, optic neuritis or uveitis at screening which precludes long-term Lzd therapy;
- have the following laboratory abnormalities (DAIDS grading of adverse events):
    - creatinine grade 2 or greater, *i.e.,* >1.5 times the upper limit of normal (ULN);
    - haemoglobin grade 4 (<6.5 gm/dL);
    - platelet count grade 3 or greater (≤ 49 999/$mm^3$);
    - absolute neutrophils count grade 3 or greater (≤ 749/$mm^3$);
    - aspartate aminotransferase (AST) grade 2 or greater (>2.5 times ULN);
    - alanine aminotransferase (ALT) grade 2 or greater (>2.5 times ULN);
    - total bilirubin grade 2 or greater (>1.6 times ULN); and
    - lipase grade 2 (with no signs or symptoms of pancreatitis) or greater (>1.5 time ULN).

*Note: If results of the serum chemistry panel, haematology or urinalysis are outside the normal reference range (including above listed parametres), the patient may still be considered if the physician judges abnormalities or deviations from normal to be not clinically significant or to be appropriate and reasonable. Hypokalaemia, hypomagnesaemia and hypocalcaemia should be corrected prior to a patient receiving BDQ.*

Patients who are not found to be eligible for a BDQ containing regimen would have DST results for the key first line and second line drugs and managed with DST guided regimens.

Bdq is indicated in adult MDR-TB patients not eligible for the newly WHO-recommended shorter regimen. [32]These may include:

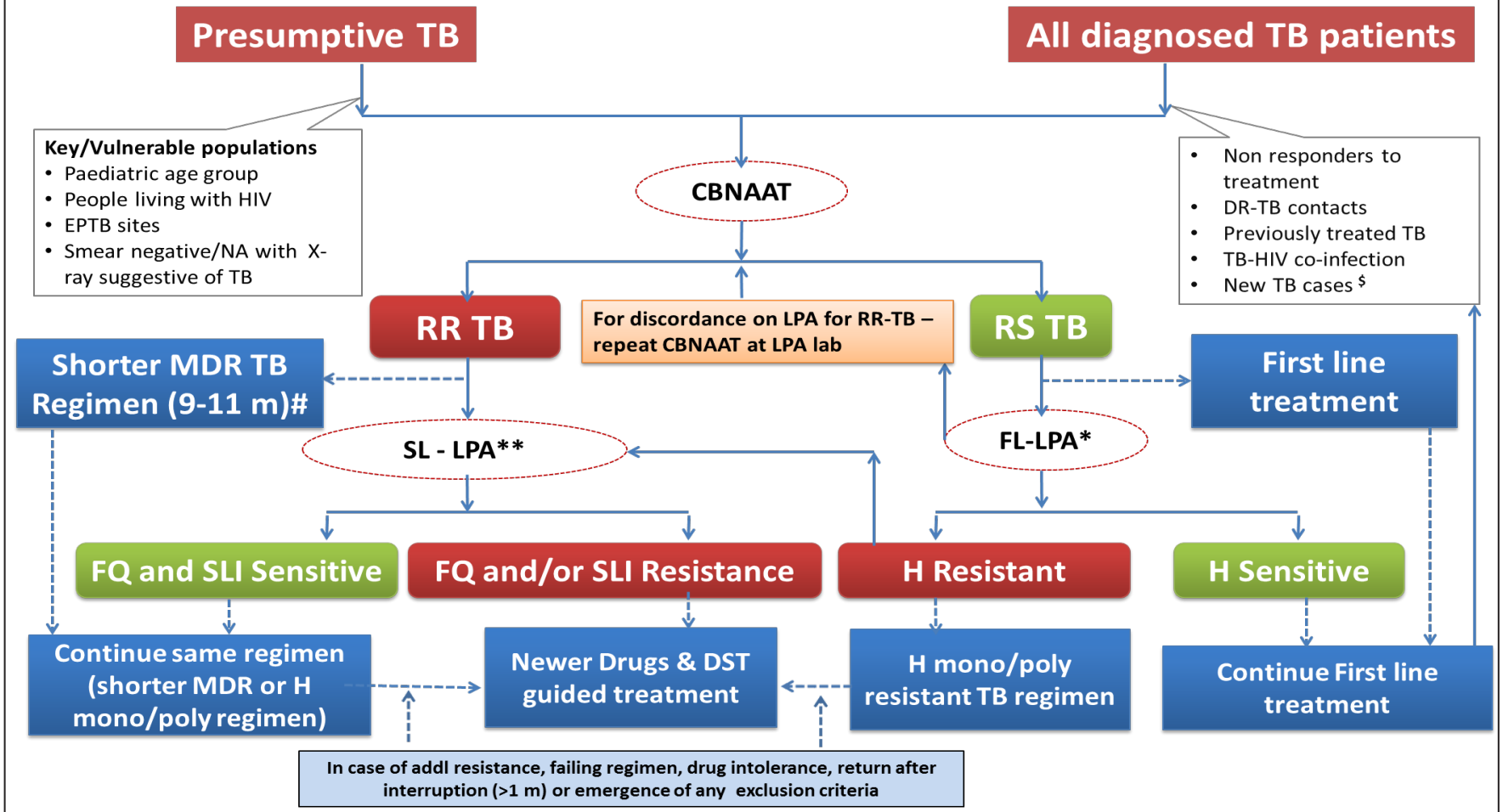
- MDR/RR-TB patients with resistance to any/all FQ OR to any/all SLI;
- XDR-TB patients;
- mixed pattern resistant TB patients;
- treatment failures of MDR-TB + FQ/SLI resistance OR XDR-TB; and
- MDR/RR-TB patients with extensive pulmonary lesions, advanced disease and others deemed at higher baseline risk for poor outcomes.[32]

Caution must be exercised that Bdq is not added to a failing regimen in any patient already on DR-TB treatment**.**

## 7.3 Integrated drug-resistant TB algorithm

The integrated DR-TB algorithm (Figure 7.1) clearly indicates the management strategies to be followed right from the day the results of CBNAAT test are available. These management strategies are described in subsequent sections of this document. The programme is improving laboratory capacity to offer DST for all TB patients (Universal DST). An integrated algorithm offers FL-LPA for the presumptive DR-TB found R sensitive on CBNAAT. The LPA will be primarily utilized to do DST for second-line drugs and to test FL- LPA to detect H mono resistance as per the available laboratory

# Figure 7.1 Integrated Drug Resistant TB Algorithm




# Conventional MDR TB Regimen (24 m) for pregnant women or for EP TB patients those who are not eligible for shorter MDR/RR regimen

*Offer molecular testing and treatment for H mono/poly resistance to TB patients prioritized by risk as per the available lab capacity

**LC DST (Mfx 2.0, Km, Cm, Lzd) will be done only for patients with any resistance on baseline SL-LPA. DST to Z, Cfz, Bdq & Dlm would be considered for policy in future, whenever available, standardized & WHO endorsed.

$ States to advance in phased manner as per PMDT Scale up plan for universal DST based on lab capacity and policy on use of diagnostics

capacity. The programme would continue building capacity of the laboratories to eventually test all R sensitive patients on LPA to detect H mono resistance that would serve as a surrogate for poly drug resistance. These patients would be managed as per the algorithm. Within the first 2-3 months, patients would receive the LC DST results, reach their final classification and treated with the appropriate regimen.

## 7.4 Treatment initiation

If CBNAAT reveals RS-TB, the patient is initiated on the first-line treatment at the PHI level.

**At DDR-TB Centres**

If the CBNAAT/ LPA reveal RR/ MDR-TB or if LPA reveals H mono resistance, the patient must be counselled by the PHI staff to visit the DDR-TBC with a family member for further management without further delay. The treatment initiation of the patients following results of CBNAAT and FL LPA on standard regimens *i.e.,* either shorter MDR-TB regimen or conventional MDR-TB regimen or H mono/poly DR-TB regimen could be undertaken by the physicians at the DDR-TBC. The first-line treatment will be stopped and the appropriate standard DR-TB regimen will be initiated at the DDR-TBC.

For all laboratory confirmed RR-TB patients, the DDR-TBC will initiate the standard shorter MDR-TB regimen (9-11 months) after careful assessment ensuring SL-LPA is conducted and management of pregnancy or if the patient has EP-TB through specialist consultation if required. Such patients, if after assessment, are found to be not eligible for shorter MDR-TB regimen will be initiated on standard conventional MDR-TB regimen (24-27 months). Less severe forms of EP-TB patients like those with lymph node TB or pleural effusion who are drug resistant and eligible based on the DRT/DST and SL-LPA, would also be offered the shorter MDR-TB regimen. Similarly, a laboratory confirmed H mono resistant TB patient is initiated on a standard regimen for H mono/poly DR-TB.

The DDR-TBC Committee can decide on a case-to-case basis, the need for admission for DR-TB patients for initiation of treatment. The patient is initiated on standard regimen at DDR-TBC on indoor or outpatient basis. The first dose is given under supervision at DDR-TBC for ambulatory patients. In patients who are admitted, the duration of indoor management would be decided by DDR-TBC committee as clinically indicated. On discharge, the patient will be provided with a maximum seven days of drug supply for the transit.

In both scenarios, DTO refers the patient to the identified treatment supporter with advance information to MO-PHI to identify and prepare the treatment supporter and provide drugs and records to the treatment supporter. The results of SL-LPA are expected to be available within a week of specimen submission. The LPA laboratory must report whether the specimen is resistant to all SLIs (*i.e.,* rrs mutation) or only Km (eis mutation). The results of LC DST are expected to be available after 6-8 weeks of specimen submission. Based on

results, if no additional resistance is detected, the patient will be continued on the same regimen at the district level.

**At NDR-TB centres**

If additional resistance to FQ and/or SLI is reported, the patient is counselled by the PHI staff and referred to the NDR-TBC immediately. The pretreatment evaluations and baseline aDSM assessment done at the DDR-TBC would be considered valid for assessment of the patient and further management, only if additional resistance is reported on the basis of SL-LPA received within a week. However, these would be repeated if the additional resistance is reported on the basis of LC-DST. The standard DR-TB regimen will be stopped and the NDR-TBC committee would re-design the regimen based on the DST pattern with or without newer drugs as appropriate. As the patient would still be in early IP, the patient would be re-classified and re-registered for a new episode of treatment and updated on Nikshay on the same ID. For monitoring the treatment outcome, the patient would be accounted for the most recent episode of treatment. However, patients who need regimen change in CP will be declared with outcome as "Treatment failed" and re-registered for the next episode of an appropriate treatment.

Apart from this, in case of emergence of additional resistance, failing regimen, drug intolerance, return after interruption (>1 months) or emergence of any exclusion criteria, the patient must be immediately referred by N/DDR-TBC. S/he should have a copy of all records or baseline pretreatment evaluation, baseline aDSM assessment and PMDT treatment book to the NDR-TBC for the committee to consider redesigning the DST guided regimen with or without newer drugs for managing the patient as applicable.

The patient will be reclassified as per DST pattern and date and type of patient's re-classification will be specified in the remarks column of the PMDT treatment register of DDR-TBC and NDR-TBC as well as updated on Nikshay. The duration of treatment, follow-up timelines, interim and final outcomes of the patient would be considered from the date of most recent reclassification. All modifications in regimens or change of the regimen class would be undertaken by NDR-TBC.

Patients eligible for newer drugs need to be offered counselling along with a patient education booklet (Annexure 9) which will give details of the nature and duration of treatment including information on the new drug Bdq; need for regular treatment; possible side-effects of these drugs; drugs to be avoided with Bdq and the consequences of irregular treatment or premature termination of treatment. Female patients will receive special counselling on family planning. The informed decision making process and its documentation is detailed in Chapter 6.

Cohort Event Monitoring (CEM) treatment initiation form needs to be completed and uploaded on Nikshay for all DR-TB patients at the time of initiation of each new regimen. All patients eligible for newer drugs containing regimen would be managed in an in-patient setting for a period of two weeks (15 days) to complete the initial two weeks of Bdq doses. The final decision of further duration of in-patient management rests with the NDR-TBC Committee and must be well-documented for every patient. After discharge, treatment will be continued on ambulatory basis with strict adherence to treatment and follow-up schedule. All measures for AIC must be implemented as per the national AIC guidelines while managing all TB patients. In exceptional patients who are not seriously ill, it is important to have all pretreatment evaluations within normal limit and for those who are ambulatory or residing close to the NDR-TBC and are willing to visit NDR-TBC for periodic ECG and clinical monitoring, the NDR-TBC Committee may decide to manage the patient on an ambulatory basis.

## 7.5 Regimen type (with or without newer drugs)

**Principles of designing a WHO recommended DR-TB regimen**

In patients with RR or MDR-TB, a regimen with at least five effective TB medicines during the intensive phase is recommended, including Z and four core second-line TB medicines - one chosen from group A, one from group B and at least two from group C. If the minimum of effective TB medicines cannot be composed as above, an agent from group D2 and other agents from D3 may be added to bring the total to five. In patients with MDR/RR-TB, it is recommended that the regimen be further strengthened with $H^h$ and/or E.[12]

Newer drugs containing regimen would contain the new drug as one of at-least four core second-line drugs considered to be effective (choice of drugs should be based on DST pattern) and Z as per the above principles. Because of GI side effects of PAS, as far as possible, will be avoided in Bdq containing regimen.

Designing a regimen is the prerogative of the DR-TBC committee. The regimen may be with or without inclusion of new drugs like BDQ and would be classified into the following types:

**At DDR-TB centres:**

- H Mono/Poly drug resistant TB
- MDR/RR-TB
    - shorter MDR-TB regimen; and
    - conventional MDR-TB regimen.

**At NDR-TB centres:**

- MDR/RR-TB with additional resistance to any/all FQ or SLI
- XDR-TB
- Mixed pattern drug resistant TB

- with H mono + FQ/SLI/Lzd resistance;
- with MDR/RR-TB + FQ/SLI+ Lzd resistance; and
- other patients who need careful regimen designing detailed later.

- Non - TB *Mycobacterium* (NTM)

### 7.5.1 H Mono/Poly drug-resistant TB

On receiving the reports showing H mono resistance on LPA that serves as a surrogate for first-line drugs mono & poly DR-TB (excluding R), the patient and their family members should be counselled and referred to the nearest DDR-TBC. Baseline SL DST will be performed as detailed above. The DDR-TBC will undertake the pretreatment evaluation (including clinical and radiological evaluation) and initiates the patient on the standard treatment regimen as in Table 7.3

### 7.5.2 MDR/RR-TB

All lab confirmed MDR/RR-TB patients will be initiated on the shorter MDR-TB regimen with special precautions in pregnant women and EPTB patients (see sections later).

**Shorter MDR-TB regimen**

The shorter MDR-TB regimen is recommended for patients in whom the diagnosis of MDR/RR-TB has been reliably confirmed by molecular (e.g. CBNAAT/ LPA) or phenotypic DST method and are found to be sensitive to both FQ and SLI by SL-LPA. All patients with confirmed R-resistant disease are treated as for MDR-TB and the shorter MDR-TB regimen could be used in these patients too. Children and PLHIV on antiretroviral therapy (ART) could receive the shorter MDR-TB regimen. [12][26] The following are the features of shorter MDR-TB regimen:

- standardized shorter MDR-TB regimen with seven drugs and a treatment duration of 9-11 months;
- indicated conditionally in MDR-TB or RR-TB, regardless of patient age or HIV status;

**Exclusion criteria**

- Second-line drug resistance (FQ and/or SLI drugs), previous exposure for >1 month to a FQ or a second-line injectable medicine which is not in the shorter MDR-TB regimen but which may generate cross-resistance is considered an exclusion criterion. However, if resistance to both of these two agents has been excluded by a reliable drug-susceptibility test (DST), then the shorter MDR-TB regimen can be used.
- It is not recommended to base treatment decisions on the DST to any other drug in the regimen (Z, $H^h$, E, Eto, Cfz) apart from those mentioned, owing to the unreliable nature of the tests.

- Pregnancy.
- Extra-pulmonary TB (Other than plural effusion & lymph node TB)

**Justification for shorter MDR-TB regimen**

It has been observed from pooling individual patient data (n=1116) of observational studies from Bangladesh, Uzbekistan, Swaziland, Cameroon, Niger and nine countries of sub-Saharan Africa that patients who met specific inclusion criteria for receiving the shorter MDR-TB treatment regimens had a statistically-significant higher likelihood of treatment success than those who received longer conventional regimens (89.9% vs. 78.3% respectively when success was compared with treatment failure/relapse/death (Table 7.3) and 83.4% vs. 61.7% when compared with treatment failure/relapse/death/lost to follow-up). The number of relapses was very low, although this may have been as a result of the relatively small number of patients followed up. As expected, the treatment success was lower in patients with additional resistance to Z and/or FQ on shorter MDR-TB regimens, even if in general it remained high and exceeded that in the patients on individualized, conventional regimens (although the differences were not statistically significant).[12]

Table 7.2: Treatment success in patients treated with shorter MDR-TB regimen vs conventional MDR-TB regimens*

| Resistance pattern | Shorter MDR-TB regimen | | Conventional MDR-TB regimen | |
|---|---|---|---|---|
| | N | % (95% CI) | N | % (95% CI) |
| All patients regardless of pyrazinamide and fluoroquinolone susceptibility | 1008/1116 | *90.3% (87.8%- 92.4%)* | 4033/5850 | *78.3% (71.2%- 84%)* |
| Pyrazinamide resistant; fluoroquinolone resistant | 19/28 | 67.9% (47.6%-84.1%) | 81/137 | 59.1% (50.6%-67.1%) |
| Pyrazinamide resistant; fluoroquinolone susceptible | 90/100 | 88.8% (47.3%-98.6%) | 840/1075 | 81.4% (71.6%-88.4%) |
| Pyrazinamide susceptible; fluoroquinolone resistant | 12/15 | 80.0% (50.0%-94.1%) | 72/120 | 64.4% (49.6%-76.9%) |
| Pyrazinamide susceptible; fluoroquinolone susceptible | 121/125 | 96.8% (77.3%-99.6%) | 890/1119 | 83.5% (75.7%-89.2%) |

**treatment success (cured or treatment completed) versus failure/relapse/death in patients not previously treated with second-line TB medications; percentages shown have been adjusted when possible*)

Hence, there is merit in immediately adopting the shorter MDR-TB regimen (9-11 months) based on the advantage of being substantially less expensive than the conventional regimen and its potential to improve treatment success and decrease deaths when compared to conventional MDR-TB regimens (24-27 months) with poor treatment outcomes in MDR-TB of about 46% in India (2013 cohort).[1][6] It should be noted the value and guidance for implementing the shorter regimen will be updated as soon as the results of the phase 3

STREAM clinical trial evaluating this regimen are reviewed by WHO, according to well established process. The shorter MDR-TB regimen is detailed in Table 7.3.

A recently published meta-analyses of individual patient data and aggregate data from five studies on effectiveness and safety of standardized shorter regimens for MDR-TB revealed 669/796 lab confirmed MDR/RR-TB patients not previously exposed to second-line drugs. They were successfully treated (83%, 95% CI 71.9-90.3%) while failure/relapse was associated with fluoroquinolone resistance (crude OR 46, 95% CI 8–273), pyrazinamide resistance (OR 8, 95% CI 2–38) and no culture conversion by month 2 of treatment (OR 7, 95% CI 3–202). The major implications for the programmatic use of shorter MDR-TB regimens are highlighted in the box below.

**The study has revealed few major implications for the programmatic use of shorter MDR-TB regimens**

- The observation that resistance to fluoroquinolones or resistance to pyrazinamide were risk factors for failure/relapse and that acquired drug resistance occurred among failures underscore that programmes implementing shorter regimens should simultaneously strive to scale-up DSTs for component drugs. Others recently suggested alternate regimens available for patients identified to have strains that are resistant at baseline or that acquire resistance during treatment. This is supported by evidence from other studies showing that successful MDR-TB treatment is determined, in part, by the number of drugs used against which the infecting strain is susceptible, including pyrazinamide.
- All shorter regimen studies used intensive treatment adherence systems along with social support. They provided ART to all PLHIV. Hence, such interventions should be considered essential components of shorter regimens.
- Programmes should ensure proper screening and management of adverse events
- Successfully treated patients should be followed for relapse for at least 1 year after completion of therapy.[27]

All these recommended strategies are well adopted in this guideline

The NDRS results revealed concerns around resistance to component first and second-line drugs included in shorter MDR-TB regimen as detailed in Chapter 1 and was discussed extensively among the national experts.[7] However, the meta-analysis detailed above with high treatment success rates excluded RR-TB patient susceptible to H, implying that H resistance do not affect the treatment success observed in shorter MDR-TB regimen. Resistance to E was not significantly affecting the treatment outcomes with shorter MDR-TB regimen. The treatment success rate was still observed to be as high as 92% in patients treated with shorter MDR-TB regimen with Z resistance but FQ susceptible.[27]

WHO recommended regimens always have Z without counting it with the other effective drugs which still remains a practice globally with inconclusive evidence.[12][26][31]With SL-LPA simultaneously introduced across India, patients with FQ and/or SLI class resistance will be clearly excluded. However, the programme would continue to simultaneously strive to scale-up DSTs for component drugs in future and help generate evidence around the use of Z, $H^h$ as component drugs are included in shorter MDR-TB regimen. Till such time that it gets accepted, the programme would continue implementing the standard shorter MDR-TB regimen as recommended by WHO.

**Conventional MDR-TB regimen**

Those patients who are not considered eligible for shorter MDR-TB regimen (particularly after assessment in pregnant women, EP-TB except patients with lymph node TB and pleural effusion) should be initiated on conventional MDR-TB regimen as in Table 7.3

As the above three regimen for treatment of MDR/RR-TB or H mono-poly DR-TB are standard regimen, these would be initiated at the DDR-TBC based on results of CBNAAT or FL-LPA. The regimen designs adhere to the principles of designing a WHO recommended DR-TB regimen [12] detailed in Table 7.3

Table 7.3: Standard regimen for initiating treatment of MDR/RR-TB or H mono-poly DR-TB at district DR-TB centre based on CBNAAT or FL-LPA

| Resistance pattern | Regimen class | Intensive phase | Continuation phase | Principle of regimen design |
|---|---|---|---|---|
| ***Regimen for H mono/poly DR-TB*** | | | | |
| H mono/poly DR-TB (R susceptible H resistant TB & DST of SEZ not known) | H Mono-poly DR-TB regimen | (3-6) Lfx Km R E Z | (6) Lfx R E Z | REZ + augment with 1 GpA + 1 GpB drug |
| ***Shorter MDR-TB regimen*** | | | | |
| R resistant + H sensitive/ unknown **or** MDR–TB | Shorter MDR-TB regimen | (4-6) $Mfx^h$ Km* Eto Cfz Z $H^h$ E | (5) $Mfx^h$ Cfz Z E | As per WHO recommendation |
| ***Regimen for MDR/RR-TB*** | | | | |
| R resistant + H sensitive/ unknown **or** MDR–TB | Conventional MDR-TB regimen | (6-9) Lfx Km Eto Cs Z E | (18) Lfx Eto Cs E | 1 GpA + 1GpB + 2 GpC + Z + add on 1 GpD1 |

**If the intensive phase is prolonged, the injectable agent is only given three times a week in the extended intensive phase.*

### 7.5.3 MDR/RR-TB with additional resistance to any/all FQ or SLI

All patients with additional resistance to FQ class or SLI class on SL-LPA would be assessed for eligibility for newer drug containing regimen. Patients who have consented and are found to be eligible would be initiated on newer drugs containing the regimen while rest of the patients would be initiated on a DST guided regimen and reclassified. The regimen design is detailed in Table 7.4. If a patient is resistant to either FQ or SLI class on SL-LPA, is found to be resistant to any of the drugs of the other class, in LC-DST, the patient will be re-classified as XDR-TB.

### 7.5.4 XDR-TB

All XDR-TB patients identified with SL-LPA or LC-DST would be assessed for eligibility for the newer drug containing regimen and reclassified as above. The regimen design is detailed in Table 7.4.

### 7.5.5 Mixed pattern drug resistant TB

Following patient/s would be operationally re-classified into mixed pattern drug-resistant TB:

- H mono-poly DR-TB patients with additional resistance to FQ and/or SLI and/ or Lzd based on LPA and/or LC-DST will be reclassified as mixed pattern DR-TB. All such patients should be subject to consultation by a thoracic surgeon for consideration of surgery at tertiary centres with surgical facilities. These patients would be initiated on a DST guided regimen as detailed in Table 7.4;
- Similarly, RR-TB patients with additional resistance to FQ and/or SLI with Lzd resistance detected on LC-DST will also be reclassified as mixed-pattern DR-TB; and
- Apart from this, MDR/RR-TB patients would also be managed with the regimen for mixed-pattern DR-TB if the following events prevent them to be managed with any of the above regimen classes:

    - patients who are failing any DR-TB regimen or
    - patients who have drug intolerance or contraindications or
    - patients who return after interruption (>1 months) or
    - patients who have emergence of any exclusion criteria
    - patients with extensive pulmonary lesions, advanced disease and others deemed at higher baseline risk for poor outcomes.

All such patients would be re-evaluated for eligibility for newer drug containing regimen and re-classified as above. The regimen design is detailed in Table 7.4. In patients who have failed an M/XDR TB regimen, the regimen proposed for mixed pattern regimen should be

designed using drugs considered to be effective-based on previous use. It is also advised to use a minimum of 5 drugs and maximum 8-9 drugs in the regimen. [9]

The regimen design will start with appropriate modification of the regimens proposed for XDR-TB with or without newer drugs as per the eligibility/consent of the patient using the guidance in the footnotes as detailed in Table 7.4.

In future, whenever standardized, WHO endorsed DST methods for Z, Cfz, Bdq, Dlm *etc.,* are available and included as programme policy, patients found to be resistant to any of the above drugs would be managed with the regimen for mixed pattern DR-TB.[12][26] [27][28]

**Replacement drugs in sequence of preference**

In case of an adverse drug reaction, poor tolerance, contraindication or non-availability of any component drug of the combination regimen warranting replacement, the following drugs would be added to replace that drug in the order of sequence, if not already used: Cs, $H^h$ E, Bdq*(*in RR-TB patients also where a WHO recommended regimen could not be formed), PAS, Amx-clv.[12]

If more than one drug is required to be replaced but this replacement is just a modification and does not change the regimen class, the patient would continue to be reported on the same regimen. However, if the replacement of drugs affects the regimen class, then these patients would need to be re-classified as mixed pattern DR-TB. This decision would be based on the judgment of the NDR-TBC Committee. As far as possible, patients need not be unnecessarily asked to travel to NDR-TBC while required changes may be carried out at DDR-TBC or at field-level as per modifications recommended in the regimen by NDR-TBC.

Table 7.4: DST guided regimen with or without newer drugs for initiating treatment of DR-TB patients with additional resistance to FQ class and/or SLI class, at NDR-TBC based on SL-LPA

| Resistance pattern | DST guided regimen class | Intensive phase | Continuation phase | Principle of regimen design |
|---|---|---|---|---|
| ***Regimen with new drugs for MDR-TB + FQ / SLI resistance*** | | | | |
| MDR/RR + resistance to FQ class / SLI[1] class | MDR/RR + res to FQ class | (6-9) Km Eto Cs Z Lzd[3]Cfz + (6) Bdq | (18) Eto Cs Lzd[3]Cfz | 0 GpA + 1GpB + 2 GpC + Z + add on 2 GpC + 1 GpD2 |
| | MDR/RR+ res to SLI[1] class | (6-9) Lfx Cm[1] Eto Cs Z Lzd[3]Cfz + (6) Bdq | (18) Lfx Eto Cs Lzd[3] | 1 GpA + 1 GpB[1] + 2 GpC + Z + add on 2 GpC + 1 GpD2 |
| ***Regimen MDR-TB + FQ / SLI resistance: (without new drugs)*** | | | | |
| MDR/RR + resistance to FQ class / SLI[1] class | MDR/RR + res to FQ class | (6-9) Mfx[h2] Km Eto Cs Z Lzd[3]Cfz | (18) Mfx[h2]Eto Cs Lzd[3]Cfz | 1 GpA[2] + 1GpB + 2 GpC + Z + add on 2 GpC |
| | MDR/RR+ res to SLI[1] class | (6-9) Lfx Cm[1]Eto Cs Z Lzd[3]Cfz | (18) Lfx Eto Cs Lzd[3] | 1 GpA + 1 GpB[1] + 2 GpC + Z + add on 2 GpC |
| ***Regimen with new drugs for XDR-TB*** | | | | |
| XDR-TB (res to both FQ and SLI[1] class) | XDR-TB | (6-12) Cm[1]Eto Cs Z Lzd[3]Cfz E + (6) Bdq | (18) Eto Cs Lzd[3]Cfz E | 0 GpA + 1 GpB[1] + 2 GpC + Z + add on 2 GpC + 1GpD1 + 1 GpD2 |
| ***Regimen for XDR-TB: (without new drugs)*** | | | | |
| XDR-TB (res to both FQ and SLI[1] class) | XDR-TB | (6-12) Mfx[h2]Cm[1]Eto Cs Z Lzd[3]Cfz E | (18) Mfx[h2] Eto Cs Lzd[3]Cfz E | 1 GpA[2] + 1 GpB[1] + 2 GpC + Z + add on 2 GpC + 1GpD1 |
| ***Regimen with new drugs for mixed pattern DR-TB:*** | | | | |
| Mixed pattern DR-TB | MDR/RR-TB + res to FQ / SLI[1] + Lzd[3] or more | Modify the regimen with new drugs for XDR-TB as per the footnotes | | |
| ***Regimen for mixed pattern DR-TB: (without new drugs)*** | | | | |
| Mixed pattern DR-TB | H mono-poly + res to FQ/SLI[1]/ Lzd[3] | (3-6) R E Z Cm[1]Eto Lzd[3] | (6) R E Z Eto Lzd[3] | REZ + augment with 1 GpB[1] + 2 GpC drug |
| | MDR/RR-TB + FQ / SLI[1] with Lzd[3] or others. | Modify the regimen for XDR-TB (without new drugs) as per the footnotes | | |

**Foot note:**

1. if only Km resistant (at eis mutation), then add Cm in IP upfront in the regimen design;
2. in patients with MDR/RR + FQ class resistance, XDR-TB and mixed pattern resistance where a new drug is not considered in the regimen for any reason, Mfx$^{h}$ would be added upfront in the regimen design and the decision to continue or replace it would be taken based on LC-DST results to Mfx (2.0) by NDR-TBC; and
3. Lzd to be replaced with a suitable drug if found to be resistant on LC-DST. In such situation the patient must be reclassified as mixed pattern DR-TB.

Once the baseline LC-DST results are available, the decision to modify the regimen in patients with FQ or SLI class resistance by adding individual drugs in FQ or SLI groups in the above DST guided regimen would be considered. This would be done as below, based on LC-DST results:

- if Mfx (2) is susceptible, then add Mfx$^{h}$ and increase the frequency of ECG monitoring if used in a regimen containing other cardio toxic drugs like Bdq, Cfz, Dlm *etc.;*
- if Mfx (2) is resistant, then remove all FQ;
- if any of the SLI are susceptible, then add one susceptible injectable in the order of Km, Cm; and
- if all SLI are resistant, then do not add the SLI.

However, the susceptible drug added from the FQ or SLI class would not be counted as an effective drug in the regimen but rather considered as an add-on drug to substantiate the strength of the regimen.

### 7.5.6 Non-TB *Mycobacterium* (NTM)

**Treatment of NTM:** [24][25]

NTM are uncommonly encountered clinical pathogens; some species, in fact, are much more likely to be isolated as a result of specimen contamination than as a result of disease. It can also be isolated from patients with lower respiratory infections especially from patients who live in areas of higher density of environmental NTM presence. This is a transient carriage and usually does not meet the criteria for NTM disease. However, even these species can, under some circumstances, cause clinical disease. The clinician, therefore, must always know the context in which an NTM isolate was obtained to assess accurately the clinical significance of that isolate. Given these complexities, the treatment of NTM will be the prerogative of the NDR-TBCs. When questions about the clinical significance of an NTM isolate arise, expert consultation is strongly encouraged. In this context, important points to note would be:

- treatment recommendations for infrequently encountered NTM are made on the basis of only a few reported patients. With that limitation in mind, unless otherwise stated, the duration of therapy for most pulmonary NTM pathogens is based on treatment recommendations for more frequently encountered species such as MAC and *M. kansasii* (e.g., continue therapy for at least 12 months after the last negative sputum culture). For disseminated disease, treatment duration for most NTM pathogens is the same as for disseminated MAC infection;
- treatment of NTM disease is generally not directly analogous to the treatment of TB. In vitro susceptibilities for many NTM do not correlate well with clinical response to anti-mycobacterial drugs. Recommendations for routine in vitro susceptibility testing of NTM isolates are limited. The clinician should use in vitro susceptibility data with an appreciation for its limitations;
- empiric therapy for suspected NTM lung disease is not recommended; and
- there are not widely accepted criteria for choosing patients with NTM lung disease for resectional surgery. In general, the more difficult an NTM pathogen is to treat medically, the more likely surgery should be considered from a risk/ benefit perspective. Expert consultation is strongly encouraged at NDR-TBC.

**Suggested treatment regimen covering maximum NTMs mainly MAC**

- Rifampicin 450-600 mg OD;
- Ethambutol 800 – 1200 mg OD;
- Clarithromycin 1gm OD (split into two doses); and.
- Add injection Amikacin 750mg-1gm thrice weekly for the first 2-3 months.

Intensive phase is for 3 months and can be extended to a maximum of 6 months with all four drugs. Continuation phase of treatment will be with the same drugs except injectable
This should be continued for 12 months after sputum culture conversion. Drugs will be given as per the standard weight bands. If the patient does not culture covert by end of 3 months, then species identification and DST is required for further management by the NDR-TBC committee based on expert consultations.

*Note:- As the proportion of the patients estimated is very low, drugs will not be available through RNTCP but will have to be made available through the general health system.*

**Points to note for treatment of NTM**

- recommended initial regimen for most patients with nodular/bronchiectatic MAC lung disease is a thrice-weekly regimen including Clarithromycin 1000 mg or Azithromycin 500 mg, E 25 mg/kg, and R 600 mg administered three times per week;
- recommended initial regimen for fibro-cavitary or severe nodular/bronchiectatic MAC lung disease includes clarithromycin 500–1000 mg/day or azithromycin 250 mg/ day, E

15 mg/kg/day, and R 10 mg/kg/day (maximum, 600 mg). An initial 2 months of E at 25 mg/kg/day is no longer recommended;

- intermittent drug therapy is not recommended for patients who have cavitary disease, or patients who have been previously treated or patients who have moderate or severe disease;
- primary microbiologic goal of therapy is 12 months of negative sputum cultures while on therapy; therefore, sputum must be collected from patients for AFB examination throughout treatment on monthly basis in IP and quarterly basis in CP after culture conversion is achieved;
- macrolides should not be used as monotherapy for MAC because of the risk for developing macrolide-resistant MAC isolates;
- macrolide with a single companion drug, E, may be adequate for nodular/bronchiectatic MAC disease but should not be used in patients with fibro-cavitary disease because of the risk of emergence of macrolide resistance;
- patients respond best to MAC treatment regimens the first time they are administered; therefore, it is very important that patients receive recommended multidrug therapy the first time they are treated for MAC lung disease; and
- expert consultation should be sought for patients who have difficulty tolerating MAC treatment regimens or who do not respond to therapy.

For further details on management of individual NTMs including EP NTMs, the physicians of NDR-TBC are encouraged to refer to the latest ATS guidelines.

## 7.6 Drug dosage and administration

The dosage of drugs would vary as per weight of the patients. Patients would be classified in weight bands of <16 kg, 16-29 kg, 30-45 kg, 46-70 kg and >70kg. All drugs in the regimen are to be given on a daily basis under observation. Injectable will be administered for six days/week (excluding Sundays). All morning doses are to be supervised by the treatment supporter. After taking the morning doses on Saturday, next day's oral drugs would be given to the patient to be taken at home on Sunday. Empty blisters of medicines taken unsupervised in the evening and on Sundays are to be collected by treatment supporter. In cases of drug intolerance – E, Cs and Na-PAS can be given in divided doses (twice a day).

The dosage for drugs used in various DR-TB regimens by weight bands for adults are enumerated in Table 7.5. These are in accordance to the WHO recommended doses of anti-TB drugs for adults and paediatric patients. [31](Annexure 10)

Table 7.5: Dosage of DR-TB drugs for adults

| S.No | Drugs | 16-29 kg | 30-45 kg | 46-70 kg | >70 kg |
|---|---|---|---|---|---|
| 1 | Rifampicin(R)[1] | 300mg | 450mg | 600mg | 600mg |
| 2 | High dose H ($H^h$) | 300 mg | 600 mg | 900 mg | 900 mg |
| 3 | Ethambutol(E) | 400 mg | 800 mg | 1200 mg | 1600 mg |
| 4 | Pyrazinamide(Z) | 750 mg | 1250 mg | 1750 mg | 2000 mg |
| 5 | Kanamycin(Km) [2] | 500 mg | 750 mg | 750 mg | 1000 mg |
| 6 | Capreomycin (Cm) | 500 mg | 750 mg | 750 mg | 1000 mg |
| 7 | Amikacin (Am) | 500 mg | 750 mg | 750 mg | 1000 mg |
| 8 | Levofloxacin(Lfx) [4] | 250 mg | 750 mg | 1000 mg | 1000 mg |
| 9 | Moxifloxacin (Mfx) [4] | 200 mg | 400 mg | 400 mg | 400 mg |
| 10 | High dose Mfx ($Mfx^h$)[4] | 400mg | 600mg | 800mg | 800mg |
| 11 | Ethionamide (Eto) [4] | 375 mg | 500 mg | 750 mg | 1000 mg |
| 12 | Cycloserine (Cs)[4] | 250 mg | 500 mg | 750 mg | 1000 mg |
| 13 | Na-PAS (60% weight/vol) [3,4] | 10 gm | 14 gm | 16 gm | 22 gm |
| 14 | Pyridoxine(Pdx) | 50 mg | 100 mg | 100 mg | 100 mg |
| 15 | Clofazimine (Cfz) | 50 mg | 100 mg | 100 mg | 200 mg |
| 16 | Linezolid (Lzd) | 300 mg | 600 mg | 600 mg | 600 mg |
| 17 | Amoxyclav (Amx/Clv) (In child: WHO 80mg/Kg in 2 divided doses) | 875/125 mg BD | 875/125 mg BD | 875/125 mg (2 morning +1 evening) | 875/125 (2 morning +1 evening) |
| 18 | Bedaquiline (Bdq) | Week 0–2: Bdq 400 mg daily<br>Week 3–24: Bdq 200 mg 3 times per week | | | |

[1]*For H mono/poly resistant TB;*

[2]*For adult more than 60 yrs of age, dose of SLI should be reduced to 10mg/kg (max up to 750 mg)*

[3]*In patient of PAS with 80% weight/volume the dose will be changed to 7.5gm (16-29Kg); 10 gm (30-45 Kg); 12 gm (46-70 Kg) and 16 gm (>70 Kg)*

[4]*Drugs can be given in two divided doses in a day in the event of intolerance*

***Bedaquiline*:** All patients eligible for Bdq will receive Tab. Bdq400 mg once daily for the first 2 weeks and 200 mg 3 times a week (with at least 48 hours between doses) for the following 22 weeks, in combination with an optimized background regimen (OBR) as detailed above. The OBR will be continued beyond 24 weeks of Bdq administration for the RNTCP recommended duration of treatment. As mentioned above, OBR will be designed as per RNTCP PMDT guidelines and WHO recommendations for designing an OBR for concomitant use with Bdq for avoiding drugs like Mfx and Cfz that are likely to cause increased toxicity when administered in combination with Bdq.

***Week 0–2***: Bdq 400 mg (4 tablets of 100 mg) daily (7 days per week) + OBR;
***Week 3–24***: Bdq 200 mg (2 tablets of 100 mg) 3 times per week (with at least 48 hours between doses) for a total dose of 600 mg per week + OBR; and
***Week 25 (start of month 7) to end of treatment***: Continue other second-line anti-TB drugs only as per RNTCP recommendations.

If taking a light meal with Bdq and other anti-TB drugs, patients should not consume milk-containing products at the same time, as the calcium in these can decrease the absorption of FQs. Also, large fatty meals should be avoided, as these can impair absorption of some of the other anti-TB drugs (Cs, H, *etc.*,). [4][10][32]

The following medications are disallowed during the 24-week administration of Bdq and up to one month after the last dose of Bdq because of potential drug–drug interactions**:**

- systemic use of moderate and strong CYP3A4 inhibitors, e.g. azole antifungals: ketoconazole, voriconazole, itraconazole, fluconazole; ketolides such as telithromycin and macrolide antibiotics other than azithromycin for more than 2 consecutive weeks;
- systemic use of strong CYP3A4 inducers, e.g. phenytoin, carbamazepine, phenobarbital, St. John's wort and rifamycins (rifampin, rifabutin, rifapentine); and
- cholesterol lowering medications of the "statin" class.

Bdq should be used with caution in PLHIV infection treated with ARVs that exhibit drug-drug interactions with Bdq (efavirenz) or prolong the QT interval (lopinavir/ritonavir) as well as in patients with comorbidities (such as diabetes) or persons with drug or alcohol use, due to limited or no information. Bdq has been used in large cohorts of patients. While experience is growing and drug monitoring is still required, concern is less, due to cohort data reviewed from South Africa.[32] However, frequent clinical and cardiac evaluation is required in these patients.

Bdq will be provided through RNTCP once the patient has been confirmed as eligible by the DR-TBC Committee and has consented for the same. The dosage of BDQ would apply to all weight bands while the dosage of other drugs in the OBR would be as per weight bands in accordance to guidelines.

## 7.7 Patient counselling for DR-TB patients

Counselling offers an opportunity to explore and heal past and present difficulties faced by patients in a confidential and supportive environment, especially while dealing with life issues. The role of counselling in the management of DR-TB is significant. A competent counsellor can sensitize patients in several key aspects of TB with particular emphasis on DR-TB care and control. It is important to understand the mode of disease transmission and its prevention.

A counsellor can help the patient to better understand importance of regular treatment as well as consequences of deviation from the treatment. His/her role is to empower patients with disease related information and enable him to take informed decision related to treatment adherence. Treatment duration of any DR-TB regimen is long enough for the patient which needs multiple sessions of counselling, preferably more frequently in the initial phase of treatment.

Documentation is an essential part of counselling. It helps the counsellor in being aware about the progress of the sessions and targets achieved in the process. It informs about the efforts made by the clients to make desired changes. It also gives space for the counsellor to record his/her observations. In case of any legal issue cropping up, the documentation stands as proof of the work done with the client. The DR-TB counselling register that is available with the counsellor serves the purpose of documentation (formats are discussed in Chapter 12). RNTCP provides a counsellor at every NDR-TB centre for this purpose.

## 7.8 Treatment duration for various DR-TB regimen

The treatment is administered in two phases, namely intensive phase (IP) and continuation phase (CP). The treatment duration for DR-TB patients would depend upon the classification of the patient and regimen designed.

### 7.8.1 H mono-poly DR-TB regimen

Total duration of H mono-poly DR-TB regimen is 9-12 months, depending on IP duration. The IP should be given for at least three months. After the third month of treatment, if the result of microscopy is negative, then CP should be initiated. If the third month smear result is positive then, IP is extended by one month. If the IP is prolonged, the injectable agent is only given three times a week. IP should be extended for a maximum of three months (*i.e.,* total duration of IP is not more than six months). Duration of CP is fixed for six months. [5]

### 7.8.2 Shorter MDR-TB regimen

Total duration of shorter MDR-TB regimen is for 9-11 months, depending on IP duration. IP should be given for at least four months. After fourth month of treatment, if the result of sputum microscopy is negative then CP should be initiated. If sputum smear does not become microscopy negative by the fourth month of treatment, the IP should be prolonged until sputum smear converts. If the intensive phase is prolonged, the injectable agent is only given three times a week. IP should be extended for a maximum of two months (*i.e.,* total duration of IP is not more than six months). Duration of CP is fixed for five months. [12] [26]

### 7.8.3 Conventional MDR-TB regimen

Total duration of conventional MDR-TB regimen is 24 – 27 months, depending on IP duration. IP should be given for at least six months. After the sixth month of treatment, the patient must be reviewed and the treatment changed to CP if the fourth or fifth month culture result in solid or liquid culture is negative. If the fourth or fifth month culture result remains positive, the treatment is extended by one month. Extension of IP beyond one month is decided based on the results of subsequent culture results and the clinical/radiographic response. If the result of the fourth month culture is still pending after six months of treatment, IP is extended until the result is available, with further treatment being decided, based on the culture result. IP should be extended for a maximum of three months (*i.e.,* total duration of IP is not more than six months).

The recommended duration for CP is 18 months and if the patient continues to remain culture positive or reverts back to culture positive after the extended IP up to a maximum of three additional months. After this, the patient is declared as "Treatment failed", re-evaluated as per integrated DR-TB algorithm, reclassified as mixed pattern DR-TB and initiated on an appropriate DST guided regimen.

### 7.8.4 XDR-TB regimen

Total duration of regimen for XDR-TB would be of 24-30 months duration with 6-12 months IP and 18 months CP. The change from IP to CP will be done only after achievement of culture conversion (two consecutive negative cultures taken at least one month apart with no subsequent positive cultures). In case of delay in culture conversion, the IP can be extended on monthly basis from 6 months up to a maximum of 12 months. In case of extension, the NDR-TRC committee, which will be responsible for initiating and monitoring the regimen for XDR-TB, can decide on administering Cm injection intermittently (3 times/week) for the months 7 to 12.

### 7.8.5 Regimen for mixed pattern resistant TB

***H mono-poly + resistance to FQ/SLI with Lzd (without newer drugs*)**: Total duration of regimen for mixed pattern resistant TB - H mono with FQ/SLI resistance (without newer drugs) is 9 - 12 months. The duration of IP is for a minimum of three months. If sputum smear does not become microscopy negative by the third month of treatment, the initial phase should be prolonged until sputum smear converts. IP can be extended on a monthly basis for a maximum period of three months. Total duration of IP is not more than 6 months. Duration of continuation phase (CP) is fixed for 6 months. [9]

***MDR/RR-TB + resistance to FQ / SLI[1] with Lzd[3] or failure of DR-TB regimen or patients who do not fit in any of the above regimen (with or without newer drugs*)**: This will be the same as for XDR-TB patients detailed above. [9]

## 7.9 Patient flow for DR-TB patients

- DDR-TBC and NDR-TBCs should be involved actively in management of all DR-TB patients;
- DDR-TBC will be the reporting unit for the respective district and will register all MDR/RR-TB and H mono/poly DR-TB patients of respective districts initiated on standard regimen based on CBNAAT or FL-LPA or culture DST results in PMDT treatment register with issue of unique PMDT number. Patient details will be entered and regularly updated on Nikshay;
- NDR-TBC will be the reporting unit for catering districts and re-register all DR-TB patients of respective districts (who need DST guided regimen redesigning with or without new drugs) in PMDT treatment register with issue of unique DR-TB number. Patient details would be entered and regularly updated on Nikshay. Overall accountability of all such patients will be shared by the concerned DDR-TBC and DTO;

- PMDT treatment card of DR-TB patients managed at the concerned DR-TBC for pretreatment evaluation will be opened by responsible staff of DR-TBC at district or nodal level;

- after pretreatment evaluation and initiation of treatment, the patient should be referred back to the PHI with up to a maximum of one week's supply of drugs, arrangements for injections in transit and a copy of the PMDT treatment book and referral form under intimation of the DTO. The respective DTO/ MO-PHI should be informed by the concerned DR-TBC on referral of patients for ambulatory care in advance, by means of the RNTCP PMDT referral for treatment form via email or mobile phone;

- drugs provided to the patients to cover for transit period may be counted as unsupervised doses. However, as much as possible, efforts should be made by the district staff to restrict these transit doses;

- DTO arranges for availability of the monthly IP drug box to the treatment supporter (via the TU staff like STS/TBHV) and the patient records at the identified treatment support centre with timely information to the respective Medical officer peripheral health institution (MO-PHI);

- MO-PHI is responsible for supplying patient records and drugs to the designated treatment supporter. The MO-PHI will need to make suitable arrangements during intensive phase of the treatment for daily injections, including free sterile needles and syringes; and

- patient's information as per PMDT treatment book and CEM treatment review form in patient on newer drugs as detailed later must be regularly updated on Nikshay (at least weekly) by the concerned field staff responsible.

Overall responsibility of monitoring the patient's progress on treatment, including follow-up is with the MO-PHI where patient is being treated with support of the respective TU team.

## 7.10 Management of DR-TB patients with treatment interruptions and lost to follow-up

All efforts must be made to ensure that DR-TB patients do not interrupt treatment or are lost to follow-up. Action should be taken to promptly retrieve patients who fail to come for their daily dose by the treatment supporter as discussed in Chapter 11. The following strategies are applicable for patients who interrupt treatment:

***Patients in IP/CP who miss doses:*** All missed doses during IP must be completed prior to switching the patient to CP. Similarly, all missed doses during CP must be administered prior to ending treatment.

***Patients who interrupt treatment for less than one month during IP:*** When the patient returns to resume treatment, IP will be continued. However, the duration of treatment will be extended to complete IP. The follow-up cultures will be done as per the schedule.

***Patients who interrupt treatment for less than one month during CP*:** When the patient returns to resume treatment, the CP will be continued, however the duration of treatment will be extended to complete CP. The follow-up cultures will be done as per the schedule.

***Patients who are "lost to follow-up" (interrupt treatment continuously for one month or more) and return back for treatment*:** Such patients will be given an outcome of "lost to follow-up". The patient would be subjected to repeat CBNAAT & FL-SL LPA and LC as per the diagnostic algorithm to restart with appropriate DST guided regimen with or without newer drug for a fresh episode of treatment.

**MDR-TB patients who do not respond to the shorter MDR-TB regimen or who interrupt treatment:**

- patients on the shorter MDR-TB regimen who do not respond need to be assessed to decide whether they need to be switched to an appropriate DST guided DR-TB regimen;
- if there are signs of impending treatment failure (no sputum smear conversion by6 months or deterioration of clinical condition despite treatment) while the patient is on a shorter MDR-TB regimen, s/he should be considered for an appropriate DST guided DR-TB regimen;
- if patients interrupt shorter MDR-TB treatment continuously for one month or more, the outcome is lost to follow-up;
- if a patient has received the shorter MDR-TB regimen for more than one month and returns for treatment after an interruption of one month or more, s/he is not restarted

on a shorter MDR-TB regimen. Rather s/he is put on an appropriate DST guided DR-TB regimen. Patients need to be made aware of this; and

- if there are interruptions of less than one month (medical indication in the patient of adverse events, patient decision) then the shorter MDR-TB regimen can be continued and missed doses added to the rest of the treatment.

**DR-TB patients on Bdq containing regimen who interrupt treatment or are "lost to follow-up" or recurrent DR-TB** [4]

***BDQ:*** If a dose is missed during the first 2 weeks of treatment, one should not make up for the missed dose but continue the usual dosing schedule. From the third week onwards, if a 200 mg dose is missed, one should take the missed dose as soon as possible and then resume the three-times-a-week regimen.

**Patients who interrupt treatment during the first two weeks of Bdq course and returns to resume the treatment:**

- if interruption is up to 7 days, Bdq containing regimen will be continued to complete the doses and the duration of treatment will be extended to complete IP. Follow-up cultures will be done as per the revised schedule; and
- if interruption is more than 7 consecutive days, Bdq course will be reloaded (started afresh) and a fresh specimen collected for culture. The culture isolate must be stored for Bdq DST in future.

**Patients who interrupt treatment during 3-24 weeks of Bdq course and return to resume treatment**:

- if interruption is up to one month, Bdq containing regimen will be continued to complete the doses and duration of treatment will be extended to complete IP. Follow-up cultures will be done as per revised schedule; and
- if interruption is more than one month, Bdq will be permanently discontinued. Such patients will be given an outcome of "Lost to follow-up" (LTFU) based on duration of LTFU and managed as per DST guided treatment and registered afresh. A sputum specimen will be collected for culture. The culture isolate must be stored for Bdq DST in future. In addition, the serum sample will be collected and transported to the concerned lab within 6 hours for Bdq levels for correlation with outcomes, wherever feasible and lab capacity is available.

Where further treatment is concerned, if the patient has any indication of a treatment failure or recurrence, the NDR-TBC Committee will be contacted to discuss whether s/he should be retreated. The decision will be made on a case-to-case basis, using all available bacteriological and clinical data.

## 7.11 Transfers of DR-TB patients

It is important to note the patient's address (current residence), native place, occupation and place of work to get a fair idea about the possible places that s/he could move to. If a DR-TB patient on treatment decides to move and informs the health care worker, the patient can be transferred out to the district where s/he wishes to migrate. Transfer out should be brought to the notice of the DDR-TBC by the concerned DTO. Transfer module of Nikshay redirects a patient to the receiving district while referring district is still able to fetch the patient details through their district/TU login.

If the patient is migrating to an adjoining TB Unit being served by the same DDR-TBC as the current district of residence, then s/he may be shifted with 7 days of drugs for the transit period to a suitable treatment supporter. This would be at that place where s/he proposes to move, in consultation with the DTO of that district and under intimation of the DDR-TBC. The patient will continue treatment on the same PMDT TB number and same patient records, including a referral for treatment form and copies of the PMDT treatment book with a transfer note sent to the district receiving the patient. Further, details of the patient will be updated in the PMDT treatment register at the DR-TBC and on Nikshay.

If the patient is migrating to any other district that is not being served by the same DR-TBC,s/he may be formally transferred out with 7 days of drugs for transit period to a suitable treatment supporter at that place where s/he proposes to move. During transfer to other N/DDR-TBC, the patient should be referred along with the referral for treatment form, the PMDT treatment card and the PMDT treatment booklet. This would be in consultation with DTO of that district and under intimation of DDR-TBC. This patient will be registered at the DDR-TBC catering to the receiving district with a new PMDT TB number mentioning the old PMDT TB number in the remarks column for future reference. However, the Nikshay ID of the patient would remain the same and transfer details with the new PMDT TB number updated online. The patient will be continued on the same treatment on the new PMDT TB number. The patient records including the referral for treatment form, copies of PMDT treatment cards with a transfer note, copy of CEM treatment initiation and review forms as applicable, pre-treatment and follow-up investigations from the concerned DR-TBC transferring the patient, will be sent to the DDR-TBC and DTO of the district receiving the patient, by the DTO who initiated the transfer-out process.

The patient must be motivated to carry the PMDT treatment book to the receiving DDR-TBC. The details of the patient will be updated in the PMDT treatment register at both the DR-TBCs for future reference. It is the responsibility of the receiving DTO and DDR-TBC to send feedback about the patient with the new PMDT TB number to the former DDR-TBC to establish a link for future exchange of information about interim reports, culture conversion and treatment outcomes of the patient.

## 7.12 Managing referrals from private/other sectors of patients for DR-TB evaluation and treatment

Some patients with previous diagnosis of DR-TB and/or treatment with second-line anti-TB drugs will wish to avail RNTCP services. RNTCP has a policy against empirical treatment of DR-TB without microbiological confirmation. Microbiological confirmation is required before initiation of treatment for DR-TB. DST results from private laboratories will be considered acceptable under the following situations:

- CB-NAAT results from labs that regularly undertake annual calibration of machines; and
- C-DST labs who participate in the annual proficiency testing through NRLs under RNTCP for the respective DST technology.

For patients who do not have results in accordance to the above, DST would be offered under RNTCP as per the integrated DR-TB algorithm. Similarly, even though some patients may have consumed variable amounts of second-line anti-TB drugs, such prior anti-TB treatment is not likely to be uniformly reliable in quality of drugs, or quantity and duration consumed. Given that uncertainty, the basic principle is that duration of the regimen for DR-TB offered under RNTCP will not be reduced. There may be exceptional circumstances that the DR-TBCs may consider where prior treatment is very well-documented, adequate and effective. The DDR-TBC/NDR-TBC committee can exceptionally adjust the duration after detailed patient review, approval and documentation of decisions taken.

## 7.13 Access to PMDT treatment services including newer drugs for the patient seeking care in private/other sector

To expand access of PMDT services including newer drugs for patients seeking care in the private sector, the following mechanisms are applicable:

- interested and potential providers/institutions are expected to fulfill the requirement to serve as a NDR-TBC and establish an MoU with state as per the partnership guidelines to avail access to PMDT services including newer drugs. The collaboration will be implemented as per conditions detailed under the MoU; [15] and
- for providers not in a position to fulfill such requirement, their patients can still be provided access to PMDT services including newer drugs, through a mechanism that entails engaging with the programme. They can approach the concerned district TB Officer for linkage with the concerned DDR-TBC for access to standard regimen (H mono/poly DR-TB, RR/MDR-TB)or the concerned NDR-TBC for access to newer drugs and DST guided regimen as per guidelines. The provider needs to consult the concerned N/DDR-TBC with an understanding to be a responsible partner with the programme for complete patient care throughout the treatment course, including decisions around patient eligibility, treatment initiation, follow-up schedule, ADR monitoring and

management, treatment outcomes and long term follow-up as per programme guidelines. To locate the nearest DTO or NDR-TBC, providers can refer to the list of all NDR-TBC in India with contact details of the nodal physician available at www.tbcindia.gov.in. For assessing eligibility for specific regimens, all relevant medical records and reports of the patient should be shared with N/DDR-TBC to facilitate a decision on the regimen design that would suit the patient's condition.

Based on mutual understanding between patient, provider and N/DDR-TBC, initial hospitalization to patients will be offered at N/DDR-TBC or private institute with requisite monitoring. After discharge, the provider can continue to be the treatment supporter and monitor the patient in close coordination with N/DDR-TBC. The monthly drug supply and follow-up investigations need to be undertaken as per guidelines. Moreover, patients would be enrolled under RNTCP, recording and reporting the system for monitoring, including long term follow-up. The state/district TB officers should provide public health support to all notified TB patients seeking care from private/other sector for DSTs, pretreatment investigations, drugs, follow-up investigations, counselling and social support.

## 7.14 Palliative care

Palliative care is a multidisciplinary approach to medical care for people with serious illnesses. It focuses on providing patients with relief from symptoms, pain, physical and mental stress of a serious illness, whatever the diagnosis. WHO defines palliative care as an approach that improves quality of life of patients and their families facing the problem associated with life-threatening illness, through prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychosocial and spiritual. The goal of such therapy is to improve quality of life for both patient and family. [33]

**Need of palliative care in DR-TB**

While high cure rates of TB are being reported by majority of the programmes across the globe, in many countries DR-TB remains a life-threatening condition with high mortality and poor cure rates. There is significant suffering associated with DR-TB illness and its treatment. This kind of burden adds to the possibility of TB patients not being able to adhere to treatment and resultantly, the treatment failing to cure them. The life-threatening nature of DR-TB and the burden of disease management in terms of symptoms, adverse treatment effects, adherence, stigma and subsequent discrimination and social isolation, clearly show the need for care that addresses physical, social and emotional suffering by patients. Thus, the need for palliative and end-of-life care is being increasingly recognized as an important part of the continuum of care for DR-TB patients. Improvement in availability of diagnostic services, have led to increased detection of people with DR-TB. Therefore, the demand for treatment and need for palliative care has also grown.

**Challenges in palliative care**

Current TB treatment strategy is based on a patient-centered approach to treatment and care and international guidelines have identified practices resulting in better treatment outcomes. However, alleviation of the patient's suffering associated with disease and its management has been restricted mostly, to physical aspects and not adequately too. Difficulties faced by patients and families affected by life-threatening diseases span across physical, psychological, social and spiritual aspects. Neither trained health workers nor local community-based palliative care resources are usually available in the settings that are most in need. Although, clinical expertise in palliative care for patients who die in respiratory distress has developed considerably, individuals with DR-TB are yet to see the benefits.

**Services under palliative care for DR-TB**

Palliative care would be necessary for care of patients who are chronically ill, with extensive drug resistance, with extensive fibro-cavitary or disseminated bilateral lung disease, who have failed regimen for XDR-TB or mixed pattern resistance and for whom a WHO recommended regimen could not be designed even with new drugs. They would be required in some patients when there are symptoms or other suffering during the treatment process. All measures to relieve the patient of suffering caused by the disease and its treatment begins at the time of diagnosis and continues, regardless of whether or not s/he is expected to be cured of or will fail the treatment.

Services under palliative care include addressing pain and symptom control (including respiratory insufficiency), nutritional support, need for medical intervention after treatment cessation (including management of psychological morbidity), ensuring appropriate place of care, preventive care, infection control and end-of-life care.

**Supportive measures in palliative care**

The details on palliative care supportive measures are summarized below:

***Respiratory rehabilitation*:** Relief from dyspnoea with oxygen may be used to alleviate shortness of breath in some patients but there is no significant evidence to generalize its practice. Physiotherapy, evaluation for surgery, respiratory rehabilitation including yoga *etc.,* need to be considered in such patients. An example of pulmonary rehabilitation is placed at Annexure 11. Morphine provides significant relief from respiratory insufficiency and should be offered according to established clinical protocols available in the medical literature.

***Relief from pain and other symptoms*:** Paracetamol or Tramadol with paracetamol gives relief from moderate pain. If possible, stronger analgesics, including morphine, should be

used when appropriate to keep the patient adequately comfortable. The WHO has developed analgesic guides, pain scales and a three step "ladder" for pain relief.

***Infection control measures*:** The patient who is taken off anti-TB treatment because of failure often remains infectious. Infection control measures should be continued with reinforcement of administrative, environmental and personal measures, including N-95 mask use for caregivers.

***Nutritional support:*** Small meals as needed are often best for a person at the end of life. It should be accepted that the intake will reduce as the patient's condition deteriorates and during end-of-life care. Nausea and vomiting or any other conditions that interfere with nutritional support, should be treated. Bowel functions should be monitored for regularity.

***Regular medical visits*:** When DR-TB treatment stops, regular visits by healthcare providers and the support team should be continued to address medical needs and ensure that infection control practices are being followed. Early identification, periodic assessment and management of post treatment sequelae could be beneficial for the patient.

***Vocational rehabilitation:*** Wherever possible, based on the interest of the patient, an appropriate linkage for vocational rehabilitation and new skill learning opportunities through various NGOs may be explored to help the patient regain his/her source of livelihood and move towards socio-economic sufficiency. This would also have an indirect impact on improving the patient's nutritional, psychological and mental wellbeing.

***Continuation of ancillary medicines:*** All necessary ancillary medications should be continued as needed. Opioids help control cough, as well as pain. Other cough suppressants can be added. Bronchospasms can be controlled with a metered dosed inhaler with a spacer or mask. Depression and anxiety, if present, should be addressed. Antiemetic may still be needed and fever treated if the patient is uncomfortable. Appropriate use of alternative medicine may be considered through expert consultation.

***Preventive measures:*** Oral care, prevention of bedsores, bathing and prevention of muscle contractures are indicated in all patients. Regular scheduled movement of the bedridden patient is very important. Additionally, patients must be encouraged to move their bodies in bed if able. Keeping beds dry and clean are also important.

***Provide psychosocial support*:** Psychological counselling to the patient and family caregivers is critical at this stage, especially to assist patients in the planning of decisions related with the end-of-life stage. Also, emotional support, especially in settings in which strong stigma is attached to the disease would be necessary.

***Respect for patient's beliefs and values at the end-of-life*:** It is common for the patient and family caregivers to develop or increase their interest in spiritual and religious matters once they perceive that the end-of-life is approaching. Healthcare providers should respect those

beliefs and should not impose personal values and practices that prevent the patient to seek and find comfort in services delivered by faith-based organizations.

It incorporates management of side-effects such as breathlessness, fatigue, cachexia and end-of-life crises such as haemoptysis and acute respiratory failure besides anxiety of patients and their families, which typically accompanies these symptoms.

When patient isolation is done, strong measures to prevent loneliness, boredom and sense of abandonment are needed to be in place. These comprise of daily access to family and friends under proper infection control conditions, interaction with staff and access to activities according to the patient's condition (radio, television, hobbies, *etc.*,).

**Approach: Need for human resources and infrastructure**

Palliative care is provided by a team of physicians, nurses and other health professionals who work together with the primary care physician and referred specialists (or for patients who don't have those, hospital or hospice staff) to provide an extra layer of support. Hence, palliative care is to be initiated by those NDR-TBCs in various states. Further, NDR-TBC staff can counsel and train family members or caretakers of the patient, so that these services are extended as home-based palliative care to patients by family members or caretakers.

In rare circumstances, institution-based palliative care may be initiated with longer duration of admission at selected NDR-TBCs developed in old TB sanatoria. Alternately, states may identify interested NGO's or faith-based organizations with indoor facilities that could be engaged through an MoU and guided by NDR-TBCs. In all such facilities, airborne infection control measures as per national AIC guidelines must be strictly implemented. Further, as soon as the patient's condition improves, s/he must be discharged with adequate counselling to the family member or caretaker for home-based palliative care and regular consultative visits to NDR-TBC as and when medically required.

All health workers must receive training in palliative care to enable them extend support to family members or caretakers providing home-based palliative care and to undertake regular contact tracing and extend support to address their problems. Existing expertise from palliative care, HIV and respiratory medicine can, therefore, translate directly to TB. Delivery of palliative care from within respiratory clinical services by existing staff with additional training, with clear criteria for referral to palliative care specialists for complex patients, is to be established. NDR-TBCs should link up with local palliative care and hospice teams from the network of Pallium India and Indian Association of Palliative Care. However, it is of paramount importance from the infection control perspective to avoid sending infectious patients to these palliative care centres where immunocompromised cancer or stroke patients could potentially be infected.

Effective control of the various problems faced by patients and their families is possible across many settings (hospital, hospice, primary care and home-based care) and flexibility in place of delivery need to be established. Palliative and end-of-life care should be delivered to the patient and their family in the setting where they are receiving care, whether an inpatient, an outpatient, or at home. Community-based workers could, be trained in palliative care to scale-up existing health care delivery to include pain and symptom control. Having a patient die at home can be difficult for the family and the other way around. Home-based care should be offered to patients and families who want to keep the patient at home, whenever appropriate infection control practices can be followed. Institution-based end-of-life care should be available to those for whom home care is not feasible or desirable.

As far as possible, institution-based palliative care should be minimized to a duration that is absolutely essential as per decision of the concerned NDR-TBC committee. Most palliative care must be home-based through a trained and counselled family member or caretaker with regular visits by health care workers and psychosocial/spiritual support through local community based self-help groups, NGOs or panchayati raj institutions.

# Chapter 8: Monitoring and outcome definitions

This chapter provides information on the clinical and laboratory monitoring for patients on treatment for M/XDR-TB. It provides the treatment outcome definitions to be used. [4] [5] [10] [19]

## 8.1 Clinical monitoring

DR-TB patients should be seen by a medical officer trained in RNTCP PMDT guidelines for clinical evaluation after discharge back to the N/DDR-TBC, at monthly intervals during the IP and at 3-monthly intervals during CP until the end of treatment. The responsible medical officer at N/DDR-TBC should assess clinical, microbiologic and radiologic response to treatment, measure weight, assess possible adverse reactions and encourage the patient to continue treatment. The follow-up visit should result in updating of treatment cards. Close monitoring of patients is necessary to ensure that adverse effects are recognized early by the treatment supporter. Patients should be encouraged to volunteer if they experience any adverse effects, though patients should not be asked any leading question to elicit any adverse reaction. However, if the patient makes any complaint, s/he should be interrogated in detail and necessary action taken.

The treatment supporter should be trained to recognize adverse reactions like nausea, vomiting, diarrhoea, skin rash, loss of hearing, reduced sensation, psychiatric symptoms and jaundice. Training should also be provided on the management of minor reactions and when the patients should be referred to the medical officer. Severe adverse reactions should be referred to an appropriate clinical facility, which may include D/NDR-TBC coordinating care for the patient. Other relevant investigations may be done as and when clinically indicated. These investigations can be done at D/NDR-TBC or any institute as per local arrangement, however patients should not be charged for these investigations. Some patients may need to be hospitalized during treatment for medical or psychosocial reasons.

## 8.2 Follow-up evaluations during treatment

The follow-up evaluation schedule during treatment for DR-TB patients managed with various regimen classes are summarized in the table 8.1 below:

Table 8.1 Follow-up evaluation schedule of DR-TB patients during treatment on various regimen classes

| Regimen class | Regimen for H Mono /Poly DRTB &with FQ/SLI/Lzd resistance | Shorter MDR-TB regimen | Conventional MDR-TB regimen | Regimen for RR-TB with resistance to FQ/SLI ± Lzd (without newer drugs) | Newer drugs containing regimen for RR-TB with resistance to FQ/SLI ± Lzd |
|---|---|---|---|---|---|
| Duration | 9 – 12 months (3-6m IP, 6m CP) | 9 – 11 months (4-6m IP, 5m CP) | 24-27 months (6-9m IP, 18m CP) | 24-27 months (6-9m IP, 18m CP) 24-30 months in XDR & MPR (6-12m IP, 18m CP) | |
| Clinical + Wt. | As suggested by treating clinician, at least monthly in IP and quarterly in CP | | | | |
| Smear mMicroscopy | Monthly till end of IP, monthly in extended IP only if previous month S+ve | | With culture at C-DST labs | | |
| Culture | End of IP, end of extended IP **&** end of treatment | | Monthly from 3m to end of IP if converted, Monthly in extended IP only if previous month culture +ve, quarterly in CP, 2 consecutive monthly if any culture +ve from 12m onwards | | |
| DST | SL-LPA if S+ve/C+ve at end of IP **&/or** extended IP **&** expanded DST if any resistance on SL- LPA | | SL-LPA if C+ve at end of IP **&/or** extended IP **or** any time in CP **&** expanded DST if any resistance on SL-LPA | | |
| S. Creatinine | Monthly till 3m, then every 3m till SLI course is completed | | | | |
| Audiometry | As and when clinically indicated till SLI course is completed | | | | |
| UPT | As and when clinically indicated | | | | |
| CBC/Hb/platelets* | As and when clinically indicated | | | Monthly in IP, quarterly in CP | |
| CXR, TSH & LFT# | As and when clinically indicated | At end of IP, as and when clinically indicated CXR also at end of treatment | | | |
| ECG$ | As and when clinically indicated | At 2 wks, monthly in IP, as and when clinically indicated | As and when clinically indicated | At 2 wks, monthly in IP, as and when clinically indicated | |
| S. Electrolytes (Na, K, Cl) | As and when clinically indicated | | | Quarterly in IP, as and when required | |
| S. Mg, Ca, Amylase, Proteins, Lipase | As and when clinically indicated | | | | Quarterly in IP, as and when required |
| Specialist cons. | As and when clinically indicated | | | | |
| Long term follow-up | at 6, 12, 18 and 24 months after completion of treatment and if found symptomatic, clinical evaluation, CXR, smear and culture. DST would be repeated if culture is positive. | | | | |

*CBC/Hb/Platelets done to rule out bone marrow suppression and anemia onlyif Linezolid is included in the regimen
# HBsAg and other viral markers (Hepatitis A, C & E) to be done on signs of jaundice during treatment
$ In patients with baseline ECG abnormalities, ECGs must be done on daily basis for the first 15 days if patient is managed with regimen containing Mfx[h], Bdq, Cfz and further frequency as advised by cardiologist. Repeat ECG after an hour if abnormal at any time to reconfirm with long lead II for one minute.

The most important evidence of response to DR-TB treatment is conversion of sputum smear and culture to negative. Good quality sputum specimen is therefore essential to get reliable results that form the basis of monitoring bacteriological response to treatment.

Smear examination would be used on a monthly basis in IP to guide the decision on moving from IP to CP only in regimen with less than one year of duration *i.e.,* H mono/poly DR-TB regimen (3-6m IP) and shorter MDR-TB regimen (4-6m IP). It would be continued on monthly basis if IP needs to be extended if the previous month smear is positive up to a maximum of six months in both regimens. Follow-up culture will be done at the end of IP, end of extended IP and end of treatment.

If smear/culture remains positive at the end of IP and/or extended IP, a fresh specimen/culture isolate of that time will be subjected to SL-LPA to check for amplification of resistance to second-line drugs. If the patient is found to be susceptible to both FQ and SLI class at fourth month of treatment, the intensive phase will be extended on monthly basis up to a maximum of six months. At end of extended IP or later, if any resistance is detected by SL-LPA **OR** if found to be culture positive, the patient will be declared as treatment failure. The patient is then re-evaluated for a newer drug containing **OR** DST guided regimen at the nodal DR-TB centre. However, the patient will be continued on CP till declared as treatment failure as detailed above.

Extended DST to Mfx (2.0), Km, Cm and Lzd will be set up on liquid culture on the LPA deposits only for patients who are found resistant to FQ or SLI class on SL-LPA if patient remains smear/culture positive at the end of IP and/or the extended IP during treatment with any DR-TB regimen. This would be done if any culture is positive any time in CP in regimen of longer duration. Long term follow-up will be done with six monthly cultures among symptomatic patients till two years after completion of any DR-TB regimen.

Smear conversion is less reliable than culture conversion, which reflects viability of tubercle bacilli even in very low bacilli per ml of sputum and is a more accurate reflection of response to treatment. Hence, for all patients put on longer term regimen, culture will be done as per table 8.1 to decide on moving from IP to CP. Patients will be considered smear converted after having two consecutive negative smears taken at least one month apart. Patients will be considered culture converted after having two consecutive negative cultures taken at least one month apart.

Time-to-culture conversion is calculated as the interval between the date of DR-TB treatment initiation and date of the first of these two negative consecutive cultures (date of sputum specimens collected for culture should be used). Arrangements can be made to collect sputum specimens at the respective DMC which will then be transported to the RNTCP-certified culture and DST laboratory, along with intimation of DTC. Necessary

arrangements for supply of conical tubes for follow-up sputum smear and culture examination should be ensured.

Follow-up sputum culture should be done using liquid culture. If liquid culture is not available, it is to be communicated to Central TB Division. In case of extension of IP, the follow-up culture months will shift by every month of extension of IP till maximum limit of IP for all regimen classes. However, it must be noted that the final treatment outcome of all DR-TB patients will be declared on the basis of follow-up culture results only.

## 8.3 Treatment outcomes

The progress of the patient on treatment would be monitored using interim as well as final treatment outcomes for various DR-TB regimens.

### 8.3.1 Interim outcomes

***Culture conversion***: Patient is considered to have culture converted when two consecutive cultures, taken at least 30 days apart, are found to be negative. In such a case, the specimen collection date of the first negative culture is used as the date of conversion.

***Culture reversion***: Patient is considered to have culture reverted when, after an initial culture conversion, two consecutive cultures, taken at least 30 days apart, are found to be positive. For the purpose of defining Treatment failed, reversion is considered only when it occurs in the continuation phase.

The term microbiological conversion/reversion can also be used either for smear or culture based follow ups.

### 8.3.2 Final outcomes

The final treatment outcome definitions would vary with the duration of treatment. The treatment outcomes are defined for patients on longer term and shorter-term regimen in following section.

**Outcomes for regimen for RR-/MDR-TB (except shorter MDR-TB regimen) and/or XDR-TB patients**

***Cure:*** Treatment completed as recommended by the national policy without evidence of failure and three or more consecutive cultures taken at least 30 days apart during CP are negative including culture at the end of treatment.

***Treatment completed*:** Treatment completed as recommended by the national policy without evidence of failure but no record that three or more consecutive cultures taken at least 30 days apart are negative after the intensive phase.

***Treatment failed*:** Treatment terminated or need for permanent regimen change of at least two or more anti-TB drugs in CP because of lack of microbiological conversion by the end of the extended intensive phase or microbiological reversion in the continuation phase after conversion to negative or evidence of additional acquired resistance to FQ or SLI drugs or adverse drug reactions (ADR).

***Died:*** A patient who dies for any reason during the course of treatment.

***Lost to follow-up*:** A patient whose treatment was interrupted for one month or more for any reasons prior to being declared as failed.

***Not evaluated*:** A patient for whom no treatment outcome is assigned.

***Regimen changed:*** A TB patient's need for permanent regimen change of at least one or more anti-TB drugs prior to being declared as failed.

**Outcomes for H mono/ poly DR-TB patients and RR-TB patients put on shorter MDR-TB regimen**

***Cure:*** Treatment completed as recommended by the national policy without evidence of failure and culture was negative at end of treatment and on at least one previous occasion.

***Treatment completed*:** A patient who has completed treatment according to guidelines but does not meet the definition for cure or treatment failure due to lack of microbiological results.

***Failure:*** Treatment terminated or need for permanent regimen change of at least two or more anti-TB drugs in CP because of lack of microbiological conversion by end of extended intensive phase or microbiological reversion in the continuation phase after conversion to negative or evidence of additional acquired resistance to FQ or SLI drugs or adverse drug reactions (ADR).

***Died:*** A patient who dies for any reason during the course of treatment.

***Lost to follow-up*:** A patient whose treatment was interrupted for one month or more for any reasons prior to being declared as failed.

***Not evaluated*:** A DR-TB patient for whom no treatment outcome is assigned, this includes former "transfer-out".

***Regimen changed*:** A TB patient's need for permanent regimen change of at least one or more anti-TB drugs prior to being declared as failed.

Treatment outcome is defined by reviewing patient PMDT treatment card. The treatment outcome and date the patient stopped treatment is written in the appropriate column in the TB treatment card. The date on which the patient stopped treatment is the date of the last dose of drugs taken. Details of treatment outcome should be updated in Nikshay. The MO of the PHI should record the treatment outcome in the treatment card and sign it. The treatment card of the patients whose outcome has been declared should be handed over to the STS during his routine monthly visits. Every patient started on treatment has to be given one and only one treatment outcome.

# Chapter 9: Treatment in special situations

Compared to drug sensitive TB, DR-TB is more demanding in terms of cost of treatment, duration of treatment, higher adverse reactions to second line drugs, resources required by treatment providers and prolonged adherence required by patients. To add to these issues, certain associated special situations make the treatment of DR-TB more difficult. [12] [31] This chapter outlines the management of DR-TB in certain special situations and conditions. These include DR-TB in pregnancy and in children; those co-infected with HIV infection; requiring surgery; in patients with renal impairment; in patients with pre-existing liver disease; seizure disorders; psychiatric illnesses; in extra-pulmonary TB patients; and management of contacts of DR-TB.

## 9.1 DR-TB in pregnancy

Pregnancy is not a contraindication for treatment of active drug-resistant TB, but poses great risk to both the mother and foetus. There is lack of experience in treating pregnant women with DR-TB. Teratogenicity has been demonstrated with only some of the drugs used to treat DR-TB. However, majority of the studies have demonstrated that it is common during the first trimester. It is prudent to solicit the opinion of an experienced gynecologist/obstetrician while treating such patients.

All women of childbearing age who are receiving DR-TB therapy should be advised to use birth control measures because of the potential risk to both mother and foetus. It should be remembered that oral contraceptives might have decreased efficacy due to vomiting and drug interactions with DR-TB drugs. Thus, for prevention of pregnancy, use of barrier methods (condoms/diaphragms), intrauterine devices or IUDs (CuT) or depot-medroxyprogesterone (Depo-provera) are recommended, based on individual preference and eligibility.

All women of childbearing age identified as presumptive DR-TB should be advised to use a reliable and appropriate contraceptive method till results of culture and DST are available.

All female DR-TB patients of childbearing age should be counselled intensively in relation to use of contraceptive methods. All women of childbearing age should be tested for pregnancy as part of the pretreatment evaluation and whilst on treatment. DR-TB patients found to be pregnant prior to treatment initiation or whilst on treatment are evaluated in consultation with a gynecologist or obstetrician, taking into account factors such as risks and benefits of DR-TB treatment; severity of DR-TB; gestational age; and potential risk to foetus. Further management of DR-TB patients who are pregnant prior to initiation of treatment or whilst on treatment are based on duration of pregnancy.

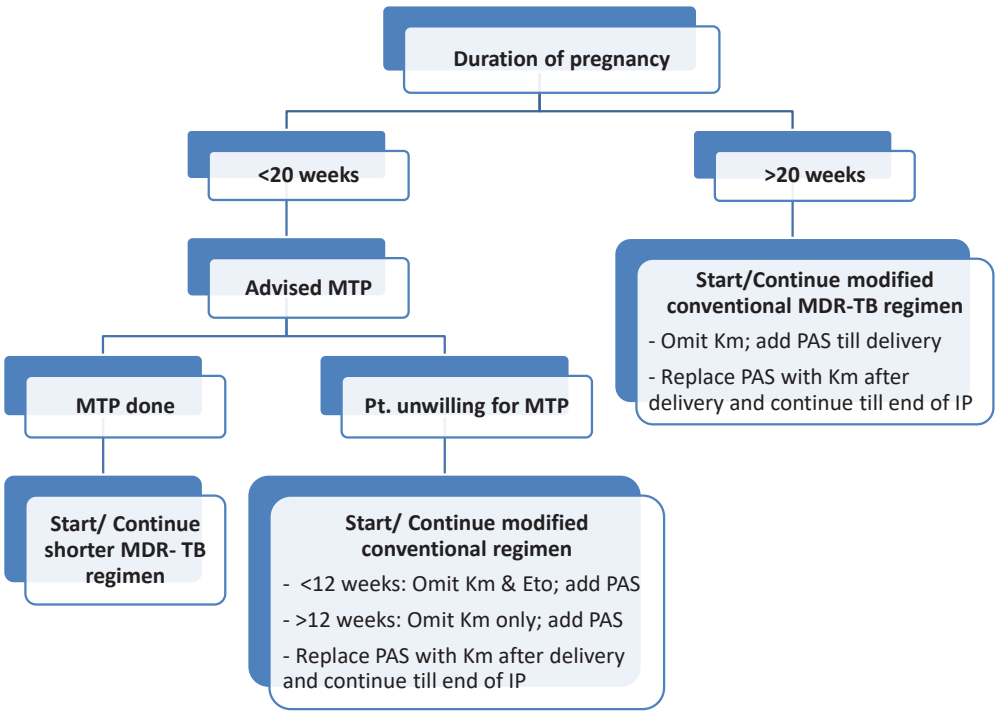
In pregnant women diagnosed with DR-TB, if the duration of pregnancy is <20 weeks, the patient should be advised to opt for a medical termination of pregnancy (MTP) in view of the potential severe risk to both mother and foetus. If the patient is willing, she should be referred to a gynecologist or obstetrician for a medically terminated pregnancy (MTP) following which a shorter MDR-TB regimen can be initiated (if the patient has not started treatment) or continued (if the patient is already on treatment) by the DR-TBC Committee.

For patients who are unwilling for MTP or have pregnancy of >20 weeks (making them ineligible for MTP), the risk to mother and foetus needs to be explained clearly and a modified conventional MDR-TB regimen started or continued as detailed below:

- for patients in the first trimester (≤ 12 weeks), Km and Eto are omitted from the regimen and PAS is added; and
- for patients who have completed the first trimester (>12 weeks), Km is replaced with PAS. Postpartum, PAS may be replaced with Km and continued until the end of the intensive phase.

In women of reproductive age who have been initiated on shorter MDR-TB regimen and who become pregnant in spite of precautions and use of contraceptives, the risk to the mother and foetus needs to be explained clearly. If the pregnancy is $\underline{\leq}$ 20 weeks, the decision on continuing shorter MDR-TB regimen would depend upon the willingness of the patient to opt for an MTP. If she is unwilling for MTP or has a pregnancy > 20 weeks duration, she needs to be shifted to a modified conventional MDR-TB regimen. Similarly, Km must be replaced with PAS in pregnant women considered for initiation or continuation of the regimen for H mono/poly DR-TB. The management of DR-TB patients with pregnancy is summarized in the flow chart in Figure 9.1 below:

## Figure 9.1: Management of DR-TB patients with pregnancy




Pregnant DR-TB patients need to be monitored carefully, both in relation to the treatment and progress of the pregnancy. This approach should lead to good results, since the patient should be smear-negative at the time of parturition and mother and infant do not need to be separated. Breastfeeding should be encouraged as long as the patient is sputum negative.

Avoid aminoglycosides as it is particularly toxic to the developing fetal ear. Moreover, Eto can increase the risk of nausea and vomiting associated with pregnancy, more so since teratogenic effects have been observed in animal studies. If injectable agents, Eto/Pto, or other drugs were withheld because of the pregnancy, they can be added back postpartum to make a more complete regimen.

For patients with mono- and poly-resistant TB but who are susceptible to R, note that the use of R interacts with the contraceptive drugs, resulting in decreased efficacy of protection against pregnancy. A woman on oral contraception while receiving R treatment may choose between two options following consultation with a physician, namely:

- use of an oral contraceptive pill containing a higher dose of estrogen (50 µg); and
- use of another form of contraception.

## 9.2 DR-TB in children

**Principles of treatment of drug-resistant TB in children**[30][12]

- always be treated in consultation with an expert;
- include at least 4-6 bactericidal medications to which the strain is known or likely to be susceptible;
- do not add a single drug to a failing regimen;
- ensure treatment is given for at least 12 months after M. tuberculosis cultures have converted to negative; and
- extend treatment to 24 months in case of HIV infection or cavitatory lesions.

**Building a treatment regimen**

The principles of designing a WHO recommended regimen detailed in the previous chapter also applies to children. However, newer drugs like Bdq are not yet recommended in children as the evidence is yet to evolve. However, they can be offered the shorter MDR-TB regimen. Thus, children would be managed with regimen designs without newer drugs as in Table 7.3 and 7.4, depending on the DST pattern and other parameters for mixed pattern DR-TB.

WHO recommends that in children with less severe forms of DR-TB, a regimen free of SLI may be designed keeping in view the aforementioned principles. [12] Thus, in such patients, the regimen design suggested in Table 7.3 and 7.4 could be applied if the child is detected as RR-TB or H mono/poly DR-TB by replacing Km with a suitable second-line drug.

The dosages for drugs used in various DR-TB regimens by weight bands for paediatric DR-TB patients are enumerated in Annexure 10. [31] The monitoring of DR-TB treatment in children would be as per the suggested schedule by regimen type in Table 8.1. However, the paediatrician of the N/DDR-TBC must regularly evaluate the progress of the child on treatment and initiate any other investigations as suggested.

## 9.3 DR-TB with HIV co-infection

The presentation of DR-TB in the HIV-infected patient does not differ from that of drug sensitive tuberculosis in the HIV-infected patient. However, the diagnosis of TB in HIV positive persons can be more difficult and may be confused with other pulmonary or systemic infections. As the HIV disease progresses and the individual becomes more immune compromised, the clinical presentation is proportionately more likely to be EP or smear-negative than in HIV-uninfected TB patients. This can result in misdiagnosis or delays in diagnosis and in turn, higher morbidity and mortality.

With the nationwide scale-up of intensified TB-HIV package, it is expected that more and more numbers of TB patients know their HIV status and if found HIV positive, must be linked to ART centres and provided co-trimoxazole preventive therapy (CPT). Early diagnosis of drug-resistant TB and HIV, prompt initiation of appropriate second-line anti-TB drugs and ART, sound patient support and infection control measures are essential components in the management of drug-resistant TB in PLHIV.

**Diagnosis of DR-TB among PLHIV**

Four symptom (4S) screening is carried out regularly during each visit among PLHIV persons approaching ART centres. The RNTCP guidelines are offering CBNAAT testing for all presumptive TB among PLHIV patients for early diagnosis of TB and DR-TB. The other possibility of identifying DR-TB PLHIV patient is HIV reactive patient found amongst the established DR-TB patient. Assessment of HIV status is one of the important TB/HIV collaborative activities, thus, majority of DR-TB patients know their HIV status and many would have been diagnosed with the help of a molecular test.

The treatment of HIV positive individuals with DR-TB is the same as for HIV negative patients. However, treatment is more difficult and adverse events more common. Deaths during treatment, partly due to TB itself and partly due to other HIV-related diseases, are more frequent in HIV-infected patients, particularly in the advanced stages of immunodeficiency. Due to increased frequency of adverse drug events, rigorous monitoring in this particular group of patients is required in order to ensure adherence to treatment, early identification and treatment of adverse events and to reduce default.

**Initiating ART in patients with DR- TB**

The use of ART in HIV infected patients with TB improves survival for both drug-resistant and susceptible disease. However, HIV infected DR-TB patients without the benefit of ART may experience mortality rates exceeding 90%. The likelihood of adverse effects could compromise the treatment of HIV or DR-TB if both treatments are started simultaneously. Second-line anti-TB drugs should be initiated first, followed by ART as soon as second-line anti-TB drugs are tolerated. Generally this should be within the first two weeks of initiating DR-TB treatment. On the other hand, undue delay in starting ART could result in significant risk of HIV related death amongst DR-TB patients. Co-trimoxazole can be provided to all patients with HIV as per WHO recommendation.

Based on the WHO Guidelines on ART for HIV infection in adults and adolescents; irrespective of CD4 cell counts, patients co-infected with HIV and TB should be started on ART as soon as possible after initiating TB treatment. The ART should be initiated as soon as possible in all HIV/TB co-infected patients with active TB (within 8 weeks after the start of TB treatment). However, if the CD4 cell count is below 50 cell/cmm, start ART

simultaneously with ATT, with strict clinical and laboratory monitoring. All new co-infected patients should be initiated on FDC of Tenofovir, Lamivudine and Efavirenz (TLE) single pill-based first-line regimen to be taken at bedtime irrespective of Hb level/ CD4 count as per ART guidelines laid down by the National AIDS Control Organization (NACO).

Patients who are already on ART at the time of DR-TB diagnosis can be continued on ART when DR-TB therapy is initiated. Occasionally, patients with HIV related TB may experience a temporary exacerbation of symptoms, signs or radiographic manifestation of TB after beginning TB treatment. This paradoxical reaction occurs in HIV infected patients with active TB and is thought to be a result of immune restitution due to the simultaneous administration of ART and tuberculosis medication (IRIS syndrome). Symptoms and signs may include high fever, lymphadenopathy, expanding intrathoracic lesions and worsening of chest radiographic findings. The diagnosis of a paradoxical reaction should be made only after a thorough evaluation has excluded other etiologies, particularly TB treatment failure. For severe paradoxical reactions prednisone (1-2 mg/kg for 1-2 weeks, then gradually decreasing doses) may be used.

If the patient is already on a sensitive TB drug regimen, s/he should be switched over to an appropriate DR-TB regimen by closing the previous treatment with a 'regimen change' outcome. Assessments of therapeutic response to both infections should be carried out regularly including close monitoring of potential adverse effects including psychiatric assessments and nutritional status. Patients with drug-resistant TB and HIV may suffer from severe wasting, diarrheal diseases and malabsorption syndromes. Wherever possible, patients with drug-resistant TB living with HIV should be offered socioeconomic and nutritional support. The programme also monitors treatment outcomes separately for HIV TB patients. Considering the risk of developing primary DR-TB among susceptible close contacts, effective TB infection control measures are mandatory. If the patient shows signs of TB treatment failure, further evaluation is warranted. In addition, the ART regimen should be evaluated for possible treatment failure as described in other WHO guidelines.

***Concomitant therapy with ART or other medicines***: Rifampicin (R) is not used in DR-TB treatment; they are used in the treatment of R-sensitive poly- and mono-resistant TB, which has drug interaction with Protease Inhibitor (PI). Newer drug like Bdq are metabolized by the CYP3A4 having multiple drug interactions with protease inhibitors and non-nucleoside reverse-transcriptase inhibitors (NNRTI). This has been detailed in section 7.6.

## 9.4 Role of surgery in management of DR-TB

In DR-TB patients with localized disease, surgery, as an adjunct to chemotherapy, can improve outcomes provided skilled thoracic surgeons and excellent postoperative care is available. When unilateral resectable disease is present, surgery should be considered in the following cases:

- absence of clinical or bacteriological response to chemotherapy despite six to nine months of treatment with effective anti-TB drugs;
- high risk of failure or relapse due to high degree of resistance or extensive parenchymal involvement;
- morbid complications of parenchymal disease e.g. haemoptysis, bronchiectasis, broncho-pleural fistula, or empyema;
- recurrence of positive culture status during course of treatment; and
- relapse after completion of anti-TB treatment.

WHO has recommended surgical procedures like wedge resections or lobectomy in patients with localized lesions. If surgical option is under consideration, at least six to nine months of chemotherapy is recommended prior to surgery to ensure culture conversion. [12] Linkages may be established with existing institutions of excellence where patients can be referred for expert opinion and decisions taken to provide surgical options after a detailed review of the patient. States need to identify institutes with capacity to conduct thoracic surgery and link up DR-TBCs to such institutes with support to the patient to cover costs involved for surgery through innovative mechanisms.

## 9.5 DR-TB in patients with renal impairment

Renal insufficiency due to longstanding TB disease itself, previous use of aminoglycosides or concurrent renal disease is not uncommon. Great care should be taken in the administration of second-line drugs in patients with renal impairment. Care has to be taken to see if DR-TB patients require aminoglycosides for 6 months or more. Other drugs, which might require a dose or interval adjustment in the presence of mild to moderate renal impairment, are E, FQ, Cs and PAS. In the presence of severe renal impairment, many other drugs may require adjustments (Table 9.3).

In DR-TB patients, blood urea and serum creatinine should be monitored prior to treatment initiation, monthly for 3 months after treatment initiation and then every three months whilst injection Km is being administered. In patients with mild renal impairment, the dose of aminoglycosides may be reduced. In the presence of severe renal failure, the aminoglycoside therapy should be discontinued and replaced with other potent non-nephrotoxic anti-TB drugs.

Table 9.3 Dose adjustment of anti-TB drugs in presence of renal impairment[a]

| Drug | Recommended dose and frequency for patients with creatinine clearance < 30 ml/min or for patients receiving haemodialysis (unless otherwise indicated dose after dialysis) |
|---|---|
| Isoniazid | No adjustment necessary |
| Rifampicin | No adjustment necessary |
| Pyrazinamide | 25-35 mg/kg per dose three times per week (not daily) |
| Ethambutol | 15-25 mg/kg per dose three times per week (not daily) |
| Rifabutin | Normal dose can be used, if possible monitor drug concentrations to avoid toxicity. |
| Rifapentine | No adjustment necessary |
| Streptomycin | 12-15 mg/kg per dose two or three times per week (not daily) |
| Capreomycin | 12-15 mg/kg per dose two or three times per week (not daily) |
| Kanamycin | 12-15 mg/kg per dose two or three times per week (not daily) |
| Amikacin | 12-15 mg/kg per dose two or three times per week (not daily) |
| Ofloxacin | 600-800 mg per dose three times per week (not daily) |
| Levofloxacin | 750-1000 mg per dose three times per week (not daily) |
| Moxifloxacin | No adjustment necessary |
| Cycloserine | 250 mg once daily, or 500 mg / dose three times per week[c] |
| Terizidone | Recommendations not available |
| Prothinamide | No adjustment necessary |
| Ethionamide | No adjustment necessary |
| PAS | 4 g/dose, twice daily maximum dose |
| Bedaquiline | No dosage adjustments required in patients with mild to moderate renal impairment (dosing not established in severe renal impairment, use with caution) |
| Linezolid | No adjustment necessary |
| Clofazimine | No adjustment necessary |
| Amoxicillin/ clavulanate | For creatinine clearance 10-30 ml/min dose 1000 mg as amoxicillin component twice daily;<br>For creatinine clearance <10 ml/min dose 1000 mg as amoxicillin component once daily |
| Imipenem / cilastin | For creatinine clearance 20-40 ml/min dose 500 mg every 8 hours;<br>For creatinine clearance <20 ml/min dose 500 mg every 12 hours |
| Meropenem | For creatinine clearance 20-40 ml/min dose 750 mg every 12 hours;<br>For creatinine clearance <20 ml/min dose 500 mg every 12 hours |

*[a]Companion Handbook to the WHO guidelines for the programmatic management of drug-resistant tuberculosis 2014. [30]*
*Estimated creatinine clearance calculations:*
*Men: Ideal Body Weight (kg) X (140 – age) / 72 X serum creatinine (mg/dl)*
*Women: 0 85 X Ideal Body Weight (kg) X (140 – age) / 72 X serum creatinine (mg/dl))*

## 9.6 DR-TB in patients with pre-existing liver disease

In various DR-TB regimen under RNTCP, Z, PAS and Eto are potentially hepatotoxic drugs. Hepatitis occurs rarely with the FQ. The potential for hepatotoxicity is increased in the elderly, alcoholics and in patients with pre-existing liver disease. In general, most second-line drugs can be safely used in the presence of mild hepatic impairment, as they are relatively less hepatotoxic than first-line drugs. However Z should be avoided in such patients.

Once a patient on second-line drugs develops hepatitis, other etiologies should be excluded such as viral hepatitis, alcoholic hepatitis, drug induced hepatitis by non-TB drugs *etc.,* Further management should be on the same line as in non- DR-TB patients.

DR-TB patients having deranged liver function test (LFT) during pretreatment evaluation should be strictly monitored through monthly LFTs while on treatment. However, routine LFT is not recommended in all patients.

## 9.7 DR-TB in patients with seizure disorders

Some patients requiring treatment for DR-TB will have a past or present medical history of a seizure disorder. The first step in evaluating such patients is to determine whether the seizure disorder is under control and whether the patient is taking anti-seizure medication to control the disorder. If the seizures are not under control, initiation or adjustment of anti-seizure medications will be needed prior to the start of DR-TB therapy. In addition, if other underlying conditions or causes for seizures exist, they should be corrected.

Among second-line drugs, Cs, Eto and FQ have been associated with seizures and hence should be used carefully amongst DR-TB patients with history of seizures. Pyridoxine should be given with Cs to prevent seizures. Cs should, however be avoided in patients with active seizure disorders that are not well controlled with medication. In patients where no other drug is appropriate, Cs can be given and the anti-seizure medication adjusted as needed to control the seizure disorder. The risk and benefits of using Cs should be discussed with the patient and the decision on whether to use Cs are made together with the patient.

Antiepileptic drugs may have drug interactions with Cs and FQ. Hence close monitoring of serum levels of anti-epileptic drugs should be done. One should remember that TB might itself involve the central nervous system and may cause seizures. However, when seizures present for the first time during anti-TB therapy, they are likely to be the result of an adverse effect of one of the anti-TB drugs.

## 9.8 DR-TB in patients with psychosis

For DR-TB patients with a concurrent psychiatric illness, it is advisable to have an evaluation carried out by a psychiatrist before the start of treatment for DR-TB. The initial evaluation documents any pre-existing psychiatric condition and establishes a baseline for comparison if new psychiatric symptoms develop while the patient is on treatment. Any identified psychiatric illness at the start or during treatment should be fully addressed. There is a high baseline incidence of depression and anxiety in patients with DR-TB, often connected with the chronicity and socioeconomic stress factors related to the disease. If a health care worker with psychiatric training is not available, the treating healthcare provider should document any psychiatric conditions the patient may have at the initial evaluation.

Treatment with psychiatric medication, individual counselling and/or group therapy may be necessary to manage the patient suffering from a psychiatric condition or adverse psychiatric effect due to medication. Group therapy has been very successful in providing a supportive environment for DR-TB patients and may be helpful for patients with or without psychiatric conditions (adequate measures to prevent infection risk should be in place for the group therapy).

FQ and Eto have been associated with psychosis. Pyridoxine prophylaxis may minimize the risk of neurologic and psychiatric adverse reactions. Cs may cause severe psychosis and depression leading to suicidal tendencies. However, the use of Cs is not absolutely contraindicated for the psychiatric patient. Adverse effects of Cs may be more prevalent in the psychiatric patient but the benefits of using this drug often outweighs the potential higher risk of adverse effects. Close monitoring is recommended if Cs is used in patients with psychiatric disorders. If patient on Cs therapy develops psychosis, anti-psychotic treatment should be started and Cs therapy should be temporarily suspended. Once symptoms resolve and patient is stabilized, Cs therapy may be resumed. Such patients may require antipsychotic treatment till anti-TB treatment is completed. When any patient on DR-TB treatment develops psychosis, other etiologies such as psychosocial stresses, depression, hypothyroidism, illicit drug and alcohol use, should also be looked for.

All health care workers treating drug-resistant TB should closely work with a psychiatrist and have an organized system for psychiatric emergencies. Psychiatric emergencies include psychosis, suicidal ideation and any situation involving the patients being a danger to him/herself or others. Mechanisms to deal with psychiatric emergencies (often inpatient psychiatric hospital admissions) should be available 24X7. Proper infection control measures must be taken for the smear-positive patient who requires hospitalization.

## 9.9 Management of DR-TB in extra pulmonary TB patients

Management of bacteriologically confirmed EP DR-TB patients will be considered by the programme provided the diagnosis is made by an RNTCP C-DST laboratory. Treatment regimen and schedule for EP DR-TB patients will remain the same as for pulmonary DR-TB. Patients must be registered in the PMDT register and the treatment outcome of treatment completed will be considered.

**Investigations and pretreatment evaluation**

Patients would be sent to the DR-TBC for pretreatment evaluation and treatment initiation. EP DR-TB patients will undergo all those pretreatment investigations as done for pulmonary DR-TB patients as part of pretreatment evaluation prior to initiating regimen for DR-TB.

In addition, ultrasound of abdomen of the patient will also be done, if necessary, to rule out involvement of other organs and abdominal nodes. If required, additional imaging investigation will be done to rule out any other conditions.

**Initiation of treatment**

After pretreatment evaluation, treatment for EPDR-TB should be initiated based on weight of the patient. Treatment regimen, weight band and schedule for EP DR-TB patients will remain the same as for pulmonary DR-TB.

**Monitoring progress during treatment and follow-up**

Clinical monitoring is the most important criteria for the follow-up of patients with EP DR-TB. Regular patient monitoring and periodic follow-up of nodes and other EP symptoms with culture from the discharging node/sinus is the key in monitoring of treatment in EP lymph nodal DR-TB.

***Bacteriological monitoring***: Two specimens from the discharging sinus/pus in the lymph node should be collected, one each for smear and culture. The specimen should be taken at the end of the third month of treatment and then every month (at least 30 days apart) in IP till there is pus/discharge from sinus (in the node). Unlike sputum smear and culture, culture from the node can be given only till the pus/discharging sinus is present from the node. The follow-up is mainly based on clinical parametres.

***Clinical monitoring:*** This is important in case of EPDR-TB. Monitoring and follow-up can be done clinically based on the following:

- weight gain;
- decrease or increase in symptoms (healing of ulcer/scrofuloderma);

- increase or regression in size of nodes {possibility of Immune Reconstitution Inflammatory Syndrome (IRIS) to be considered and differentiated from disease progression};
- appearance of new nodes;
- if chest symptomatic, monthly sputum for AFB and chest X-ray (to rule out pulmonary involvement);
- other EP sites to be monitored (USG abdomen, if necessary);
- serum creatinine-monthly for first three months of treatment and then quarterly till the patient receives Km and further when clinically indicated
- liver function test, as clinically indicated;
- USG-abdomen, if necessary; and
- monitoring for drug adverse reactions.

Same outcome definitions would be used as for pulmonary DR-TB patients. Treatment outcome will depend on availability of culture reports of specimens taken from discharging sinuses, treatment completion and clinical improvement of the patient.

## 9.10 Management of contacts of DR-TB

'Close contacts' of drug-resistant TB (DR-TB) patients are defined as people living in the same household as the index patient, or spending many hours a day together with the patient in the same indoor space. Contact tracing is an underutilized strategy that can stop the transmission of multidrug-resistant strains. Studies have shown that contact investigation is a high-yield strategy that, in many high-burden TB countries, probably merits more resources even for regular, drug-susceptible TB patients. [34]

All close contacts of DR-TB patients should be identified through contact tracing and evaluated for active TB disease as per RNTCP guidelines.

If the contact is found to be suffering from pulmonary TB disease irrespective of the smear based microbiological confirmation, s/he will be identified as a "Presumptive DR-TB". The patient will be initiated on regimen for new or previously treated patient based on their history of previous anti-TB treatment. Simultaneously, two sputum specimens will be transported for culture and DST to an RNTCP-certified C&DST laboratory where s/he will be evaluated as per DR TB diagnostic algorithm (Figure 5.2).

If the patient is confirmed as having DR-TB, appropriate DR-TB treatment must be initiated. Further, the patient will be sent to the DDR-TBC ward for pre-treatment assessment and initiation of appropriate regimen. Among asymptomatic contacts of patients with DR-TB, it would be important to rule out active TB by appropriate clinical examination and investigation. Although alternative prophylaxis treatments have been suggested, there is no consensus regarding the choice of drug(s) and the duration of treatment. Prompt treatment

of DR-TB in index case is the most effective way of preventing the spread of infection to others. There are multiple opportunities to investigate contacts of DR-TB patients, namely:

***Patient:*** Contact investigation starts with education of the DR-TB patient. Patients should be educated about the infectiousness of their disease and the high risk of transmission to contacts who share the same living space. While they should not be unduly alarmed, they should be informed that their family members are likely already infected with DR-TB, so the most important intervention is to monitor them closely for symptoms of active TB.

***Family***: One of the most important reasons to do a home visit for every DR-TB patient at the initiation of DR-TB treatment is to do contact investigation. A community nurse or health care provider should educate the family that they are all likely already infected with DR-TB, and explain the importance of notifying the community or clinical team quickly about family members who develop symptoms of active TB.

***Clinical team***. The clinical team has multiple opportunities to inquire about the health of the DR-TB patient's family contacts. At every clinical evaluation, doctors and nurses should ask the patient whether any family member has developed TB symptoms.

***Community nurses or health care providers educated on DR-TB:*** During home visits to check adherence or assess the social situation, the community nurse should inquire if there are any family members who have developed symptoms of active TB. The community nurse may directly interview family members at their home as they are best suited to address fears or doubts about the health system or other social barriers to treatment for DR-TB contacts.

***Community health workers:*** In community-based programmes that incorporate home-based treatment support, community health workers are the closest to the family and are most likely to identify family members with TB symptoms. This is particularly true for members of the extended family who visit periodically. The following measures should be taken to prevent spread of DR-TB infection:

- early diagnosis and appropriate treatment of DR-TB patients;
- counselling patient and family on infection control measures like cough etiquette and sputum disposal;
- screening of contacts as per RNTCP guidelines; and
- further research into effective and non-toxic chemoprophylaxis in areas of high DR-TB prevalence.

# Chapter 10: Managing adverse reactions

DR-TB can be deadly but the drugs used to treat the disease can be harmful in many ways. This chapter focuses on the measures to promote patient safety that contribute to improving quality of care during the treatment of drug-resistant TB, relieving unnecessary suffering. This section also includes ADR recording and reporting mechanism with Pharmacovigilance Programme of India (PvPI) :

- monitoring for early detection of ADR;
- commonly encountered ADRs with the regimen used;
- strategies for managing and reporting ADRs; and
- documentation and reporting of ADR (pharmacovigilance).

The timely and intensive monitoring for and management of, adverse effects caused by second-line drugs are essential components of DR-TB programmes.

Poor management of adverse effects increases the risk of default or irregular adherence to treatment and may result in death or permanent morbidity. The ability to monitor patients for adverse effects daily is one of the major advantages of having a treatment supporter over self-administration of drug-resistant TB treatment.

Table 10.1 Notable adverse reactions to drugs used for DR-TB patients [31]

| Common or relevant adverse effects of drug-resistant TB therapy | | |
|---|---|---|
| Nausea/vomiting | Abdominal pain | Visual disturbances |
| Diarrhoea | Anorexia | Seizures |
| Arthralgia | Gastritis | Hypothyroidism |
| Dizziness/vertigo | Peripheral neuropathy | Psychosis |
| Hearing disturbances | Depression | Suicidal ideation |
| Headache | Tinnitus | Hepatitis (hepatotoxicity) |
| Sleep disturbances | Allergic reaction | Renal failure (nephrotoxicity) |
| Electrolyte disturbances | Rash | QT prolongation |

Adverse effects are easy to recognize and are usually reported by patients when they experience it. However, few effects may not be reported by patients in the presence of other major adverse effect/s. All treatment supporters, including hospital, clinic or community health workers should be trained to screen patients regularly for symptoms of common adverse effects. They should be trained in simple adverse effect management and on when to refer patients to a nurse or physician. Certain laboratory investigations are required on routine basis during the course of treatment to monitor the ADR.

Table 10.2 Additional monitoring for patients on DR-TB regimens [31]

| Monitoring Evaluation | Recommended action in specific situations beyond routine follow up evaluation schedule (refer Table 8.1) |
|---|---|
| Serum creatinine | Every one to three weeks in HIV infected patients, diabetics and other high-risk patients. |
| Serum electrolytes (Na, K, Cl) | Every one to three weeks in HIV infected patients, diabetics and other high-risk patients. |
| Serum magnesium and calcium | Check magnesium and calcium blood levels whenever hypokalaemia is diagnosed. Repeat if any electrocardiogram (ECG) abnormalities develop (prolonged QT interval). |
| Thyroid stimulating hormone (TSH) | Every three months if receiving Eto/Pto and *p*- aminosalicylic acid (PAS). Every six months if receiving Eto/ Pto or PAS, but not both together. TSH is sufficient for screening for hypothyroidism and it is not necessary to measure hormone thyroid levels. Monthly monitoring for clinical signs/symptoms of hypothyroidism is also necessary. |
| Liver serum enzymes (SGOT, SGPT) | Periodic monitoring every six months if receiving Eto/ Pto for extended periods or for patients at risk for or with symptoms of hepatitis. For HIV-infected patients monthly monitoring/for patients on Bdq, monthly monitoring/or patients with viral hepatitis, monitoring every one to two weeks for the first month and then every one to four weeks is recommended. |
| HIV testing | Repeat if clinically indicated. |
| Haemoglobin and white blood cell count, Platelets | If on Lzd, monitor weekly at first, then monthly or as needed based on symptoms; there is little clinical experience with prolonged use of Lzd. For HIV-infected patients on Zidovudine, monitor monthly initially and then as needed based on symptoms. |
| Amylase, Lipase and S. protein (Albumin:Globulin) | Indicated for work-up of abdominal pain to rule out pancreatitis in patients on Lzd, Bdq, Dlm, D4T, ddI or ddc. |
| Lactic acid | Indicated for work up of lactic acidosis in patients on Lzd or antiretroviral treatment (ART). |
| Serum glucose | If receiving Gfx, monitor fasting blood glucose at baseline and monitor monthly. Educate/remind patients on signs and symptoms of hypoglycaemia and hyperglycaemia monthly. |
| Audiometry (hearing test) | Ask patients about changes in hearing at every clinic visit and evaluate their ability to participate in normal conversation. |
| Vision tests | For patients on long-term E or Lzd perform at least a visual acuity test with Snellen charts and colour vision test at baseline (as a small percentage of the population has colour blindness). Repeat the test for any suspicion of change in acuity or colour vision. |
| Educational, mental health and social consultation | At baseline by personnel trained in health education, mental health and social issues relevant to TB management; during treatment and repeat as indicated. Refer to social worker, psychologist or psychiatrist when indicated. |

## 10.1 Management of adverse drug reactions

The treatment supporter will monitor and record all adverse events routinely. Laboratory screening tests will be done on a routine basis as per national guidelines. The initial evaluation serves to establish a baseline and may identify patients who are at increased risk for adverse effects or poor outcomes. Laboratory screening is invaluable for detecting certain adverse effects that are more occult and before serious harm is done. [31]

Training of all the health staff will be done to identify and manage ADRs.[34] Close monitoring of patients is necessary to ensure that adverse effects of drugs are recognized quickly by healthcare personnel. The ability to monitor patients on a daily basis for adverse effects is one of the major advantages of having a treatment supporter as opposed to self-administration of treatment. It is important for the treatment supporter to be trained to screen patients regularly for symptoms of common adverse effects such as rashes, toxic epidermal necrolysis, gastrointestinal symptoms (nausea, vomiting, diarrhoea), psychiatric symptoms (psychosis, depression, anxiety) jaundice, ototoxicity, peripheral neuropathy, symptoms of electrolyte wasting (muscle cramping, palpitations) and convulsions. Treatment workers should also be trained to identify ADRs and must refer the patient to the MO PHI for minor ADRs and to the DTO for major ADRs. Most ADRs can be managed by the DTO/chest physician of the district hospital. If required, hospitalization can be done at the districts where inpatient facility is available or referred to a referral hospital for admission. A symptom-based approach should be followed to manage minor ADR where the patient is usually able to tolerate ATT drugs and continue medication with symptomatic treatment. Patients with major adverse effects should be managed at the hospital level (they may require admission).

The N/DDR-TBC Committee would be consulted to take decisions regarding reduction/termination of any drug. If any drug is withheld/ terminated due to ADR, it would be replaced with an appropriate substitute drug as per DR-TBC committee. Before starting treatment, the patient should be instructed in detail about potential adverse effects that could be produced by the prescribed drug regimen and if and when they occur, to notify a healthcare provider. Proper management of adverse effects begins with pretreatment patient education. Depending on the severity of ADRs, following actions may be indicated:

- if adverse effect is mild and not serious, continuing the treatment regimen, with the help of ancillary drugs if needed, is often the best option;
- most adverse effects of a number of second-line drugs are dose-dependent. Reducing the dosage of the offending drug or terminating it is another method of managing adverse effects; and

- psychosocial support is an important component of the management of adverse effects. This may be provided through patient education and motivation by treatment supporter, patient support groups like patients association/organization or through group discussions while in the hospital.

The recommended strategies for ADR management are detailed below. [31]

### 10.1.1 Rash, allergic reaction and anaphylaxis

**Suspected agent (s):** Any drug

**Suggested management strategies**

- hypersensitivity reactions such as pruritus or rash, can occur with any of the drugs used and are commonly managed with antihistamines;
- at the PHC/ field level, if the case is of a mild reaction, the patient will be assured and managed symptomatically by the PHC MO;
- for serious allergic reactions, it will be important to stop all therapies pending resolution of reaction and refer the patient to Nodal DR-TB centre/ tertiary centre for further management. In case of anaphylaxis, the condition will have to be managed with standard emergency protocols;
- other potential causes of allergic skin reactions (like scabies or other environmental agents) will need to be eliminated;
- for minor dermatologic reactions, various agents can be helpful. In this case medication must be continued. This could include antihistamines, Hydrocortisone cream for localized rash, Prednisone in a low dose of 10 to 20 mg per day for several weeks if other measures are not helpful and phototoxicity (may respond to sunscreens, but can also cause rash). Dry skin may cause itching (especially in diabetics). In which case, a liberal use of moisturizing lotion is recommended, since dry skin is a common and significant problem with Cfz;
- once rash resolves, reintroduce remaining drugs, one at a time with the one most likely to cause the reaction last. The order of reintroduction will be H, R, Z, Eto, Cs, E, PAS, FQ and Km;
- consider not reintroducing even as a challenge any drug that is highly likely to be the cause; and
- suspend permanently any drug identified to be the cause of a serious reaction.

**Points to note**

- history of previous drug allergies should be carefully reviewed. Any known drug allergies should be noted on the treatment card;
- flushing reaction to R or Z is usually mild and resolves with time. Antihistamines can be used. Hot flushes, itching, palpitations can be caused with H and tyramine containing foods (cheese, red wine). If this occurs, advise patients to avoid foods that precipitate reaction;
- any of the drugs can cause hives (urticaria). To identify the drug, introduce drugs one at a time; and
- in case of hives, a desensitization attempt can be made. Any drug that results in anaphylaxis or Stevens–Johnson syndrome should never be reintroduced (not even as a challenge).

### 10.1.2 Gastrointestinal symptoms (nausea and vomiting)

Gastrointestinal symptoms may be due to the bulk of drugs. Patients who complain of nausea or vomiting can be advised to take drugs embedded in a banana.

**Suspected agent(s):** Eto, Pto, PAS, Z, E, Bdq

**Suggested management strategies**

- if vomiting persists, drugs will be administered one hour after one tablet of Domperidone and/or a course of proton pump inhibitor (Omeprazole) or H2 receptor inhibitor (Famotidine, Ranitidine);
- other antacids are usually not given, since they interfere with absorption of FQ;
- in case of severe vomiting, the hydration status of the patient should be monitored and rehydration therapy initiated if required;
- if the offending drug is Eto, the drug is more acceptable if administered with milk, or after milk, or at bedtime to avoid nausea;
- if vomiting is severe, drugs can be withheld temporarily and tests can be conducted to rule out other causes of vomiting like hepatitis;
- assess for danger signs including dehydration, electrolyte disturbance and hepatitis. Initiate rehydration therapy if indicated and correct any electrolyte disturbance. If there is blood in the vomit, check haemoglobin and treat for possible bleeding ulcers; and
- with Bdq, ancillary drugs need to be chosen as mentioned in Section 10.2.

Initiate the following 3-step approach to manage nausea and vomiting:

***Step 1:*** Adjust medication and conditions without lowering overall dose. Give Eto/Pto at night; Eto or PAS twice or thrice daily; light snack (biscuits, bread, rice, tea) before medication; and PAS two hours after other anti-TB drugs.

***Step 2:*** Start antiemetic(s) like Metoclopramide 10 mg, 30 minutes before anti-TB medication; Ondansetron 8 mg, 30 minutes before anti-TB drugs and again eight hours after. Ondansetron can either be used on its own or with metoclopramide (if ondansetron is not available, promethazine can be used). For refractory nausea give 24 mg, 30 minutes before the dose.

***Step 3***: Decrease dose of suspected drug by one weight class if this can be done without compromising the regimen. It is rarely necessary to suspend the drug completely.

**Points to note**

- nausea and vomiting are universal in early weeks of therapy and usually abate with time on treatment and adjunctive therapy. Some nausea and even vomiting may need to be tolerated at least in the initial period;
- creatinine and electrolytes should be checked if vomiting is severe. Give intravenous fluids and replace electrolytes as needed;
- another strategy is to stop the responsible medicine for 2-3 days and then add it back, gradually increasing the dose (advice the patient that the medicine will be increased back to a therapeutic dose in a way that is better tolerated);
- odansetron prolongs QT interval; avoid use of odansetron with Bdq or Dlm; and
- for patients particularly anxious about nausea, (and with "anticipatory nausea and vomiting") a small dose of an anti-anxiety medicine (5 mg of diazepam) can help when given 30 minutes prior to intake of anti-TB drugs.

### 10.1.3 Gastrointestinal symptoms (gastritis & abdominal pain)

Abdominal pain is often associated with serious adverse effects, such as pancreatitis (Lzd, Bdq), lactic acidosis and hepatitis. If any of these are suspected, it is important to obtain appropriate laboratory tests to confirm and suspend the suspected agent.

**Suspected agent(s):** PAS, Eto, Pto, Cfz, FQs, H, E, and Z

**Suggested management strategies**

- if symptoms are associated and consistent with gastritis (epigastric burning or discomfort, sour taste in mouth associated with reflux) initiate medical therapy with the

use of H2-blockers (ranitidine 150 mg twice daily or 300 mg once daily) or proton-pump inhibitors (omeprazole 20 mg once daily). Avoid use of antacids as they decrease absorption of FQ;

- for severe abdominal pain, stop suspected agent(s) for short periods of time (1-7 days);
- lower the dose of the suspected agent, if this can be done without compromising the regimen; and
- discontinue suspected agent if this can be done without compromising the regimen.

**Points to note**

- severe gastritis, as manifested by blood in the vomit or stool is relatively rare, but should always be treated to facilitate adherence to treatment;
- if antacids must be used, they should be carefully timed so as to not interfere with absorption of FQ (take two hours before or three hours after anti-TB drugs);
- stop any non-steroidal anti-inflammatory drugs the patient may be taking;
- diagnose and treat for *Helicobacter pylori* infections; and
- severe abdominal distress has been reported with use of Cfz. Although these reports are rare, if this occurs, Cfz should be suspended.

### 10.1.4 Diarrhoea and/ or flatulence

**Suspected agent(s):** PAS, Eto/Pto

**Suggested management strategies**

- motivate patients to tolerate some degree of loose stools and flatulence;
- encourage fluid intake;
- treat uncomplicated diarrhoea (no blood in stool and no fever) with loperamide 4 mg by mouth initially followed by 2 mg after each loose stool to a maximum of 10 mg per 24 hours;
- check serum electrolytes (especially potassium) and dehydration status if diarrhoea is severe; and
- fever and diarrhoea and/or blood in the stools indicate that diarrhoea may be secondary to something other than the simple adverse effect of anti-TB drugs

**Points to note**

- consider other causes of diarrhoea;
- pseudo-membranous colitis related to broad-spectrum antibiotics (such as FQ) is a serious and even life threatening condition. Fever, bloody diarrhoea, intense abdominal pain and increased white blood cells are warning signs of possible pseudomembranous colitis;
- parasites and common waterborne pathogens in the area should be evaluated in the patient and treated;
- lactose intolerance, especially if patient has been exposed to new foods in a hospital not normally part of their diet; and
- loperamide can be used in children over two years of age.

### 10.1.5 Hepatitis

**Suspected agent(s):** Z, H, R, Pto / Eto, PAS, FQ, Bdq

**Suggested management strategies**

- in cases where patient is very sick *i.e.,* meningitis, sputum smear grade 3+, give ATT e. g. Streptomycin, FQ and Cs. Where patient is not seriously ill and one can wait, introduction of ATT can be done once enzyme levels are near normal;
- if enzymes are more than five times the upper limit of normal, stop all hepatotoxic drugs and continue with at least three non-hepatotoxic medications (for example, the injectable agent, FQ and Cs). If hepatitis worsens or does not resolve with the three-drug regimen, then stop all drugs;
- eliminate other potential causes of hepatitis (viral hepatitis and alcohol induced hepatitis being the two most common causes) and treat any that are identified; and
- once enzyme level improves, reintroduce remaining drugs, one at a time with the least hepatotoxic agents first, while monitoring liver function by testing enzymes every three days. If the most likely agent is not essential, consider not reintroducing it.

**Points to note**

- history of previous drug hepatitis should be carefully analyzed to determine the most likely causative agent(s); these drugs should be avoided in future regimens;
- viral serology should be done to rule out other etiologies of hepatitis if available, especially to hepatitis A, B and C;
- alcohol use should be investigated and alcoholism addressed; and
- generally, hepatitis due to medications resolves upon discontinuation of the suspected drug.

### 10.1.6 Giddiness

**Suspected agent (s)-**Aminoglycosides, Eto, FQ and/or Z

**Suggested management strategies**

- whenever a patient complains of giddiness, over sleepiness or poor concentration, s/he will have to be counselled;
- if severe, the offending drug should be identified by administering drugs individually and observing response;
- the dose of the offending drug identified may be adjusted or the offending drug terminated if required; and
- aminoglycosides, especially in elderly age group must be kept in mind for giddiness as it may be early sign of 8$^{th}$ nerve toxicity.

**Point to note**

In cases of severe giddiness, the patient may be referred to the neurologist for further management as per standard protocol.

### 10.1.7 Haematological abnormalities

**Suspected agent(s):** Lzd

**Suggested management strategies**

- stop Lzd if myelosuppression (suppression of white blood cells, red blood cells or platelets) occurs;
- consider restarting with a lower dose of Lzd (300 mg instead of 600 mg) if myelosuppression resolves and if Lzd is considered essential to the regimen;
- consider nondrug related causes of haematological abnormality; and
- consider blood transfusion for severe anaemia.

**Points to note**

- haematological abnormalities (leukopenia, thrombocytopenia, anaemia, red cell aplasia, coagulation abnormalities and eosinophilia) can rarely occur with a number of other anti-TB drugs; (see individual drug sheets, Part 3)
- there is little experience with prolonged use of Lzd.

### 10.1.8 Hypothyroidism

**Suspected agent(s):** Eto/Pto, PAS

**Suggested management strategies**

In cases of hypothyroidism, opinion of general physician/ endocrinologist may be taken.

**Points to note**

- symptoms of hypothyroidism include fatigue, somnolence, cold intolerance, dry skin, coarse hair and constipation, as well as occasional depression and inability to concentrate;
- in cases with abnormal weight gain, hypothyroidism may be ruled out;
- it is completely reversible upon discontinuation of PAS and/or Eto/Pto; and
- combination of Eto/Pto with PAS is more frequently associated with hypothyroidism than when each individual drug is used.

### 10.1.9 Arthralgia

**Suspected agent(s):** Z, FQ, Bdq

**Suggested management strategies**

- initiate with paracetamol in the beginning;
- therapy with nonsteroidal anti-inflammatory drugs (indomethacin 50 mg twice daily or ibuprofen 400 to 800 mg three times a day);
- lower the dose of the suspected agent (most commonly Z) if this can be done without compromising the regimen; and
- discontinue the suspected agent if this can be done without compromising the regimen.

**Points to note**

- symptoms of arthralgia generally diminish over time, even without intervention;
- uric acid levels may be elevated in patients on Z. There is little evidence to support the addition of allopurinol for arthralgia. However, if gout is present it should be used; and
- if acute swelling, redness and warmth are present in a joint, consider aspiration for diagnosis of gout, infections, autoimmune diseases, *etc.,*

### 10.1.10 Peripheral neuropathy

**Suspected agent(s):** Cs, Lzd, H, S, Km, Amk, Cm, FQ, rarely Pto/Eto, E

**Suggested management strategies**

- to prevent occurrence of such adverse reaction, all patients on an RNTCP regimen for MDR-TB should receive daily Pyridoxine. If peripheral neuropathy develops, an additional 100mg Pyridoxine will be given;
- the commonest offending agent is Lzd, almost 60-70% of the patients on Lzd 600 mg/day may develop neuropathy and pyridoxine does not help in preventing Lzd induced neuropathy. Early recognition of neuropathy symptoms and early dose reduction of Lzd helps to prevent the progression. If there is no improvement or symptoms worsen, Amitriptylline 25mg will be added and if still there is no improvement, patient should be referred to a neurologist;
- correct any vitamin or nutritional deficiencies and increase pyridoxine to the maximum daily dose (200 mg per day);
- consider whether the dose of Cs can be reduced without compromising the regimen. If H is being used (especially $H^h$), consider stopping it and if possible, switching the aminoglycoside to Cm may also be helpful;
- initiate medical therapy:

  - non-steroidal anti-inflammatory drugs or acetaminophen may help alleviate symptoms;
  - therapy with tricyclic antidepressants such as amitriptyline (start with 25 mg at bedtime, the dose may be increased to a maximum of 150 mg) can be tried. Do not use tricyclic antidepressants with selective serotonin reuptake inhibitors and antidepressant drugs;
  - in such a case, refer to specialist for further management; and
  - rarely, medication may be discontinued, but only if an alternative drug is available and the regimen is not compromised.

**Points to note**

- patients with comorbid disease (diabetes, HIV, alcohol dependence) may be more likely to develop peripheral neuropathy, but these conditions are not contraindications to the use of agents listed here; and
- neuropathy associated with Lzd is common after prolonged use and often permanent though it may be irreversible. Thus, suspension of this drug should be strongly considered when neuropathy persists despite above measure.

### 10.1.11 Headache

**Suspected agent(s):** Bdq, Cs

**Suggested management strategies**

- Rule out more serious causes of headache, including meningitis and other infections of the central nervous system (HIV co-infected patients should receive a head computed tomography scan and cerebrospinal fluid analysis).
- Start analgesics like ibuprofen or paracetamol. Also, encourage good hydration and consider low dose tricyclic antidepressants for refractory headaches.

**Points to note**

- headaches are common during the initial months of DR-TB therapy and can present as migraine or cluster headaches;
- to minimize headaches at the start of therapy, Cs can be started at lower doses of 250–500 mg and gradually increased over 1-2 weeks to achieve the target dose;
- headaches due to Cs and Bdq are usually self-limited; and
- Pyridoxine (Vitamin B6) should be given to all patients receiving Cs to help prevent neurotoxicity.

### 10.1.12 Depression

**Suspected agent(s):** Psychological and socioeconomic circumstances, chronic disease, Cs, FQ H, Eto/Pto

**Suggested management strategies**

- assess and address underlying emotional and socioeconomic issues (see Chapter 12 on social support);
- assess patients for coexisting substance abuse and refer to treatment if appropriate;
- initiate individual counselling (or group counselling if the patient is sputum smear and culture negative);
- when depression is more significant, initiate antidepressant therapy (amitriptyline, fluoxetine or similar);
- tricyclic antidepressants and selective serotonin reuptake inhibitors should be given together and not given to patients on Lzd;
- lower the dose of the suspected agent if this can be done without compromising the regimen (reducing dose of Cs and Eto to 500 mg daily to see if depression is lessened is a common strategy); and
- discontinue suspected agent if this can be done without compromising the regimen.

**Points to note**

- socioeconomic conditions and chronic illness should not be underestimated as contributing factors to depression;
- depressive symptoms may fluctuate during therapy and improve as illness is successfully treated;
- history of previous depression is not a contraindication to use of agents listed but may increase the likelihood of depression developing during treatment. If significant depression is present at the start of treatment, avoid a regimen with Cs, if possible; and question the patient regarding suicidal ideation any time the depression is judged to be more than mild.

### 10.1.13 Psychotic symptoms

**Suspected agent(s):** Cs, H, FQ,

**Suggested management strategies**

- stop the suspected agent for a short period (1–4 weeks) while psychotic symptoms are brought under control;
- the most likely drug is Cs followed by H[h]
- if moderate to severe symptoms persist, initiate antipsychotic therapy (haloperidol);
- hospitalize in a ward with psychiatric expertise if patient is at risk to himself/herself or others;
- increase pyridoxine to the maximum daily dose (200 mg per day);
- lower the dose of the suspected agent (most commonly Cs to 500 mg a day) if this can be done without compromising the regimen;
- discontinue suspected agent if this can be done without compromising the regimen; and
- once all symptoms resolve and patient is off Cs, antipsychotic therapy can be tapered off. If Cs is continued at a lower dose, antipsychotic therapy may need to be continued and any attempts of tapering off should be done after referring to a psychiatrist trained in the adverse effects of second-line anti-TB drugs.

**Points to note**

- some patients will need to continue antipsychotic treatment;
- previous history of psychiatric disease is not a contraindication to Cs, but its use may increase the likelihood of psychotic symptoms developing during treatment;
- some patients will tolerate Cs with an antipsychotic drug but this should be done in consultation with a psychiatrist, as these patients will need to be under special observation; this should be done only when there is no other alternative;
- psychotic symptoms are generally reversible upon completion of DR-TB treatment or cessation of the offending agent; and
- always check creatinine in patients with new onset psychosis. A decrease in renal function can result in high blood levels of Cs, which can cause psychosis.

### 10.1.14 Suicidal ideation

**Suspected agent(s): CS**, H, Eto/Pto

**Suggested management strategies**

- hospitalize the patient and put under 24-hour surveillance;
- discontinue Cs;
- request psychiatric consultation;
- initiate antidepressant therapy; and
- lower the dose of Eto/Pto to 500 mg daily until the patient is stable.

**Points to note**

- keep the patient in the hospital until risk of suicide has passed; and
- if no improvement occurs after holding Cs, hold H and/or Eto/Pto.

### 10.1.15 Seizures

**Suspected agent(s):** Cs, H, FQ

**Suggested management strategies**

- hold Cs, FQ and H pending resolution of seizures;
- initiate anticonvulsant therapy (carbamazepine, phenytoin or valproic acid are most commonly used);
- increase pyridoxine to maximum daily dose (200 mg per day);
- check serum electrolytes including potassium, sodium, bicarbonate, calcium, magnesium and chloride; and
- when seizures have resolved, restart medication, one at a time. Cs should not be restarted unless it is absolutely essential to the regimen. If Cs is reinitiated, start a dose one weight band lower.

**Points to note**

- an anticonvulsant is generally continued until DR-TB treatment is completed;
- history of previous seizure disorder is not a contraindication if a patient's seizures are well controlled and/ or the patient is receiving anticonvulsant therapy;
- do not include Cs if an alternative drug is available;
- patients with history of previous seizures may be at increased risk for developing seizures during DR-TB therapy; and
- always check creatinine in patients with new onset seizures. A decrease in renal function can result in high blood levels of Cs, which can cause seizures. Adjusting the dose of Cs in the presence of low creatinine may be all that is needed to control the seizures.

### 10.1.16 Tendonitis and tendon rupture

**Suspected agent(s):** FQ

**Suggested management strategies**

- if significant inflammation of tendons or tendon sheaths occur, consider stopping FQ;
- give a non-steroidal anti-inflammatory drug (ibuprofen 400 mg four times daily)
- rest the joint;
- if treatment failure is likely without FQ:

  - reduce dose if possible;
  - ensure joint is strictly rested; and
  - inform patient of the possible risk of tendon rupture and discuss the risks and benefits of ongoing use of FQ.

**Points to note**

- tendon rupture with FQ use is more likely in patients doing new physical activities and more common among older patients and diabetics; and
- tendon rupture is relatively rare in patients on DR-TB regimens with FQ.

### 10.1.17 Nephrotoxicity (renal toxicity)

Prior to starting treatment, all patients will have renal function evaluated. During treatment of DR-TB, if the patient presents with symptoms and/or signs of renal impairment (oliguria, anuria, puffiness of face, pedal oedema), all the drugs should be withheld, renal function tests done and, if required, opinion of nephrologist sought. Reintroduction of drugs will be undertaken by the DR-TBC committee in consultation with a nephrologist, along with frequent monitoring of renal parametres.

**Suspected agent(s):** S, Km, Am, Cm

**Suggested management strategies**

- discontinue the suspected agent;
- consider using Cm if an aminoglycoside had been the prior injectable drug in the regimen;
- consider other contributing etiologies (non-steroidal anti-inflammatory drugs, diabetes, other medications, dehydration, congestive heart failure, urinary obstruction, *etc.*,) and address as indicated;
- follow creatinine (and electrolyte) levels closely, every 1-2 weeks;
- consider dosing the injectable agent 2-3 times a week if the drug is essential to the regimen and the patient can tolerate (close monitoring of creatinine). If creatinine continues to rise despite twice/thrice a week dosing, suspend the injectable agent; and

- adjust all TB medication according to creatinine clearance in consultation with nephrologist. Also, note that renal impairment may be permanent.

**Points to note**

- during treatment, blood urea and serum creatinine should be done every month for the first three months after treatment initiation and then every three months thereafter whilst injection Km is being administered;
- silent renal toxicity may be picked up by these routine follow-up biochemical examinations;
- if at any time, the blood urea or serum creatinine becomes abnormal, treatment should be withheld and further management decided upon in consultation with the DR-TBC committee;
- an example of how to calculate a creatinine clearance based on the serum creatinine is provided in Chapter 7, Box 7.2; and
- history of diabetes or renal disease is not a contraindication to the use of agents listed here, although patients with these comorbidities may be at increased risk for developing renal failure.

### 10.1.18 Vestibular toxicity (tinnitus and dizziness)

**Suspected agent (s):** S, Km, Am, Cm, Cs, FQs, H, Eto, Lzd

**Suggested management strategies**

- if early symptoms of vestibular toxicity appear, there may be a need to change dosing of the injectable agent to twice/thrice a week. Also, consider using Cm if an aminoglycoside had been the prior injectable in the regimen; and
- if tinnitus and unsteadiness worsen with the above adjustment, stop the injectable agent. This is one of the few adverse reactions that may cause permanent intolerable toxicity and can necessitate discontinuation of a class of agents.

**Points to note**

- ask the patient about tinnitus and unsteadiness every week especially in elderly patients;
- fullness in the ears and intermittent ringing are early symptoms of vestibular toxicity; and
- a degree of disequilibrium can be caused by Cs, FQs, Eto/Pto, H or Lzd. Some clinicians will stop all drugs for several days to see if symptoms are attributed to these drugs. Symptoms of vestibular toxicity generally do not improve on withholding medications.

### 10.1.19 Hearing loss

**Suspected agent (s):** S, Km, Am, Cm

**Suggested management strategies**

- document hearing loss and compare with baseline audiogram if available (some degree of hearing loss occurs with most patients starting with high frequency loss);
- if early symptoms of hearing loss are documented, change closing of the injectable agent to twice/thrice a week. Also, consider using Cm if an aminoglycoside had been the prior injectable in the regimen; and
- discontinue injectable agent if hearing loss continues despite dose adjustment and add additional drugs to reinforce the regimen. Even when additional drugs are not available, stopping the injectable agent can be considered based on the patient's desire to maintain hearing.

**Points to note**

- patients with previous exposure to aminoglycosides may have baseline hearing loss. In such patients, audiometry may be helpful at the start of DR-TB therapy;
- hearing loss is almost always permanent. Continuing the injectable agent despite hearing loss almost always results in irreversible deafness;
- while benefit of hearing aids is minimal to moderate in auditory toxicity, consider a trial use to determine if a patient with hearing loss can benefit from its use; and
- cochlear transplant can also be considered.

### 10.1.20 Optic neuritis

**Suspected agent(s):** E, Lzd, Eto/Pto, Cfz, Rifabutin, H, S

**Suggested management strategies**

- stop E and Lzd. Do not restart; and
- refer patient to an ophthalmologist.

**Points to note**

- the most common drugs are E and Lzd; the condition usually reverses with cessation of the drug; and
- improve diabetes control in diabetic patients.

### 10.1.21 Metallic taste

**Suspected agent(s):** Eto/Pto, FQs

**Suggested management strategies:** Encourage the patient to tolerate this side effect.

**Point to note:** Normal taste returns when treatment is stopped.

### 10.1.22 Electrolyte disturbances (hypokalaemia and hypomagnesaemia)

**Suspected agent(s):**Cm, Km, Am, S

**Suggested management strategies**

- evaluate potassium levels;
- if potassium is low, check for magnesium and calcium (if unable to check for magnesium, consider empiric treatment with magnesium in all patients of hypokalaemia);
- replace electrolytes as needed. Dose oral electrolytes apart from FQ as they can interfere with FQ absorption; and
- manage electrolyte disturbances

**Points to note**

- if severe hypokalaemia is present, consider hospitalization;
- amiloride, 5–10 mg daily, or spironolactone, 25 mg daily, may decrease potassium and magnesium wasting and thus useful in refractory patients; and
- oral potassium replacements can cause significant nausea and vomiting and oral magnesium may cause diarrhoea.

### 10.1.23 QT prolongation

**Suspected agent(s):** Bdq, Dlm, FQ, clarithromycin, Cfz

**Suggested management strategies**

- any patient found to have a QTc value greater than 500ms,stop all suspected QT prolonging drugs;
- repeat ECG and confirm the prolongation;
- check potassium, calcium and magnesium levels;
- electrolyte levels should be maintained in the normal range in any patient with an elevated QT interval;

- avoid other drugs that increase the QT interval. Monitor the patient's renal and hepatic function and adjust the dose of FQ if impairment is present;
- consider suspension of FQ if risk of torsades de pointes outweighs benefits of the drug; and
- also see relevant section for more information on QT interval monitoring with Bdq and Dlm.

**Points to note**

- QT interval is measured from the end of the QRS complex to the beginning of the T wave on a standard ECG. The QT is corrected for heart rate, which is referred to as the QTc and calculated by most ECG machines;
- values above QTc 450 ms are referred to as prolonged. Patients with prolonged QTc are at risk for developing cardiac arrhythmias like torsades de pointes, which can be life threatening;
- FQ may cause prolongation of the QTc. Mfx and Gfx cause the greatest QTc prolongation, while Lfx and Ofx have a lower risk; and
- currently, ECG monitoring prior to initiation and during DR-TB therapy is only required with the use of Bdq, Dlm, or when two drugs known to prolong QT (e.g.Mfx, Cfz) are combined in the same regimen.

### 10.1.24 Gynaecomastia

**Suspected agent(s):** Eto/Pto

**Suggested management strategies**

- breast enlargement can be a troublesome side effect of Eto/Pto therapy, especially for male patients. Galactorrhoea has also been reported; and
- encourage patients to tolerate this side effect.

**Point to note:** Resolution occurs after treatment is stopped.

### 10.1.25 Alopecia

**Suspected agent(s):**H, Eto/Pto

**Suggested management strategies**

- hair loss can occur or there can be significant thinning of the hair, but this is temporary and not progressive during treatment; and
- encourage patients to tolerate this side effect.

**Point to note:** Significant cosmetic change has not been reported.

### 10.1.26 Superficial fungal infection and thrush

**Suspected agent(s):** FQ

**Suggested management strategies**

- topical antifungal agents or short course oral antifungal drugs are helpful; and
- Exclude other diseases if response to treatment is not prompt (such as HIV).

| |
|---|
| **Point to note:** Vaginal or penile candidiasis, oral thrush or cutaneous candidiasis in skin folds may occur with antibiotic treatment. |

### 10.1.27 Lactic acidosis

**Suspected agent(s):** Lzd

**Suggested management strategy:** Stop Lzd if lactic acidosis occurs.

| |
|---|
| **Point to note:** Lactic acidosis can be managed at the Nodal DRTB centre as per standard protocol and monitored with a blood test that measures lactic acid. |

### 10.1.28 Dysglycaemia and hyperglycaemia

**Suspected agent (s):** Gfx, Eto/Pto

**Suggested management strategies:** Replace the offending drug with a suitable drug.

| |
|---|
| **Point to note:** Treat diabetes as needed. Good glucose control is important during treatment. |

## 10.2 Specific toxicities due to Bedaquiline

Monitoring for specific toxicities is based on target organs defined in preclinical toxicity studies. For monitoring the specific toxicities related to second-line TB drugs, the RNTCP guidelines should be followed. (eye care, audiometry, *etc.,*)

Management of patients with AST and/or ALT elevations, amylase and/or lipase elevations, musculoskeletal system and cardiac muscle abnormalities, cardiac rhythm disturbances, gastrointestinal system disorders or other toxicities is enumerated below. [4] [32] [10]

**AST and/or ALT elevations**

Management will be at the discretion of the physician, according to generally accepted medical practice standards.

In Grade 1 (>1.0 to <2.0 x ULN) or Grade 2 (>2.0 to <3.0 x ULN) AST or ALT elevation, patients may continue Bdq. Patients should be followed until resolution (return to baseline) or stabilization of AST/ALT elevation; and

In Grade 3 (>3.0 to <8.0 x ULN) or Grade 4 (>8.0 x ULN) AST or ALT elevation, patients are allowed to temporarily discontinue treatment of the suspected causative agent (usually Eto, Z or PAS). AST, ALT and serum bilirubin should be monitored as frequently as necessary to manage the patient's condition.

If ALT and AST do not return to baseline, Bdq may be temporarily withheld for up to two weeks. Additional tests should be performed to evaluate the cause of hepatitis (e.g. hepatitis A, B, C). Liver enzymes, including serum bilirubin should be monitored as frequently as necessary to manage the patient's condition. If LFT improves, then the rest of the dosages of Bdq can be given. For patients who fail to show improvement in the clinical course and to return to baseline values of AST and ALT, it is recommended that the patient discontinue Bdq.

**Amylase and/or lipase elevation**

Management will be at the discretion of the physician, according to generally accepted medical practice standards.

In Grade 1 (>1.0 to <1.5 x ULN) or Grade 2 (>1.5 to <2.0 x ULN), patients may continue BDQ and should be carefully evaluated and followed closely.

In Grade 3 (2.0 to <5.0 x ULN) or Grade 4 (> 5.0 x ULN), for asymptomatic grade 3 amylase elevations with no history or concomitant disease of pancreatitis, patients may continue Bdq but should be carefully evaluated and followed closely.

For confirmed grade 4 elevations of amylase and confirmed grade 3 or 4 elevations of lipase, it is recommended that the patient discontinue BDQ.

**Musculoskeletal system and cardiac muscle abnormalities like Myalgia**

In Grade 1 (mild with no limitation of activity), patients may continue Bdq and be carefully evaluated and followed closely.

In Grade 2, there may be muscle tenderness at site other than injection site or with moderate impairment of activity).

In Grade 3, there could be severe muscle tenderness with marked impairment of activity.

In Grade 4 (Frank myonecrosis), it is recommended that the patient discontinue Bdq.

**Cardiac rhythm disturbances**

***QT interval monitoring***: An ECG should be obtained before initiation of treatment and daily for the first two weeks, then every two weeks for three months and then monthly. ECGs should be done at least weekly throughout the Bdq course if other QT prolonging drugs like FQ (Mfx, Gfx), Cfz or macrolide antibacterial drugs (erythromycin, clarithromycin, azithromycin) are included in the regimen. Other drugs with additive or synergistic QT prolongation observed when Bdq is co-administered are those with serotonin 5-HT3 receptor antagonist (ondansteron), prokinetics (Cisapride), azole agents (ketoconazole, itraconazole, fluconazole), common ART drugs, antimalarials (chloroquine and quinine sulfate), some drugs used for psychiatric disorders (chlorpromazine, haloperidol, thioridazine) and drugs known to lower serum electrolytes. If possible, avoid use of QT prolonging drugs with Bdq. If it is absolutely necessary to include a QT prolonging drug and increase ECG monitoring as described earlier.

QT prolongation can result in ventricular arrhythmias (Torsades de Pointes) and sudden death. It is therefore imperative that ECGs be used to monitor the QT interval regularly during Bdq use.

In Grade 1 (asymptomatic) or Grade 2 (asymptomatic, transient rhythm abnormality not requiring any treatment) there could be cardiac rhythm disturbances. Patients may continue BDQ and be carefully evaluated and followed closely.

In Grade 3 (recurrent, persistent, symptomatic arrhythmia requiring treatment) or Grade 4 (unstable dysrhythmia requiring hospitalization and treatment) there could be cardiac rhythm disturbances. It is recommended that the patient discontinue Bdq.

A normal value for the corrected QTcF interval is less than 0.44 seconds (440 ms). Whenever an abnormal QTc value is found, the ECG and calculations should be repeated.

- value greater than 440 ms is considered prolonged but does not need action until >450 ms;
- value between 450 – 480 ms: Rule out other causes of prolonged QTc, before deciding to withhold Bdq;

- value greater than 480 ms (or an increase of greater than 60 ms from baseline) should trigger the following actions:

    - repeat ECG to confirm prolongation;
    - check for serum K, Mg and Ca and correct the levels if found to be abnormal. Withhold BDQ until electrolytes have normalized;
    - if the QTc interval is between 480 and 500 ms, the patient is stable and electrolytes are within normal values, repeat weekly ECG to confirm that the QTc interval is stable; and
    - if the QTc interval is > 500 ms (confirmed by repeat ECG), discontinue Bdq and all other QTc-prolonging drugs in the regimen.

Bdq and all other QTc-prolonging drugs are to be discontinued if the patient develops a clinically significant ventricular arrhythmia. If Bdq is stopped for QTc prolongation, monitor ECG at least weekly to confirm that QTcF interval has returned to baseline**.** If syncope occurs, obtain an ECG to detect QT prolongation. Because of the long half-life of Bdq, if the ECG has QTc prolongation at week 24, ongoing weekly monitoring should take place until QTc interval normalizes (even though the drug is no longer being given).

If a QTcF of greater than 500 ms is recorded and confirmed by a repeat ECG, it is recommended that Bdq and all other QTc-prolonging drugs be discontinued. Such patients must be closely monitored until the resolution of the prolonged QTcF. The physician should rule out other causes of QTc prolongation such as electrolyte imbalances and steps should be taken to remedy any underlying causes of such prolongation.

**Gastrointestinal system disorders**

Patients with Grade 4 elevation of gastrointestinal parametres should be hospitalized and monitored closely. In case of Grade 4 nausea (hospitalization required) or Grade 4 vomiting (physiologic consequences requiring hospitalization or requiring parenteral nutrition), the patient's Bdq treatment should be discussed with the DR-TBC Committee.

**Other toxicities**

***Grade 1 or 2:*** Patients who develop grade 1 or 2 AE or laboratory toxicity may continue intake of Bdq.

***Grade 3 or 4:*** Patients who develop grade 3 or 4 AE or laboratory toxicity should be carefully evaluated by the physician. Patients may discontinue intake of Bdq if, in the opinion of the physician, AE or laboratory toxicity possess significant risk for the patient in case of continued treatment. Patients should be followed as appropriate until resolution of the AE or toxicity. Refer DAIDS criteria for grades [36].

Patients should be monitored for the common side effects of concomitant TB therapy, including decreased hearing, tinnitus, vision changes, dizziness, psychosis, depression, tremors, nausea, vomiting, diarrhoea, joint pain and renal function.

ADR management is crucial to improve treatment compliance of DR-TB patients. Majority of the side effects and ADR management is possible with a simple intervention which can be easily executed even at peripheral level. Following drugs can be used for common side effects or ADR reported by patients.

Table 10.3 Drugs used in management of adverse event[30]

| ADRs | Suggested drugs to manage the ADR |
|---|---|
| Nausea, vomiting, upset stomach | Metoclopramide, dimenhydrinate, prochlorperazine, promethazine, bismuth subsalicylate, domperidone |
| Heartburn, acid indigestion, sour stomach, ulcer | H2-blockers (ranitidine, cimetidine, famotidine, *etc.*,), Proton pump inhibitors (omeprazole, lansoprazole, *etc.*,) Avoid antacids because they can decrease absorption of fluoroquinolone eg. alumnium hydroxide |
| Oral candidiasis (non-AIDS patient) | Fluconazole, clotrimazole lozenges, Nystatin suspension, itroconazole liquid |
| Diarrhoea | Loperamide |
| Depression | Selective serotonin reuptake inhibitors (fluoxetine, sertraline), tricyclic antidepressants (amitriptyline) |
| Severe anxiety | Lorazepam, diazepam, clonazepam |
| Insomnia | Any hypnotic |
| Psychosis | Haloperidol, thorazine, risperidone (consider benzotropine or biperiden to prevent extrapyramidal Effects), Buromazine, thioridazine |
| Seizures | Phenytoin, carbamazepine, valproic acid, phenobarbital |
| Prophylaxis of neurological complications of cycloserine | Pyridoxine (vitamin B6) |
| Peripheral neuropathy | Amitriptyline |
| Vestibular symptoms | Meclizine, dimenhydrinate, prochlorperazine, Promethazine |
| Musculoskeletal pain, arthralgia, headaches | Ibuprofen, paracetamol, codeine, diclofenac |
| Cutaneous reactions, itching | Hydrocortisone cream, calamine, caladryl lotions |
| Systemic hypersensitivity Reactions | Antihistamines (diphenhydramine, chlorpheniramine, dimenhydrinate), corticosteroids (prednisone, dexamethasone) |
| Bronchospasm | Inhaled beta-agonists (albuterol, *etc.*,), inhaled corticosteroids (beclomethasone, *etc.,*), oral steroids(prednisone), injectable steroids (dexamethasone, methylprednisolone) |
| Hypothyroidism | Levothyroxine |
| Electrolyte wasting | Potassium, magnesium and calcium replacement therapy (oral and intravenous formulations) |

## 10.3 Role of DR-TB centre committee in management of adverse reactions

Whenever a patient has serious adverse reactions to any of the second-line anti-TB drugs, ideally, s/he is admitted at the DR-TBC and the committee decides on further management of the patient. This may require withholding or discontinuing the offending drug in the treatment regimen. The committee will be responsible for arranging the drugs to be given for managing these reactions. Timely and intensive monitoring for identifying and managing adverse reactions are essential components of the PMDT services. This will help improve the patient's adherence to treatment, reduce mortality, ensure timely management of ADR and obtain better treatment outcomes. Ancillary drugs for the management of adverse reactions should be made available to the patient free of cost. Proper training of staff and support to the patient are other important activities that are required.

## 10.4 Pharmacovigilance in TB programme

Pharmacovigilance is defined by the World Health Organization (WHO) as the "science and activities relating to the detection, assessment and prevention of adverse effects or any other drug-related problem". The overall objective of the pharmacovigilance programme is to improve patient care by assessing both the harm and benefit received from the drug. More than 200 ADR Monitoring Centres (AMCs) are currently collecting ADR related information at various levels (which is usually at the Medical College level). The Pharmacovigilance Programme of India (PvPI) is a national programme for surveillance of ADR related information. DR-TB management is inclusive of newer drugs which require a robust system of monitoring and reporting of ADR related information to build the guideline for safe use of drugs. [37]

Since newer drugs like Bdq, Dlm and not frequently used drug formulation like $Mfx^h$, $H^h$ would be there in various regimen, aDSM is essential. [17] [12]Hence, RNTCP in collaboration with PvPI and support from WHO India developed the comprehensive cohort event monitoring system for aDSM for DR-TB patients. This platform is initially used at DR-TB centers involved in Bdq (newer drugs) containing regimen. Now, this mechanism of CEM is expanded to all DR-TB centres. A drug safety monitoring committee periodically monitors the occurrence of AE/SAE including deaths of patients while on newer drugs containing regimen for necessary signaling and guidance to the programme on their safety and efficacy.

The PvPI recommends reporting of any adverse event (AE) or serious adverse event (SAE) to be done using a suspected ADR reporting format. This has been adopted by RNTCP for drug susceptible TB patients put on first-line treatment using existing mechanisms available under PvPI. In addition, ADR reporting is also possible through the use of mobile app ADR available on android/IOS platforms and toll-free number (18001803024) [36] [37]

However, based on a consensus between RNTCP and PvPI, reporting of SAE is done using cohort event monitoring (CEM) forms for all forms of DR-TB patients initiated on treatment in India. At the time of initiation of any DR-TB treatment regimen, CEM – treatment initiation form (Annexure 12A) is filled in hardcopy and same is entered in Nikshay within 24 hours. During the course of treatment CEM – treatment review form (Annexure 12B) should be filled for any SAE that is reported. Also, any SAE should be managed at the appropriate health facility level and reported in Nikshay within 24 hours by the health facility managing the SAE/ parent health facility. For the patients under the conditional access programme, CEM – treatment review form (Annexure 12B) will be filled as per the defined follow-up schedule.

## 10.5 Adverse event monitoring and reporting

Timely, accurate and complete reporting and analysis of adverse events are required to be reported under the programme. This is crucial for the protection of patients.

**Adverse event definitions and classifications**[36][37]

***Adverse event*:** An AE is any untoward medical occurrence in a clinical study subject administered a medicinal (investigational or non-investigational) product. An AE does not necessarily have a causal relationship with the treatment. An AE can therefore be any unfavourable and unintended sign (including an abnormal finding), symptom or disease temporally associated with the use of a medicinal (investigational or non-investigational) product, whether or not related to that medicinal (investigational or non-investigational) product (definition as per the International Conference on Harmonization [ICH]). This includes any occurrence that is new in onset or aggravated in severity or frequency from the baseline condition or abnormal results of diagnostic procedures including laboratory test abnormalities.

***Serious adverse event:*** A serious adverse event (SAE) based on ICH is any untoward medical occurrence that at any dose:

- results in death;
- is life-threatening (subject was at risk of death at the time of the event. It does not refer to an event that hypothetically might have caused death if it more severe.)
- requires inpatient hospitalization or prolongation of existing hospitalization;
- results in persistent or significant disability/incapacity;
- is a congenital anomaly/birth defect;
- is a suspected transmission of any infectious agent via a medicinal product; and

– is medically important.[1]

***Non-serious adverse drug reaction (ADR)*** (associated with use of the drug): Any untoward medical occurrence that does not meet the above criteria to be serious and considered associated with use of the drug.

***Life-threatening*****:** Any event in which the patient was at risk of death at the time of the event; does not refer to an event, which hypothetically might have caused death if it were more severe.

***Associated with use of the drug*****:** An AE is considered associated with use of the drug if attribution is possible, probable or very likely.

**Attribution definitions**

Causality assessment will be done by the physician at DR-TBC and are divided in five categories (mentioned below). The drug safety monitoring committee (DSMC) for BDQ conditional access will review and confirm causality of all serious events/reactions in relation to therapy.

***Not related*****:** An AE that is not related to the use of the drug.

***Doubtful:*** An AE for which an alternative explanation is more likely, e.g. concomitant drug(s), concomitant disease(s) or the relationship in time suggests that a causal relationship is unlikely.

***Possible*****:** An AE that might be due to the use of the drug. An alternative explanation, e.g. concomitant drug (s) or concomitant disease (s) is inconclusive. The relationship in time is reasonable; therefore, the causal relationship cannot be excluded.

***Probable:*** An AE that might be due to use of the drug. The relationship in time is suggestive, e.g. confirmed by de-challenge. An alternative explanation is less likely, e.g. concomitant drug (s), concomitant disease (s).

***Very likely*****:** An AE that is listed as a possible adverse reaction and cannot be reasonably explained by an alternative explanation, e.g. concomitant drug (s), concomitant disease (s). The relationship in time is suggestive, e.g. confirmed by dechallenge and rechallenge.

[1]Medical and scientific judgment exercised in deciding whether expedited reporting is appropriate in other situations, such as important medical events that may not be immediately life threatening or result in death or hospitalization, but may jeopardize the subject or require intervention to prevent one of other outcomes listed in the definition above. These is usually considered serious.

**Severity criteria**

An assessment of severity grade will be made using the following general categorical descriptors:

***Mild:*** Awareness of symptoms that are easily tolerated, causing minimal discomfort and not interfering with everyday activities.

***Moderate*****:** Sufficient discomfort is present to cause interference with normal activity.

***Severe:*** Extreme distress, causing significant impairment of functioning or incapacitation. Prevents normal everyday activities.

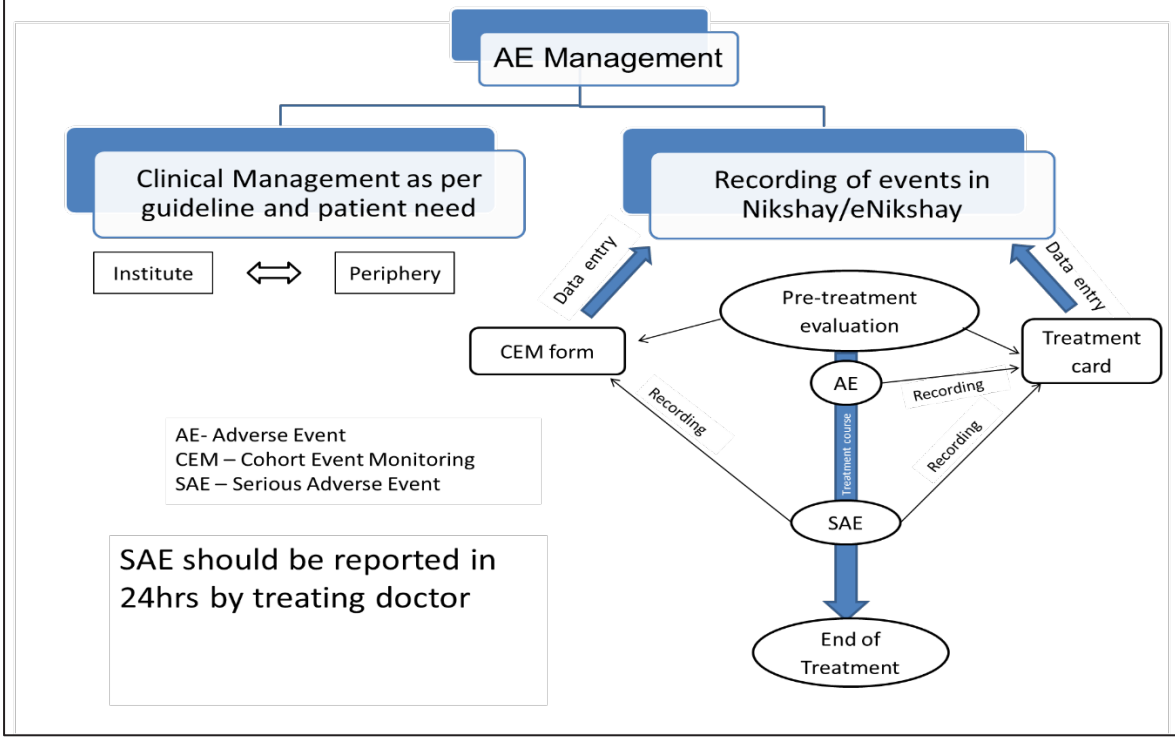
The investigator should use clinical judgment in assessing the severity of events not directly experienced by the subject, e.g. laboratory abnormalities.

**Reporting of AE, SAE and pregnancy**

All SAEs and AE's (non-serious adverse events which are possibly, probably or very likely related to administration of BDQ) that fit the definition as detailed later relate to detailed formats for AE reporting and pregnancy occurring during the programme. They must be reported by the physician to RNTCP as they occur. If pregnancy occurs during BDQ treatment, BDQ must be stopped and OBR modified as per RNTCP PMDT guidelines. Any death of a patient occurring during treatment in a BDQ-containing regimen, regardless of causality, must be reported as SAE and a verbal autopsy (Annexure 13) should be undertaken. It is recommended that the patient be questioned before commencement of treatment and at each subsequent consultation in order to obtain a detailed description of

any sign of toxicity or adverse drug reaction, which they might have experienced. The standard WHO formats for cohort event monitoring in the prescribed formats need to be maintained for every patient. RNTCP will ensure that strict aDSM is implemented by all NDR-TBC and district physicians for ambulatory patients. Figure 10.1 on ADR management and data capturing format related to Pharmacovigilance is shown below.

**Figure 10.1 ADR management recording and reporting**




The treating physician at N/DDR-TBC and medical practitioners at periphery will observe patients for any adverse events (spontaneous reporting by patient and active screening) and manage as per laid down criteria in document. The SAE will be reported to ADR monitoring centre (AMC) and CTD within 24 hours. Cohort Event Monitoring (CEM) will follow the patient pathway from registration to treatment outcome. Patient details will be captured as baseline (before starting treatment) and will get updated at regular monthly intervals till completion of treatment. The primary responsibility of filling up above forms will be with the treating physician. The Nodal officer will be responsible for data entry in Nikshay with help of SA at NDR-TBC and DR-TB supervisor at DDR-TBC centre. Concerned DTO should make necessary arrangements for data entry in absence of abovementioned support staff

Once relevant forms CEM are filled in Nikshay, information is directly communicated to PvPI through Vigiflow connecting bridge for signal generation. The centers need to ensure reporting of SAE within 24 hours to the Central TB Division using Nikshay followed by email to ddgtb@rntcp.org. Records need to be maintained in hard copies at respective centers.

The ADR data submitted to Vigiflow will be analyzed by PvPI and shared with CTD on regular basis. The data with action required on immediate basis will be shared with CTD by PvPI.

Safety assessment measure is the proportion of patients experiencing a Grade 3 or greater adverse event, as defined by DAIDS (Division of AIDS) criteria during treatment and follow-up. [37]

# Chapter 11: Treatment support

Tuberculosis, including the drug-resistant forms are completely curable with early detection and complete treatment. However, even the "short course" chemotherapy for drug sensitive TB might not be perceived as really short by the patient. Treatment of DR-TB, poses a greater challenge due to its longer course. It is observed that approximately 50% of DR-TB patients do not complete treatment, often due to loss to follow-up or death.

Early detection of drug resistance through rapid diagnostics and early initiation of treatment favours good prognosis. Adverse drug reactions, comorbidities, social neglect and catastrophic expenditures are major deterrents to successful treatment completion and relapse-free cure. A patient-centric approach ensuring adequate medical, social, psychological and nutritional support is essential for good quality of life. These supports are to be built into a well-tailored treatment support programme customized for each DR-TB patient. [13] [3]

Treatment support is the compendium of services provided to the patient enabling them to successfully complete the treatment. These services include therapeutic, emotional, social and financial support.

## 11.1 Principles of treatment support

The principles of treatment support entail minimizing patient travel, delay in treatment initiation and follow-up, catastrophic expenditure and infection in transit. It also includes maximizing patient satisfaction, adherence to treatment and transparency in operations.

Treatment support is not applied to all patients at all times in a uniform fashion. Some patients may not need additional support other than diagnosis and treatment TB including DR-TB. Some may need total support including social, nutritional, financial and residential support and special support for management of substance abuse, comorbidity *etc.,* Even though the patient is in need of support, s/he may not request for it since s/he may not be aware that such support could be availed through the healthcare team. Hence the need for support should be assessed and process of providing support initiated by the health system. However, health system may not be equipped to provide all forms of support with its own resources. In such situations, health system personnel should be able to link the patient to appropriate sources of support. There are successful examples in many states for such coordination that resulted in preventing unfavourable outcomes.

## 11.2 Patient-friendly approach

All staff in the DR-TB service delivery institutions and field should be considerate and behave humanely to patients suffering from DR-TB. They need to be trained and monitored for the same. Some of them may have concerns about working close to a DR-TB patient and getting infected with DR-TB; these need to be addressed in their training. All patients must be provided with a feedback form with rating scales for staff behaviour, health facility cleanliness and details of costs incurred. This feedback should be collected on a monthly basis and monitored at the district level. Initially these may be done on a paper-based system which will eventually be migrated into an ICT-based system. Once migrated to ICT-based evaluation, it could get linked to the annual performance report of health staff for further assessment.

## 11.3 Diagnosis and treatment initiation

Presumptive DR-TB patients are identified at the PHI/ private health facility and counselled by the MO-PHI/ treating physician regarding their status as presumptive DR-TB and subsequent course of action. Patients then need to present themselves to a DMC for specimen collection and transport to the relevant laboratory for DST. Some of the patients may need support to travel to the specimen collection centre. The MO/treating physician will ensure they receive the eligible financial support from the programme provision.

The lab technician should communicate the result of DST to the MO-PHI/ treating physician, patient and DR-TBC at the earliest possible time. In case the result indicates drug resistance, the patient will be asked to visit the nearest DR-TBC. In the DR-TBC, the PTE and any inpatient care should be completely free of cost. In case the service is chargeable, the same needs to be reimbursed to the patient. The DR-TB Counsellor/equivalent at the DR-TBC is the nodal person for ensuring that appropriate services are provided to the patients.

If the patient has not reached the DR-TBC within a week of the result being announced, tracking action will have to be initiated by the DR-TBC Counsellor in coordination with the District DR-TB Coordinator, STS and MO-PHI. For this purpose, the DR-TB Counsellor should have an electronic directory of field staff, who will assist in tracking. Patients referred for treatment initiation or further management to DR-TBC, should be advised to report to the DR-TB Counsellor. The patient should be provided with contact details of the necessary staff at the DR-TBC. Once the patient reports to the DR-TB Counsellor, s/he should counsel the patient about DR-TB drugs and treatment regimen, adverse drug reactions, follow-up and access to services.

In the DR-TBC, all patients needing in-patient care should be admitted under a committee member designated as nodal clinician responsible for DR-TB management. Patients who do not need in-patient care will be attended to by the nodal clinician who will coordinate

consultations and referral to other members of the committee or other clinical care. In case, services needed are not available under the same institution, the DR-TB counsellor should coordinate the same under the guidance of the nodal clinician. To minimize delays in pretreatment clinical evaluation, individual clinical specialists may evaluate the patient separately and convey their findings or comments in writing in the Treatment book. The nodal clinician, after considering the remarks or recommendations of members, will initiate treatment on the appropriate regimen. In no case, should the patient be kept waiting for beds, consultation with specialists and enablers.

## 11.4 Domiciliary, ambulatory and inpatient care

Patients on ambulatory care and those stabilized after inpatient care will be referred to the concerned PHI through a filled PMDT- referral for treatment form. The patient will carry his/her copy of the Treatment book. The DR-TB counsellor should counsel the patient on the various drugs that have been initiated and their adverse effects. Patients should be provided with a list of adverse drug reactions along with necessary actions to be taken. The Counsellor should train the patient about the possible adverse event. The DR-TB counsellor should also intimate the referral and relevant details of treatment in advance to the MO-PHI or treating clinician to arrange for drugs and necessary support.

All patients may not need inpatient care. However, patients such as those who are seriously ill, those to be initiated on newer drug regimens, patients who need management of drug reactions and patients from distant geographical locations, will preferably be managed as inpatients. Facilities such as diet and stay for patient and one attendant, ancillary drugs and any other related procedures should be provided free of cost.

## 11.5 Treatment supervision

In view of the potential adverse reactions of second-line drugs, the MO-PHI should arrange for provision of daily treatment supervision by a trained treatment supporter. Treatment supervision may be provided based on the collective decision of the patient and the MO-PHI through any of the various options available, such as health care worker, community volunteer, private provider, family observation of treatment Electronic Pill Box (Annexure 14), ICT support *etc.,*

## 11.6 Follow-up of treatment

The DR-TB Counsellors, DR-TB-HIV Coordinators, DR-TBC SA should track pending follow-up cultures and consultations and the information update at various centres. They should coordinate among themselves to ensure that their records are currently valid. Timely alerts should be provided to health staff and patients for prompting follow-up testing and consultations through ICT when available. ICT should also support real time record updating

at various levels. The need for counselling would be reassessed and done on each visit of the patient to their health facility by the MO-PHI and or DR-TB Counsellor. The patient should be reviewed monthly by the MO-PHI for treatment response, adverse drug reaction and follow-up specimen collection. Also, details of the same should be recorded in the relevant sections of the treatment book.

At no point during the diagnosis, pretreatment, treatment and follow-up, should the patient incur any direct cost to avail any service. The patient may incur some indirect costs such as travel expenses and loss of wages while accessing services. These are expected to be at least partially supported through the financial enablers provided to the patient at various stages post notification. These financial enablers are expected to be provided to the patient by a Direct Beneficiary Transfer (DBT) system.

## 11.7 Patient support for comorbidities

Many patients undergoing treatment for DR-TB might also be diagnosed with additional ailments such as hypertension, diabetes, COPD. The MO-PHI should ensure that treatment services for these comorbidities are made available.

## 11.8 Social support at the community level

During ambulatory care the MO/treating physician and staff of the PHI should identify the need for various forms of social support such as travel, nutrition, de-addiction, social security schemes and link the patient to sources that provide these supports. Occasionally this support may have to be locally initiated and financed. Social interventions with stewardship of the community have been found to be effective in promoting treatment adherence.

## 11.9 Treatment support for detection and management of ADRs

Adverse drug reaction (ADR) is a major cause for fatality and loss to follow-up. Patients should be carefully watched to detect and manage ADRs at its onset. Nodal Clinician and DR-TB Counsellor should educate the patients on potential adverse reactions, their signs and symptoms, necessity of timely reporting and their recording in the ADR section of treatment book. The STS responsible for supervision of the patient at home should actively search for signs and symptoms of ADR on every visit. Additionally, they need to keep in touch with the patient over phone to actively enquire for ADRs. Patient should be encouraged to report even the mildest symptom and record them in the ADR diary. On self-reporting by the patient or on elicitation by health worker, the MO PHI/treating physician should clinically examine and investigate for the cause. Some of the ADRs like a drug induced gastritis or itching might be mild and managed at the PHI level. Moderate or severe forms of ADRs such as toxicities to liver, kidneys or nervous system, psychiatric

abnormalities may warrant stopping of drugs and referral to the DR-TBC immediately. The MO should fill a referral for the treatment form with detailed history of the ADR and refer the patient with advance intimation to the nodal clinician and DR-TB Counsellor. Based on the ADR management guidelines, modifications in the present regimen may be made at the DR-TBC. Once the patient is referred back, the MO PHI should ensure that the patient has understood the modifications and educate the treatment supporter for the same. DTO needs to review ADR management at PHIs during the district review. [35] [36]

## 11.10 Support for airborne infection control

Principles of good DR-TB management include early detection, appropriate treatment and practice of infection control measures. All TB patients should receive counselling on prevention of airborne infection at home and at work place. Patients should be provided with spittoon, disinfectant and reusable masks and educated on their use. During house visits, the peripheral health workers should observe patient practicing cough hygiene and reinforce AIC messages. [14]

## 11.11 Patient helpdesk

At all service provision levels there are possibilities where the patient needs may not be adequately addressed. Patients should provide such feedback in the form of a help request or query to the system through a help desk mechanism. These can be submitted to the DTO on paper or email or through a website or ICT when available. Once such feedback is received, the District TB officer should categorize it based on the level at which the feedback is related and whether it needs action or not. Even if it does not require action, the message should be responded to. If it needs further action, the DTO must assign a competent authority to address the request/grievance along with a suitable time to review it. All feedback should be registered to a central state-wide registry. If the request/grievance is not addressed at a particular level it will be escalated to the next higher level. At all critical points, the patient should be informed of the status of his/her submission and have the ability to confirm that the grievance has been addressed. The turnaround time of various types of grievances and successful resolution should be monitored and included as a part of the quality assessment of TB care in the country.

# Chapter 12: RNTCP PMDT reporting and recording system

This chapter describes the information system for patients that fall under RNTCP PMDT. It also provides details on both paper-based as well as electronic reporting and recording system.

## 12.1 Aims of the information system

- to allow managers at different levels in RNTCP to follow overall programme performance through the distribution and trend in DR-TB notification; and the response to treatment in DR-TB patients treated with RNTCP regimen;
- to aid the staff in the treatment units in providing adequate management of the individual patient; and
- to enable real-time monitoring of the patients and real-time generation of reports.

There is the eNIKSHAY platform, which envisages that there will be different e-Health and m-Health solutions, that will ensure that the data is captured at the origin with web and mobile platforms for all levels. Data thus collected would then be analysed and shared with concerned all authorities in real time for actionable course correction.

## 12.2 Scope of the information system

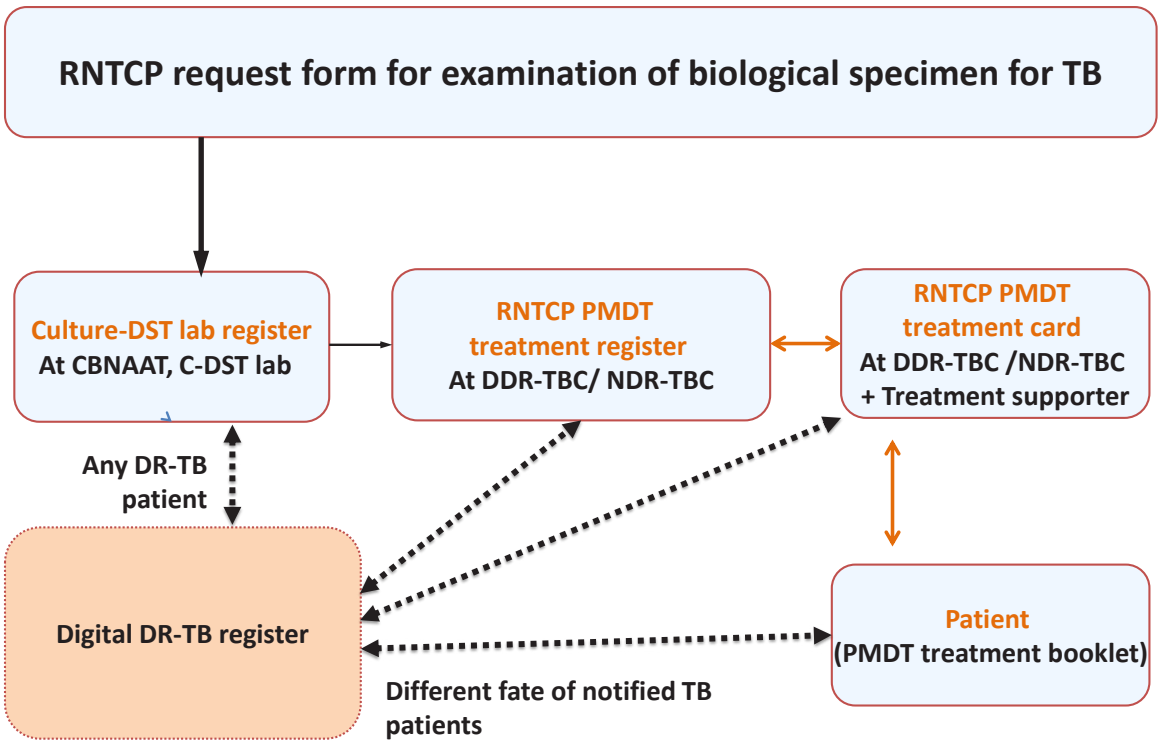
The information system for RNCTP PMDT is based upon and is an extension of, the basic RNTCP information system. The forms are therefore made as similar as possible to the existing forms in RNTCP. This chapter defines the minimum instruments and variables of the information system, necessary to satisfactorily implement and monitor treatment with various RNTCP regimens for DR-TB. This information system does not include all the detailed information that the treatment units may need to manage the individual patient. This is contained in the clinical record and other special forms used in the wards or clinics and depends on the local requirements and practices.

## 12.3 Records, reports and flow of information

The following section describes the forms, registers and reports that will be used for RNTCP PMDT to enable proper recording of diagnosis, monitoring and care, in addition to the reporting of outcomes. [5] [32]

The overall flow of information for data management in PMDT with hard copies and NIKSHAY/e-NIKSHAY is depicted in Figure 12.1 below:

**Figure 12.1 Data management in PMDT**




### 12.3.1 RNTCP Request Form for examination of biological specimen for TB (Annexure 15A)

All individuals who are presumptive TB or presumptive DR-TB are required to have a sputum or an appropriate EP specimen examination for diagnosis. The comprehensive Request form for examination of biological specimen for TB is to be used for requesting for microscopy, CBNAAT or culture DST or chest X-ray or TST or any other tests. It is essential to record patient details, reason for testing and type of tests requested. The front page of the form is for recording patient details, reason for testing (diagnosis or follow-up), test requested, result of sputum smear microscopy and NIKSHAY ID (expected in case of notified TB patient subjected for DR-TB screening). This form should be filled by the staff of the health facility that is sending the specimen with a request for examination and expected to be transported along with specimen to the concerned laboratory or health facility for the requested test. Back page is to report the results of CBNAAT, LPA, DST and any other tests requested.

For all specimens sent from peripheral health institutes, test requests should be initiated from the specimen collecting health facility and entered in NIKSHAY/e-NIKSHAY on the spot in real time. This will enable instant online intimation about the upcoming specimen at the health facility (CBNAAT site or C-DST laboratory) where these tests are requested prior to

receipt of the specimen while it is in transit. The result will be updated at the testing health facility in NIKSHAY/e-NIKSHAY on real time to save time required to scan results and send it to the concerned health facility. Final report of the test may be printed out through NIKSHAY at any health facility. Real time data entry helps the programme to disseminate information at all levels like N/DDR-TBC, District, TU and PHI to initiate reflex actions for further patient management on real time basis.

### 12.3.2 RNTCP laboratory register for culture, CBNAAT and drug susceptibility testing (Annexure 15L)

The RNTCP laboratory register for culture, CBNAAT and DST is used to record CBNAAT, LPA and culture and DST examination results. This will be maintained by the concerned laboratory staff. Results of all specimens tested at these sites are expected to be entered in NIKSHAY/ e-NIKSHAY. For all presumptive DR-TB patients, test results should be updated in existing NIKSHAY ID to maintain continuity while presumptive TB patients tested at these laboratories, NIKSHAY ID may be generated, after ensuring that it is not generated elsewhere. The individual should be notified if diagnosed with TB and NIKSHAY ID should be shared with along with the report. Results of subsequent tests carried out will be entered in the same ID. This gives an opportunity to easily extract the test results of all specimen provided by the patient and thereby track his/her response to the treatment. All the follow-up investigations carried out will be entered periodically using the same NIKSHAY ID.

### 12.3.3 RNTCP PMDT referral for treatment form (Annexure 15H)

The RNTCP PMDT referral for treatment form has to be filled for all confirmed DR-TB patients that are referred from one centre to another. The form has to be filled by the doctor of the referring centre in duplicate and one copy sent along with the copy of the current treatment card to the referred centre. This form can be used for referring the patient at various points in time during the management of the patient between the PHI, DTC and DR-TBC for reasons like initiation of treatment, adverse drug reaction, transfer out, ambulatory treatment or any other reason. In patients that are transferred out, a copy of the updated PMDT treatment card must be sent along with the referral for treatment form.

Referral module in NIKSHAY facilitates access to patient information through NIKSHAY ID. Provision of shifting of patient from one district to another is possible if the patient has changed his/her residence permanently. In NIKSHAY, the referring health facility must update details under ‘request for transfer’ section to intimate receiving health facility about the transfer even before the patient reaches. In addition to information mentioned in ‘request for transfer’ section, the receiving health facility is able to access all other patient information as well. The accountability of transferred patient is now with the receiving health facility and the treatment initiating facility.

### 12.3.4 RNTCP PMDT treatment card (Annexure 15 E)

The RNTCP PMDT Treatment card is a key instrument for the treatment supporter administrating drugs daily to the patient. The card will be initiated at D/NDR-TBC when the patient is initiated on treatment either on outpatients or indoor basis. The original treatment card will be maintained at the respective DR-TBC and copy kept by the treatment supporter. The card should be updated daily, documenting the administration of drugs by the treatment supporter. The card is the source to update periodically the PMDT register and adherence details on NIKSHAY. Accountable systems have to be developed locally for updating cards at all levels. When or if the patient moves from DR-TBC to his/her district of residence a copy of the card, must follow the patient. Once the patient has switched to other regimen at NDR-TBC, a new set of treatment cards along with new treatment booklet should be prepared for the patient and the same shared with the field. A copy of this card may be used as a notification form and to inform final outcome of treatment.

Newer ICT solutions (MERM *etc.,*) to promote information management about the patient's treatment adherence are being tested at field level for DR-TB patients. This will help in auto updation of treatment adherence details in NIKSHAY/e-NIKSHAY as well as flag patients with frequent treatment interruptions to prompt visits of health care providers and supervisors to intervene and retrieve the patient on treatment. If the patient is under the ICT monitoring system, treatment card in hard copy should be maintained at the treatment supporter level. A treatment supporter will be relieved from the regular updation of treatment card in NIKSHAY during such condition. The card contains the following sections:

**Page 1 of the treatment card:**

***Basic demographic information*:** Name, sex, age, address, telephone number, state, DR-TBC, district, TU, PHI and details of the treatment supporter. Some part of this information will be auto populated from available information while other information can be added at district or N/DDR-TBC level which should be supplemented by treating PHI.

***PMDT TB number***: This is a new unique patient identification number given to the patient at the DR-TBC on initiation of treatment. The PMDT TB number should include S.No./Nodal or District/Name of DR-TBC code/year of initiation of treatment. E.g. PMDT TB number of first patient started on treatment at Nagpur DDR-TBC during 2017 will be 1/D-NGP/2017. Every year PMDT TB number will be started at 1.The Nodal DR TB center will maintain its PMDT number separately starting from 1$^{st}$ of every year. e.g. PMDT number of 1$^{st}$patient initiated at Nodal DR-TB center belong to Nagpur District will be 1/N-NGP/2017.The district where the Nodal DR-TB center is located has to maintain separate District DR-TB PMDT register if center is also functioning as DDR-TBC. This would remain as a transitory system till the time most DR-TB patients are tracked with the help of NIKSHAY. This will be done till it is time to

completely transition from paper-based registers to autogeneration of electronic registers from NIKSHAY PMDT modules directly.

***NIKSHAY ID*:** NIKSHAY ID refers to the unique ID which is generated at the time of testing for TB. As per RNTCP TOG, all TB patients are to be notified at the time of diagnosis. This information is recorded in the TB Notification register. Pool of all DR-TB patients (RR/ MDR/H Mono/poly *etc.,*) diagnosed either at CBNAAT lab or C& DST labs or in private lab will be updated digitally. The NIKSHAY ID will capture all subsequent events that are taking place for diagnostic and treatment pathways. This creates a lifecycle approach where multiple events happen for a single patient pertaining to diagnosis and treatment pathways which are captured in a single NIKSHAY ID. All relevant information of a particular patient would be available with the same NIKSHAY ID which can be printed at all accessible users' levels. Once the patient is initiated on treatment, details are captured in PMDT treatment register that is maintained at D/NDR-TBC and PMDT number is assigned. The NIKSHAY ID has a provision to capture more than one PMDT number generated at different D/NDR-TBC while patient transfer is done. This provides a log of patient information that helps in tracking all updates starting from DR-TB diagnosis to long-term treatment outcomes.

***Reason for testing***: This section lists and describes details of the reason for testing. This includes types of patient that have to be ticked as applicable for new, previously treated, presumptive TB, private referral, presumptive NTM, criteria for presumptive DR-TB, presumptive H mono/poly and criteria for presumptive XDR-TB.

**Drug susceptibility test (DST) results at diagnosis**

Detail DST information for each patient is captured with date of specimen collection. All DST results for CBNAAT, LPA and LC-DST are captured here once the result is received. SL LPA result is available as FQ and/or SLI class resistance. Resistance is captured as R against all drugs that belong to that group. Once LC-DST result is available, it is mentioned in lower section of DST preferably.

***HIV testing***: This section lists the date of testing, PID number, date of starting CPT and ART (wherever applicable). As per the national policy, information sharing on HIV status of patients should be restricted within health care facilities based on the concept of "shared confidentiality". Hence, this information must not be written on the copy of the card held by the treatment supporter and patient booklet.

***Contact tracing***: This section details number of household contacts screened, number of presumptive TB patients identified, number of presumptive TB patients evaluated, number diagnosed with TB and number of DR-TB diagnosed. First contact screening should be carried out at the time of treatment initiation, however, this should be a continuous process during the course of treatment as any close contact may present with TB symptoms at any

point of time during the course of treatment for index patients. In pediatric DR-TB patients, reverse contact tracing to search for an adult index patient in the household must be considered by health care workers.

**Page 2 of the treatment card:**

***Treatment regimen*:** The RNTCP regimen for MDR/RR-TB, XDR-TB, H mono/poly resistant TB, modified regimen for MDR/RR + FQ/SLI resistant, modified regimen for mixed pattern resistant and regimen containing newer drugs are recorded in this section.

***Drugs and dosages:*** This section details drugs and dosages used including status of eligibility and consent for regimen containing newer drug. In NIKSHAY/e-NIKSHAY, default value of drugs and doses based on treatment regimen and weight will be auto populated. However, modification in doses and drug included in regimen may be possible if required.

***DR-TB centre Committee meetings*:** There should be periodic meetings of the DR-TBC committee, with caregivers involved with DR-TB patients, in which the progress of the individual patient is reviewed. This section provides space to record major changes by the committee, like extension of IP; change of IP to CP; completion of treatment; severe adverse reactions; change of treatment, declaring treatment outcome *etc.,* Date of decision taken is an important parameter which needs to be captured in this section. Some decisions may be taken at peripheral level in consultation with DR-TBC like discontinuation of medicine due to serious ADR.

**Page 3 of the treatment card:**

***Monitoring of culture/other investigations:*** Record the date, specimen number and result of the monitoring culture examinations and other investigations like Serum Creatinine, LFT, ECG/QTC interval, Electrolyte (K, Mg, Ca) and UPT. The culture date is the date on which the sputum was collected from the patient for these tests. Follow-up investigation carried are also captured in the relevant month of follow-up.

***Blood sugar***: This section details date of testing of Random Blood Sugar (RBS), Fasting Blood Sugar (FBS) and Date of starting treatment of diabetes (wherever applicable).

***Thyroid function test:*** This section details date and level of testing of T3, T4 and TSH.

***CXR***: Details of the report of Chest X rays performed should be entered in relevant section.

**Page 4 of the treatment card:**

***Patient detail:*** This details name of patient, initial weight and height, date of starting intensive phase, date of stopping Bdq and date of starting continuation phase.

***Detail of change*:** This portion details date of change of regimen and reason for change.

***DST:*** Record the date, type of culture test used and results of all DST performed on the treatment card. Enter 'R' for resistant and 'S' for sensitive under drugs for which DST has been performed at the RNTCP-certified laboratory. Drugs which have not been tested will remain blank.

**Page 5 and 6 of the treatment card:**

***Record of administration of drugs***: One line per month which makes it easy to assess adherence. One box is checked for each day the treatment is administered. Regimen is going to modify based on SL-LPA and LC-DST results. A down arrow is placed in the cell of regimen modification date to indicate day when regimen modification was done. Details information regarding drug modification is written in the place where DR-TB committee decision is captured. Any modification made should be indicated in patient adherence record as well. The CP should be documented on new line.

**Page 7 of the treatment card:**

***Date and details of retrieval action***: These should be taken and recorded in relevant section.

***Date and details of adverse drug reactions*:** These actions should be taken and recorded in the relevant section.

***Patient follow up visit details:*** All DR-TB patients need to be counselled about monthly clinical follow up with doctor and finding of this visit along with date should be documented in this

***Outcome of treatment***: At the end of treatment, the outcome should be recorded on the treatment card. The outcome definitions are given in Chapter 10.

***Post treatment follow-up clinical & sputum*:** This section detail 6, 12, 18 and 24 post treatment follow up months of clinical, sputum and chest X-ray follow-up of the patients.

### 12.3.5 TB Notification register (Annexure 15I)

A TB notification register is maintained at each peripheral health facility. This register contains records of all patients diagnosed with TB and eligible for TB treatment, regardless of initiation of treatment. Registration data is based on date on which a TB patient is diagnosed. All newly diagnosed TB patients and newly diagnosed DR-TB patients need to be notified in TB notification register at all diagnosing health facilities. All newly diagnosed DR-TB patients should be considered as new episode and patient details should be entered in

new row in TB notification register with the same NIKSHAY ID. Majority of DR-TB diagnostic labs (CBNAAT and C& DST) are located at health facility (PHI) where TB notification register is already maintained. Same TB notification register needs to be used to notify diagnosed DR-TB patients unless there is need of separate TB notification register at DR-TB diagnosing health facility. If NIKSHAY ID is available, where the patient is already notified to the programme at the time of TB diagnosis, same ID will be used to notify it for DR-TB diagnosis. If presumptive TB patient diagnosed as DR-TB patient and NIKSHAY ID was not generated any time in the past, a diagnostic health facility will notify this patient and generate NIKSHAY ID.

Once the patient is notified in TB notification register (hard copy) for DR-TB episode, status of treatment must be updated against that episode. Treatment outcome of previous episode should be updated in TB notification register and in NIKSHAY at the concerned health facility. For every patient, the status of treatment should be recorded and this could be any one of the following:

- initiated on first-line treatment in the same health facility;
- initiated on treatment outside the health facility;
- initiated on second-line treatment;
- treatment initiated outside RNTCP;
- incomplete/ incorrect address;
- died;
- migrated & untraceable;
- refusal for treatment;
- repeat diagnosis;
- patient already on treatment/ follow-up patient;
- wrong diagnosis;
- referred for treatment with pending feedback; and
- other

In each health facility, TB notification register is maintained by its staff. STS of the respective TB units will support updating and coordination for completing the information.

Once, the patient is initiated on DR-TB treatment, the patient will be entered in RNTCP-PMDT treatment register maintained at the DDR-TBC. TB notification register kept at the peripheral health facility where DR-TB patient is residing does not need to be updated as treatment and follow-up schedule for DR-TB patient is not the purview of TB notification register. However, if needed, PHI may keep a separate RNTCP- PMDT register for the same purpose. Information about DR-TB patients notified from private sector and RNTCP diagnostic labs can be extracted from NIKSHAY.

### 12.3.6 RNTCP PMDT treatment register (Annexure 15J)

This register is maintained at NDR-TBC and DDR-TBC. Districts where DDR-TBC is away from DTC or have more than one DDR-TBCs, additional register should be maintained at DTC. While TB notification register captures the details about the notified TB patients, the RNTCP PMDT TB register is restricted to patients who have actually started on a second-line TB treatment regimen. Patients should be entered consecutively by their date of treatment initiation. The following is recorded in the PMDT treatment register:

**LEFT side of page:**

***PMDT TB No*:** This is a unique identification number for patients initiated on DR-TB treatment and has been described earlier. Every year the PMDT TB number will be started from 1. DR-TB patients transferred in from the other DR-TBC will also find the place in this register with new PMDT number of receiving district.

***Other personal details:*** Name, sex, age, address, RNTCP district and TU of residence and name of PHI providing treatment support***.***

***Reason for testing*:** This includes various types of patients that are coded as applicable for various reasons for testing the patient. Only codes to be used as mentioned below the page.

***Site of disease*:** Whether pulmonary or EP is mentioned in this column.

***Type*:** Type of TB patient whether new, recurrent, Treatment after Lost to Follow-up (TALFU), Failure or Others is mentioned in this column.

***DST details:*** Type of test (L J / LC / LPA/ CBNAAT), date of DST and results of DST for various drugs are mentioned in this section. The drug whose DST is not done is kept blank. Extra column is added here for recording name of drug for which DST result might be available in the future within the programme.

**RIGHT PAGE**

***Type of DR-TB patient*:** Type of patient, whether H mono-/poly DR-TB,RR-TB, MDR-TB or XDR-TB is mentioned here

***Type of DR-TB regimen*:** Type of regimen prescribed for patients as per code mentioned below the page is recorded in this column.

***Microscopy examination, culture & DST results during follow-ups*:** Culture result with date during initiation (0 month) and subsequent follow-up culture results are mentioned in this

section without any delay under the specific month of follow-up until the end of treatment. 'Pos' for positive and 'Neg' for negative result is mentioned.

***TB-HIV collaborative activities*:** Date of testing the DR-TB patient, PID no, HIV status ('Pos' for positive and 'Neg' for negative result) and date of initiation of CPT and ART (wherever applicable) is recorded in this section.

***Final treatment outcome*:** The final treatment outcome as described earlier is mentioned in this column. There will be only one outcome for each patient.

***Remarks*:** This column is reserved for any additional information that may be need to be given in the register.

### 12.3.7 TB Laboratory Register (Annexure 15K)

This register need to be maintained at microscopy center. All the specimens examined for diagnosis or follow up on microscopy should be entered in the register. It is essential to capture patient information such as contact details and reason for testing, which will aid in approaching the patient subsequently. If any TB patient is eligible for the further DRT/DST, detail of specimen collection and result needs to be updated in respective column of this register. These details are to be captured in TB notification register as well. In case the specimen is send for DST from the other health facility, date of specimen sent and name of health facility sending the specimen is be updated here.

### 12.3.8 RNTCP Laboratory Register for culture, CBNAAT and drug susceptibility testing (Annexure 15L)

All the labs doing DST/DRT/Culture either by genotypic (CBNAAT/ FL-LPA/ SL-LPA) or phenotypic (Solid/ liquid Culture) technology need to maintain this register. All specimens tested by any method need to be entered in this register. This register is used to capture details of specimen tested for diagnosis as well as follow up.

### 12.3.9 RNTCP PMDT treatment book (Annexure 15M)

RNTCP PMDT Treatment Book is the document that must be always available with the patient. When a patient is diagnosed as having DR-TB and is placed on a regimen for DR-TB, RNTCP PMDT patient Treatment book should be filled out by the health care provider at the same time when the PMDT treatment card is filled out at the time of initiation of treatment. This Treatment book will be kept by the patient and should be brought whenever s/he comes to DR-TBC or DTC or PHI for clinical follow-up or for ADR management. The Treatment book contains the following section:

- name, sex, age, complete address, marital status, contact number and *Aadhar* ID of the patient;
- name & designation and contact number of Treatment supporter;
- name of TB Unit, PHI, DR-TBC, District and State;
- information about Initial home visit, by whom it has been done and on what date;
- information about NIKSHAY ID and PMDT TB numbers;
- essential information about Reason for Testing, which are as follows:

    - whether new or previously treated patient;
    - whether presumptive TB, private referral or presumptive NTM;
    - whether presumptive DR-TB at diagnosis or due to contact of DR-TB patients or follow-up smear positive at end of IP or it is a private referral;
    - whether presumptive H mono/poly resistant TB; and
    - whether presumptive XDR-TB patient due to MDR/RR-TB at diagnosis or four months culture positive or culture reversion or failure of MDR/RR-TB regimen or recurrent patient of second-line treatment.

- information about DST results for different anti-TB drugs;
- information about contact investigation (number of members screened, out of it number of presumptive TB identified, out of which number of presumptive TB patients evaluated, and out of which number of TB patients & DR-TB patient diagnosed);
- information about DR-TB Committee meetings with dates and decisions;
- information about TB site (whether pulmonary or extra-pulmonary) and the different types of treatment regimen under which patient has been provided the treatment, with date of initiation of treatment and date of registration;
- information about weight (in kg) and height (in cms) of the patient;
- information about different weight-bands for DR-TB regimen;
- information about different types of anti-TB drugs prescribed to patients and its dosages;
- information about eligibility and consent of patient if a new drug has been prescribed;
- information about culture results and other investigations (serum creatinine, liver function tests, ECG, complete blood count, serum electrolytes, urine test for pregnancy) done at different interval of continuation of treatment till the end of treatment;
- information about DST results for different first and second-line anti-TB drugs done in different months [with date of specimen collection & type of DST (L J/LC/LPA/CBNAAT);]
- information about blood sugar testing (random blood sugar and fasting blood sugar) with date and initiation of anti-diabetic treatment;
- information about thyroid function test done at initiation of treatment and at end of six months of treatment;
- information about X-ray test done at different interval;
- information about dates of starting intensive and continuation phase;

- information about change of regimen and reason for the same;
- information about monthly administration of drugs with weight of patient for full duration of treatment;
- information about retrieval action taken for a patient who has missed his doses;
- information in detail about any adverse drug reaction taken place and action taken for its remedy
- clinical notes made by physicians during visit by DR-TB patient for any complaint, which includes the following:

    - date of visit;
    - chief complaints made by the patient;
    - major findings of clinical examination;
    - different types of investigations done;
    - what treatment provided; and
    - counselling notes.

- Information about treatment outcome of the patient with date; and
- Information about post-treatment follow-up clinical & sputum examination (result with date) done at interval of 6, 12, 18 and 24 months after end of treatment.

### 12.3.10 Patient counselling register (Annexure 15N)

DR-TB counsellor is supposed to update and maintain this register on a regular basis. Counselling session carried out with the patient and family members at the time of pretreatment evaluation to long-term follow-up are expected to be captured in this register. Priority should be given for counselling of the DR-TB patient, however, counselling services utilized for the DS-TB patient also needs to be captured in the same register.

### 12.3.11 Additional records and reports for Bedaquiline

In addition to the above, the following two new forms would be introduced for patients treated with BDQ containing regimen:

**Patient education booklet for BDQ-containing regimen**

A detailed patient education booklet has been developed for educating the patient on the use of BDQ (Annexure 9). The patient education booklet must be provided to the patient that contains the list of drugs contraindicated or to be used with caution with BDQ along with the PMDT treatment book. The patient must be motivated to carry these documents at every visit to any health care provider throughout the treatment course.

**Cohort event monitoring form**

The standard formats for cohort event monitoring (Annexure 12A, 12B) would be used at all the sites for aDSM. [10] [17] [4] Patient initiated on any DR-TB regimen must be covered under cohort event monitoring mechanism from the time of initiation of treatment. Treatment initiation form should be filled at the time of initiation of treatment for any SAE reported during the course of treatment. The review form of CEM is filled by health facility managing SAE. Any SAE needs to be promptly reported by the concerned health facility to RNTCP using the standard form on a real-time case-to-case basis in NIKSHAY.

Hard copies of the CEM forms need to be maintained at the respective health facility. If the patient's treatment regimen is changed due to any reason, irrespective of duration of treatment, a new CEM - treatment initiation form is filled by the treatment initiating health facility for the new class of regimen the patient is initiated on. The process of data entry and analysis for aDSM and event-based ADR reporting using NIKHSAY and Vigiflow has been described in detail in the Chapter on Adverse Drug Reaction & Monitoring.

## 12.4 Periodic reporting

There would be no periodic reporting based on paper or excel. This can be prepared as an output of the NIKSHAY/e-NIKSHAY module of PMDT, *i.e.,* laboratory form, treatment card and patient booklet. This can only be possible, if real time entries are made by all labs, DDR-TBC/ NDR-TBC as well as field staff. The dash board will give the process and output indicators as required. However, if states do not ensure real time data entry on NIKSHAY Lab and PMDT module, there remains a threat that incomplete and incorrect data entries may not reflect the real performance of the districts and states in PMDT. For the patient diagnosed from 01 January 2017, all laboratories and DR-TBC are expected to enter the updated RNTCP form for examination of biological specimen for TB and PMDT treatment on real time basis in NIKSHAY for all level of monitoring. However, for historical cohorts, this would be for only six months/12 months. Treatment outcome reports would be prepared for those quarters for which case finding was given till Q4, 2016.

## 12.5 Computerized systems

All information will be available in both paper and electronic versions. To facilitate better quality of information as well as data analysis, NIKSHAY for real time monitoring of DR-TB patients through Dashboards and monitoring indicators will be used. States are expected to monitor all relevant indicators of diagnosis and treatment initiation cohort. All relevant users will be able to monitor auto calculated monitoring indicator from Nikshay and gradually the whole MIS system will migrate to e-NIKSHAY over a period of time.

## 12.6 Training in data management

The information system requires knowledge of the RNTCP basic information system, with additional training on the specifics of the RNTCP PMDT MIS. Regular supervisory visits by the central team to the PMDT treatment centres using the information system, are fundamental to maintaining good quality of information. The digital DR-TB register generated by NIKSHAY consists of all DR-TB patients diagnosed by any laboratory including the private sector laboratory available within the district. The status of DR-TB treatment initiation similar to the TB notification register is updated by the district for each DR-TB patient diagnosed within the district. Once the patient is initiated on treatment, all follow-up investigation details must be uploaded by C-DST laboratory while treatment adherence details must be updated by PHI, TU or district in coordination with the respective DR-TBC and C- DST lab.

## 12.7 Cohort analysis

Patient diagnosed with any type of drug-resistance irrespective of diagnostic technology in specified period is considered in the DR-TB diagnostic cohort for that period. This pool of patients includes DR-TB patients diagnosed and notified from the private sector located in the catering area of the health facility. Reasons for not initiating patients on any treatment regimen need to be updated in NIKSHAY by the concerned health facility where the patient is residing.

Similarly, all patients identified with DR-TB and to be treated with a regimen for DR-TB, should be entered into the RNTCP PMDT register maintained at the DDR-TBC / NDR-TBC. A DR-TB treatment cohort is defined as a group of patients registered for treatment during a specified time period (e.g., one quarter of the year). The date of registration for regimen for DR-TB determines what case finding cohort the patient belongs to. However, since it would take around 2-3 months for most DR-TB patients to be either continued on standard regimen or reclassified to a DST guided regimen with/without newer drugs based on SL-LPA and LC-DST detailed earlier, for analytical purposes, the cohort definition would be done three months after the last day of the quarter for the case finding report. For example, for Q1 (Jan – March), the cohort definition will be done in the 1$^{st}$ week of July to enable accounting of patients in the respective standard or DST guided regimen with/without newer drugs only once and avoid double counting of DR-TB patients in the same quarter due to change in regimen. This will enable monitoring of interim and final outcomes of patients stratified by the specific regimen that the patient was eventually put on.

The cohort analysis would be done electronically as outputs from NIKSHAY which has a feature to maintain a log of all changes made in treatment. For example, based on the CBNAAT result, if the patient is initiated on a shorter regimen and is found to be FQ resistant on SL-LPA, s/he switches over to newer drugs containing regimen and is accounted in a system as a patient on newer drug containing regimen along with cohort. However, the

shorter regimen treatment related detail is still available in NIKSHAY for future reference. Reports for specific periods or cohorts could be generated online through NIKSHAY. Further, auto-generated indicators and interactive maps would be made available to facilitate supervision, monitoring and evaluation at all levels.

Cohort analysis should be performed on all registered DR-TB patients, using the date of DR-TB registration to define the cohort. Cohort analysis of treatment outcomes should also be performed on all patients who receive DR-TB treatment, regardless of treatment duration. The recommended time-frame for DR-TB treatment cohort analyses reflects the long duration of DR-TB treatment regimens. Final analysis should be performed 36 months after the last patient enrolment date in the cohort.

Patients still on treatment at the end of a designated cohort treatment period must be explicitly identified as such irrespective of whether they were culture-positive or negative at the time of cohort analysis. This is an interim status until a final outcome is available. The interim status must be assessed at six months and 12 months of treatment to monitor patient progress.

# Chapter 13: Supervision, monitoring and evaluation on PMDT

In this chapter, participants will learn about guidelines for supervision, monitoring and evaluation systems that need to be rationalized in all states and districts to ensure quality of care of DR-TB patients enrolled under the programme. An effort has been made by the programme to standardize these mechanisms along with requisite tools to do so which have been shared in this section.

RNTCP has a robust recording and reporting system in place along with multiple internal/external checks to ensure good quality data generation which forms the basis for existing RNTCP supervision and monitoring strategy.

However, in view of the expansion in programe activities the strategy needs to be more comprehensive with transition from target-focused monitoring of performance to analysis of key process and outcome indicators. Establishing a reliable monitoring and evaluation system with regular communication between the central and peripheral levels of the health system is vital. This requires standardized recording of individual patient data, including information on treatment outcomes, which are then used to monitor indicators.

The strong supervision, monitoring and evaluation ensure that activities are implemented as planned and that the data recorded and reported is accurate and valid; incorporating a system which leads to remedial action to improve performance; serving as a tool to facilitate commitment of higher authorities at different levels, ensuring equitable provision of services to all sections of the community, including vulnerable areas and populations such as urban slums, scheduled caste/tribal/minority pockets *etc.;* and above all, bringing transparency and accountability.

## 13.1 Organization of SME for PMDT

The SME will be driven by NIKSHAY/e-NIKSHAY with user-based/institute-based logins, user-based task lists and reminders, escalation matrix enabling prioritization, ICT-based adherence support, including ICT-based adherence monitoring systems like MERM boxes and other modalities. A more comprehensive SME plan is required for use at the state, district, sub-district and field levels. Various registers and reports on PMDT related services will be extracted from NIKSHAY/ e-NIKSHAY. Adherence score of individual patients based on an automated ICT enabled system like the pill box (MERM) or that which is manually updated, will be auto calculated and various relevant escalations generated for different users like NDR-TBC, DDR-TBC, C-DST Labs, CBNAAT labs, District and TU level.The organization of SME activities and the requisite tools are classified in the table below:

Table 13.1: Organization of SME in PMDT

| Stage | Tools |
|---|---|
| Supervision | • Supervisory checklists for various levels (CBNAAT, C&DST labs, N/DDR-TBC, District, TU, DMC, and patient) |
| Monitoring | • PMDT outputs and dashboards from NIKSHAY / e-NIKSHAY<br>• Monitoring indicators on lab services & PMDT implementation |
| Evaluation | • IE formats to include section for implementing states/ districts with access * Progress on scale-up plan, visits to D/NDR-TBC, CBNAAT Lab, C-DST Lab, DDS, SDS, patients' interview *etc.,* |

## 13.2 Supervision of PMDT services

RNTCP has a robust built in system for supervision. PMDT supervision will be an extension of this system. Similarly, the built in M&E system of RNTCP will be customized according to the levels of implementation and scale-up plans. Further, to guide field level staff to prioritize their activities, there are supportive systems developed under NIKSHAY/ e-NIKSHAY. Use of this supportive mechanism could improve their capacity and quality of services. It is very important to remember that supervision promotes successful implementation of programme policies and processes and M&E ensures that implementation progresses in the right direction to achieve desired targets, objectives and goals.

**Objectives of supervision**

- build capacity of health staff to implement PMDT procedures correctly;
- ensure that the data recorded and reported is accurate and valid;
- incorporate a system of analysis and review aimed at improving quality of programme implementation;
- increase involvement and commitment of staff at different levels;
- ensure field staff respond to NIKSHAY/ e-NIKSHAY tasks, lists activities and updates missing information promptly;
- provide actionable and timely feedback;
- evaluate impact of training on performance of health staff;
- assess retraining needs; and
- assess stocks and replenishment of supplies.

**Preparation for supervisory visit**

Since PMDT is not a standalone activity within RNTCP, all functionaries responsible for implementation of RNTCP are bound to supervise and in turn, be supervised in PMDT. A

checklist of activities to be supervised in a centre proposed to be visited is to be prepared in advance. Priority actionable points should be identified from NIKSHAY/ e-NIKSHAY (refer to PMDT supervisory checklists for various levels). Actions taken and pending from the previous supervisory visit must be reviewed during every visit. Review of previous reports is useful for identifying priority areas to be focused during supervision. Existing documents like RNTCP Supervisory registers placed at all health institutions may be used for recording observations on DS-TB as well as DR-TB.

During field visits by State-level supervisors to districts implementing PMDT, a selection of patients on various regimen for DR-TB and their Treatment supporters are to be interviewed. In addition, processes involved in the recording and reporting; drugs and logistics and supply chain management, tracking of transportation of sputum specimens to CBNAAT and C-DST laboratory, status of data update on NIKSHAY/ e-NIKSHAY, referral of diagnosed DR-TB patients to D/NDR-TBC *etc.,* have to be examined in detail.

**Modalities of supervision**

Though supervision of PMDT must, ideally, be linked to supervision of DS-TB, additional supervisory checkpoints pertaining to PMDT are discussed below. Recommended modalities for supervision by different levels of supervisory staff are presented in Table 13.2. All other RNTCP staff are to follow their terms of reference (ToR) ensuring that diagnosis and care of presumptive DR-TB and patients is taken care of on priority basis. Extensive checklists and monitoring tools have been developed for use by all supervisory staffs and must be put to use. Moreover, all visits to the district and subdistrict levels by district and State level officials have to, mandatorily include supervision of PMDT activity. All Central and State level appraisals have to review PMDT activities using standard PMDT supervisory checklists.

Table 13.2: Recommended modalities for supervision by different level of supervisors

| Supervisor | Methodology | Frequency |
|---|---|---|
| DTO | • Conduct interview with health staff and RNTCP key staff involved in PMDT;<br>• Interact with community and local opinion leaders and mobilize their support to help DR-TB patients with diagnosis/treatment;<br>• Randomly interview patients on treatment, their Treatment supporter, family members and community leaders;<br>• Inspect records of DTC, TU, DMCs, PHI and Treatment centre, NIKSHAY entries and stock of drugs;<br>• Check status of card updating and ensure original card at DR-TBC is updated at least once a month with digitalization of adherence data in NIKSHAY manually or by ICT supported mechanism; | • Visit all TUs every month and all DMCs every quarter;<br>• Visit all CHCs and Block PHCs in the district every quarter, one sub-centre from each Block PHC and proportion of treatment observation centres every quarter;<br>• Conduct supervisory visit at least 3-5 days a week; and |

| | | |
|---|---|---|
| | • Ensure prompt identification of presumptive DR-TB and transport of sputum specimens to lab as per guidelines maintaining cool chain;<br>• Physically verify stock of PWBs at District, TU and PHI stores;<br>• Ensure uninterrupted supply of medicines; and<br>• Liaise with State TB Cell, DR-TBC and designated C-DSTlab | • Visit at least three patients at their homes per visit including one DR-TB patient on treatment. |
| MO –DTC | • Conduct interview with health staff and RNTCP key staff involved in PMDT;<br>• Interact with community and local opinion leaders and mobilize their support to help DR-TB patients with diagnosis/treatment;<br>• Randomly interview patients on treatment, their Treatment supporter, family members and community leaders;<br>• Inspect records of DTC, TU, DMCs, PHI and Treatment centre, NIKSHAY entries and stock of drugs; and<br>• Check status of card updating at District, TU, PHI and Treatment supporter. | • Visit all TUs every month and all DMCs every quarter;<br>• Visit all CHCs and Block PHCs in the district every quarter, one sub-centre from each Block PHC area and proportion of treatment observation centres every quarter;<br>• Conduct supervisory visit at least 3-5 days a week; and<br>• Visit at least 3 patients at their homes per visit, including one DR-TB patient on treatment. |
| MO-TC | • Interview MO I/C Block PHC/CHC/PHC/private/NGO hospitals regarding implementation of PMDT activities;<br>• Randomly interview patients, their Treatment supporter, family members and community leaders;<br>• Interact with community and local opinion leaders and mobilize their support to help DR-TB patients with diagnosis/treatment;<br>• Inspect records of TU, DMCs, PHI and Treatment centre, NIKSHAY entries and stock of drugs;<br>• Check status of updating treatment card, booklet and NIKSHAY / e-NIKSHAY entries; and<br>• Ensure drug availability for all DR-TB patients during treatment course. | • Visit all DMCs every month;<br>• Visit all CHCs/ BPHCs/ PHCs and a proportion of treatment observation centres at least once every quarter;<br>• Conduct supervisory visits 7 days a month; and<br>• Visit at least 3 patients at their homes per visit, including one DR-TB patient on treatment. |
| Senior DR-TB TB-HIV Supervisor (this section deals only with PMDT responsibilities) | • Assist DTO in organizing direct observation of treatment for DR-TB patients and DR-TB drug logistics management;<br>• Facilitate MOs, STSs, STLSs, LTs and other health system staff to subject all Presumptive DR-TB to appropriate diagnostic tests for diagnosis of DR-TB at an RNTCP-certified laboratory;<br>• Identification and training of Treatment supporters for DR-TB patients and maintenance of a directory of such Treatment supporters at TU and district levels;<br>• Maintain district level PMDT records and reports;<br>• Ensure that records and NIKSHAY entries of all DR-TB patients in the district are updated regularly; | • Visit all TUs every month and all DMCs every quarter.<br>• Visit all treatment observation centres in the district once in every quarter |

| | | |
|---|---|---|
| | • Supervise all PMDT treatment observation centres once a quarter;<br>• Coordinate with field staff to ensure drug availability for all DR-TB patients receiving various regimen;<br>• Update treatment cards and NIKSHAY entries at DDR-TBC;<br>• Assist DTO in providing training to staff of health facilities under his/ her jurisdiction to carry out PMDT related activities;and<br>• Establish liaison with private practitioners, NGOs and other sector dispensaries/ hospitals to provide PMDT services as per programme guidelines. | |
| STS | • Interview MPHS/ MPWs at PHC sub-centre regarding implementation of PMDT activities;<br>• Interview Treatment supporter of patients on DR-TB treatment;<br>• Help DTO in identifying/training suitable Treatment supporters for diagnosed DR-TB patients to be initiated on DR-TB treatment;<br>• Verify records, cards and TB lab register; ensure treatment cards at DTC, TU and PHI as well as NIKSHAY entries are updated at least once monthly;<br>• Visit and interview all patients eligible for DST who haven't been tested so far and who are on treatment; ensure they are diagnosed at the earliest and complete treatment as per guidelines; facilitate for follow-up sample collection of all DR-TB patients;<br>• Ensure drugs and logistics management for patients on treatment;<br>• Interview health staff of identified private/NGO/other sector health care centres; and<br>• Impart hands-on training and guidance to Treatment supporters on proper administration of treatment, recording in treatment card and prompt identification of adverse drug reactions. | • Visit all PHIs at least once a month and all Treatment Centres once a quarter;<br>• Visit all diagnosed DR-TB patients at their home within a month of treatment initiation; and<br>• Conduct supervisory visits at least 5 days a week |
| STLS | • In consultation with DTO and MO-TC, put systems in place to ensure all presumptive DR-TB are diagnosed at the earliest; facilitate transport of sputum specimens of these presumptive DR-TB to the designated RNTCP-certified lab for C-DST;<br>• Visit all microscopy centres, review lab records, check stocks of conical tubes, packing materials, lab form and specimen transport boxes and ensure that cool chain is maintained; ensure NIKSHAY entries from all labs; and<br>• Impart hands-on training and guidance to LTs on identification of presumptive DR-TB and transport of their sputum specimens to the lab as per guidelines with proper documentation. | • Visit all microscopy centres in the jurisdiction of the TU at least once a month; and<br>• Visit all sputum collection centres at least once a month. |

## 13.3 Surveillance

Well-performed surveillance is an instrument for informing healthcare workers, public health experts and decision makers in order to guide and prioritize their action. It is a basic component in the control and elimination of TB and provides information on the epidemiology of disease, evolution of trends and description of groups in the population at increased risk of TB and unfavourable prognosis. It is an essential element in monitoring effectiveness of interventions aimed at control and elimination of the disease.

A good TB surveillance system would require timely notification of all TB patients in the population and should be able to capture necessary variables for demographic, clinical, socioeconomic, geographic and spatial characteristics to enable better understanding of local epidemiology and trend of TB.

TB surveillance should include data from laboratories as they play a pivotal role in TB diagnostics and patient ascertainment. This will help ensure completeness of reporting. Surveillance of TB should address current challenges of the disease. In that sense, surveillance of drug resistance and treatment outcome monitoring are essential tools for evaluation of TB programme. Reliable case-based notification systems are vital for a good surveillance system and surveillance should be enhanced for vulnerable groups.

NIKSHAY/ e-NIKSHAY will play a pivotal role in surveillance of DR TB patients by capturing important information at source and disseminating it at various levels and thereby acquire patient-wise information.

**Objectives**

- evaluate epidemiological characteristics of TB in the population over time and geography, within states, regions and across the country as a whole;
- monitor performance of TB management activities and feed this information into the decision making cycle to allow for appropriate interventions to upgrade districts, state and national TB plans; and
- identify and describe vulnerable populations at increased risk of TB and unfavourable prognosis to which targeted public health activities should be addressed.

**Strategies/actions**

***Evaluating epidemiological characteristics of TB:*** Doing this by strengthening nationwide surveillance systems and other sources of data collection; and reinforcing use of standard reporting and definitions including DR-TB patients to gather reliable data that is comparable within and between states, and internationally over time;

***Developing use of enhanced lab techniques***: This could include DNA fingerprinting and molecular typing to evaluate spread of DR-TB patients and identifying outbreaks; integrating lab, clinical and epidemiological data on TB patients, at district, state and national levels; creating algorithms for detection of local outbreaks and clusters.

***Monitoring TB elimination activities***: This can be achieved by expanding drug-resistance surveillance activities to monitor and improve patient management; expanding drug-resistance surveillance activities to monitor and improve patient management; collecting TB patients with lab information on comorbidity status to improve care such as joint management of TB/HIV coinfected patients, TB/DM management *etc.;* and enhancing collection of information on case notification, monitoring treatment adherence, social support and treatment outcomes at all levels in order to monitor and improve patient management

***Identifying and describing vulnerable populations for TB***: Analyzing routine surveillance data and performing ad-hoc surveys to identify vulnerable populations; and enhancing or implementing TB surveillance in migrants, prisoners and other vulnerable populations according to the particular situation in the district/state.

***Establishing TB surveillance system from district to national levels***: Doing this by setting up TB surveillance units at district level in DTC, at state level in STDC and national level at NTI; - setting up sentinel surveillance units at medical colleges and lab surveillance units at all IRLs and NRLs; and using e-NIKSHAY as the major data source with analytical outputs readily available at all levels.

## 13.4 NIKSHAY

NIKSHAY is the platform for the National Tuberculosis Programme Surveillance System. It envisages establishing ICT enabled state-of-art surveillance system with system utilization by 100% stakeholders, ensuring 100% notification of TB patients at diagnosis (microbiologically confirmed and clinically diagnosed). The programme envisions continuous monitoring and treatment adherence for all TB patients registered with e-NIKSHAY, enabling tracking of all registered TB patients across TB elimination lifecycle, geographies, transfers and referrals.

The first step is to ensure complete a entry in all formats of R&R. Dashboard functions to track activities and online monitoring indicators with graphical and geomapping displays in NIKSHAY/e-NIKSHAY is helpful in programme monitoring provided completeness of data entered is ascertained. Primarily, the source of information for all monitoring indicators will be NIKSHAY. Thus validation of actual report and the entered report should be done at each level to sustain quality of information available digitally. Patient-wise details and aggregated monitoring indicators for specified period is available within NIKSHAY.

## 13.5 e-NIKSHAY

In an attempt to reduce the TB burden in the country, it is envisaged that an ICT system such as e-NIKSHAY could help with coordinated planning and action that is required at various levels. This would be from improving patient awareness and ease of access to TB care to time bound and effective diagnosis and treatment by the service providers. The goal would be to develop a common integrated platform as an open system to engage ecosystem stakeholders towards effective, timely and quality assured diagnosis and effective treatment of TB. It would essentially use a case-based approach to the TB lifecycle, enabling patient-based tracking and monitoring allowing for stakeholder integration, as well as timely and accurate reporting and real time decision support.

RNTCP shall roll-out e-NIKSHAY with support to states on logistics and trainings. This platform will serve as a web-based, case-based recording and reporting system which will also feed into routine surveillance of drug-resistant TB in the country. e-NIKSHAY will be rolled-out initially in the states of Gujarat and Maharashtra and gradually scaled-up nationally by end of 2018.

**Role of e-NIKSHAY in recording, reporting and SME**

- e-NIKSHAY will gradually replace the cumbersome paper-based system of recording and reporting. Options shall be provided to capture the event at origin through provision of mobile/web applications, tablets and call centres on real-time basis. This will change and improve reporting structures for more agility and efficiency by providing relevant reports at relevant levels without delay due to paper-based collection, collation and compilation of reports.
- e-NIKSHAY will generate user specific task lists on real time basis according to their job responsibilities aiding staff to prioritize tasks at various levels. ICT enabled adherence mechanisms (99 DOTS, MERM, *etc.,*) will feed treatment related information from patients and providers into e-NIKSHAY enriching it as a patient care and support platform to prioritize patients for differential care.
- e-NIKSHAY will aid the referral and feedback mechanism under the programme by providing real time information on referral of patients for treatment initiation and ADR management. This will aid the programme to manage referral and provide feedbacks thereby decreasing lost to follow-up and improved tracking of patients nationally.
- e-NIKSHAY will populate dashboards for all supervisory staff at various levels to aid in their day-to-day work by providing dashboards for different facilities, categories of staff and geographies. This will aid supervisors and programme managers in identifying areas, both thematic and functional for intensive supervision.
- e-NIKSHAY shall generate interactive tables of real time monitoring indicators to aid review at various levels. The platform will generate alerts in the form of SMS, emails and call from call centres for patients, providers and supervisors for actions to promote

favourable outcomes and improve programme management efficiency. The indicators shall be projected as interactive tables/maps to promote information driven monitoring at all levels.

## 13.6 Programme monitoring

Monitoring is the process of observing whether an activity or service is occurring as planned. It implies systematic and purposeful observation, aiming to identify any diversion from the planned course of action. It is a routine tracking of programme using input, process, output and outcome data collected on a regular and ongoing basis. This helps identify the need for more formal evaluation of activities and find timely solutions to the problems.

Monitoring in TB programmes is of paramount importance for ongoing programme planning and implementation.

A good monitoring strategy moves beyond the widely used case detection and treatment outcome indicators and applies the concept of input, process, output, outcome and impact indicators for measurement of key programme activities.

***Monitoring indicators:*** Various components of programme service delivery are fed in NIKSHAY from where various input, process and outcome indicators are drawn for different levels of health facilities. Analysis of these indicators will help in monitoring improvement in programme performance. List of monitoring indicators is placed at Annexure 16.

## 13.7 Online clinics (Project ECHO)

Project ECHO (Extension of community health care outcomes) is an innovative continuing education model that uses Zoom video conferencing to link specialist teams ("hubs") with primary care providers and educators ("spokes") in rural and underserved areas. The benefits of the ECHO model™ are numerous, including ongoing education, greater access to high quality care and specialists for patients, improved provider networks, and reduced costs from inappropriate and untimely care.

Project ECHO aims to have better integration into health systems across the country. Zoom is the application which provides a platform for linkages either through website or mobile application. Zoom platform is downloaded through https://zoom.us/download and once it is installed in the system, it can be accessed via https://zoom.us/signin for web-link while mobile application is downloadable from play store as 'ZOOM cloud meetings'.

This platform is intended to be used for clinical and programmatic discussions to improve quality of care for DR-TB patients in India.

# Chapter 14: Supply chain management in PMDT

This chapter outlines the guidance regarding procedures for inventory management of second-line drugs used in the treatment of drug-resistant TB.

## 14.1 Overview of drug distribution flow

All drugs used in the various DR-TB regimens shall be supplied through a centralized procurement system at Central TB Division, MoHFW, GoI. Supplies of the second-line drugs shall be from the respective Government Medical Store Depot (GMSD)/ Central Medical Services Society (CMSS) to the state drug store (SDS). An advance intimation of all drug supplies shall be communicated to the States for SDS to make available requisite space in the drug store. The State/ SDS shall be supplied only loose form of second-line anti-TB drugs (SLD). On receipt of drugs, the SDS shall acknowledge the receipt to CTD.

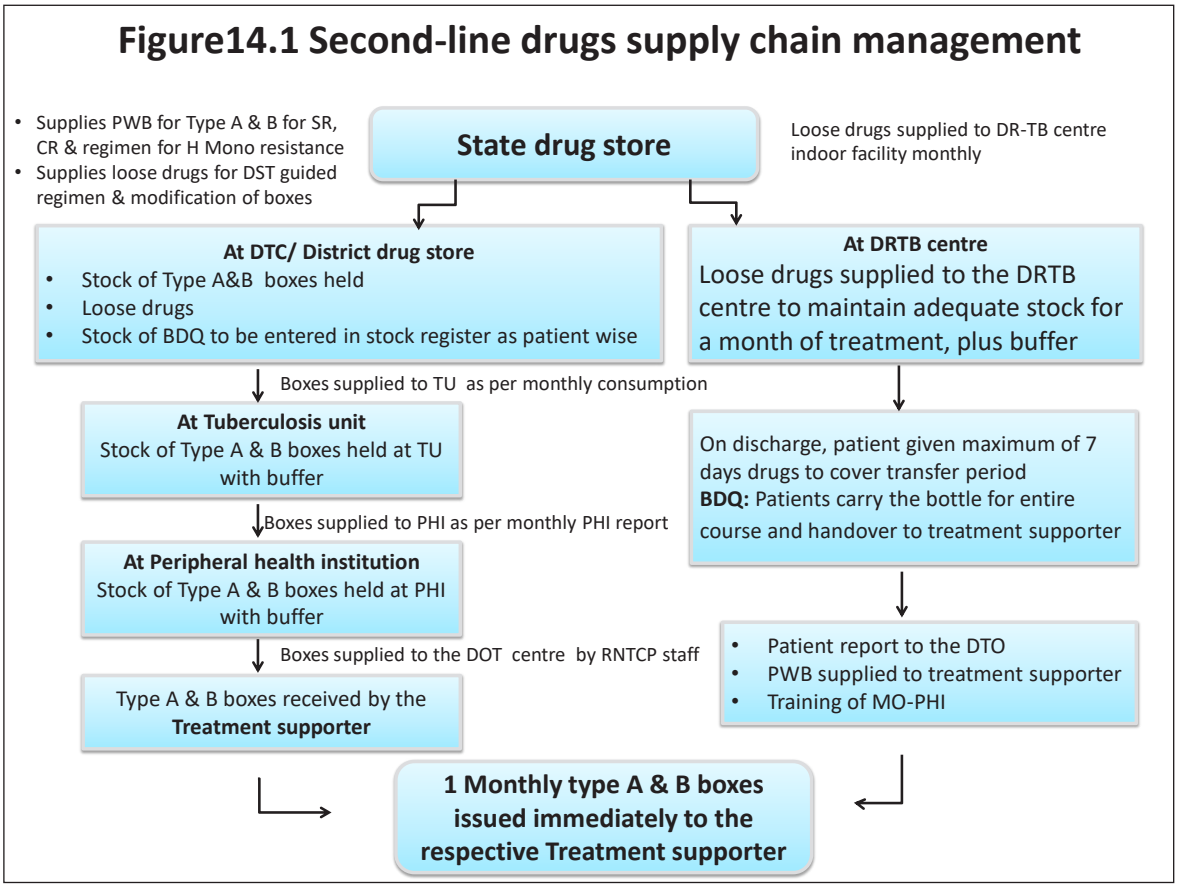
The SDS shall repack the loose drugs into one-monthly patient-wise boxes of Type A (oral drugs common in IP and CP), Type B (IP Plus boxes) and supplies to districts for treatment. SDS shall be preparing 'standardized drug boxes' for standard regimen and supplies to districts, namely for shorter MDR/RR TB regimen, conventional MDR-TB regimen and regimen for H mono/poly DR-TB. SDS shall supply additional loose quantity of SLD to districts for constituting modification. Drug requirement for treatment initiation at DDR-TBC/NDR-TBC, SDS shall supply loose SLD to all DDR-TBC and NDR-TBC based on consumption pattern. However, when the state shall build the capacity of the districts, including having dedicated full-time human resource support at DDS level, entire exercise of preparation of patient-wise boxes shall need to be conducted at DDS under the guidance and supervision of DTO.

Modification/ change in regimen may be required during the course of treatment based on the decision of NDR-TBC. In such patients, the supply of the monthly box would be as per the regimen-class to which the patient is reclassified to. For regimen containing Bdq, on discharge, the patient will carry the bottle containing BDQ and hand it over to the treatment supporter under supervision of the DR-TB/TB-HIV supervisor. This will be included in the first-monthly Type B box for the intensive phase containing SLI and oral drugs in which are issued to the treatment supporter. The bottle will remain under custody of the treatment supporter up to 24 weeks, while the Type B box will be issued on a monthly basis.

For maintaining inventory of BDQ and its accountability, once BDQ bottle is issued to patients from NDR-TBC with written information to concerned DDS, the stock is entered in the stock register at DDS. The bottle provided to the patient must be handed over to the treatment supporter. District DR-TB TB-HIV Supervisor shall be the link for passing

information to DDS keeper regarding BDQ stock supplies to treatment supporter. DDS keeper shall keep a record of the patients' stock of BDQ in the district drug stock register.

**Figure14.1 Second-line drugs supply chain management**




## 14.2 Technical specification of patient-wise box

The technical specification of the monthly patient-wise box for DR-TB patients is detailed in Annexure 17. The patient on intensive phase (IP) shall be put on Type A and Type B boxes in each month. During the continuation phase (CP), the patient will be put on only Type A box for the entire duration.

***For IP***: Type A box + Type B box of same weight band; and
***For CP***: Type A box of same weight band.

These drug boxes will be prepared at the SDS for all standard weight bands. For <16 kg patient, boxes will be prepared from the loose drug provided.

## 14.3 Constitution of patient-wise boxes and role of various drug stores

**At State drug store level**

SDS shall formulate drug boxes (Type A and B) for shorter MDR-TBregimen, conventional MDR-TB regimen and regimen for H mono/poly DR-TB and supplies to linked districts.

Table 14.1 Drug boxs for standard DR-TB regimen

| Type A (Oral drug box) | Type B (IP plus box) |
|---|---|
| **Shorter MDR-TB regimen** | |
| Moxifloxacin – high dose ($Mfx^h$) | Kanamycin (Km) |
| Clofazimine (Cfz) | Isoniazid – high dose ($H^h$) |
| Pyrazinamide (Z) | Ethionamide (Eto) |
| Ethambutol (E) | |
| Pyridoxine (Pdx) | |
| **Conventional MDR-TB regimen** | |
| Levofloxacin (Lfx) | Kanamycin (Km) |
| Ethionamide (Eto) | Pyrazinamide (Z) |
| Cycloserine (Cs) | |
| Ethambutol (E) | |
| Pyridoxine (Pdx) | |
| **H mono/poly DR-TB regimen** | |
| Levofloxacin (Lfx) | Kanamycin (Km) |
| Ethambutol (E) | |
| Pyrazinamide (Z) | |
| Rifampicin (R) | |

**At District drug store level**

When modification in regimen suggested by DR-TBC, DDS keeper shall take a call and prepare modified boxes from loose SLD supplied from SDS. The state shall provide necessary support for capacity building of DDS for carrying out entire exercise of preparing standardized patient-wise drug boxes decentralized at DDS level. Dedicated full time DDS keeper is mandatory to be recruited/ placed for successful decentralized system of preparation of drug boxes at DDS level.

The drug dosage for DR-TB for adults by weight band and for children (less than 30 kg body weight) by mg per kg body weight are detailed earlier in the chapters on treatment and special situation respectively.

Patient initiated on DST guided regimen will receive monthly box as per the regimen decided by NDR-TBC prepared by DDS. If the regimen is modified for the patient or patient is under the DST guided regimen, DDS needs to ensure that the change in regimen should be incorporated in supply of subsequent boxes.

## 14.4 Packing instructions

- packaging of loose drugs into Type A & B boxes should be done under guidance of the STO/Medical Officer/Drug logistics In-charge at State level and district level;
- one monthly pouch of Cap. Cs & Tab. E each should be made from plastic bag with ziplock facility in which 1 gm. pouch of silica gel desiccant should be kept. In each Type A box, one pouch of silica gel desiccant of 4 gm. weight should also be kept;
- each Type A & B box should be numbered consecutively at SDS. The record of the serial no. of the box should be maintained at the State, District & Sub-district (TU) Drug Stores. This would be of help while tracking a particular box. Later on, this exercise will be taken care of by Nikshay-Aushadhi software;
- label on the boxes to clearly mention the following:
    - item-wise name of drugs with quantity of each drug in the box;
    - batch no. & DOE of individual drugs;
    - DOE of boxes-expiry date of the drug having shortest expiry;
    - date of issue of the box from SDS;
    - serial number of the box;
    - storage instructions on the box for ensuring adequate precautions in storage of the drugs, especially at Treatment supporter level. Some suggested messages are: “store in a cool and dark place, preferably in a clean cupboard”; “do not expose to direct sunlight”; “keep away from children/unauthorized persons”; and or “box to be closed properly every time after withdrawal of drugs”.

Prototype of label is suggested in Annexure 18. The Nikshay-aushadhi software helped in preparing boxes by estimating number of boxes of required regimen that can be prepared from available stock. Label for prepared boxes to be printed directly from the system. For each cycle of box preparation, a label which contains expiry-wise drug content of these boxes to be generated through the system. This label should be used to paste over the boxes prepared during that cycle only. Other steps include:

- ***Repackaging process for utilizing Medication Event Reminder Monitors (MERM) system***. ICT enabled monitoring systems for DR-TB patients are currently under feasibility testing at various sites. Majority of these are capable of monitoring pill-in-hand status for DR-TB patient. It is one such system where 'a specialized container' capturing a set of information is directly sync with the server. Drugs are kept inside this battery operated specialized container and when it's opened, it records date and time of each medication taking event. Usually it holds drugs for one month which should be replenished with the drugs for the next month. The monthly box prepared for standardized regimen or loose drugs is available at the district and/or TU level and should be used to refill the box.
- ***Barcoding and real time tracking system:*** First-line drugs provided under the programme are barcoded and programme is in plan to have second-line drugs also made available with barcode. The NIKSHAY-AUSHADHI software has an in-built feature of tracking drug stock with the help of barcode. The barcode may be printed at state or district level as per the programme guideline and pasted over the boxes where it is prepared.

## 14.5 Drug management cycle of second-line anti-TB drugs

The management cycle of second-line anti-TB drugs comprises six elements, namely, drug selection; quantitative assessment of drug requirements; management of procurement; distribution protocol; assurance of drug quality; and ensuring rational drug use. Accurate demand forecasting of second-line anti-TB drugs, (correct quantification of drug needs for a specific period of time) is one of the elements guaranteeing an uninterrupted drug supply.

***Inventory management*:** Procedures for ongoing tracking and replenishment of the inventory of second-line anti-TB drugs at SDS and all subordinate stocking points ensures these are maintained at or close to the stocking norms presented below:

Table 14.2 Standard drug box for regimen for H mono/poly DR-TB

| Level | Stock for utilization | Reserve stock | Drug requirements |
|---|---|---|---|
| PHI | 1 month | 1 month* | (Monthly consumption x 2) – (existing stock in PHI at end of the month) |
| TU drugstore | 0 months | 1 months | (Quarterly consumption/ 3) x 3 – (existing stock in TU including PHI drug stores at end of the quarter) |
| DTC drugstore | 0 month | 2 months | (Quarterly consumption/ 3) x 5 – (existing stock in DTC drug store including TU &PHI drug stores at end of the quarter ) |
| SDS | 0 months | 3 months | (Quarterly consumption/ 3) x 8 – (existing stock in SDS including stocks at all districts at end of the quarter) |

**All PHIs may not have a reserve stock. Only PHIs where patient/s are initiated or on treatment will have reserve stock of second line drugs.*

## 14.6 Monitoring of drug distribution and supply chain management

***Distribution from Centre to SDS*:** as mentioned in overview of this chapter, loose SLD shall supply to SDS directly from centre. The SDS pharmacist shall prepare a Monthly Stock Statement (MSS) providing details of receipts, issues and opening/ closing balance of loose drugs as well as details of monthly Type A& B boxes, as at the last day of each calendar month in the prescribed format. The MSS shall be sent to STO by the 7$^{th}$of every month, by all SDSs, in the state. The statement shall facilitate determination of drug stocks available with SDS (s) within the state. MSS shall thereafter be forwarded to CTD through the STO, by the 10$^{th}$of every month. In the event of more than one/ multiple SDSs within the state, all the MSSs shall be forwarded to CTD within the timelines stated above.

***Distribution from SDS to District*:** The SDS will supply drugs to the DTC in the form of monthly patient-wise Type A and Type B drug boxes. It shall review monthly consumption report received from linked districts and issue Type A & B boxes as well as loose medicine to the district. Format for monthly consumption report to be sent by the district is attached as Annexure 19. The DTC shall send the boxes to its implementing TU in a similar manner on monthly basis and then monitor through the TU monthly SLD requirement report. Buffer stocks of both Type A & B boxes of all weight bands shall be held at all levels as per stocking norms. DTO will ensure arrangement for supply of monthly drug boxes of Type A & B from

the respective PHI to the Treatment centre. The STS shall identify the Treatment supporter in consultation with MO-PHI and the patient. Considering roll-out of DST guided treatment, SDS will supply loose core oral SLD to the DDS for modification in regimen.

***Distribution of SLD to DR-TBC*****:** Issuance of loose drugs to DR-TBC (including DDR-TBC) from SDS shall be based on the monthly stock statement submitted by DR-TBC, to ensure adequate stocks for a month of treatment, plus a buffer of one month. On discharge from DR-TBC, patient shall be given drugs for 7 days to cover the transit period. During this time, it is expected that the patient shall reach home for the ambulatory treatment and commence therapy from monthly IP box which has by then been issued to the respective treatment supporter. For patient put on BDQ treatment, BDQ bottle contains entire course of treatment for one patient which shall be earmarked for each enrolled patient and handed over to treatment supporter in supervision of District DR-TB TB-HIV Supervisor.

***Distribution from DDS to TU drug stores*****:** Buffer stock equivalent to one month will be kept at the TU. The drug boxes will be supplied from the TU to PHI. It will then be transferred from the TU to respective PHI on instruction of DTO as per monthly consumption report submitted by TU.

***Distribution from TU drug store to PHI:*** Buffer stock equivalent to two months will be kept at the PHI at the beginning of each month. The drug boxes will be supplied from PHI to Treatment centre/ Treatment supporter. All PHIs may not have reserve stock. Only PHIs where patient/s are initiated or on treatment will have reserve stock of second-line drugs.

Drug stock register (Annexure- 20) is to be maintained in the drug stores at all levels. Details such as drugs received, distributed and balance stock are to be entered in this register. Drug distribution mechanism is supported by Nikshay-aushadhi where drugs stock, dispatch and return of drug can be reported through tracking of the drug. The system is able to capture information log of boxes allotted to the patients who are provided by each health facility who is using this software for drug management purpose. Hardcopy reporting will be discontinued once all health facilities embark on the Nikshay-aushadhi.

***Scenario 1 - Modification in regimen:*** If N/DDR-TBC committee decides on modification of regimen, DDS shall prepare modified Type A or B boxes from available standard boxes using loose SLD available at district level and arrange supply to treatment supporter.

***Scenario 2 - Extension of intensive phase:*** If IP of the patient is required to be extended; the respective N/DDR-TBC committee shall inform DTO who will intimate the same to the MO-PHI and the respective TU. The PHI will release 1 Type A and Type B to the respective treatment centre from where the patient is taking treatment. When the patient is switched to CP, the DTO shall intimate the same to the MO-PHI and the respective TU. On instruction of DTO, the PHI will release 1 Type A box only to the respective Treatment centre from

where the patient is taking treatment. During the period between when the DTO has been notified of the decision to change over to CP and the delivery of drug box from the PHI to the Treatment centre, the patient's IP shall be continued. All patients who are given an extended IP must complete the monthly box of a full month of extension.

***Scenario 3 - Change in regimen:*** If DR-TBC Committee decides to change the regimen then DDS shall arrange supply of new treatment regimen box from PWB supplied from the SDS.

## 14.7 Reconstitution: Repackaging and use of partially used IP/CP boxes

- in case of default/death/transferred-out/treatment stopped patients, unconsumed boxes shall be brought back from Treatment centre to PHI to TU to DTC within the shortest possible time. All loose drugs remaining in the boxes received back shall be accounted for in the Stock register and Nikshay-aushadhi at the DDS and issued as per FEFO principles to either DR-TBC or for repackaging into monthly Type A or B boxes;
- partially used BDQ bottle shall be sent back to SDS where repackaging will also be done. Remaining tablets in the bottle received back shall be accounted for in the reconstitution register (Annexure-21) and Nikshay-aushadhi at SDS. Upon reconstitution, the bottle shall be accounted for in the Stock register (loose tablets to be mentioned in remarks column) to be issued as per FEFO principles. When reconstitution is done, tablets of same expiry can be considered using same container to a maximum of 188 tablets. These reconstituted containers shall be used for treatment of subsequent patients found eligible for Bdq. All such drugs that are taken from the new containers shall be collected as a group of 188 tablets of same expiry and put in a light resistant container as per advice from the manufacturer. The actual expiry of tablets should be mentioned over the container;
- in the event of SDS falling short of 188 tablets from an expiry batch, reconstitution can still be done using number of tablets to complete 188 tablets with another expiry batch. In such a case, tablets of the respective expiry should be retained in their same respective containers and issued to patients and providers with counselling to consume the tablets with the nearest expiry first; and
- if expiry of remaining tablets is less than six months, the same shall be consumed at NDR-TBC for admitted patients. It will be adjusted from the new long expiry bottle on discharge.

## 14.8 Guidelines for storage of second-line anti-TB drugs for State and District drug store

**Storage space**

Requirement of space for various levels of drug stores should be based on the estimated number of DR-TB patients likely to be placed on treatment in the concerned state for whom

the maximum quantity of drug stocks are to be maintained at the concerned stocking unit. The recently completed national DRS and the programme surveillance data on PMDT would form the basis for quantification and storage space that will need to be worked out separately for each SDS and DDS.

**Specifications for drug stores**

- should preferably comprise one large room; where multiple rooms already exist, they should be contiguous or proximate to each other;
- preferably separate space for storage, handling and repacking into Type A & B boxes;
- ceiling to have a height of at least 3 metres;
- lockable door and at least one window with grill;
- proper lighting;
- even-level, 'pukka' floor;
- plastered walls and ceiling with whitewash without any kind of seepage in the room;
- in a situation where separate room for storing second-line drugs is not possible, an attempt to demarcate and enclose a specified area for storing the drugs should be made within the larger store. This will ensure required temperature control for the drugs;
- architects should be consulted for making suitable modifications in the existing drug store/construction of a new drug store for the same;
- signage board with instructions in local language should be put near the entrance of the store to remind concerned officials regarding good storage practices;
- ideally, vacuum dewatered flooring (VDF) should be used; however depending on the feasibility, such flooring may be done at SDS level;
- in case it is feasible at SDS level, separate areas should be demarcated for receiving and dispatching the drugs;
- contract for pest control should be entered into by the State to ensure drug stores being free from pests, rodents *etc.;* and
- the SDSs shall have adequate mechanism to continuously monitor temperature and humidity of the stores as detailed below:

**Shelves, racks & storage arrangements**

- if sufficient space is available on the existing storage shelves in the SDS, these shelves made of 40 mm bore medium quality (external diameter - 48.3 mm) mild steel pipes should continue to be used as per existing RNTCP guidelines. New shelves, if required, are to be made from prefabricated slotted angles ensuring sufficient 'gap' between cartons from ceiling, floor and walls, facilitating ventilation and free movement of air;
- shelves to be positioned so that there is no possibility of seepage into cartons;
- typically, five rows of shelves to be fabricated, one on top of the other into racks. A single rack to be long enough to accommodate up to a minimum of three cartons on each shelf;

- in case of a broad room, there shall be multiple rows of racks, all parallel to one another. There should be sufficient space between parallel blocks of racks and walls, to facilitate free movement of men and trolleys for smooth stacking and removal of cartons. In case of a long and narrow room, racks to be positioned in a such a way that there is sufficient space between them and the walls;
- drug cartons to rest on shelves and not on each other, to prevent eventual sagging of the cartons in the bottom row;
- rows and columns, where drugs are stored should be defined and locations to be assigned a unique identification number; and
- in future, if SDS of a particular state has to handle a large volume of drugs and occupy larger space, the walkway space (between racks across the storeroom) can be of 3 metres. In such a situation, material handling equipment will be required.

**Stacking arrangement**

- name of the drugs along with their expiry dates to be indicated on stickers pasted on the face of cartons/ drug boxes. These should be written again by hand, in large easily visible characters using a coloured, permanent marker pen;
- each drug should be stored in a single location within the store in order of date of expiry;
- additionally, drugs of the same expiry should be stored together at the same location;
- recognizing the above rules, drugs expiring earliest should be so stored that they are issued first. For example, in case IP (> 70 Kgs) boxes are placed on multiple shelves in a single part of the store, boxes expiring earlier should be stored at ground level and fresher boxes (which shall expire later) on elevated shelves. This method of stacking shall ensure that drugs that shall expire first shall automatically be issued first, based on the principle of FEFO (First Expiry First);
- Expired drugs should be segregated, sealed and stored in a separate area (quarantine) of the store eliminating the possibility of their issue to patients. Expiry dates should be highlighted in these boxes; and
- bin cards at SDS level to be displayed, providing details of receipts, issues, closing balance (quantity) and expiry dates of drugs. Only Na-PAS is a slow moving drug and should be stored at higher level shelves. Rest all other second-line drugs are fast moving, hence, should be stored on lower shelves.

**Control of humidity and temperature**

- monitoring of humidity and temperature is important**.** Hygro-thermometers are to be installed up to TU drug store levels to monitor humidity and temperature regularly. The record of both these variables should be maintained in charts properly and checked on a daily basis by the concerned store in-charge. This should be reviewed by STO/ Officer in-charge of SDS and necessary corrective measures taken immediately;

- control of humidity must be ensured at all times. In order to keep humidity levels below the maximum 60% recommended for storage of drugs, following measures to be taken:
    - *Ventilation:* Open windows or air vents of the store to allow air circulation. Ensure all windows have screens/ wire mesh to keep out insects and birds and have metallic grills/ iron bars. Drug boxes/cartons to be placed on shelves ensuring there is sufficient space between shelves and walls of the storeroom;
    - *Packaging*: The cartons/drug boxes should not be opened unless necessary;
    - *Circulation*: Use fans to circulate fresh air from outside;
    - *Protection from sunlight:* To protect drugs from sunlight, following measures may be taken: Shade the windows or use curtains if they are in direct sunlight. Keep products in cartons/drug boxes. Do not store or pack products in sunlight. Maintain trees around the premises of the drug store to help provide shade and cooling. Check their condition regularly to prevent any untoward incident;
    - *Control of temperature*: The second-line anti-TB drugs should preferably be stored below $25^0$ C. In the area specified for storing second-line drugs, temperature of about $20^0$ C should be maintained with help of airconditioners (tonnage would depend on size of the room); and
    - *Power supply***:** Regular power supply should be available for air conditioning in the SDS. Arrangements for backup power supply should also be made through solar panels/ fuel-based power generators.

The purpose of information provided above is to emphasis that drugs be stored in cool and dark place for proper efficacy. However, after drug boxes are moved outside temperature controlled (AC) environment till it is consumed by the patients. All efforts must be made to store drug boxes in cool and dry place.

## 14.9 Quality assurance of drugs

The quality assurance component of the RNTCP drug supply system ensures that each drug used by a patient is safe, efficacious and has appropriate standards of quality. As per the protocol developed by Central TB Division (CTD), random samples of second-line Anti TB Drugs shall be picked from all stocking points in the field and sent for testing by an independent drug testing laboratory contracted by CTD to ensure quality of drugs is continuously maintained and remains the same throughout the supply chain of the drugs. This should be done based on communication sent by CTD to concerned states and districts.

## 14.10 Waste disposal guidelines

If any drug expires due to reasons beyond control, it should be disposed off after writing off the loss as may be required under the rules and thereafter procedures laid down in the

Rules under Drugs & Cosmetics Act and Biomedical Waste (management and handling) Rules of Government of India.

## 14.11 Guidelines for recording and reporting of SLD

The recording and reporting system for drug stock management from SDS to the DR-TBC and to Districts, TB Units and PHIs has been recently revised to suit the 1 monthly patient-wise boxes system. Formats for drug logistics management of second-line drugs under PMDT are described in Annexure 19. An ICT based recording and reporting systems up to PHI level for the real time data, inventory management, demand generation, analytical tools, drugs accountability, forecasting & anticipation have been identified and will soon be implemented under RNTCP. Bar code reader, scanner and printer too shall be provided up to the district level.

## 14.12 Nikshay-aushadhi: Supply chain management software solution

The Nikshay-aushadhi, Centre For Development of Advanced Computing (C-DAC)'s web-based TB drug supply management system is a major step towards adapting technology to improve supply chain management of TB drugs in State level and district stores, TB Unit (TU), peripheral health institutions (PHIs), DR-TBC and treatment supporter centres. These applications deal with demand and distribution of all essential anti-TB drugs and consumables required at State, district, TU, DMC and PHI level.

It also caters to data from Government Medical Stores Depot (GMSD) and the procurement agency, currently CMSS. The advantage of inventory automation is to get the complete detail of stock on hand at various levels, supplies in pipeline and distribution and consumption patterns in the state, up to higher management level. This will help in anticipation of future requirement while ensuring dramatic improvement in performance along with reducing costs. Integration of Nikshay-aushadhi system with current Nikshay-aushadhi enhances its user-friendliness.

# Chapter 15: Infection control measures

Evidence indicates that DR-TB is similar in transmissibility to DS-TB. Thus, infection control policies and strategies are not much different for DR-TB. All facilities treating DR-TB patients must comply with adequate infection control measures. Ensuring implementation of infection control policy in all healthcare facilities, at public/private/household level and in congregate settings (correctional facilities, military barracks, homeless shelters, refugee camps, student dormitories and nursing homes, among others) is essential. This will help prevent transmission before diagnosis up to initial stages of treatment till the patient has culture converted and turned noninfectious. Actions are required at national and subnational level to provide managerial direction and at health facility level to implement airborne infection control measures. [14]

## 15.1 Activities for national and sub-national airborne infection control (AIC)

- identify and strengthen a coordinating body for TB infection control, ensuring that TB infection control is part of a general infection prevention and control programme;
- develop a comprehensive budgeted plan for implementation of TB infection control at all levels;
- ensure that health facility design, construction, renovation and use are as per prescribed criteria;
- conduct surveillance of TB disease among health workers, and conduct assessment at all levels of the health system and in congregate settings;
- address TB infection control advocacy, communication and social mobilization;
- monitor and evaluate the set of TB infection control measures; and
- enable and conduct operational research.

## 15.2 Measures for facility-level TB infection control

- identify and strengthen local coordinating bodies for TB infection control as part of the facility wide comprehensive infection prevention and control programme. In addition, develop a facility plan (including human resources and policies and procedures to ensure proper implementation of the controls listed below) for implementation:

    - adequate ventilation (ACH >12 per hrs) should be maintained for high priority health facility like DR-TB wards, ART centre, TB containment laboratory *etc.;*
    - beds should be arranged with 6 feet distance in between them;
    - staff should be trained for standard precaution, airborne infection control precaution, use of personal protective equipment (PPEs) and importance of educating patient to follow the same;
    - availability of different size of N 95 particulate respirators; and

- renovating existing facilities and constructing new ones to optimize implementation of infection controls. Seating rearrangements within the room or ward, patient flow within health facility premises could have significant impact on airborne infection control.

- conduct on-site surveillance of TB disease among health workers and assess the facility;
- address advocacy, communication and social mobilization for health workers, patients and visitors;
- monitor and evaluate the set of TB infection control measures; and
- participate in research efforts.

National guideline for infection control in all health care settings including DR-TB care settings includes having an integrated implementation plan for administrative, environmental and personal protective equipment.

### 15.2.1 Administrative controls

- promptly identify people with TB symptoms (triage), separate infectious patients, control spread of pathogens (cough etiquette and respiratory hygiene) and minimize time spent in healthcare facilities;
- Ensure linen and waste management;
- clean and disinfect patient-care equipment;
- provide a package of prevention and care interventions for health workers; and
- Observe standard universal precautions, body substance isolation and airborne precautions.

### 15.2.2 Environmental controls

- use ventilation systems including mechanical ventilation system; and
- useof ultraviolet germicidal irradiation fixtures, at least when adequate ventilation cannot be achieved.

### 15.2.3 Personal protective equipment

- use particulate respirators (N95) for staff; and
- use of surgical mask or cloth to cover the cough at the patient level.

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# Annexures

## Annexure 1: Composition and ToR of State PMDT Committee, Nodal DR-TB Center and District DR-TB Center

**State PMDT Committee:**

Composition of State PMDT Committee

| Title | Designated officials |
|---|---|
| Chairperson | Principal Secretary / Secretary (Health) |
| Vice Chairperson | Director of Health Services |
| Member | Mission Director (NRHM) |
| Member | Director of Medical Education |
| Member Secretary | State TB Officer |
| Member | Director STDC |
| Member | Chairperson State Task Force |
| Member | Microbiologist-IRL |
| Members | Chairperson of NDR-TBC Committees in the state |
| Members | 2 Eminent Chest Physicians (Medical Colleges/Private sector) |
| Members | Representative of PvPI |
| Member | State Headquarter RNTCP Consultant |
| Member | 1 representative each from any 2 NGOs working in TB programme. |
| Member | 1 representative from professional bodies like IMA. |

Terms of Reference of State PMDT Committee:

1. Develop plan of action for implementation, expansion, and maintenance of PMDT in the respective state.
2. Select, facilitate up-gradation, and designate institutes to serve as DR-TB Centres within each state.
3. Periodically review the implementation status of PMDT in the respective state to ensure that RNTCP PMDT policies and guidelines are being followed.
4. In co-ordination with the respective STO, ensure that drug ordering and distribution is managed in a timely and appropriate manner.

**Nodal DR-TB Centre Committee:**

Composition of Nodal DR-TB Centre Committee:

| Title | Designated officials |
|---|---|
| Chairperson | Medical Superintendent or Director of the institute |
| Vice Chairperson | HOD Respiratory Medicine or General Medicine |
| Nodal Officer | Senior Doctor from the department hosting the NDR-TBC |
| Member Secretary | Senior Medical Officer – DR-TB Centre |
| Member | HOD Microbiology or IRL Microbiologist |
| Member | HOD Psychiatry* |
| Member | HOD Gynaecology* |
| Member | HOD ENT* |
| Member | HOD Pharmacology |
| Member | Cardiologist* or physician |
| Member | 1 Eminent Pulmonologist from NGO/Private sector |
| Member | WHO RNTCP Consultant supporting the NDR-TBC |
| Member | DTO of the district where NDR-TBC is located |
| Special Invitees | DTOs of the districts linked (as and when needed) |

*** To be consulted as and when required**

*Note: The chairperson can co-opt other specialist as required. The routine clinical decisions can be taken by the available doctor and informed to the NDR-TBC in subsequent meetings.*

**Terms of Reference of NDR-TBC**

1. Periodically review the implementation status of PMDT in the respective Nodal DR-TB Centre to ensure that RNTCP PMDT policies and guidelines are being followed.
2. Receive the DST results from the IRL and enter the details in the PMDT register of the DR-TB Centre.
3. Arrange for examination of DR-TB patients referred for their treatment eligibility, open treatment book and start PMDT regimen for all eligible patients.
4. Admit all DR TB patients in the indoor facilities of the DR-TB Centre for an initial period of up to seven days.
5. Ensure respective DTOs are informed of patients discharge in a timely manner
6. Submit regularly reports to the State TB Cell and State PMDT committee.
7. Review record and report, Nikshay entries.

In co-ordination with the respective STO and DTOs, ensure that drug ordering and distribution is managed in a timely and appropriate manner.

**District DR-TB Centre Committee**

Composition of District DR-TB Centre Committee:

| Title | Designated official |
|---|---|
| Chairperson | Chief Medical officer/ Director/ Head of the institute |
| Co-chairperson | Medical Superintendent |
| Nodal person | Physician in-charge of DDR-TBC |
| Member Secretary | DTO of the district |
| Member | MO Medical college/MO-DTC |
| Member | All specialist available in the institute |
| Member | Any other invited member, if required |

1. Ensure arrangement for patient treatment initiation as per the PMDT guidelines
2. Periodic review of treatment initiation, aDSM and patient monitoring activities carried out at DDR-TBC
3. Coordinate with DTO, TUs, PHIs, other department and NGO to ensure patient is linked with all essential services required.
4. Review record, report and Nikshay entry updated timely.

## Annexure 2: Check-list for preparation of District DR-TB Centre (DDR-TBC):

| | | |
|---|---|---|
| 1 | Name of the DDR-TB Centre under appraisal | |
| 2 | Does the DDR-TB Centre have a DDR-TB Centre Committee constituted? | ☐ Yes ☐ No |
| 3 | If 'No' expected date of constitution of the Committee | |
| 4 | No. of doctors trained for DR-TB from the district? | |
| 5 | Number of DR patients anticipated for this DR-TB Centre annually? | |
| 6 | Is there separate 2 beds identified for male and female DR patients and/or Out-patient facility with well-ventilated waiting are developed? | ☐ Yes ☐ No |
| 7 | Infection Control Measures at the DDR-TB Centre (Refer to Infection Control Guidelines) | ☐ Adequate<br>☐ Inadequate |
| 8 | Does the DDR-TB Centre have all the facilities for undertaking pre-treatment evaluation for expected type DR-TB patient received for treatment initiation? | ☐ Yes ☐ No |
| 9 | If 'No" have an alternative arrangements been made | ☐ Yes ☐ No |
| 10 | Are there emergency drugs available? **Or** access to emergency services nearby? | ☐ Yes ☐ No |
| 11 | Is there ECG machine (12 leads with automatic QTc reader available? | ☐ Yes ☐ No |
| 12 | Is latest PMDT guideline available at site? | ☐ Yes ☐ No |
| 13 | Does the DDR-TB Centre have a computer with internet connection | ☐ Yes ☐ No |
| 14 | Are printed and electronic copies of the following recording and reporting formats available | Printed |
| i | PMDT TB Register | ☐ Yes ☐ No |
| ii | PMDT treatment Cards | ☐ Yes ☐ No |
| iii | Referral for treatment forms | ☐ Yes ☐ No |
| iv | Treatment booklet | ☐ Yes ☐ No |
| v | CEM treatment initiation and review form | ☐ Yes ☐ No |
| 15 | Who is assigned responsibility to maintain record at District DR TB center | ☐ Senior DR-TB TB-HIV Supervisor<br>☐ Other _____ |

Note: This check list should be filled by evaluator and submit it to state for further approval at the time of site evaluation for initiating District DR TB center

Additional Remark:

1.
2.
3.

## Annexure 3: Airborne Infection Control – Recommendations for DR-TB Wards & Outpatients Area

**Airborne Infection Control - recommendation for DR-TB Wards**

**Key Recommendations:**

- Located away from the other wards, with adequate facilities for hand washing and good maintenance and cleaning.
- Adequate ventilation (natural and/or assisted ventilated) to ensure >12 ACH at all times.
- Adequate space between 2 adjacent beds , at least 6 feet
- Cough hygiene should be promoted through signage, equipment and practice ensured through patients and staff training, on-going reinforcement by staff
- Adequate sputum disposal, with individual container with lid, containing 5% phenol, for collection of sputum
- All staff should be trained on standard precautions, airborne infection control precautions, and the proper use of personal respiratory protection. A selection of different sizes of re-usable N95 particulate respirators should be made available for optional use by staff.

DR-TB wards are inpatient for initiation of treatment and managing clinical complications during treatment. This includes but is not limited to all RNTCP DR-TB Centres.

Location and Design:

- The facility should be located at away from the other wards with preferably a separate passage for the patients to access the toilets.
- The facility should have adequate ventilation (natural and/or assisted ventilated) to ensure >12 ACH at all times, preferably >15 ACH. This would be possible only if adequate fixed unrestricted openings, e.g. ventilator windows, are open at all times during the day and night in all seasons. Similarly if assisted ventilation is being used (e.g. exhausts) to maintain the adequate ACH it should ensure that these are kept switched on at all times.

- In case of frequent power cuts in a setting requiring mechanical or UVGI to maintain safety, a power back-up facility (*i.e.,* generator set) is recommended along with adequate provision for fuel and maintenance.
- The distance between 2 adjacent beds should be optimal (at least 6 feet)
- Visitors should be restricted to the greatest extent practical.

General Hygiene:

- Hand washing facility (Universal Precaution) shall be in place for doctors, health care workers and patients.
- Running water, soap and alcohol hand rub solution shall be provided.
- Frequent wet mopping of the ward shall be undertaken.
- Lavatory shall be kept clean.

Patient education should be conducted on the following at each admission, and reinforced frequently by staff.

- Cough hygiene
- Cough etiquettes
- Sputum disposal
- Proper use of surgical masks.
- Restricted visitor entry

Cough Hygiene:

- Display sign boards in the ward demonstrating cough hygiene.
- All patients admitted in the ward should be issued surgical masks.
- Adequate measures for safe collection and disposal of sputum

Sputum Disposal:

- Patients should be provided with individual container with lid, containing 5% phenol, for collection of sputum
- Patients should be instructed on spitting the sputum directly in the container or in a tissue paper which is then thrown in the container.
- The container should be emptied daily and the sputum should be disposed of as per the infection control guidelines

Avoid posting of HCW working in MDR wards if they are immuno-compromised or are on immuno-suppressants for any indication

Training of MDR TB Ward staff:

- All the staff of the MDR TB ward shall be trained in Universal Workplace Precaution, Waste segregation and disposal and Air borne Infection Control Practices, with special reference to tuberculosis.

Personal Respiratory Protection:

– *i.e.,* N95 particulate respirators must be made available for optional use by any staff working in the ward area if the desired ACH cannot be reached through optimized ventilation. Regardless of whether or not staffs choose to use the respirator, all staff should be provided sensitization and appropriate training on how to choose, use and maintain the respirator.

**Airborne Infection Control - recommendation for Outpatient Area**

TB patient like other patients, have to wait for long periods before they are actually examined by the physician at outpatient facility. During this period, these patients remain **source for spread of disease to others.**

**Key recommendation**

1. Reduce overall duration of stay in outpatient department by screening, separation and fast tracking of coughing patients and other means.
2. Location of OPD area should be properly ventilated. Adequate ventilation (natural and/or assisted ventilated) to ensure >6 ACH at all times.
3. Education on cough etiquette and respiratory hygiene in waiting area
4. Training of institute staff to identify symptomatic and provide mask and segregate them if fast tracking is not possible

**Reducing the patient stay in outdoor department:** Reducing the overall stay of such patients in the health care facility is likely to prove the single-most effective measure of reducing airborne disease transmission. This can be achieved by fast tracking these patients, which can be accomplished by several measures that are not mutually exclusive. Fast-tracking will also depend upon the type of health care facility. At a chest center/ hospital, most patients are chest symptomatic where fast-tracking has no real application, but the process will be more useful for general hospitals. Identification of patient coughing in waiting area and providing mask or segregating them in such situation is possibly more feasible and effective way. Implementation of key administrative intervention would vary from facility to facility.

**Screening**: Screening for TB symptoms can be done at registration counter itself or at designated cough corners within the facility and those suspected of TB can be given priority. This screening can be performed by physicians, nurse, paramedical staff or volunteer specially deputed for this purpose. A simple three layer surgical mask must be provided to all coughing patients with directions to cover the cough throughout the time they are in the facility and practice the same at home and community level.

**Education on cough etiquette and respiratory hygiene:** This education can easily be imparted to patients through posters and other audio-visual means, as well as by actual discussion by paramedical staff or volunteers in the waiting area. Outpatient setting should make easy availability of tissue papers and bins with disinfectants for sputum disposal.

**Patient segregation:** Segregation of patients with respiratory symptoms can be achieved by separate waiting areas for chest-symptomatic. Outpatient area should be well ventilated to reduce overall risk of airborne transmission.

**Fast Tracking:** Patients identified at the registration counter or designated cough corners within the institute must be fast – tracked *i.e.,* allowed to jump the queues at all levels to minimize the time they spent in the facility and thereby the risk of transmission. Mechanisms like stamping the outpatient form for fast –tracking could be instituted with adequate education materials displayed in the facility on Airborne Infection Control.

**Location and Design:**

- The outpatient facility with an open shaded waiting area should be located at away from the other OPDs of vulnerable immunosuppressed diseases and paediatric OPD with preferably a separate passage for the patients to access the toilets.
- The outpatient facility should have adequate ventilation (natural and/or assisted ventilated) to ensure >12 ACH at all times, preferably >15 ACH. This would be possible only if adequate fixed unrestricted openings, e.g. ventilator windows, are open at all times during the day in all seasons. Similarly if assisted ventilation is being used (e.g. exhausts) to maintain the adequate ACH it should ensure that these are kept switched on at all times.
- In case of frequent power cuts in a setting requiring mechanical or UVGI to maintain safety, a power back-up facility (*i.e.,* generator set) is recommended along with adequate provision for fuel and maintenance.
- The seating between patient and provider must be such that the direction of air is passing between them to create an air curtain between patient and the provider.
- Overcrowding should be restricted to the greatest extent practical.

## Annexure 4: Technical specification of ECG machines for PMDT- RNTCP

### A. PREAMBLE

This guidance is provided in regard to the requirements that should be considered when procuring ECG machines for monitoring of patients being treated either with the shorter treatment regimen (STR) or the new drugs for drug-resistant TB treatment, including Bedaquiline and Delamanid under Programmatic Management of Drug Resistant TB (PMDT).

The ECG machine can be 1/3/6 channel; however, it should fulfil all the requirements as per Technical Specifications given below:

### B. TECHNICAL SPECIFICATIONS

| S. N. | Features | Technical Specifications & Operational/Functional Requirements |
|---|---|---|
| 1. | Size & Weight | • Sturdy & light weight machine <5kg,<br>• Should be compact<br>• Should have carry handle for portability |
| 2. | Power Supply | • Compatibility with mains 220-240 V (normal), 50/60 Hz power supply<br>• High performance Li-ion rechargeable battery with built-in charger.<br>• Equipment should have sufficient battery backup for taking minimum 100 ECGs without AC power.<br>• Digital filtering to remove interference from power line, muscle tremor *etc.,* |
| 3. | ECG recording | • ECG recording with 12 leads<br>- Standard Leads (the limb leads or bipolar Limb leads: I, II & III)<br>- Augmented Limb Leads- (aVL, aVR and aVF)<br>- Chest Leads ( the unipolar or V leads)- from V1 to V6<br>• Simultaneous acquisition from 12 leads<br>• Recording speed selection of 25 mm/ sec and 50 mm/ sec with facility for speed selection<br>• Automatic adjustment of baseline for optimal recording<br>• Should have different filters like Baseline Filter, EMG Filter & AC Filter<br>• Multiple operating modes -automatic, manual and rhythm<br>• Common Mode Rejection Ratio >90dB |
| 4. | Built-in ECG Parameters measurement and interpretation | • Built-in ECG auto-measurement including: HR, P-R interval, P-Duration, QRS duration, Q-T interval, Q-TcF (Friedericia), P Axis, QRS Axis, T Axis, R(V5), S(V1), R(V5)+S(V1)<br>• QTcF interval reading/ measurement should also be available with Limb leads alone. |

| | | |
|---|---|---|
| 5. | Printing and Communication | • High-resolution thermal printing array system<br>• Built in printer should work with standard universal thermal printer paper |
| 6. | Standard Accessories | • The machine should be supplied with<br>• power cord , patient cable, user manual and warranty card, Operation Manual with user demonstration video CD, interpretation manual & 10 thermal recording paper rolls , 5 bottles of jelly,<br>Two sets each of :<br>- patient cable<br>- chest electrodes – Both adult and paediatric (2 sets each)<br>- limb electrodes – Both adult and paediatric (2 sets each) |
| 7. | Safety Profile | • Should be provided with terminal for good earth connection to preclude electric disturbances while recording-<br>• Must have a safety certificate or valid detailed electrical and functional safety test report from a recognised competent authority<br>• Copy of the certificate / test report shall be produced along with the technical bid. |
| 8. | Installation & Training | • The firm should install the instrument at the designated location and provide one-day training/ demonstration of operation of ECG machine. |
| 9. | Warranty | • Performance warranty of at-least one year from date of installation + additional two years comprehensive warranty,<br>• In case of breakdown of the machine, the supplier shall make the machine functional by repair (including replacement of parts) free of cost at the user site, within 3 (three) days of the receipt of complaint, or replace the machine (if necessary). |
| 10. | After Sales Services | • The suppliers should have adequate after sales service facilities covering all districts of the country.<br>• They should have infrastructure and trained manpower to attend to any complaints within 3 days of receipt of complaints |

## Annexure 5: Bio-medical Waste Management

| Category | Type of Waste | Type of Bag or Container to be used | Treatment and Disposal options |
|---|---|---|---|
| Yellow | (a) Human Anatomical Waste: Human tissues, organs, body parts and fetus below the viability period | Yellow coloured non-chlorinated plastic bags | Incineration or Plasma Pyrolysis or deep burial* |
| | (b)Animal Anatomical Waste : Experimental animal carcasses, body parts, organs, tissues etc. | | |
| | (c) Soiled Waste: Items contaminated with blood, body fluids like dressings, plaster casts, cotton swabs, specimen container, packing material and bags containing residual or discarded blood, blood components or body secretions. | | Incineration or Plasma Pyrolysis or deep burial* |
| | (d) Expired or Discarded Medicines: Pharmaceutical waste like antibiotics, cytotoxic drugs including all items contaminated with cytotoxic drugs along with glass or plastic ampoules, vials etc. | Yellow coloured non-chlorinated plastic bags or containers | Expired `cytotoxic drugs and items contaminated with cytotoxic drugs to be returned back to the manufacturer or supplier for incineration at temperature >1200$^{0}$C |
| | (e) Chemical Waste: Chemicals used in production of biological and used or discarded disinfectants. | Yellow coloured containers or non-chlorinated plastic bags | Disposed of by incineration or Plasma Pyrolysis or Encapsulation in hazardous waste treatment, storage and disposal facility. |
| | (f) Chemical Liquid Waste : Liquid waste generated due to use of chemicals in production of biological and used | Separate collection system leading to | After resource recovery, the chemical liquid waste shall be pre-treated before |

| Category | Type of Waste | Type of Bag or Container to be used | Treatment and Disposal options |
|---|---|---|---|
| | or discarded disinfectants, Silver X-ray film developing liquid, discarded Formalin, infected secretions, aspirated body fluids, liquid from laboratories and floor washings, cleaning, house-keeping and disinfecting activities etc. | effluent treatment system | mixing with other wastewater. The combined discharge shall conform to the discharge norms given in Schedule III. |
| | (g) Discarded linen, mattresses, beddings contaminated with blood or body fluid. | Non-chlorinated yellow plastic bags or suitable packing material | Non- chlorinated chemical disinfection followed by incineration or Plazma Pyrolysis or for energy recovery. |
| | (h) Microbiology, Biotechnology and other clinical laboratory waste: Blood bags, Laboratory cultures, stocks or specimens of microorganisms, live or attenuated vaccines, human and animal cell cultures used in research, industrial laboratories, production of biological, residual toxins, dishes and devices used for cultures. | Autoclave safe plastic bags or containers | Pre-treat to sterilize with non-chlorinated chemicals on-site as per National AIDS Control Organisation or WHO guidelines thereafter for Incineration. |
| Red | Contaminated Waste (Recyclable)<br>(a) Wastes generated from disposable items such as tubing, bottles, intravenous tubes and sets, catheters, urine bags, syringes (without needles and fixed needle syringes) and vacutainers with their needles cut) and gloves. | Red coloured non-chlorinated plastic bags or containers | Autoclaving or micro-waving/ hydroclaving followed by shredding or mutilation or combination of sterilization and shredding. Treated waste to be sent to authorize recyclers. Plastic waste should not be sent to landfill sites. |

| Category | Type of Waste | Type of Bag or Container to be used | Treatment and Disposal options |
|---|---|---|---|
| White (Translucent) | Waste sharps including Metals: Needles, syringes with fixed needles, needles from needle tip cutter or burner, scalpels, blades, or any other contaminated sharp object that may cause puncture and cuts. | Puncture proof, Leak proof, tamper proof containers | Autoclaving or Dry Heat Sterilization followed by shredding or mutilation or encapsulation in metal container or cement concrete and sent for final disposal to iron foundries or sanitary landfill. |
| | This includes both used, discarded and contaminated metal sharps | | |
| Blue | (a) Glassware: Broken or discarded and contaminated glass including medicine vials and ampoules except those contaminated with cytotoxic wastes. | Cardboard boxes with blue colored marking | Disinfection (by soaking the washed glass waste after cleaning with detergent and Sodium Hypochlorite treatment) or through autoclaving or microwaving or hydroclaving and then sent for recycling. |
| | (b) Metallic Body Implants | Cardboard boxes with blue colored marking | |

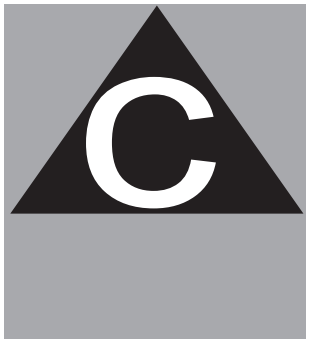
*Disposal by deep burial is permitted only in rural or remote areas where there is no access to common bio-medical waste treatment facility. This will be carried out with prior approval from the prescribed authority and as per the Standards specified in Schedule-III. The deep burial facility shall be located as per the provisions and guidelines issued by Central Pollution Control Board from time to time.

**SCHEDULE IV [See rule 8(3) and (5)] Part A**

**LABEL FOR BIO-MEDICAL WASTE CONTAINERS or BAGS**

HANDLE WITH CARE

CYTOTOXIC HAZARD SYMBOL




HANDLE WITH CARE

Part B

LABEL FOR TRANSPORTING BIO-MEDICAL WASTE BAGS OR CONTAINERS

Day ..... Month .....
Year .....
Date of generation .....

Waste category Number .....
Waste quantity.....

Sender's Name and Address:
Phone Number .....

Receiver's Name and Address:
Phone Number .....

Fax Number.....

Fax Number .....

Contact Person .....

Contact Person .....

In case of emergency please contact:

Name and Address:

Phone No.

Note: Label shall be non-washable and prominently visible.

## Annexure 6: Standard Operative Procedure for collection, transport and processing and inoculation of Extra-pulmonary specimens

### *1. Introduction:*

Mycobacteria may not be suspected as the causative agent of an extra pulmonary disease because the chest X-ray or the tuberculin test is negative or both. However, based on clinical symptoms and because mycobacteria can infect almost any organ in the body, the laboratory should expect to receive a variety of extra pulmonary specimens such as body fluids, surgically excised tissues, aspirates or draining pus and urine.

Extra pulmonary specimens are divided in to two groups based on the site and mode of collection and the extent of contamination.

- Aseptically collected specimens, usually free from other microorganisms (sterile) – fluids like spinal, pleural, pericardial, synovial, ascitic, blood, bone marrow, tissues (lymph node, tissue biopsies) and fine needle aspirates (FNAs)

- Specimens contaminated by normal flora or specimens not collected aseptically (not sterile) – gastric lavage, bronchial washings, urine , pus and stool (in case of disseminated TB in HIV infected patients and infants)

### *2. Collection of extra pulmonary specimens*

Body fluids (spinal, pleural, pericardial, synovial, ascitic, bone-marrow) should be aseptically collected in a sterile container by the physician using aspiration techniques or surgical procedures. Specimens should be transported to the laboratory as quickly as possible.

**2.1 Pleural fluid**

Considered as a suboptimal specimen, as tubercle bacilli are mainly in the pleural wall and not within the fluid. The minimum volume for pleural fluid required for processing for culture is 20–50ml. The fluid is collected using pleural tap or thoracocentesis.

**2.2 Pericardial fluid**

Should be collected using ultra sonogram

**2.3 Blood**

Blood as a specimen for isolating *M. tuberculosis* should be generally discouraged for the low diagnostic yield and high possibility of contamination with respect to the technique required for its culture. However, if there are specific indications when a physician suspects disseminated TB in a HIV infected patient, blood can be collected provided, the culture systems for recovery of mycobacteria is available in that laboratory (BacTAlert, MB Bact or Mycolytic F medium on BACTEC 9050 systems)

### 2.4 Tissues

The aseptically collected tissues are placed by the physician in sterile containers preferably without fixatives or preservatives. If the specimen is to be shipped, it should be protected from drying by adding sterile saline or ideally in selective Kirchner's liquid medium and maintaining a temperature of 4- $15^0$C. Specimens should be transported to the laboratory as quickly as possible.

### 2.5 Swabs

Swabs are always sub optimal specimens and not recommended because of risk of infection for specimen collector. They may be useful in children and patients who cannot produce sputum or may swallow it. A sterile absorbent cotton swab should be used for collection. The best time for the collection is early morning before food and drinks are taken. The swab should be placed in a screw capped container containing normal (0.9%) saline to prevent drying. Swabs except for laryngeal swabs or from discharging sinus should be avoided.

### 2.6 Urine

Among specimens expected to be contaminated, urine is the most common. To minimize excessive contamination of urine specimens, special instructions for collecting urine with adequate cleansing of external genitalia to prevent contamination by commensals should be given. Early morning sample should be collected in 500 ml screw capped sterile containers. Once received in the laboratory, urine must be immediately processed or centrifuged and the pellet refrigerated for further processing. As excretion of tubercle bacilli in urine is intermittent, three early morning specimens must be collected on different days.

### 2.7 Bronchial secretions

Other respiratory specimens that can be submitted to the laboratory for mycobacteria culture are bronchial secretions (minimum volume: 2- 5ml) and bronchial alveolar lavage (BAL) (minimum volume of 20 – 50 ml). Trans-bronchial and other biopsies should be collected under sterile conditions and placed in 0.5-1.0 ml of sterile normal (0.9%) saline to prevent drying during transportation to the laboratory.

### 2.8 Gastric Lavage

In children, who rarely produce sputum, the aspiration of the early morning (gastric content) may be used for TB diagnosis. This is done as an inpatient procedure. This should be transported immediately to the lab and processed (nor more than 4 hours) to prevent the killing action of the acid content in the gastric lavage on the tubercle bacilli. In the event of delay, the sample can be neutralised using 1-2 ml of sterile 10 % sodium bicarbonate solution depending on the volume of gastric aspirate. Trisodium phosphate at a final concentration of 25% can be used but it may affect the viability of tubercle bacilli with prolonged storage.

**NOTE:**

- Samples for culture should never be collected in formalin.
- If histo-pathological examination is required, two samples should be collected

- No preservative should be used for any extra-pulmonary specimen for culture. Necessary instructions are to be given to the concerned staff for sending the biopsy specimen in normal saline for culture and NOT IN FORMALIN as it will kill the bacilli.
- Extra pulmonary specimens should never be collected or transported in CPC.

## *3. TRANSPORTATION OF EXTRA PULMONARY SPECIMENS*

As for pulmonary samples, extra pulmonary specimens will need to be transported in cool boxes which maintain temperatures below $20^0$C for specimens to be compatible for solid, liquid culture systems as well as molecular methods. Triple packing system should be utilized for transportation. All precautions that are followed for transporting pulmonary samples should be followed. For sending material across international or state boundaries this container may have to be packed in the same way with an additional outer container; in such cases, special administrative arrangements with postal authorities and/or airlines may be necessary.

When sending out specimens or when receiving them, check that:

- Request forms are located separately from the specimen containers
- Containers are labelled not on the cap but on the wall of the container
- Each transport box has an accompanying list which identifies the specimens and the patients; the information on the specimen containers should correspond to that on the accompanying list.
- Accompanying list contains the necessary data for each patient
- Date of dispatch and particulars of the health centre are on the accompanying list.

### 3.1 Specimens and request forms

All specimen transported to the laboratory must be accompanied by the request form for C & DST in hard and soft copy formats (See C & DST request form). For quality control reasons, the tests must be performed only upon written request of authorized persons and oral requests without follow up written instructions should not be allowed. It is also important that specimen request forms are kept separate from the specimens themselves. Forms that have been contaminated by specimens should be sterilized by autoclaving. If mistakes in filling request forms and labelling of specimens are found, reject specimens and do not register them. Document the arrival of specimens in the laboratory and note any delays in

delivery in the remarks column of the specimen register and on the report form, particularly for negative/contaminated results. The packaging material should be autoclaved before discarding.

## *4. REGISTRATION OF SAMPLES*

### 4.1 Receipt of incoming specimens

For safety and work-flow reasons, specimens should be received in the office area of the laboratory and delivery boxes should be opened using all the applicable biosafety procedures inside the lab.

To minimize risk of infection, the following procedures should be applied:

1. The specimen package received should be opened only in a biosafety cabinet which may be located in a small area within the reception or in the culture room, as they could potentially be MDR or XDR Tuberculosis. (DO NOT OPEN ON AN OPEN BENCH AT THE LAB RECEPTION)

2. Before opening the packet, inspect the delivery box for signs of leakage; if there is gross leakage evident, discard the box by autoclaving or burning; do not try to open and retrieve any specimen.

3. If on gross inspection there is no leakage, disinfect the outside of the delivery box using cotton wool or paper towels saturated with a suitable disinfectant (5% phenol)

4. Open carefully and check for cracked or broken specimen containers or leakage within the packaged container. If there is minimal leakage without any gross loss of specimen, they may be processed with an asterix that leakage was noted on receipt. (This will assist in identifying reasons for contamination used in lab performance indicators). In case of gross leakage, with only very little sample being available, accept the sample and process after carefully making a note of the same – as extra-pulmonary specimens are precious and repeat collection may not be possible.

5. Check labelling of specimens with individual identification numbers and correspondence with numbers on the accompanying list or Clinical information forms (CIF) that are accompanying the specimens.

6. Disinfect the inside of the delivery box, wash hands after handling specimen containers

7. Autoclave the packaging material before discarding.

8. Assign unique lab serial number to each patient.

9. Evaluate the quality of specimens and make a note as to volume (in case of fluids), leakage, blood mixed *etc.,* Always register the incoming specimen in the laboratory register; each specimen receives a serial number that should be used to label every test for the specimen. Other data that should be reported on the laboratory register are: the date of the receipt of the specimen, patients name, age, sex and address, the name of the referring health centre, the reason for DST. The signature (with the name in capitals) of the person requesting the examination should always be present.

### 4.2 Decontamination of extra pulmonary samples

Most of the extra pulmonary specimens are paucibacillary in nature. Hence, they require milder decontamination. When using solid culture for primary isolation of tubercle bacilli from these specimens, it is preferable to use multiple media including one liquid medium made selective by the use of specific antibiotics that inhibit the growth of other micro-organisms. The media include, LJ, LJ with sodium pyruvate (LJ–P) and selective liquid Kirchner's medium (SK).

Sodium pyruvate facilitates the growth of *M. bovis*. Antibiotics incorporated in the liquid medium include polymixinB, amphotericin B, carbenicillin and trimethoprim (PACT) and vancomycin.

**Preparation of media**

**LJ MEDIUM WITH SODIUM PYRUVATE**

LJ medium is enriched with 0.5% sodium pyruvate. In the preparation of the mineral salt solution, glycerol is omitted and 8.0g sodium pyruvate is added for every 600 ml. This is added to 1 litre of egg fluid, mixed well and distributed.

**SELECTIVE KIRCHNER'S MEDIUM** (For culture of extra-pulmonary specimens**)**

| | |
|---|---|
| Disodium hydrogen phosphate, $Na_2HPO_4.12H_2O$, A.R. 19.0 g (7.5g of anhydrous salt) | |
| Potassium dihydrogen phosphate, $KH_2PO_4$, A.R. | 2.0 g |
| Magnesium sulphate, $MgSO_4.7H_2O$, A.R. | 0.6 g |
| Sodium citrate | 2.5 g |
| L-asparagine | 5.0 g |
| Casein hydrolysate (Bactocasitone) | 0.5 g |
| Glycerol | 20.0 ml |
| Phenol Red, 0.4% solution | 3.0 ml |
| Distilled water, to | 1 litre |

Check pH to 6.9 – 7.2

Autoclave at 15 lbs/15 minutes

Then add aseptically the following:

| | |
|---|---|
| Polymyxin B (20,000 units) | 31 mg |
| Carbenicillin | 100 mg |
| Trimethoprim | 10 mg |

Amphotericin B, solubilised 10 mg

Dissolve the above in 5 ml sterile distilled water before addition

Also, add sterile calf serum 100 ml

Mix the above carefully and distribute, under sterile conditions, in 10 ml amounts. Check sterility by overnight incubation at $37^{o}C$ and store in the cold.

## *5. CULTURE BY SOLID CULTURE METHODS*

### 5.1 CSF and pericardial fluid smear:

1. Label a clean dry slide with the lab number and place the slide and the sample container inside the cabinet
2. Mix well and aseptically remove one loopful of the fluid and place in the centre of the slide; close the container and allow the drop to air-dry
3. Place one more drop of the CSF on the same spot and let dry.
4. Place the third drop after processing the sample as below:

**Culture:**

Culture of CSF is done in two steps:

1. Direct inoculation in media
2. Inoculation after decontamination

**Direct**

1. Place one loopful of CSF on to one slope each of LJ and LJ-P
2. Add 0.2 ml of CSF in to one bottle containing SK medium
3. Label the set as 'A'

**Decontamination**

1. Add 1ml of 5% $H_2SO_4$ to CSF
2. Mix well and let stand for 15 minutes
3. Fill the container with sterile distilled water and centrifuge at 3000 x g for 15 minutes
4. Aspirate the supernatant carefully without disturbing the deposit or discard carefully in to a disinfectant bin containing 5% phenol or any other mycobactericidal solution

5. Inoculate one slope each of LJ and LJ-P with one loopful of deposit for each slope
6. Transfer the remaining deposit in to one bottle of SK
7. Label the set as 'B'
8. Incubate both set A and B at $37^{O}C$

**5.2 BAL**

1. Make a direct smear
2. Process using 5% $H_2SO_4$ as in CSF
3. Inoculate two slopes each of LJ and LJ-P with one loopful of deposit using 5mm twisted wireloop
4. Transfer the remaining deposit in to one bottle of SK
5. Incubate the slopes and SK medium at $37^{O}C$

**5.3 Gastric Lavage**

1. Gastric Lavage should be processed immediately upon arrival in the lab to prevent the killing action of the gastric pH (due to HCl) on the tubercle bacilli
2. Make a direct smear and process by modified Petroff's method
3. Place one drop of the final pellet on the direct smear
4. Inoculate two slopes each of LJ and LJ-P with one loopful of deposit for each slope
5. Transfer the remaining deposit in to one bottle of SK
6. Incubate the slopes and SK medium at $37^{O}C$

**5.4 Tissue / Biopsy**

1. Ideally, biopsy specimens should be collected and transported in SK medium
2. Carefully place the tissue inside a sterile petriplate inside the BSC
3. Using sterile scissors and forceps, cut the tissue in to tiny pieces
4. Transfer to a sterile tissue grinding tube – add a little water to the petriplate to facilitate transferring
5. Add sterile distilled water to the tube (not more than 5 ml)

6. Homogenise using a sterile Teflon grinding rod using a foot operated tissue grinder

7. Make a direct smear from the homogenate

8. Centrifuge the homogenate at 3000 x g for 15 minutes

9. Decant the supernatant carefully in to the disinfectant bath

10. To the deposit add 1 ml of sterile distilled water

11. Add one drop to the direct smear, air dry, fix and stain

12. To the remaining pellet, add 1ml of 5% $H_2SO_4$

13. Proceed as for CSF

14. Inoculate two slopes each of LJ and LJ-P with one loopful of deposit for each slope

15. Transfer the remaining deposit in to one bottle of SK

16. Incubate the slopes and SK medium at 37$^{O}$C, along with the SK medium used for transporting

**5.5 Fine Needle Biopsy specimen**

1. Fine needle specimens should be collected and transported only in SK medium or any other liquid medium

2. The medium is incubated as such at 37$^{O}$C, since only a very tiny piece of the tissue is obtained as sample

If the sample is received without SK

1. Add the contents of two SK medium bottles to the specimen

2. Shake vigorously and let stand for 10 minutes

3. Divide the medium in to two aliquots and incubate both at 37$^{O}$C

**5.6 Pus**

1. Make a direct smear, air dry, fix and stain

2. <u>If the pus is thick or purulent</u>, process by modified Petroff's method using 4% NaOH

3. Inoculate two slopes each of LJ and LJ-P with one loopful of deposit for each slope

4. Transfer the remaining deposit in to one bottle of SK
5. Incubate the slopes and SK medium at $37^{O}C$
6. If the pus is thin or dilute, proceed with decontamination using 5% $H_2SO_4$
7. Inoculate two slopes each of LJ and LJ-P with one loopful of deposit for each slope
8. Transfer the remaining deposit in to one bottle of SK

**5.7 Urine / Ascitic fluid**

1. Distribute the entire specimen in to 20 or 40 ml volumes in Universal containers / Falcon tubes inside a BSC
2. Centrifuge at 3000 x g for 15 minutes

Process the supernatant and deposit independently as follows:
Supernatant:

3. Aspirate carefully 1ml of the top layer from each tube and pool
4. Process by 5% $H_2SO_4$ as for CSF
5. Transfer 1ml of the final supernatant on to two bottles of SK each – Label the set as DSS (Decontaminated Supernatant Supernatant)
6. Decant the supernatant carefully in to the disinfectant bath
7. From the deposit transfer about 0.2 ml and the remaining in to 2 bottles of SK respectively – Label as DSD (Decontaminated Supernatant Deposit)

Deposit:

8. Pool all the deposit in to one tube
9. Process using 5% $H_2SO_4$ as for CSF
10. Inoculate two slopes each of LJ and LJ-P with one loopful of deposit for each slope
11. Transfer the remaining deposit in to one bottle of SK

**5.8 Swabs:**

If two swabs are available, use one for smear and one for culture; if only one is available do only culture

1. Immerse the swab in 5 ml of 4% $H_2SO_4$ for 1 minute

2. Transfer the swab to another tube containing 5 ml of 4% NaOH
3. Directly inoculate two slopes each of LJ, LJ-P
4. Transfer the swab finally to a tube containing SK medium
5. Incubate all tubes at 37$^{o}$C

**5.9 Culture Reading**

1. Read all cultures used for isolating *M. tuberculosis* from extra-pulmonary specimens every week for up to 8 weeks using the same methodology used for pulmonary samples
2. If the solid media show typical growth report immediately after confirmation
3. Read SK medium up to 6 weeks
4. MTB appears as whitish granular or flaky growth that settles down at the bottom
5. If the SK medium shows growth or contamination (in the form of turbidity) within 6 weeks, decontaminate as sputum by modified Petroff's method and inoculate deposit on LJ medium alone and read up to 8 weeks
6. Even if the SK medium shows no growth within 6 weeks, proceed with decontamination using modified Petroff's method and inoculate deposit on LJ medium alone and read up to 8 weeks
7. If LJ shows typical MTB growth within 8 weeks, report immediately after confirmation
8. Report as negative only after LJ completes 8 weeks (a total of 14 weeks)

***6. Processing of extra pulmonary samples for MGIT960***

Isolation of M. tuberculosis by MGIT system requires the final inoculum to be in an ideal condition that will not interfere with the fluorescence.

**6.1 Pus and other muco-purulent specimens**

1. Thick pus of volume >10 ml is decontaminated using the NALC – NaOH method as sputum
2. If the volume is < 10 ml, either aliquot and process only 10 ml by NALC – NaOH method or concentrate the initial volume by centrifugation for 15 – 20 minutes and resuspend the pellet in 5 ml of sterile distilled water. If the pus is too thick, add about 50-100 mg of NALC powder; mix well and decontaminate using NaOH. Resuspend the final pellet in buffer to reduce the pH

3. If the pus is not thick, decontaminate using 2-4% NaOH. The concentration of NaOH can be changed based on the expected level of contamination in the specimen which depends on the site of collection

### 6.2 Gastric aspirates

1. Distribute the volume in smaller aliquots and centrifuge the tubes at 3000 x g
2. Pool the deposits, add 5ml distilled water and decontaminate it using NALC-NaOH or 2-4% NaOH

### 6.3 Bronchial washings

1. Process using NALC-NaOH like sputum
2. If the specimen is >10 ml in volume, process the whole specimen.
3. If <10ml, concentrate the specimen by centrifugation (3000x g, 15-20 minutes)
4. Add 5 ml sterile water to the pellet and decontaminate as for sputum

### 6.4 Laryngeal swabs

1. Transfer the swab into a sterile centrifuge tube and add 2 ml sterile water.
2. Add 2 ml of NaOH-NALC solution and mix well in a vortex mixer.
3. Let stand for 15 minutes. Remove the swab with forceps, squeezing the liquid out of the swab and discarding it.
4. Fill the tube with phosphate buffer and mix
5. Centrifuge at 3000xg for 15-20 minutes.
6. Discard the supernatant fluid and resuspend the sediment in 1-2 ml sterile buffer. Use this suspension for smear and culture.

### 6.5 Tissue

1. Homogenize the tissue in a tissue grinder with a small quantity of sterile saline or water (2-4 ml).
2. Decontaminate the homogenized specimen using NALC-NaOH procedure as in sputum.
3. Resuspend the sediment with phosphate buffer
4. If the tissue grinder is not available, use a mortar and pestle.

5. Tissue may also be placed in a Petri dish with sterile water (2-4 ml) and be torn apart with the help of two sterile needles.

### 6.6 Urine

Isolation of mycobacteria from urine specimens using MGIT has not been validated.

1. Aliquot the entire volume in several centrifuge tubes
2. Concentrate the specimen by centrifugation for at least 20-25 minutes
3. Resuspend the pellet in each tube with 1-2 ml of sterile water and pool together
4. Decontaminate using 4% NaOH as for sputum

### 6.7 Other body fluids (CSF, synovial fluid and pleural fluid)

As these fluids are collected usually under aseptic conditions, they require only milder decontamination

1. If the specimen volume is more than 10 ml, concentrate by centrifugation at about 3000x g for 15-20 minutes
2. Liquefy thick or mucoid specimens prior to centrifugation by adding NALC powder (50-100 mg).
3. Resuspend the sediment in about 5 ml of saline
4. Mix and decontaminate as for sputum

### 6.8 Blood

Isolation of mycobacteria from blood specimens by MGIT 960 has not been evaluated thoroughly. A few published studies have used blood after lysis centrifugation. Ideally BACTEC Myco/F Lytic medium is recommended for isolation of mycobacteria from blood samples.

## Annexure 7: Laboratory Processes for Diagnosis of NTM to be followed at C-DST laboratories

**Specimen Processing**

To minimize contamination or overgrowth of cultures with bacteria and fungi, digestion and decontamination procedures should be performed on specimens collected from non-sterile body sites.

Specimens from contaminated sites contain other organisms that may grow more rapidly than NTM and interfere with the recovery of mycobacteria.

Because NTM, especially rapidly growing mycobacteria (RGM), are much more susceptible to decontamination than M. tuberculosis, these procedures should not be so severe as to eliminate the mycobacteria potentially present. Tissue specimens or fluids from normally sterile sites do not require decontamination.

Tissues should be ground aseptically in sterile physiological saline or bovine albumin and then directly inoculated onto the media.

The most widely used digestion–decontamination method uses N-acetyl-l-cysteine–sodium hydroxide (NALC-NaOH). This method is often used in conjunction with a 5% oxalic acid procedure ("double processing") for specimens from patients with Cystic Fibrosis (CF) or bronchiectasis whose sputa are known to be contaminated with aerobic gram-negative rods, especially Pseudomonas aeruginosa.

Because NTM may be sensitive to oxalic acid decontamination, with reduced yield on culture, another option is to use a two-step decontamination approach reserving oxalic acid only for those specimens overgrown by bacteria other than NTM.

**Smear Microscopy**

The recommended method for staining clinical specimens for AFB, including both M. tuberculosis and NTM, is the fluorochrome technique, although the Ziehl-Neelsen method or Kinyoun stain are acceptable but less sensitive alternatives.

The Gram stain is not adequate for detection of mycobacteria. In many patients, the NTM, especially the RGM, may be more sensitive to the AFB decolourisation procedure and may not stain at all with fluorochrome stains.

Therefore, if RGM are suspected, it may be prudent to use a weaker decolorizing process. It is also noteworthy that negative smears do not necessarily mean that NTM, especially RGM, are not present in a clinical specimen.

Semi-quantitative analysis of smears can be useful for diagnostic purposes. Fluorochrome smears are graded from 1+ (1–9 organisms per 10 high-power fields) to 4 + (90 organisms per high-power field).

The burden of organisms in clinical material is usually reflected by the number of organisms seen on microscopic examination of stained smears. Environmental contamination, which usually involves small numbers of organisms, rarely results in a positive smear examination.

Previous studies have indicated that specimens with a high number of mycobacteria isolated by culture are associated with positive smears and, conversely, specimens with a low number of mycobacteria isolated by culture are less likely to have positive smears

**Culture Techniques**

All cultures for mycobacteria should include both solid and broth (liquid) media for the detection and enhancement of growth.

However, broth media cultures alone may not be satisfactory because of bacterial overgrowth. Cultures in broth media have a higher yield of mycobacteria and produce more rapid results than those on solid media.

The advantages of solid media over broth media are that they allow the observation of colony morphology, growth rates, recognition of mixed (more than one mycobacterial species) infections, and quantitation of the infecting organism, and serve as a backup when liquid media cultures are contaminated.

**Broth media:** One of the most widely used broth systems is the non-radiometric mycobacteria growth indicator tube (MGIT) (Becton Dickinson, Sparks, MD), which contains a modified Middlebrook 7H9 broth in conjunction with a fluorescence quenching–based oxygen sensor to detect mycobacterial growth. As the mycobacteria grow and deplete the oxygen present, the indicator fluoresces when subjected to ultraviolet light.

**Solid media:** Recommended solid media include either egg based media, such as Lowenstein-Jensen agar or agar-based media such as Middlebrook 7H10 and 7H11 media.

Semiquantitative (0–4+) reporting of NTM colony counts on solid media is recommended by the CLSI. A single positive respiratory specimen with a low colony count (e.g., broth culture

positive only) is less likely to be clinically significant than a specimen with a high colony count (e.g., growth on both solid and broth media). This approach also helps in the assessment of response to therapy.

After successful therapy of MAC lung disease and at least 10 months of negative cultures during therapy, a single positive culture that is AFB smear negative and of low culture positivity (<10 colonies on solid media and/or positive in the broth media only) generally represents either contamination (false-positive culture) or transient new infections and not relapse of the original infecting strain.

M. haemophilum, M. genavense, M. avium subsp. Paratuberculosis (formerly M. paratuberculosis), and M. ulcerans are examples of fastidious NTM that require special supplementation for recovery on culture. M. haemophilum grows only on media supplemented with iron-containing compounds such as ferric ammonium citrate, hemin, or hemoglobin.

Because M. haemophilum has a predilection for skin and the body's extremities, all specimens from skin lesions, joints, or bones should be cultured in a manner suitable for recovery of this species.
M. genavense and M. avium subsp. paratuberculosis require mycobactin J, and M. ulcerans may be optimally recovered with egg yolk supplementation.

**Incubation of NTM Cultures**

The optimal temperature for most cultures for NTM is between $28^0$ and $37^0$C. Most clinically significant slowly growing mycobacteria grow well on primary isolation at $35^0$ to $37^0$C with the exception of the following:

The newly described M. conspicuum, which requires temperatures from $22^0$ to $30^0$C for several weeks and only grows at $37^0$C in liquid media, M. haemophilum, which prefers temperatures from $28^0$ to $30^0$ C, M. ulcerans, which grows slowly at $25^0$ to $33^0$C, and some strains of M. chelonae, which grows at temperatures of $28^0$-$33^0$C.

Cultures for RGM and M. marinum should be incubated at $280^0$ to $300^0$ C. All skin, joint fluid, and bone specimens should be cultured at $280^0$ to $300^0$ C and at 350 to $370^0$ C. Optimal recovery of all species may require duplicate sets of media at two incubation temperatures.

Most NTM grow within 2 to 3 weeks on subculture. To detect M. ulcerans or M. genavense, cultures should be incubated for at least 8 to 12 weeks. Rapidly growing mycobacteria usually grow within 7 days of subculture.

Earlier detection of NTM can be expected with liquid-based systems. When stated on the laboratory report, the time in days to the detection of mycobacterial growth can be helpful to clinicians as an indication of isolation of a rapidly growing species.

**NTM Identification**

Because of differences in antimicrobial susceptibility that determine treatment options, species-level identification of the NTM is becoming increasingly clinically important. Several factors increase the likelihood of clinical significance of NTM isolates, including the recovery from multiple specimens or sites, recovery of the organism in large quantities (AFB smear–positive specimens), or recovery of an NTM isolate from a normally sterile site such as blood.

For initial clinical mycobacterial isolates, however, it is sometimes difficult to determine the clinical significance of the isolate without species identification.

Therefore, identification of most mycobacterial isolates to the species level and not merely as groups, such as “M. chelonae/abscessus group” is strongly recommended. If, after consultation between the clinician and the laboratorian and in the event that a specific laboratory does not have the necessary technology for species identification of an NTM isolate, the isolate could be sent to a reference laboratory for further analysis.

It is equally important to recognize that not all clinically obtained NTM isolates, especially from sputum, will need extensive identification efforts. For instance, a pigmented rapidly growing mycobacterium recovered in low numbers from only one of multiple sputum specimens from a patient undergoing therapy for MAC lung disease would not need an extensive effort for identification of that isolate as it would not likely be clinically significant.

Awareness of the context from which an NTM isolate is obtained can be critically important in determining the need for speciation of that isolate. Again, communication between the clinician and laboratorian is critical for making this type of determination.

**Phenotypic testing:**

Preliminarily, NTM can be categorized by growth rate and pigmentation, which will help guide in the selection of proper testing procedures, including appropriate media selection and incubation temperatures. Recent studies have shown, however, that identification using only conventional biochemical analysis is both time consuming and increases turnaround time, leading to significant delays in diagnosis.

Isolates of NTM, which form colonies on subculture in 7 days or fewer, are referred to as "rapidly growing mycobacteria" or RGM. Conversely, those isolates of NTM that require more than 7 days to form mature colonies on subculture are termed "slowly growing mycobacteria." Although many of the conventional and traditional
laboratory tests for mycobacterial evaluation are not routinely used, the rate of growth is still useful for preliminary broad classification of a nontuberculous mycobacterium.

Traditionally, NTM have also been divided into three groups based on the production of pigment. These growth characteristics are rarely detailed in modern mycobacterial laboratories, but the presence of pigmentation and smooth colony morphology quickly exclude the isolate as belonging to the M. tuberculosis complex that forms non pigmented and rough colonies.

Biochemical testing of NTM uses a battery of tests, again based on the growth rate of the NTM. The use of conventional testing alone does not allow identification of many of the newly described NTM and thus newer methods, including HPLC and molecular methods, must be used.

**Chemotaxonomic testing:**

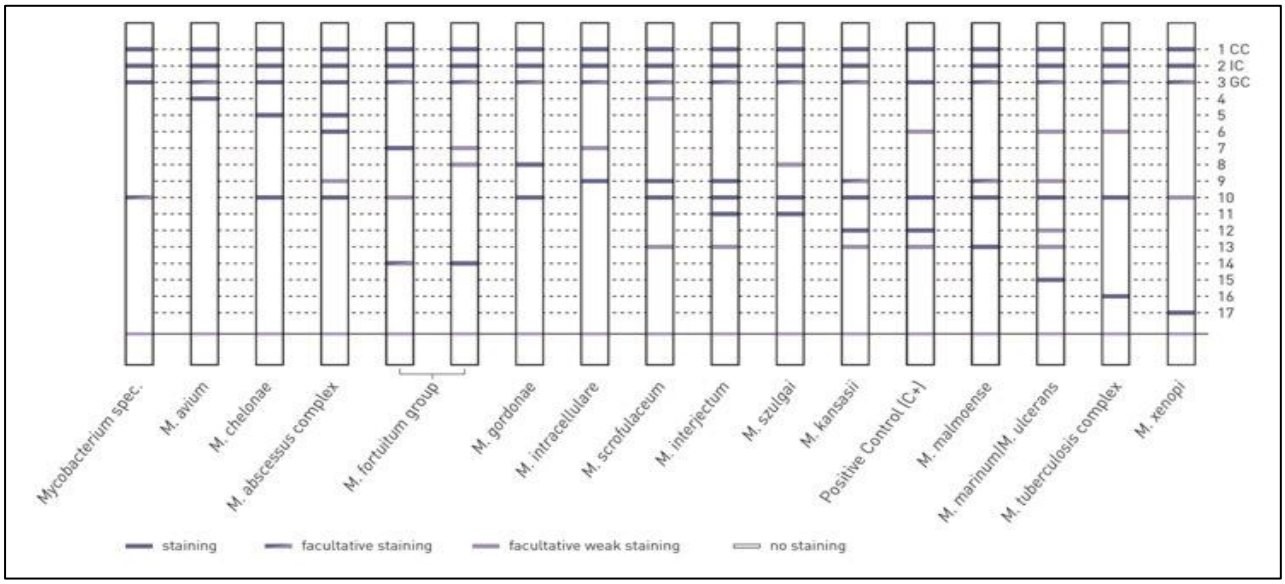
HPLC is a practical, rapid, and reliable method for identifying many slowly growing species of NTM. HPLC can also be used in the direct analysis of primary cultures of mycobacteria grown in MGIT 960 culture and the identification of MAC directly from specimens with AFB smear–positive results. However, HPLC has limitations. Recognition of some newer species and species within the M. simiae complex is difficult. It has also been reported that some species within the M. fortuitum group and the M. smegmatis group are difficult to separate, and distinction between M. abscessus and M. chelonae may not be possible.
HPLC analysis will be less useful in the future, as identification of NTM species will be accomplished by molecular methods.

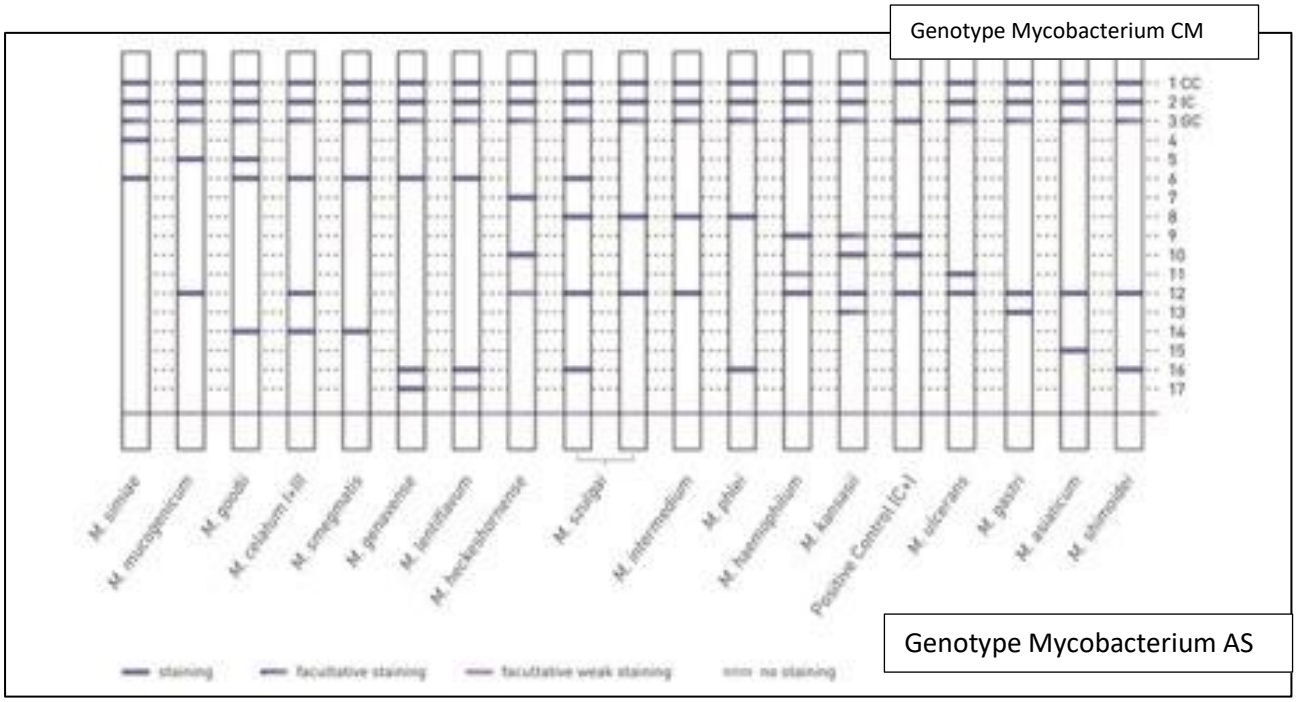
Genotypic methods for identification of NTM. Commercially available molecular probes. Acridium ester–labeled DNA probes specific for MAC (or separately for M. avium and M.intracellulare), M. kansasii, and M. gordonae have been approved by the U.S. Food and Drug Administration (FDA) and are currently used in many clinical laboratories (AccuProbe; Gen-Probe, Inc., San Diego, CA) for the rapid identification of NTM. This technique is based on the release of target 16S rRNA from the organism. Testing can be performed using isolates from solid or liquid culture media and identification of these species can be achieved within 2 hours.

Recently the Hain's LPA for NTM has become available which has two sets of LPA strips; Genotype Mycobacterium CM (detects *M.tb* and other common NTM species) and

Genotype Mycobacterium AS which detects additional NTM species. The two LPAs can be run in parallel to get the maximum advantage. This is what the programme is planning to use for diagnosis of NTM species.

The species that can be identified is as below:




**Key points to be considered for quality diagnosis of NTM:**

- As much material as possible for NTM culture should be provided with clear instructions to the laboratory to culture for mycobacteria using RNTCP request form for examination of biological specimen for TB

- All cultures for NTM should include both a rapid detection broth (liquid) media technique and solid media cultures
- Quantification of the number of colonies on plated culture media should be performed to aid clinical diagnosis
- Supplemented culture media and special culture conditions (lower incubation temperatures) should be used for material cultured from skin lesions, joints, and bone
- The time (in days) to detection of mycobacterial growth should be stated on the laboratory report
- Clinically significant NTM isolates should be routinely identified to the species level. An important exception is MAC because the differentiation between M. avium and M. intracellulare is not yet clinically significant. Although not routinely recommended, this differentiation may be important epidemiologically and, in the future, therapeutically
- The RGM (especially M. chelonae, M. abscessus, and M. fortuitum) should be identified to species level using a recognized acceptable methodology, HPLC with Molecular methods
- Clarithromycin susceptibility testing is recommended for new, previously untreated MAC isolates. Clarithromycin is recommended as the “class agent” for testing of the newer macrolides because clarithromycin and azithromycin share cross-resistance and susceptibility. No other drugs are recommended for susceptibility testing of new, previously untreated MAC isolates. There is no recognized value for testing of first-line anti-TB agents with MAC using current methodology.
- Clarithromycin susceptibility testing should be performed for MAC isolates from patients who fail macrolide treatment or prophylaxis regimens.
- Previously untreated M. kansasii strains should be tested in vitro only to rifampicin. Isolates of M. kansasii that show susceptibility to rifampicin will also be susceptible to rifabutin
- M. kansasii isolates resistant to rifampicin, should be tested against a panel of secondary agents, including rifabutin, ethambutol, isoniazid, clarithromycin, fluoroquinolones, amikacin, and sulfonamides.
- M. marinum isolates do not require susceptibility testing unless the patient fails treatment after several months
- There are no current recommendations for one specific method of in vitro susceptibility testing for fastidious NTM species and some less commonly isolated NTM species
- Validation and quality control should be in place for susceptibility testing of antimicrobial agents with all species of NTM.
- Susceptibility of RGM for eight agents, including amikacin, cefoxitin, clarithromycin, ciprofloxacin, doxycycline, linezolid, sulfamethoxazole, and tobramycin, can also be used to facilitate identification of M. abscessus, M. chelonae, and M. fortuitum.
- Communication between the clinician and laboratorian is essential for determining the importance and extent of the identification analysis for a clinical NTM isolate.

**Antimicrobial Susceptibility Testing for NTM**

There is ongoing debate about the role of in vitro susceptibility testing for managing patients with NTM disease. Susceptibility breakpoints have been defined in the laboratory to distinguish populations of mycobacteria that are labeled susceptible and resistant. DST has been recommended for certain groups of NTM only. Clarithromycin is the only drug for which susceptibility testing for MAC isolates is recommended.

**Slow Growing Mycobacteria:** There is no single method recommended.

**For MAC,** the macro or micro-broth dilution method is acceptable. Initial isolates from patients with previously untreated MAC lung disease should be tested to clarithromycin to establish baseline values. The isolate may also be saved for future testing, especially to determine if future isolates represent relapse or new infections by comparative DNA fingerprinting of baseline and subsequent isolates. Other isolates to be tested include the following:

1. Isolates from patients who previously received macrolide therapy to determine whether or not the isolates are still macrolide susceptible.
2. Isolates from patients with MAC pulmonary disease on macrolide-containing regimens who relapse or fail after 6 months of macrolide-containing therapy.
3. Isolates from patients with AIDS who develop bacteremia on macrolide prophylaxis.
4. Positive blood cultures after 3 months of treatment with macrolide-containing regimens for patients with disseminated MAC.

Untreated MAC isolates usually have minimum Hibitory concentrations (MICs) of 4µg/ml or less to clarithromycin and are considered susceptible. In contrast, relapse strains after treatment inevitably have a clarithromycin MIC of 32µ /ml or greater and no longer respond to treatment with macrolides. Isolates of MAC have only a single copy of the ribosome, and hence, macrolide monotherapy carries a significant risk of the development of mutational resistance. Not unexpectedly, all high-level clarithromycin-resistant isolates have mutations in the adenine at position 2058 or 2059 of the 23S rRNA gene, which is the presumed macrolide binding site on the ribosomal unit.

**M. Kansasii:** Only rifampicin susceptibility routinely needs to be tested, if rifampicin is resistant then additional DST to H, Ethambutol, Amikacin, Clarithromycin, rifabutin, sulfonamides, streptomycin, Levofloxacin and Moxifloxacin. The concentrations (0.5 to 5.0µg/ml) for H used for NTM are not the same as for M. tuberculosis.

1. The preferred staining procedure is the fluorochrome method.

2. Specimens should be cultured on both liquid and solid media. Species that require special growth conditions and/or lower incubation temperatures include M. haemophilum, M. genavense, and M. conspicuum. These species can cause cutaneous and lymph node disease.

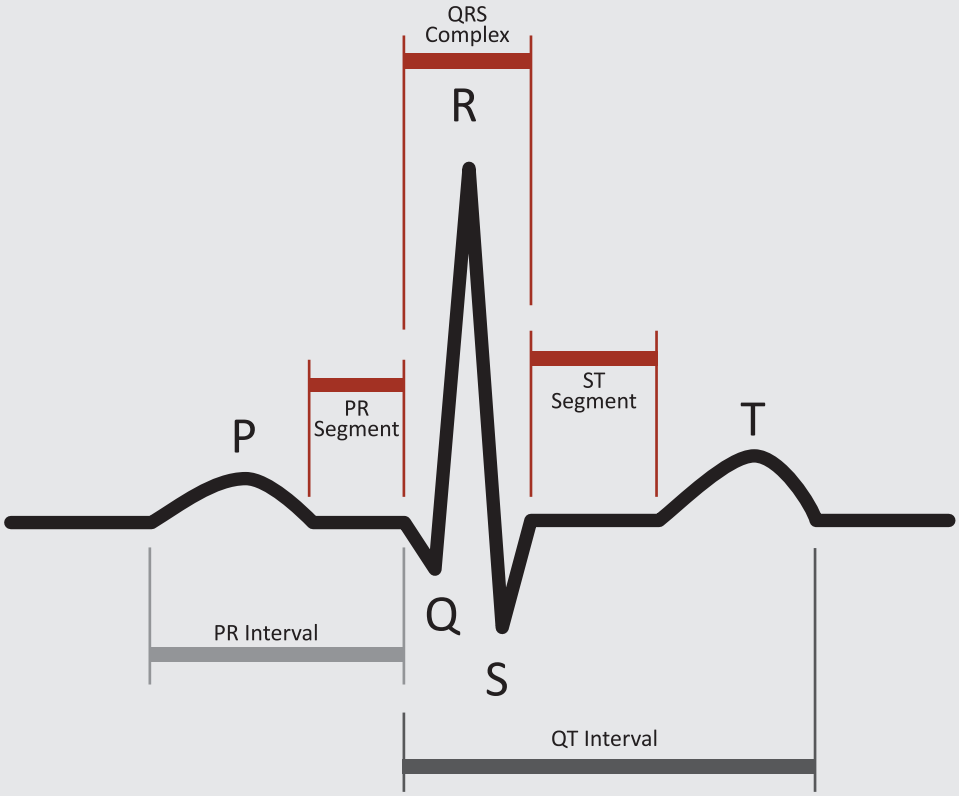
3. In general, NTM should be identified to the species level. Methods of rapid species identification include commercial DNA probes (MAC, M. kansasii, and M. gordonae) and high-performance liquid chromatography (HPLC). For some NTM isolates, especially rapidly growing mycobacterial (RGM) isolates (M. fortuitum, M abscessus, and M. chelonae), other identification techniques may be necessary including extended antibiotic in vitro susceptibility testing, DNA sequencing or polymerase chain reaction (PCR) restriction endonuclease assay (PRA).

4. Routine susceptibility testing of MAC isolates is recommended for clarithromycin only. Routine susceptibility testing of M. kansasii isolates is recommended for rifampicin only.

5. Routine susceptibility testing, for both taxonomic identification and treatment of RGM (M. fortuitum, M abscessus, and M. chelonae) should be with amikacin, imipenem (M. fortuitum only), doxycycline, the fluorinated quinolones, a sulfonamide or trimethoprim-sulfamethoxazole, cefoxitin, clarithromycin, linezolid, and tobramycin (M. chelonae only).

## Annexure 8: Definition of QTc interval

- The QT interval in an ECG is measured from the start of the Q wave to the end of the T wave (see diagram below.




- When monitoring the effect of bedaquiline, the QT interval needs to be adjusted (corrected) for the heart rate. Many ECG machines today provide an output of the corrected QT interval (QTc) automatically. If you are using a machine that does not, the following instructions can help you make the necessaru correction.
  - The preferred way to calculate the QTc is the Fredericia method (QTcF), which is derived by dividing the QT interval by the cubed root of the interval in seconds between the peak of two successive R waves (RR) read from the ECG strip:

$$QT_{cF} = \frac{QT}{\sqrt[3]{RR}}$$

- Whenever an abnormal QTc value found, the ECG and calculations should be repeated.
- A normal value for the corrected QTcF interval is equal to or less than 0.45 seconds (450 ms) in males or 0.47 seconds (470 ms) in females.

## Annexure 9: Patient Information Booklet on Bedaquiline

Revised National Tuberculosis Control Programme
Central TB Division, Directorate General of Health Services
Ministry of Health & Family Welfare,
Nirman Bhawan, New Delhi

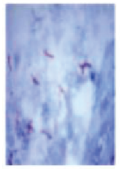
### Patient Information Booklet on TB, Drug-resistant TB and Bedaquiline

This guide is meant for MDR-TB patients to understand all you need to know about tuberculosis and MDR-TB. We hope this will help you clarify all your doubts and fears about this disease as well as enable you to cope with this illness, complete treatment as required and help you lead a healthy lifestyle during the treatment and thereafter.

TB is a curable disease and treatment is available free of cost. Yet in India we have 2.2 million cases of TB. MDR-TB has emerged as a public health problem. India is one of the countries with highest burden of MDR-TB. India accounts for 64 000 MDR-TB patients of the 300 000 MDR patients estimated in the world.

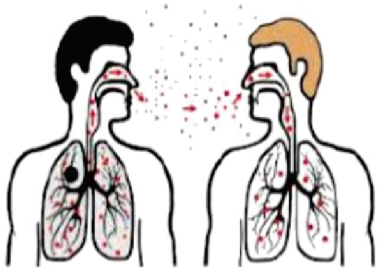
### What causes tuberculosis?

Tuberculosis (TB) is caused by bacteria (*Mycobacterium tuberculosis*) that most often affect the lungs.



### How does it spread?

It spreads through the air when a person with TB (whose lungs are affected) coughs, sneezes, spits, laughs or talks.



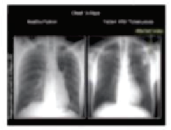
### Symptoms of tuberculosis:

The symptoms of active TB include any of the following:

- *Cough for 2 weeks*
- *Fever for more than a month*
- *Blood in sputum anytime*
- *Loss of weight*
- *Loss of appetite*
- *Night sweats.*

### Diagnosis of tuberculosis:

Tuberculosis is diagnosed by finding *Mycobacterium tuberculosis* bacteria in sputum which can be seen with the help of a microscope.

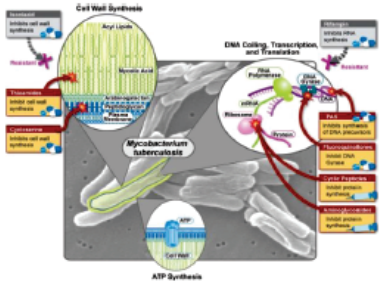


### Tuberculosis treatment:

- Treatment is available is all government health facilities.
- Treatment is free of cost.
- Duration of treatment is as short as 6 to 8 months.
- Treatment has to be continued for the prescribed period.

## What is multidrug-resistant TB?

- **Multidrug-resistant** tuberculosis **(MDR-TB)** is defined as a form of TB infection caused by bacteria that is resistant to treatment with at least two of the most powerful first-line anti-TB drugs, isoniazid (H) and rifampicin (R).




## How does drug resistance develop?

- Use of inadequate regimen and inappropriate treatment adherence leads to increase in drug resistance levels in the community.
- MDR-TB is treated by second-line drugs. Incomplete and erratic treatment for MDR-TB leads to worsening of resistance and XDR-TB.
- All patients of MDR-TB cough out bacteria that are drug-resistant. These can infect another person with the same resistant bacteria. This is how MDR-TB spreads fast.

## Who is at risk for getting MDR TB?

Drug-resistant TB is more common in people:

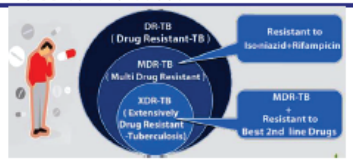
- who do not take their TB drugs regularly;
- who relapse with TB after being irregularly treated for TB in the past and are initiated on category 2 treatment of TB which is for 8 months;
- who are exposed to drug-resistant TB patients such as in families with an MDR patient;
- who are health workers working among MDR-TB patients;
- TB patients who are dependent on alcohol and habituated smokers and do not complete treatment.

## Conditions to suspect MDR-TB:

- When people continue to have symptoms of TB even after 2 months of initiating treatment;
- Those TB patients who do not complete the full course of treatment;
- When a patient becomes sputum smear positive again after initial conversion at 2 months.
- Symptomatic contacts of an MDR TB patient

## How to diagnose multidrug-resistant TB?

- Multidrug-resistant TB can be detected using special laboratory tests which test the bacteria for sensitivity to the drugs or detect resistance patterns.
- These tests can be molecular in type or else culture-based.




## Management of MDR-TB:

- The total duration of treatment is about 24 months. This includes 6 months of injections and 6 drugs daily. The drugs are kanamycin, levofloxacin, ethionamide, cycloserine, ethambutol and pyrazinamide.

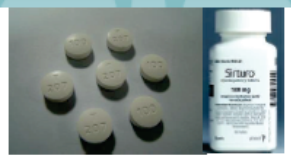
## Challenges faced by MDR-TB patients:

**Treating MDR-TB is a challenging task, and there are challenges at every level.**

## Patients have to follow:

1. Patients must fully adhere to the treatment for TB
2. Family members of an MDR patient have to be investigated for TB
3. Children below 6 years of age in the same house of an MDR patient if negative for TB need to be initiated on drugs to prevent TB.
4. Patients have to take timely, affordable and nutritious food.
5. Patients have to avoid consumption of alcohol, smoking and other addictive substances
6. Patients have to follow preventive steps to avoid spreading the infection, such as covering the mouth while coughing.
7. Patients need to follow healthy lifestyle practices like exercise, yoga and positive thinking.
8. If a patient experiences any side-effects to the drugs, e.g. nausea, vomiting, giddiness, irritability, depression, nightmares, suicidal thoughts, etc. they should inform the doctor immediately. These side-effects can be managed and there is no need to panic or discontinue treatment.
9. The patient MUST complete the full course of 24 months of treatment and cannot afford to be irregular. The consequences of irregular treatment can be dangerous and unmanageable.

# INFORMATION ON BEDAQUILINE FOR PATIENTS



## Some facts about this drug:

- The name of this drug is bedaquiline.
- It is in the form of tablet and is easy to swallow.
- This drug has shown promise in treating patients that are infected with the bacterium *Mycobacterium tuberculosis* especially for the drug-resistant TB variety.
- It might have potential value in combination with other anti-TB medications in the treatment of TB.

## How do you take bedaquiline?

- It is in the form of a tablet and is easy to swallow.
- You will receive bedaquiline for 24 weeks (6 months) treatment of MDR-TB.

Take 400 mg (4 tablets) 1 time each day. x2 weeks

Take 200 mg (2 tablets) a day 3 times a week. x 3-24 weeks

## What are the possible side effects?

- Any drug can cause unwanted, unpleasant and sometimes harmful effects on the body.
- Drugs that are especially used for the treatment of MDR-TB are called second-line anti TB drugs.

## The most common-side-effects are:

- Headache
- Common cold and sore throat
- Dizziness, fainting, lightheadedness
- Joint pain
- Nausea
- Vomiting
- Diarrhoea.
- Palpitation ( feeling an irregular, fast or slow heartbeat)
- Heart rhythm may change (Qtc)

## Benefits:

- There is a greater chance that you will be cured of tuberculosis
- You will possibly become better very much sooner than if you only took the standard medicines for treatment of resistant TB.
- Also, it is probably less likely that the drugs you are taking will develop resistance if you are taking bedaquiline.

## What do I do when I have problems?

- You should tell your health-care provider immediately about any side-effect that you experience while taking bedaquiline.

## What do women need to know?

All women must avoid getting pregnant while taking bedaquiline. If you are a woman able to become pregnant (i.e. not sterilized or less than 2 years since menopause), you should use 2 methods of birth control.

## What do I do in case of pregnancy during treatment?

- Inform the health care provider immediately
- After evaluation in consultation with Obstetrician/ gynecologist you may be required to either terminate the pregnancy (MTP), get modified regimen without bedaquiline. You and the baby both may be evaluated for longer duration post treatment

## What do men need to know?

- All men should avoid fathering a child while on treatment with bedaquiline. This is advised as the effects of the medication on your sperm are unknown; the effect of the medication also persists in the body for a period of many months.

## What should I avoid while taking bedaquiline?

- You should not drink alcohol while taking bedaquiline.
- There are some medications that cannot be taken safety with bedaquiline.
- Make sure to inform your doctor if you are taking medicines or if medicines are recommended to you by a health-care practitioner for some other illness while you are on treatment for TB with bedaquiline.
- If you do not know the names of the medicines please ensure that you show the prescriptions to the TB doctor.

## Can I take other medicines along with Bedaquiline?

- As there are many medicines that should be avoided with Bedaquiline for safety reasons, if any other medicine is required to be taken, it should be taken with permission from the concerned DR-TB Center doctor.

## Drugs not be taken along with Bedaquiline

- Class 1a or Class III antiarrhythmic drugs (such as amiodarone, sotalol, procainamide, dysopyramide and quinidine)
- tricyclic antidepressants, including amitriptyline, doxepin, desipramine, imipramine, clomipramine
- the non-sedating antihistamines astemizole and terfenadine
- the neuroleptics-phenothiazines thioridazine, haloperidol, chlorpromazine, trifluoperazine, percycline, prochlorperazine, fluphenazine, sertindole, and pimozide
- serotonin 5-HT3 receptor antagonist (ondansteron)
- the prokinetic cisapride
- Quinoline antimalarials (e.g., chloroquine and quinine sulphate)

**Name & Contacts of DR-TB Center**

_____

_____

**MDR-TB can be cured provided regular and uninterrupted treatment is taken by the patient**

**END THE GLOBAL TB EPIDEMIC**

**Let us all work together to overcome the scourge of MDR-TB**

**A WORLD FREE OF TB**

DOTS

## Annexure 10: WHO Recommended doses of Anti TB drugs for adult and pediatric

| DRUGS | Daily Dose (Pediatric) | DAILY DOSE (Adult) |
|---|---|---|
| Isoniazid[1] | 7–15 mg/kg for patients less than 30 kg; max dose 300 mg daily | 4–6 mg/kg once daily<br>High –dose: 16–20 mg/ kg once daily |
| Rifampicin | 10–20 mg/kg for patients less than 30 kg; max dose 600 mg daily | 8–12 mg/kg once daily |
| Pyrazinamide | 30–40 mg/kg for patients less than 30 kg; max dose 2000 mg daily | 20–30 mg/kg once daily |
| Ethambutol | 15-25 mg/kg once daily | 12–18 mg/kg once daily |
| Levofloxacin | 5 years and under: 15–20 mg/kg split into two doses (morning and evening) Over 5 years: 10–15 mg/kg once daily | 10-15 mg/kg once daily |
| Moxifloxacin | 7.5–10 mg/kg | 400 mg once daily |
| Ethionamide/ Protionamide | 15–20 mg/kg | 15-20 mg/kg/day in 2 divided doses |
| Cycloserine | 10–20 mg/kg | 10-15mg/kg/ day in 2 divided doses |
| p-aminosalicylic acid | 200–300 mg/kg for patients less than 30 kg | 8- 12 g/day in 2 divided doses |
| Linezolid | 10 mg/kg given three times daily (pyridoxine should also be given) | 600 mg once daily |
| Clofazimine | Limited data, but 1 mg/kg once daily has been given | 200–300 mg daily (2 first months) then reduce to 100 mg daily (alternative dosing 100 mg daily) |
| Amoxicillin clavulanic acid 7/1 | 80 mg/kg (based on the amoxicillin component) in two divided doses | 80 mg/ kg/ day in 2 divided doses |
| Kanamycin | 15–30 mg/kg once daily (Max 1000mg) | 15–20 mg kg once daily |
| Amikacin | 15–30 mg/kg once daily (Max 1000mg) | 15–20 mg kg once daily |
| Capreomycin | 15–30 mg/kg once daily (Max 1000mg) | 15–20 mg kg once daily |
| Imipenem cilastatin | Meropenem is preferred in children | 1000 imipenem/ 1000 mg cilastatin twice daily |
| Meropenem | 20–40 mg/kg intravenous every eight hours | 1000 mg three times daily (alternative dosing is 2000 mg twice daily) |
| Bedaquiline | Dose not yet determined in children | 400 mg once daily for 2 weeks then 200 mg 3 times per week for next 22 weeks |
| Delamanid | Dose not yet determined in children | 100 mg twice daily (200 mg) for 24 weeks. |

1. Children at risk for peripheral neuropathy (e.g. malnutrition or HIV co-infection) should also receive pyridoxine 5–10 mg/day as therapeutic dose.

## Annexure 11: Pulmonary Rehabilitation

**Pulmonary Rehabilitation Programme**

Pulmonary rehabilitation aims to restore patients to an independent, productive and satisfying life and prevent further clinical deterioration to the maximum extent compatible with the stage of the disease. This goal may be accomplished, without materially improving lung function, by helping the patients to become more aware of their disease, more actively involved in their own health care and more independent in performing daily care activities, attempting to reverse the disability from disease.

**Aims & Objectives:**

The aims of pulmonary rehabilitation are:

1) Decrease of physical and psychological impairment due to the disease,
2) Increase in physical and mental fitness and performance and
3) Maximal social reintegration of the patient to lower the handicap.

**Expected Results of Pulmonary Rehabilitation Program:**

1. A program of exercise training of the muscles of ambulation is recommended as a mandatory component of pulmonary rehabilitation for patients with COPD.
2. Pulmonary rehabilitation improves the symptom of dyspnea in patients with COPD.
3. Pulmonary rehabilitation improves HRQOL in patients with COPD.
4. Pulmonary rehabilitation reduces the number of hospital days and other measures of health-care utilization in patients with COPD.
5. Pulmonary rehabilitation is cost-effective in patients with COPD.
6. There are psychosocial benefits from comprehensive pulmonary rehabilitation programs in patients with COPD.
7. Six to twelve weeks of pulmonary rehabilitation produces benefits in several outcomes.
8. Longer pulmonary rehabilitation programs (beyond 12 weeks) produce greater sustained benefits than shorter programs.

**Pulmonary Rehabilitation Program Components:**

The goal of PRP is achieved by general conditioning, upper body exercise, lower body exercise and reduction in dyspnoea and meeting educational, psychological and medical needs. Major components of PRP include exercise training, education, psychological and behavioral interventions and outcome assessments. Patient education about the disease and its management is a central feature of PRP but is not effective alone. However, the patient can become more skilled at collaborative self-management and more adherent to the treatment plans. Nutritional supplements can improve fat free mass and muscle strength and nutritional counseling is an integral component of PRP. Psychological intervention may be useful in motivating the patient, since identification of readiness to change may improve compliance with physical training.

**Pulmonary Rehabilitation Center for Chronic Resipratory Diseases – The Maharashtra State Anti-TB Association, outside Group of TB Hospitals, Sewri, Mumbai**

**Requirements for the PR Centre:**

*Space:* Around 500 sq. ft.
*Equipment*:
1. Treadmill
2. Multi gym
3. Cycle
4. Pulse oxymeters
5. Disposables
*Human Resources*:
1. Two trained physiotherapists (preferable one male and one female)
2. Nurse
3. Attendant.

*Methods*:
· Day 1. Assessment with structured questionnaire – 30 mins. to 1 hour
· Sessions on alternate days lasting 1.5 hours each for 6 to 12 weeks.
· Home program with follow up assessment every month.

# Annexure 12A Cohort Event Monitoring (CEM) - Treatment initiation form

| **Patient details** | | | Interview Date: _____(DD/MM/YYYY) |
|---|---|---|---|
| Patient Name ..... | Age: ..... | PMDT No./ File No:_____ | Nikshay ID: |

| **Medical Details** | | |
|---|---|---|
| Type of TB<br>☐ Pulmonary<br>☐ Extra-pulmonary TB site/s: _____ | Type of drug resistance<br>☐ H mono/ poly<br>☐ RR/MDR-TB | ☐ RR/MDR-TB + any FQ/SLI<br>☐ XDR-TB |
| Pregnancy status (UPT) ☐ Yes ☐ No ☐ Unknown | Date of LMP: DD/MM/YYYY | or estimated current gestation (weeks): |
| | If PREGNANT record patient details for follow –up | |
| Breastfeeding an infant | ☐ Yes ☐ No ☐ Unknown | |

| **Addiction or substance abuse** | | | |
|---|---|---|---|
| Injectable Drug abuse Within Past Year | ☐ Yes ☐ No ☐ Unknown | Excessive alcohol use in the past year | ☐ Yes ☐ No ☐ Unknown |
| Tobacco Use Within Past Year | ☐ Yes ☐ No ☐ Unknown | | |
| Prior exposure to anti-TB medicines | ☐ Yes ☐ No ☐ Unknown | | |

| List the current and past medical conditions & Events (Diabetes, Hypertension, QT prolongation, LFT deranged, Hepatitis, Hypo/hyper thyroidism, Rash, allergic reaction, anaphylaxis, nausea, vomiting, gastritis, diarrhoea, arthralgia, nephrotoxicity, depression, psychotic syndrome, seizures, gynecomastia etc) | Date of Onset | Date of recovery | Still Continue |
|---|---|---|---|
| | | | Yes/No |
| | | | Yes/No |
| | | | Yes/No |
| | | | Yes/No |
| | | | Yes/No |
| | | | Yes/No |

**Medicines**

| Prior exposure to anti-TB medicines | ☐ Yes ☐ No ☐ Unknown | | | | | | |
|---|---|---|---|---|---|---|---|
| Medicines & traditional medicines taken at any time in PAST 30 DAYS | Indication | Dosage (µg/mg/g/ml) | Frequency (OD/BD/ TID) | Route (Oral/IV/ IM/Topical/other) | Start Date | Stop date | Continues (tick appropriate) |
| | | | | | | | ☐ |
| | | | | | | | ☐ |
| | | | | | | | ☐ |
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| | | | | | | | ☐ |
| | | | | | | | ☐ |
| | | | | | | | ☐ |

| Any medicines other than anti-TB drugs prescribed at this interview | Indication | Dosage (µg/mg/g/ml) | Frequency (OD/BD/ TID) | Route (Oral/IV/ IM/Topical/other) | Start Date | Anticipated Stop date |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |
| | | | | | | |

**Health facility and Reporter information**

Name of treatment initiating health facility:

Name of treating clinician/ team:

Name of the Reporter:

Signature:

Date:

# Annexure 12B Cohort Event Monitoring (CEM) - Treatment Review Form

**To be filled at per schedule and when any adverse event reported by the patient on newer drug containing regimen and for the other DR-TB patients when serious adverse event (DAIDS grade 3 or 4) is reported**

| PATIENT DETAILS | | Interview Date: _____(DD/MM/YYYY) | |
|---|---|---|---|
| Patient Name ..... | Age: ..... | PMDT No./ File No:_____ | Nikshay ID: |
| Patient Address: | | Weight (kg):..... | Sex: ☐ Male ☐ Female ☐ Others |
| | | Height (cm):..... | |

| MEDICAL DETAILS | | |
|---|---|---|
| Type of TB<br>☐ Pulmonary TB<br>☐ Extra-pulmonary TB site/s: _____ | Type of drug resistance<br>☐ H mono/ poly<br>☐ RR/MDR-TB | ☐ RR/MDR-TB + any FQ/SLI<br>☐ XDR-TB |
| Prior exposure to anti-TB medicines ☐ Yes ☐ No ☐ Unknown | | |
| Pregnancy status (UPT) ☐ Yes ☐ No ☐ Unknown | Date of LMP: DD/MM/YYYY | or estimated current gestation (weeks): |
| | If PREGNANT record patient details | |
| Breastfeeding an infant | ☐ Yes ☐ No ☐ Unknown | |

ADR description: Reporter's Narrative (Describe the course of events, timing and suspected causes of ADR):

ADR Terminology/Adverse event

For patient under CAP during follow up visit (Fill the details of any event happened since the last follow up visit)
**(eg. Minor ADR, Accident, Travel, other medication etc)**

DAIDS grading (Tick appropriate checkbox)

☐ GRADE 1: Mild symptoms causing no or minimal interference with usual social & functional activities with intervention not indicated

☐ GRADE 2: Moderate symptoms causing greater than minimal interference with usual social & functional activities with intervention indicated

☐ GRADE 3: Severe symptoms causing inability to perform usual social and functional activities with intervention or hospitalization indicated

☐ GRADE 4: Potentially life threatening symptoms causing inability to perform basic self-care functions with interventions indicated to prevent permanent impairment, persistent disability or death

| | |
|---|---|
| **Date of Onset** | DD/MM/YY |
| **Date Resolved** | DD/MM/YY |
| **ADR/SAE Seriousness** | ☐ Death<br>☐ Life threatening<br>☐ Hospitalization required<br>☐ Permanent disability<br>☐ Congenital anomaly/ birth defect<br>☐ Other medically important condition<br>☐ Required intervention to prevent permanent impairment/ damage |
| **Outcome** | ☐ Recovered/ resolved<br>☐ Recovered/resolved with sequel<br>**Recovery date……**<br>☐ Fatal<br>☐ Recovering/resolving<br>☐ Not recovering/ not resolved<br>☐ Unknown |
| **For Death/SAE** | Date of Death…….<br>Primary cause of death (if known): ……<br>Was autopsy performed?<br>☐ No<br>☐ Yes (If yes, attach copy of report if available)<br>Hospital Admission Date ……<br>Hospital Discharge Date…… |

**Causality /Relation to medicine**

Assign Causality with following grading for suspected drug

| | Certain | Probable | Possible | Doubtful or Unlikely | Conditional or Unclassified | Unassessable |
|---|---|---|---|---|---|---|
| H | | | | | | |
| R | | | | | | |
| E | | | | | | |
| Z | | | | | | |
| Km | | | | | | |
| Am | | | | | | |
| Cm | | | | | | |
| Lfx | | | | | | |
| Mfx | | | | | | |
| Cs | | | | | | |
| Eto | | | | | | |
| PAS | | | | | | |
| Lzd | | | | | | |
| Cfz | | | | | | |
| Amx-Clv | | | | | | |
| Clr | | | | | | |
| Bdq | | | | | | |
| Dlm | | | | | | |
| | | | | | | |

| | | | | |
|---|---|---|---|---|
| **Suspected drugs name** | | | | |
| **Dechallenge** | ☐ No Dechallenge<br>☐ Reaction Abated<br>☐ Dose Reduced<br>☐ Result unknown | ☐ No Dechallenge<br>☐ Reaction Abated<br>☐ Dose Reduced<br>☐ Result unknown | ☐ No Dechallenge<br>☐ Reaction Abated<br>☐ Dose Reduced<br>☐ Result unknown | ☐ No Dechallenge<br>☐ Reaction Abated<br>☐ Dose Reduced<br>☐ Result unknown |
| **Rechallenge** | ☐ No Rechallenge<br>☐ Recurrence of event<br>☐ No recurrence<br>☐ Result unknown | ☐ No Rechallenge<br>☐ Recurrence of event<br>☐ No recurrence<br>☐ Result unknown | ☐ No Rechallenge<br>☐ Recurrence of event<br>☐ No recurrence<br>☐ Result unknown | ☐ No Rechallenge<br>☐ Recurrence of event<br>☐ No recurrence<br>☐ Result unknown |
| **Expectedness** | ☐ Expected<br>☐ Unexpected | ☐ Expected<br>☐ Unexpected | ☐ Expected<br>☐ Unexpected | ☐ Expected<br>☐ Unexpected |

| | |
|---|---|
| **If newer drugs (Bdq or Dlm) is suspected**<br><br>☐ Bdq ☐ Dlm | Start Date of newer drug…… Stop Date of newer drug ……<br>Action taken for suspected drug<br>☐ No adjustment<br>☐ Dose Adjusted<br>☐ Temporary Stop<br>☐ Permanent Stop<br>If permanent stop,<br>it was provider decision ☐ please specify reason _____ or<br>it was patient decision ☐ please specify reason _____)<br>Batch/Lot No/ Expiry…… Dose……Frequency…… Route…… |

| **LABORATORY & OTHER TESTS** | | | | | |
|---|---|---|---|---|---|
| **Test** | **Date** | **Result (units)** | **Test** | **Date** | **Result (units)** |
| Sputum smear | | | ALT (SGPT) | | |
| Sputum Culture | | | AST (SGOT) | | |
| Line probe assay | | | Lactic acid | | |
| Nucleic acid testing | | | Lipase | | |
| Tuberculin Test | | | Chest X-Ray | | Cavities (Y/N) Pl specify Xray findings_____ |
| HIV Antibody | | | ECG | | QTc<br>Any other changes |
| CD4 Count | | | Audiometry | | |
| ESR | | | Visual acuity | | |
| Total WBC | | | Bilirubin- Direct<br>-Indirect | | |
| Haemoglobin | | | Hepatitis markers | | |
| Creatinine | | | TSH | | |
| Creatinine Clearance | | | Other | | |
| Glucose | | | | | |
| Drug susceptibility | | | | | |

| **Medicines** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Anti-TB medicines taken since last interview | Dosage (µg/mg/ g/ml) | Frequency (OD/BD/ TID) | Route (Oral/IV/IM /Topical/ other) | Start Date | Continues | If No. Stop date | Reason (s) for stopping # | Action** |
| | | | | | Yes/ No | | | |
| | | | | | Yes/ No | | | |
| | | | | | Yes/ No | | | |
| | | | | | Yes/ No | | | |
| | | | | | Yes/ No | | | |
| | | | | | Yes/ No | | | |
| | | | | | Yes/ No | | | |
| | | | | | Yes/ No | | | |
| | | | | | Yes/ No | | | |

| Other medicines & traditional medicines taken since last interview | Indication | Dosage (µg/mg/ g/ml) | Frequency (OD/BD/ TID) | Route (Oral/IV/IM /Topical/ other) | Start Date | Continues | Stop date | Reason (s) for stopping# | Action** |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Yes/ No | | | |
| | | | | | | Yes/ No | | | |
| | | | | | | Yes/ No | | | |
| | | | | | | Yes/ No | | | |
| | | | | | | Yes/ No | | | |

| All NEW medicines (anti-TB & other) prescribed at this interview | Indication | Dosage (µg/mg/ g/ml) | Frequency (OD/BD/ TID) | Route (Oral/IV/IM /Topical/ other) | Start Date | Expected Stop date | Indication |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

**Any other relevant clinical information**

**REPORTERS INFORMATION**

Name of the Reporter:

Signature:

Date:

** Action taken by clinician in case of suspected adverse event linked to a drug

Dose not changed
Drug withdrawn
Not applicable
Dose reduced
Drug interrupted

## Annexure 13: RNTCP Verbal Autopsy Form

PMDT Number: PMDT NIKSHAY ID: YEAR:

Name of the head of household :

Full Name of the deceased:

Section 1: Details for respondent and deceased

**Details of respondent**

1. Name of respondent

2. Relationship of respondent with deceased

- 1. Wife/Husband
- 2. Brother/Sister
- 3. Son/Daughter
- 4. Mother/Father
- 5. Grandchild
- 6. Son-in-law/ Daughter-in-law
- 7. Brother-in-law/ Sister-in-law
- 8. Parent-in-law
- 9. Grandfather/Grandmother
- 10. Other relative
- 11. Neighbour/No relation
- 99. Unknown

3. Did the respondent live with the deceased during the events that led to death?

1. Yes 2. No 99. Unknown

4. Respondent's age in completed years

5. Respondent's sex 1. Male 2. Female

6. What is the highest standard of education the respondent has completed?

- 0. Illiterate and literate with no formal education
- 1. Literate, Primary or below
- 2. Literate, Middle
- 3. Literate, Matric Class-X
- 4. Literate, Class XII
- 5. Graduate and above
- 99. Unknown

7. Religion of the head of the household

- 1. Hindu
- 2. Muslim
- 3. Christian
- 4. Sikh
- 5. Buddhist
- 6. Jain
- 7. No religion
- 8. Other
- 99. Unknown

**Details of deceased**

8. Deceased's Sex 1. Male 2. Female

9. Age of Deceased Years:

10. Relationship of the deceased with the head of the household

- 1. Wife/Husband
- 2. Brother/Sister
- 3. Son/Daughter
- 4. Mother/Father
- 5. Grandchild
- 6. Son-in-law/ Daughter-in-law
- 7. Brother-in-law/ Sister-in-law
- 8. Parent-in-law
- 9. Grandfather/Grandmother
- 10. Other relative
- 11. Neighbour/No relation
- 12. Self 99. Unknown

11. What is the highest standard of education the deceased had completed?

- 0. Illiterate and literate with no formal education
- 1. Literate, Primary or below
- 2. Literate, Middle
- 3. Literate, Matric Class-X
- 4. Literate, Class XII
- 5. Graduate and above
- 99. Unknown

12. What was the occupation of the deceased?

- 1. Non-worker
- 2. Salaried
- 3. Wage earner
- 4. Profession/Business
- 5. Cultivator/farmer
- 6. Agricultural wage labour
- 7. Non agricultural wage labour
- 8. Student
- 9. Other
- 99. Unknown

13. Date of death D D / M M / Y Y

14. How many years did the deceased live at this address?

15. Place of death?

- 1. Home
- 2. On way to health facility
- 3. PHC/CHC/Rural Hospital
- 4. District Hospital
- 5. Private Hospital
- 6. Other place
- 99. Unknown

16A. House address of the deceased

16B. PIN

17. What did the respondent think that this person died of?
(Allow the respondent to tell the illness in his or her own words)

Section 2: Past History

**Had a doctor EVER stated that the deceased had the following diseases?**

| | 1. Yes | 2. No | 99. Unknown |
|---|---|---|---|
| 18. Hypertension | ☐ | ☐ | ☐ |
| 19. Heart disease | ☐ | ☐ | ☐ |
| 20. Stroke | ☐ | ☐ | ☐ |
| 21. Cholesterol problem | ☐ | ☐ | ☐ |
| 22. Diabetes | ☐ | ☐ | ☐ |
| 23. Tuberculosis | ☐ | ☐ | ☐ |
| 24. HIV/AIDS | ☐ | ☐ | ☐ |
| 25. Cancer (write site in narrative) | ☐ | ☐ | ☐ |
| 26. Asthma | ☐ | ☐ | ☐ |
| 27. Other chronic illness (specify in narrative) | ☐ | ☐ | ☐ |

28. Was the deceased taking any medications regularly during the last five years? (Record up to three in Hindi or English only).

1. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐
2. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐
3. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐

First, ask the following questions for the deceased (First Column), and then ask them for the main respondent (second column)

| Tobacco, alcohol and diet | Deceased (Ask first) | Respondent (Ask second) |
|---|---|---|
| 29A. Did s/he smoke tobacco within the last 5 years? | 1. Definite Yes **2. Definite No** 99. Unknown | 1. Definite Yes **2. Definite No** 99. Unknown |
| 29B. If yes, how many bidis per day? | | |
| 29C. If yes, how many cigarettes per day? | | |
| 29D. Any other tobacco smoked? | 1. Definite Yes **2. Definite No** 99. Unknown | 1. Definite Yes **2. Definite No** 99. Unknown |
| 30A. Did s/he chew tobacco within the last 5 years? | 1. Definite Yes **2. Definite No** 99. Unknown | 1. Definite Yes **2. Definite No** 99. Unknown |
| 30B. Did s/he apply tobacco within the last 5 years? | 1. Definite Yes **2. Definite No** 99. Unknown | 1. Definite Yes **2. Definite No** 99. Unknown |
| 31A. Did s/he normally drink alcohol (use local term) at least once a week during most weeks in the last 5 years? | 1. Definite Yes **2. Definite No** 99. Unknown | 1. Definite Yes **2. Definite No** 99. Unknown |
| 31B. If yes, what was the average no. of days per week s/he drank? | _____days OR Unknown | _____days OR Unknown |
| 31C. If yes, what type of alcohol was most commonly consumed? | 1. Country liquor **3. Indian made foreign liquor** 2. Toddy **4. Beer** 5. Other | 1. Country liquor **3. Indian made foreign liquor** 2. Toddy **4. Beer** 5. Other |
| 32. Was s/he a pure vegetarian (consumed no egg, meat or fish) for the last 5 years? | 1. Definite Yes **2. Definite No** 99. Unknown | 1. Definite Yes **2. Definite No** 99. Unknown |

*For female deaths aged 15-49 ask:*

33. Was she either known or suspected to be pregnant or within 42 days of delivery or abortion?

1. Yes **2. Definite No** *If YES to question Q33 then DO NOT complete narrative below. Instead complete Form 10D and copy the Form 10D number here*

34. Key symptoms preceding death (check all that apply, and then use symptom list for narrative)

| | | |
|---|---|---|
| 1. Coughing of blood | 2. Severe shortness of breath | 3. Fainting or giddiness |
| 4. Vomitting, loss of appetite, pain in abdomen | 5. Yellowish discolouration of eyes and urine | 6. Sudden chest pain |
| 7. Irrelevant behavior or talk | 8. Palpitations | 9. Seizures/fits |
| 10. Fever | 11. Weight loss | 12. Paralysis/stroke |
| 13. Urinary problems | 14. Diarrhoea/dysentery | 15. Odeme (swelling) |
| 16. Severe Weakness | | |

35. Narrative language code

## Section 3: Written narrative in local language

Please describe the symptoms in order of appearance, doctor consulted or hospitalization, history of similar episodes, enter the results from reports of the investigations if available.

Respondent's cooperation: 1. Good 2. Poor

Interviewer name: Desig:

Date: D D / M M / Y Y

*Signature/Impresion*

Respondent

Interviewer

## Annexure 14: Real-Time Medication Event Reminder - Monitor Device (RT-MERM)

**Overview:**

Despite its success, DOT as implemented currently in India's Revised National TB Control Programme (RNTCP) has limitations. New, cost-effective adherence monitoring strategies are needed that can be utilized by the RNTCP and by private sector clinics with a high volume of TB patients. The use of electronic monitoring technologies as a method for dose and refill reminding as well as adherence monitoring has proven to be an effective and pragmatic intervention in both developed and developing countries. Electronic reminding-monitoring of actual dosing helps patients know when and how to take their medications. Moreover, such reminder-monitors generate patient-specific dosing histories that support highly impactful patient counseling. Finally, these reminder-monitors provide reliable and actionable data that providers can use to determine whether and to what degree a specific patient is adherent to the prescribed regimen and thus can direct relatively more or less care to those relatively less adherent (higher risk) or more adherent (lower risk) patients respectively.

**Brief Description of the RT-MERM Device:**

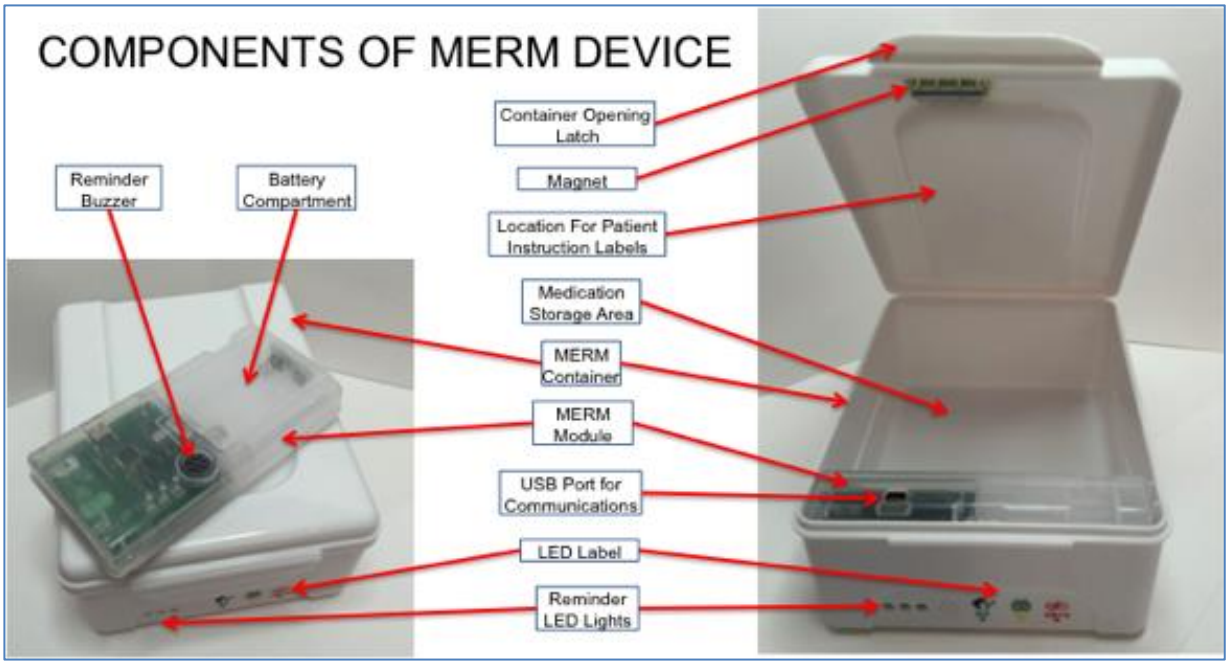
The RT-MERM (i) is highly accurate, affordable, re-usable, and suitable for TB medications, (ii) provides programmable visual and audible reminders of daily dosing and of monthly refill, and (iii) compiles and transmits automatically detailed, and patient-specific information regarding medication taking and medication adherence.

This reminder-monitor utilizes an innovative two-part, consisting of a container (the "Container") that will hold the patient's medications and a small electronic module housed within the Container (the "Monitoring Technology") that will transmit information. When the Container is opened, it records the date and time of each such medication taking event, store the date/time data, and automatically transmit (via integrated, affordable 2G data transmission capability) such date/time dosing information for centralized collection, analysis, and use by health care providers via systems such as eNikshay or 99DOTS. The components of the RT-MERM are shown in Figure 1 below:

**Core Functionalities of the RT-MERM:**

The RT-MERM is designed to be a highly robust (the water resistant corrugated Container should be usable for at least six months and the Monitoring Technology can be re-used and should have a useful life of 3-4 years) and extremely low patient-burden (the patient does not need to re-charge the RT-MERM or make any phone calls – all he or she needs to do is open the RT-MERM and take medications in order to record and transmit dosing data)

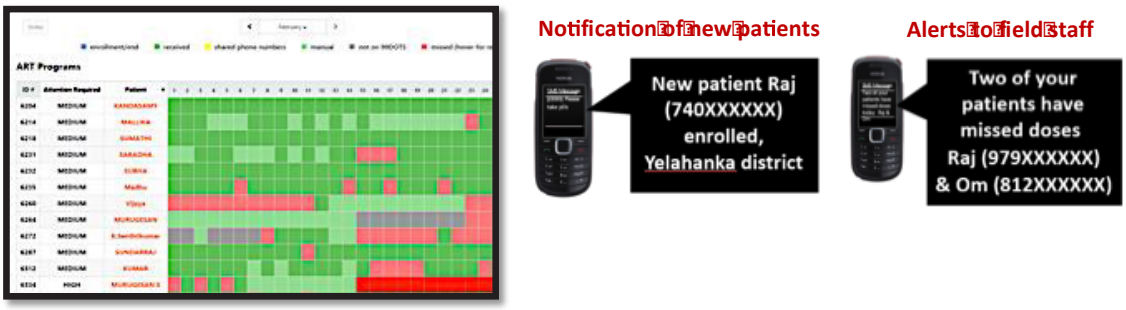
monitoring solution. The other core functionalities of the RT-MERM are described below:
Figure 1. Components of the RT-MERM.




- Captures (via integrated magnets) and stores more than 1500 medication events,
- Captures, stores, and transmits a daily “heartbeat” indicating operational status and battery life,
- Container sizes can vary (current designs include a month-wise DS-TB Container and a month-wise MDR-TB Container) and are very affordable (less than USD $1.50 per unit),
- Containers are high quality, aqueous-coated corrugated that is highly durable and water resistant,
- Electronic Module is made of highly durable plastic and electronics are resistant to vibration, heat, humidity, and impact,
- Audible (buzzer) and visual (colored LED lights) reminders of both dosing (green light) and refill (amber light),
- Reminder and refill times and dates are configurable,
- Large “billboard” space on the inside and outside lid of the Container for more detailed patient instructions and pictograms,
- Utilizes small, rechargeable lithium polymer battery that should last at least 4 months without recharging,
- Fully integrated with 99DOTS (complete) and eNikshay (in process) for both issuance/registration of the RT-MERM with the patient and for monitoring data collection

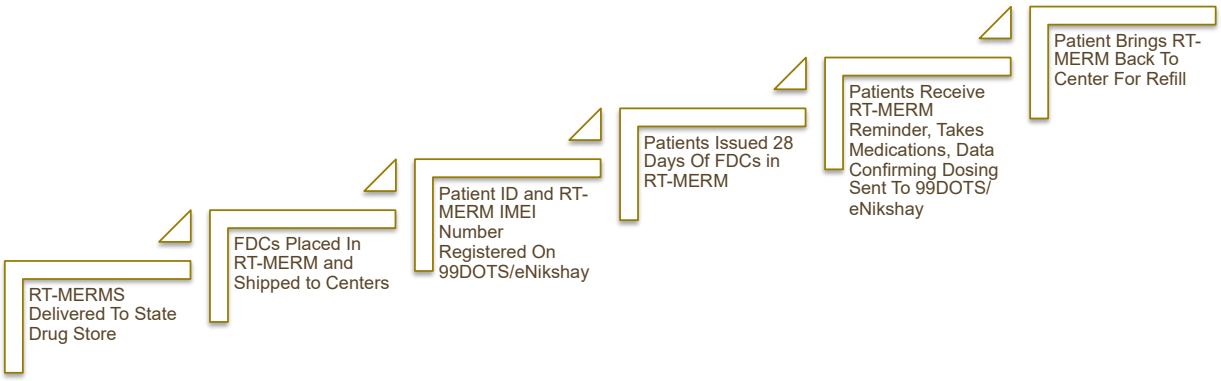
**Data Deliveries from the RT-MERM:**

The RT-MERM is fully integrated into the 99DOTS system and will generate all of the data required and support all reports generated by the 99DOTS system. A picture of the 99DOTS dashboard, which is fully supported by the RT-MERM is shown below in Figure 2. Similar integration with eNikshay reporting is in process.




**Use of the RT-MERM:**

The following figures outline the usage of RT-MERM for patient/provider workflow for both DS-TB and MDR-TB patients.




**Pictures of Prototype RT-MERM Devices:**

DS-TB MERM

MDR-TB MERM

# Annexure 15A:

# RNTCP Request Form for examination of biological specimen for TB

(Required for Diagnosis of TB, Drug susceptibility Testing and follow up)

| Patient Information | | | |
|---|---|---|---|
| **Patient name** | | **Age (in yrs):____** | **Gender: ☐M ☐F ☐TG** |
| **Patient mobile no. or other contact no.** | | **Specimen date of collection** (DD/MM/YY) _____ | **☐ Sputum ☐Other (specify)_____** |
| **Aadhaar no.**(If available ) | | | |
| **Patient address with landmark** | | **HIV Status: ☐Reactive ☐Non-Reactive ☐Unknown** | |
| | | **Key populations: ☐Contact of known TB Patient ☐Diabetes ☐Tobacco ☐Prison ☐Miner ☐Migrant ☐Refugee ☐ Urban slum ☐Health-care worker ☐Other(specify) _____** | |

| | |
|---|---|
| **Name and Type of referring facility** (PHI/DMC/TU/ DTC/ICTC/ART/Medical College/DR-TB Centre/Private/ Others, specify): _____ **Health Facility ID (HFID):** _____ | NIKSHAY ID: _____ |
| **State: _____ District:_____ Tuberculosis Unit (TU): _____** | |

**Reason for Testing**

| Diagnosis and follow up of TB | | |
|---|---|---|
| Diagnosis | | Follow up (Smear and culture) |
| H/O anti TB Rx for >1 month: ☐ Yes ☐ No | | |
| ☐ Presumptive TB<br>☐ Repeat Exam<br>☐ Private referral<br>☐ Presumptive NTM | Predominant symptom _____<br>Duration _____ days | Regimen: ☐ New ☐ Previously Treated<br>Reason: ☐ End IP ☐ End CP<br>Post treatment: ☐ 6m ☐ 12m ☐ 18m ☐ 24m |

| Diagnosis and follow up Drug-resistant TB | | |
|---|---|---|
| Drug Susceptibility Testing (DST) | | Follow up (Culture/Microscopy) |
| Presumptive MDR TB | ☐ New ☐ Previously treated<br>☐ At TB diagnosis<br>☐ Contact of MDR/RR-TB<br>☐ Follow up Sm+ve<br>☐ Private referral<br>☐ Discordance resolution | PMDT TB No _____<br>Regimen:<br>☐ Regimen for INH mono/poly resistant TB<br>☐ Regimen for MDR/RR TB<br>☐ Shorter MDR-TB regimen<br>☐ Modified Regimen for MDR/RR-TB + FQ /SLI resistance<br>☐ Regimen for XDR TB<br>☐ Modified Regimen for mixed pattern resistance<br>☐ Regimen with New Drug for MDR-TB Regimen + FQ/SLI resistance<br>☐ Regimen with New Drug for XDR-TB<br>☐ Regimen with New Drug for failures of regimen for MDR TB<br>☐ Regimen with New Drug for failures of regimen for XDR-TB<br>☐ Regimen with New Drug for mixed pattern resistance<br>Treatment Week :_____Month:_____ |
| ☐ Presumptive H mono/poly | | |
| Presumptive XDR TB | ☐ MDR/RR TB at Diagnosis<br>☐ ≥ 4 months culture positive<br>☐ 3 monthly for persistent culture positives (treatment month _____)<br>☐ Culture reversion<br>☐ Failure of MDR/RR-TB regimen<br>☐ Recurrent case of second line treatment<br>☐ Discordance resolution | |

**Test requested:**

☐Microscopy ☐TST ☐IGRA ☐Chest X-ray ☐Cytopathology ☐Histopathology ☐ CBNAAT ☐ Culture ☐DST ☐FL -LPA ☐SL -LPA ☐Gene Sequencing ☐Other (Please Specify) _____

Requested by (Name, Designation and Signature): _____

Contact Number:_____ Email ID:_____

**Results:** **NIKSHAY ID** Generated: _____

| **Microscopy** (☐ ZN ☐ Florescent) | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Lab Sr. No | Visual appearance | Result | | | | |
| | | | Negative | Scanty | 1+ | 2+ | 3+ |
| Sample A | | | | | | | |
| Sample B | | | | | | | |

**Date tested: _____ Date Reported:_____ Reported by:_____**

**Laboratory Name: _____ (Name and Signature)**

**Cartridge Based Nucleic Acid Amplification Test (CBNAAT)** Lab. sr. No._____

| | | | | |
|---|---|---|---|---|
| **Sample** | ☐ A | ☐ B | | |
| **M. Tuberculosis** | ☐ Detected | ☐ Not Detected | ☐ N/A | |
| **Rif Resistance** | ☐ Detected | ☐ Not Detected | ☐ Indeterminate | ☐ N/A |
| **Test** | ☐ No Result | ☐ Invalid | ☐ Error – Error Code_____ | (Please arrange for fresh sample) |

**Date tested:** _____ **Date Reported:**_____ **Reported by:**_____
**Laboratory Name:** _____ **(Name and Signature)**

**Culture (☐ LJ ☐ LC)**

| Lab Sr. No | Negative | Positive for M.tb. | NTM (write species) | Contamination |
|---|---|---|---|---|
| | | | | |

**Date Result:** _____ **Date Reported:**_____ **Reported by:**_____
**Laboratory Name:** _____ **(Name and Signature)**

**Line Probe Assay (LPA)**

☐ Direct ☐ Indirect Lab serial _____

**First line LPA**

**RpoB: ---- locus control:** ☐ present ☐ absent

**WT1:** ☐ present ☐ absent **WT2:** ☐ present ☐ absent **WT3:** ☐ present ☐ absent **WT4:** ☐ present ☐ absent

**WT5:** ☐ present ☐ absent **WT6:** ☐ present ☐ absent **WT7:** ☐ present ☐ absent **WT8:** ☐ present ☐ absent

**MUT1 (D516V):** ☐ present ☐ absent **MUT2A (H526Y):** ☐ present ☐ absent **MUT2B (H526D):** ☐ present ☐ absent **MUT3 (S531L):** ☐ present ☐ absent

**Kat G: ----- locus control:** ☐ present ☐ absent

**WT1 (315):** ☐ present ☐ absent
**MUT1 (S315T1):** ☐ present ☐ absent
**MUT2 (S315T2):** ☐ present ☐ absent

**Inh A:----- locus control:** ☐ present ☐ absent

**WT1 (-15, -16):** ☐ present ☐ absent **WT2 (-8):** ☐ present ☐ absent
**MUT1 (C15T):** ☐ present ☐ absent **MUT2 (A16G):** ☐ present ☐ absent
**MUT3A (T8C):** ☐ present ☐ absent **MUT3B (T8A):** ☐ present ☐ absent

**Second line LPA**

**gyrA:--**

**locus control:** ☐ present ☐ absent

**WT1 (85-90):** ☐ present ☐ absent
**WT2 (89-93):** ☐ present ☐ absent
**WT3 (92-97):** ☐ present ☐ absent

**MUT1 (A90V):** ☐ present ☐ absent
**MUT2 (S91P):** ☐ present ☐ absent
**MUT3A (D94A):** ☐ present ☐ absent
**MUT3B (D94N/Y):** ☐ present ☐ absent
**MUT3C (D94G):** ☐ present ☐ absent
**MUT3D (D94H):** ☐ present ☐ absent

**gyrB:----**

**locus control:** ☐ present ☐ absent

**WT1 (536-541):** ☐ present ☐ absent

**MUT1 (N538D):** ☐ present ☐ absent
**MUT2 (E540V):** ☐ present ☐ absent

**rrs:-----**

**locus control:** ☐ present ☐ absent

**WT1 (1401-02):** ☐ present ☐ absent
**WT2 (1484):** ☐ present ☐ absent

**MUT1 (A1401G):** ☐ present ☐ absent
**MUT2 (G1484T):** ☐ present ☐ absent

**eis:-----**

**locus control:** ☐ present ☐ absent

**WT1 (37):** ☐ present ☐ absent
**WT2 (14, 12, 10):** ☐ present ☐ absent
**WT3 (2):** ☐ present ☐ absent

**MUT1 (C-14T):** ☐ present ☐ absent

**Final LPA Interpretation: ---**
**MTB result** ☐ MTB detected ☐ MTB not detected ☐ Invalid
**RIF Resi rpoB** ☐ Detected ☐ Not detected **H Resi** ☐ KatG Detected ☐ InhA Detected ☐ Not detected
**FQ Resi gyrA/B** ☐ Detected ☐ Not detected **SLID Resi** ☐ eis Detected ☐ rrs Detected ☐ Not detected
**Date Result:** _____ **Date Reported:**_____ **Reported by:**_____
**Laboratory Name:** _____ **(Name and Signature)**

**Drug Susceptibility Test (DST) results** (☐ LJ ☐ LC)

| Lab Sr.No | 1st line drugs | | | | | | SLI | | | FQ | | | Other | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | R | H (0.1) | H (0.4) | S | E | Z | Km | Cm | Am | Lfx | Mfx (0.5) | Mfx (2) | Eto | PAS | Lzd | Cfz | Clr | Azi | Bdq | Dlm | | | |
| | | | | | | | | | | | | | | | | | | | | | | | |

**Date Result:** _____ **Date Reported:**_____ **Reported by:**_____
**Laboratory Name:** _____ **(Name and Signature)**
*R: Resistant; S: Susceptible; C: Contaminated; -- Not done*

**Other tests** for TB diagnosis

**Test** (Please Specify): _____
**Result:**_____
**Date reported:**_____ **Reported by:**_____
**Laboratory Name:** _____ **(Name and Signature)**

# Annexure 15E: RNTCP PMDT Treatment Card

| NIKSHAY ID | PMDT TB No | PMDT TB No |
|---|---|---|
| | | |

Patient's name: _____

Age: _____yrs Gender: ☐ Male ☐ Female ☐ Transgender

Address: _____

_____

Marital status: _____

Occupation: _____

Contact No: _____

Aadhar ID _____

Name, designation of treatment supporter: _____

_____

Contact no:_____

State:_____ District: _____

TB Unit: _____ PHI:_____

Initial home visit: Date _____ By:_____

DR TB Centre:_____ District _____State _____

| **Reason for Testing** | |
|---|---|
| ☐ New ☐ Previously Treated | |
| ☐ Presumptive TB ☐ Private Referral ☐ Presumptive NTM | |
| Presumptive MDR TB | ☐ At diagnosis<br>☐ Contact of MDR/RR TB<br>☐ Follow up Sm+ve at end IP<br>☐ Private referral |
| ☐ Presumptive H mono/poly | |
| Presumptive XDR TB | ☐ MDR/RR TB at diagnosis<br>☐ ≥ 4 months culture positive<br>☐ 3 monthly, for persistent culture positives (treatment month _____)<br>☐ Culture reversion<br>☐ Failure of MDR/RR-TB regimen<br>☐ Recurrent case of second line treatment |

| Drug Susceptibility Test (DST) results at Diagnosis | | | | | | | | | | | | | | Spec. coll. Date: | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R | H (inhA/0.1) | H (katG/0.4) | S | E | Z | Km | Cm | Am | Lfx | Mfx (0.5) | Mfx (2) | FQ class | SLI class | SLI (eis) | Eto | PAS | Lzd | Cfz | Clr | Azi | Bdq | Dlm |
| | | | | | | | | | | | | | | | | | | | | | | |
| Name of the lab | | | | | | | | | | Lab.sr. No._____ | | | | | Result report date | | | | | | | |

*R: Resistant; S: Susceptible; C:* ***Contaminated****;* ***--*** *Not done/available*

| | |
|---|---|
| No of household contacts | |
| No of members screened | |
| No of presumptive TB cases identified | |
| No of presumptive TB cases evaluated | |
| No diagnosed with TB | |
| No of DR-TB diagnosed | |

HIV Testing: Date:_____

□ Reactive □ Non reactive □ UN

PID number:_____

CPT start date:_____

ART start date:_____

| Pre-treatment investigation | | | | | |
|---|---|---|---|---|---|
| **Test** | **Date** | **Result (units)** | **Test** | **Date** | **Result (units)** |
| ALT (SGPT) | | | Chest X-Ray findings<br>Cavities (Y/N) | | |
| AST (SGOT) | | | | | |
| Bilirubin- Direct<br>-Indirect | | | | | |
| S. Albumin | | | | | |
| WBC (TC/DC) | | | ECG (QTc & other findings) | | |
| Haemoglobin | | | Creatinine | | |
| ESR | | | Creatinine Clearance | | |
| Platelet count | | | Blood Urea | | |
| Lactic acid | | | Visual acuity | | |
| Amylase | | | Audiogram | | |
| Lipase | | | | | |
| CD4 Count | | | Psychiatric evaluation | | Yes/ No |
| Hepatitis markers | | | Surgical evaluation | | Yes/ No |
| TSH | | | Ophthalmic evaluation | | Yes/ No |
| Urine (R/M) | | | | | |
| UPT | | | | | |
| Potassium | | | | | |
| Magnesium | | | | | |
| Calcium | | | | | |

**TB Site:** ☐ Pulmonary ☐ Extra Pulmonary
If extra pulmonary, please specify _____

**Treatment regimen**

☐ Regimen for H mono/poly resistant TB
☐ Conventional regimen for MDR/RR TB
☐ Shorter MDR-TB regimen
☐ Modified Regimen for MDR/RR-TB + FQ /SLI resistance
☐ Regimen for XDR TB
☐ Modified Regimen for mixed pattern resistance
☐ Regimen with New Drug for MDR-TB Regimen + FQ/SLI resistance
☐ Regimen with New Drug for XDR-TB
☐ Regimen with New Drug for failures of regimen for MDR TB
☐ Regimen with New Drug for failures of regimen for XDR-TB
☐ Regimen with New Drug for mixed pattern resistance

**Drugs and Dosages**

| Drugs | H | R | E | Z | Km | Am | Cm | Lfx | Mfx | Cs | Eto | PAS | Lzd | Cfz | Amx | clv | Clr | Bdq* | Dlm* | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| IP Dose (mg) | | | | | | | | | | | | | | | | | | | | |
| CP Dose (mg) | | | | | | | | | | | | | | | | | | | | |

**If patient is eligible for new drug, consent given ☐ Yes ☐ No**
**If No, reason**_____

**Name & Signature of Treating Physician:**_____

**DR-TB Centre Committee meetings – dates and decisions**

| Date | Decision |
|---|---|
| | |
| | |
| | |
| | |
| | |
| | |

| Month of Treatment | Smear/Culture Results | | | | Other Investigations | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Date | Lab No | Smear | Culture | S. Cr | LFT | ECG*-QTc Interval | CBC/ Platelets | Electrolyte (K, Mg, Ca) | other |
| 1st week | | | | | | | | | | |
| 2nd week | | | | | | | | | | |
| 3rd week | | | | | | | | | | |
| 1 | | | | | | | | | | |
| 2 | | | | | | | | | | |
| 3 | | | | | | | | | | |
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| 20 | | | `````` | | | | | | | |
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| 29 | | | | | | | | | | |
| 30 | | | | | | | | | | |

**Blood Sugar Testing:**

Date:_____

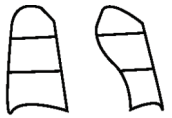
RBS:_____

FBS:_____

ADT*

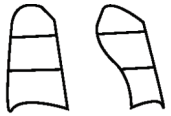
(*write date of starting)

| Thyroid Function Test | | |
|---|---|---|
| Month | Zero | Six |
| Date | | |
| T3 | | |
| T4 | | |
| TSH | | |



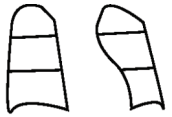
Date of X-ray

Findings_____



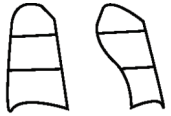
Date of X-ray

Findings_____



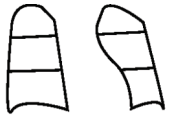
Date of X-ray

Findings_____



Date of X-ray

Findings_____



Date of X-ray

Findings_____

Patient name :_____

Initial Weight: _____kgs Height:_____cms

Weight band: ☐ <16 Kg ☐ 16-29 Kg ☐ 30-45 Kg ☐ 46-70 Kg ☐ >70 Kg

Date of starting intensive phase: _____

Date of starting continuation phase:_____

| Details of change in regimen drugs and dosages | | |
|---|---|---|
| Date | Changed regimen drugs and dosages | Reason for change |
| | | |
| | | |
| | | |
| | | |

| Drug Susceptibility Testing (DST) Results | | | | | |
|---|---|---|---|---|---|
| **Lab name** | | | | | |
| **Lab. Sr. No** | | | | | |
| **Type of DST** (L J/LC/FL-LPA/SL-LPA/CBNAAT) | | | | | |
| **Specimen collection Date** | | | | | |
| **Result report date** | | | | | |
| follow up month | | | | | |
| DST results | | | | | |
| R | | | | | |
| H (inhA) | | | | | |
| H (katG) | | | | | |
| S | | | | | |
| E | | | | | |
| Z | | | | | |
| Km | | | | | |
| Cm | | | | | |
| Am | | | | | |
| Lfx | | | | | |
| Mfx (0.5) | | | | | |
| Mfx(2.0) | | | | | |
| FQ class | | | | | |
| SLID class | | | | | |
| SLI (eis) | | | | | |
| Eto* | | | | | |
| PAS* | | | | | |
| Lzd | | | | | |
| Cfz* | | | | | |
| Bdq* | | | | | |
| Dlm* | | | | | |

**Whenever available*

**ADMINISTRATION OF DRUGS (one line per month)**

**Patient's Name:_____**

**ADMINISTRATION OF DRUGS (one line per month)** **Patient's Name:_____**

| Month/Yr) | Day | | | | | | | | | | | | | | | | | | | | | | | | | | | | | Weight in kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | |
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Mark in the boxes: ✓= directly observed; (✓) = Unsupervised; ◯ = drugs not taken; X = initiation of new box; Recording of CP should start from fresh line.

| Month/Yr | Day | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | Weight in kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | |
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Mark in the boxes: ✓= directly observed; (✓) = Unsupervised; ◯ = drugs not taken; X = initiation of new box; Recording of CP should start from fresh line.

| Date of retrieval action | By whom | Who contacted | Reason for missed doses | Outcome of retrieval action |
|---|---|---|---|---|
| | | | | |
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| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |

| Date of clinical follow up visits | | |
|---|---|---|
| 1st visit:_____ | 6th visit:_____ | 11th visit:_____ |
| 2nd visit: _____ | 7th visit: _____ | 12th visit: _____ |
| 3rd visit:_____ | 8th visit:_____ | 13th visit:_____ |
| 4th visit:_____ | 9th visit:_____ | 14th visit:_____ |
| 5th visit: _____ | 10th visit: _____ | 15th visit: _____ |

| **Treatment outcome** ( Cured, Treatment completed, Died, Failure, Lost to follow up, Regimen Change, Not evaluated) | **Date** | **Remarks (**provide cause of death, reason for lost to follow up/Failure (culture non conversion, reversion, ADR, Additional drug resistance), latest TB no. in case of failure and put on treatment further) |
|---|---|---|
| | | |

| Date of adverse drug reaction | Details of symptoms | Action taken |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |

**Comments:**_____

_____

_____

_____

**Name & Signature of MO-PHI:** (In consultation with DR-TBC)

_____

| Post treatment follow up clinical & sputum (Result with date) | | | | | |
|---|---|---|---|---|---|
| **Follow up** | **Clinical** | **Smear** | **Culture** | **CXR** | **Impression** |
| **6 months of Rx** | | | | | |
| **12 months of Rx** | | | | | |
| **18 months of Rx** | | | | | |
| **24 months of RX** | | | | | |

# Annexure 15H: RNTCP PMDT Referral for treatment form

(Fill in duplicate. Send one copy to the concerned facility receiving the patient, and file the duplicate)

Name and address of referring unit (District TB Centre/DR TB Centre):_____
E-mail address of referring unit: _____
Name of the facility where patient is referred: _____
Name of patient: _____ Age: _____ Gender: _____
Complete address: _____

**Patient detail**

**Disease classification**: ☐ Pulmonary
☐ Extra pulmonary (site_____)

**Type:** ☐ New ☐ Recurrent ☐ Treatment after Lost to follow-up ☐ Treatment after Failure
☐ Others, previously treated

**Reason for testing:**
☐ New ☐ Previously Treated

☐ **Presumptive TB**
☐ **Private referral**
☐ **Presumptive NTM**
☐ **Presumptive MDR-TB**
- ☐ At diagnosis
- ☐ Contact of MDR/RR TB
- ☐ Follow up Sm +ve
- ☐ Private referral

☐ **Presumptive H mono/poly**
☐ **Presumptive XDR-TB**
- ☐ MDR/RR TB at diagnosis
- ☐ ≥ 4 months culture positive
- ☐ 3-monthly for persistent culture positives (treatment month _____)
- ☐ Culture reversion
- ☐ Failure of MDR/RR-TB regimen
- ☐ Recurrent case of second line treatment

**Latest PMDT-TB No**:_____

**Latest regimen**:
- ☐ Regimen for INH mono/poly resistant TB
- ☐ Regimen for MDR/RR TB
- ☐ Shorter regimen
- ☐ Modified Regimen for MDR/RR-TB + FQ /SLI resistance
- ☐ Regimen for XDR TB
- ☐ Modified Regimen for mixed pattern resistance
- ☐ Regimen with New Drug for MDR-TB Regimen + FQ/SLI resistance
- ☐ Regimen with New Drug for XDR-TB
- ☐ Regimen with New Drug for failures of regimen for MDR TB
- ☐ Regimen with New Drug for failures of regimen for XDR-TB
- ☐ Regimen with New Drug for mixed pattern resistance

**Sputum, culture and DST details**

Date of culture result: __/__/___
Date of DST/LPA/CBNAAT result: __/__/___
DST/LPA/CBNAAT result* :
☐ R ☐ H(katG) ☐ H(inhA) ☐ S ☐ E ☐ Z
☐ SLI Class ☐ SLI (eis) ☐ Km ☐ Cm ☐ Am
☐ FQ Class ☐ Lfx ☐ Mfx (0.5) ☐ Mfx (2.0)
☐ Eto ☐ PAS ☐ Lzd ☐ Cfz ☐ __ ☐__ ☐ __
(*Tick the drugs to which resistance is demonstrated)

**DR TB treatment details**

NIKSHAY ID: _____
DR TB Centre:_____
Date of DR TB regimen initiation: : __/__/__
Number of doses:_____

Date of regimen change and details of change: _____
Past exposure to second-line anti TB drugs: Drugs (duration)_____
HIV Status: ☐ Pos ☐ Neg ☐ Not known Date of CPT initiation:_____ Date of ART initiation:____
Date of referral to DR-TB Centre / DTC: Day _____ Month _____ Year _____

**Referred for:**
- ☐ Initiation of treatment
- ☐ Adverse drug reaction (give details) _____
- ☐ Transfer out (give details)_____
- ☐ Ambulatory treatment (if the patient is referred to DTC)
- ☐ Any other (give details)_____

Name and designation of the referring doctor_____

**Reminder for the health facility where the patient has been referred**

Please send an e-mail to the referring unit, informing the referring doctor of the date that the above-named patient reported at the receiving health facility.

# Annexure 15I: Revised National Tuberculosis Control Programme – TB Notification Register

Year _____ PHI_____ Health Facility ID _____

| TB notification no. (NIKSHAY) | Name (in full) | Age | Sex (M/F/TG) | Complete Address *(including district / state)* | Pin code | Mobile / Landline Number | Aadhaar No. | Key population* | Type of patient** | Site (P/EP) | Case Definition£ | Microbiological confirmation test results | | | | | Results of Other tests (X-Ray/Histo patho/ FNAC/ Clinical/ /Other, specify) | HIV Status± | Diabetes Status^ | Date of sample sent for DST *(NO if not sent, NA if not applicable)* | Result of DST@ | Status of treatment *** | Health facility for treatment (Details) | Date of initiation of treatment |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | Date | Lab Name | Lab no. | Test ¥ | Results of Test# | | | | | | | | |
| | ..... | | | ..... | | | | | | | | | | | | | | | | | | | | |
| | ..... | | | ..... | | | | | | | | | | | | | | | | | | | | |
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***Key population**
1. Contact of TB/DRTB case, 2. Tobacco, 3. Prison inmates, 4. Miner, 5. Migrant, 6. Refugee, 7. Urban slum, 8. Health-care worker, 9. PLHIV 10. Other (specify)

** **Type of patient (use complete words)** → New, Recurrent, Treatment after Failure, Treatment after Lost to Follow up, Other Previously Treated, Transferred in

**£ Case Definition** → Microbiologically Confirmed, Clinically Diagnosed

**¥ Test** → ZN, FM, Culture, CBNAAT

**# Result of test**
For Smear result – Grades for smear positive (Scanty with no. of bacilli, +1, +2, +3), NEG for smear negative
For GX result – MTB detected Rif Resistance, MTB detected Rif sensitive, MTB detected Rif Indeterminate, MTB not detected, Error, Invalid, No result
For Culture result – Grades for culture positive, NEG for culture negative

**± HIV Status**
HIV status as reported before or during TB treatment R – Reactive, NR – Non-Reactive, U – Unknown.

**^Diabetes Status → D=**Diabetes**, N=**NonDiabetes**, U =** Unknown

**@** Sensitive= if sensitive to tested drugs, Name of drug if resistant to any – R= Rifampicin, H=Isoniazide, E=Ethambutol, Z=Pyrazinamide, Sm=Streptomycin Lx=Levofloxacin, Mx=Moxifloxacin, Km=Kanamycin, Cm=Capreomycin

*****Status of treatment-**
1. Initiated on First line treatment in the same Health Facility
2. Initiated on treatment outside Health Facility
3. Initiated on second line treatment
4. Treatment initiated outside RNTCP
5. Incomplete/ incorrect address
6. Died
7. Migrated & untraceable
8. Refuse for treatment
9. Repeat diagnosis
10. Patient already on treatment/ Follow up patient
11. Wrong diagnosis
12. Referred for treatment with pending feedback
13. Other

**Revised National Tuberculosis Control Programme – TB Notification Register** **Year _____ PHI_____ Health Facility ID _____**

| Type of regimen (N / PT / Outside RNTCP) | Weight at beginning of treatment | Dosage Frequency (Daily / Intermittent) | Follow-up smear examinations | | | | | | | | | | Treatment Outcome# | | If HIV-Reactive | | Post treatment follow up | | | | | | | | | | | | | | Treatment supporter details | | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | End of IP | | | | | End of Treatment Exam | | | | | | | | | At 6 months Date_____ | | | | At 12 months Date_____ | | | | At 18 months Date_____ | | | | At 24 months Date_____ | | | | | | |
| | | | Date | DMC Name | Smear result | Date of sample collected for DST | Result of DST@ | Date | DMC Name | Smear results | Date of sample collected for DST | Result of DST@ | Outcome | Date | CPT (y/n) date | ART (y/n) date | Symptoms* | CXR | Smear | Culture | Symptoms | CXR | Smear | Culture | Symptoms | CXR | Smear | Culture | Symptoms | CXR | Smear | Culture | Name | Design ation | |
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
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# **Treatment Outcome –**
Cured, Treatment Completed, Died, Lost to follow up, Failure, Not evaluated or Treatment change

± **Additional treatments if patient HIV-Reactive**
Required only for patients known to be HIV Reactive. If provided by any source during TB treatment, enter "Y" and approximate date. If not provided / unknown, enter "N".
***Symptoms**- Mention predominant system- Cough-C, Fever-F, Haemoptysis-H, Weight loss-W, Night Sweat - N Others-O, No symptoms - NS

# Annexure 15J: RNTCP PMDT Treatment Register

**Month _____Quarter _____ Year _____ District: _____ CBNAAT site: _____ C-DST Lab: _____ DDR-TB Centre:_____ NDR-TB Centre:_____**

| PMDT TB No | NIKSHAY ID | Patient's name in full | Gender (M/F/TG) | Age in yrs | Complete address & mobile number | Name of health facility, TU, District | @ Reason for Testing | Site of Disease (P/EP) | Type (New, Recurrent, TALFU, Failure, Others) | DST/ DRT Details | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Lab Name & Sr. No. | Type of DST/DRT (LJ/LC/ FL-LPA/SL-LPA/CBNAAT) Date and result of DRT/DST (R-Resistance detected, S- resistance not detected) | | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | R | H (inhA) | H (katG) | S | E | Z | Km | Cm | Am | Lfx | Mfx (0.5) | Mfx (2) | FQ class | SLID class | SLID (eis) | Eto | PAS | Lzd | Cfz | Bdq | Dlm |
| | | | | | | | | | | | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY |
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY |
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY |
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| | | | | | | | | | | | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY |
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY | DD/MM/YY |
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

@ Presumptive TB – 1; Private referral – 2; Presumptive NTM – 3;
@ Presumptive MDR TB, At diagnosis–4; Contact of MDR/RR TB – 5; Follow up Sm +ve at end IP – 6; private referral – 7; Discordance resolution – 8; Presumptive H mono/poly – 9; MDR/RR TB at diagnosis – 10; ≥ 4 months culture positive –11; 3-monthly for persistent culture positives –12; Culture reversion –13: Failure of MDR/RR-TB regimen –14; Recurrent case of second line treatment –15

| Type of DR TB Patient RRTB/MDRTB/XDR TB | | |

## Annexure 15K: TB Laboratory Register

| Lab. Serial No. | Date of collection of first specimen | Name (in full) | Age | Sex M/ F/TG | Complete address (for diagnosis patients) & Phone No. | Key Population[1] | Name and type of referring health facility[2] | Reasons for Examination | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Presumptive TB / RE / Presumptive NTM/ Private | Predominant symptom[3] & its duration[4] | History of >1 month ATT (Yes/No) | Follow-up | | | |
| | | | | | | | | | | | Nikshay ID | Regimen New (N) / Previously Treated (PT) | Month | Post Treatment follow up month |
| | | | | | | | | | | | | | | |
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[1] Key population – 1. Contact of TB/DRTB case, 2. Tobacco, 3. Prison inmates, 4. Miner, 5. Migrant, 6. Refugee, 7. Urban slum, 8. Health-care worker, 9. PLHIV 10. Other (specify)

[2] Name of referring health facility-PHI/DMC/TB/DTC/ICTC/ART/Medical College/DR-TB centre / Private/ Others, specify

[3] Predominant symptoms: Cough-C, Fever-F, Haemoptysis-H, Weight loss-W, Night Sweat - N Others-O, No symptoms - NS

[4] Duration of predominant symptoms should be recorded in days

| Type of specimen | Visual appearance[5] | | Results | | Date of Result | HIV status (Reactive/ Non Reactive/ Unknown) | Diabetic status (Diabetic/ Non Diabetic/ Unknown) | Sample for DST sent (Y/N) with date | DST result[7] (write the drugs to which resistance is demonstrated) | NIKSHAY ID (notification no.) | Treatment initiation details (TB No. & TU details) / Referral for treatment | Signature | Remarks[8] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | a[6] | b[6] | a[6] | b[6] | | | | | | | | | |
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[5]Visual appearance- mention M, B, or S., Mucopurulent, Blood stained or Saliva

[6]a- stands for supervised spot sample, b- stands for early morning sample

[7] Sensitive= if sensitive to tested drugs, Name of drug if resistant to any – R= Rifampicin, H=Isoniazide, E=Ethambutol, Z=Pyrazinamide, SM=Streptomycin Lx=Levofloxacin, Mx (0.5) or (2) =Moxifloxacin, Km=Kanamycin, Cm=Capreomycin , Am=Amikacin, Eto=Ethionamide, Lzd=Linezolid, Cfz=Clofazimine

[8] Remarks column can include date of starting treatment, treatment regimen, TB no., referral details with date, remarks on un b linded rechecking, etc.

## Annexure 15L: RNTCP Laboratory Register for culture, CBNAAT and drug susceptibility testing

| S No | NIKSHAY ID | Patient's full name (Address/contact details) | Age | Gender (M/F/TG) | Key Population[1] | Name and type (PHI / DMC / TU/ DTC / ICTC / ART / Medical College / DR-TB Centre / Private Others, specify) of referring health facility | Reason for testing | | | | | Date | | | Type (Sputum/other-specify) | Specimen condition # (M/B/S/C) | C&DST Lab Microscopy Result |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Diagnosis/DST | | | Follow up | | | | | | | |
| | | | | | | | New/ PT | @ | Predominant symptom[2] and duration[3] | PMDT TB No | Month of FU | Specimen collection | Specimen sent to lab | Specimen receipt at laboratory | | | |
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@ Presumptive TB – 1; Private referral – 2; Presumptive NTM – 3;

@ Presumptive MDR TB, At diagnosis–4; Contact of MDR/RR TB – 5; Follow up Sm +ve at end IP – 6; private referral – 7; Discordance resolution – 8; Presumptive H mono/poly – 9; MDR/RR TB at diagnosis – 10; ≥ 4 months culture positive –11; 3-monthly for persistent culture positives –12; Culture reversion –13: Failure of MDR/RR-TB regimen –14; Recurrent case of second line treatment –15

# M–Mucopurulent; B– Blood stained; S– Saliva; C– Contaminated

[1] Key population – 1. Contact of TB/DRTB case, 2. Tobacco, 3. Prison inmates, 4. Miner, 5. Migrant, 6. Refugee, 7. Urban slum, 8. Health-care worker, 9. PLHIV, 10.Other (specify)
[2]Predominant symptoms: Cough-C, Fever-F, Haemoptysis-H, Weight loss-W, Night Sweat - N Others-O, No symptoms - NS
[3]Duration of predominant symptoms should be recorded in days

## RNTCP Laboratory Register for Culture, CBNAAT and Drug Susceptibility Testing

| Rapid DST Results | | | | | | | | | | Culture Results | | | | Standard DST Results (R/S) | | | | | | | | | | | | | | | | | | | | | | | | | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Test performed (FL-LPA/SL-LPA/CBNAAT) | Date of reporting Rapid DST Results | Valid* (Y/N) | TB † (Y/N) | RIF ‡ ( R / S / I / N A) | H (KatG) (R/S/NA) ‡ | H (inhA) (R/S/NA) ‡ | FQ class (R/S/NA)‡ | SLI class (R/S/NA)‡ | SLI (eis) (R/S/NA)‡ | Type (L J/LC) | ICT (+ve/-ve/NA)€ | Final culture result§ | Date of reporting of culture result | Type (LJ/LC) | Rifampicin | Isoniazid (0.1) | Isoniazid (0.4) | Streptomycin | Ethambutol | Pyrazinamide | Kanamycin | Capreomycin | Amikacin | Levofloxacin | Moxifloxacin (0.5) | Moxifloxacin (2.0) | Ethionamide | PAS | Linezolid | Clofazimine | Clarithromycin | Azythromycin | Bedaquiline | Delamanid | | Date of reporting DST result | | |
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
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* **Valid = Y** if both Amplification Control (AC) band & Conjugate Control (CC) band present; if either are missing, record **N**, and record no additional LPA results for this specimen.

**† TB = Y if** M. tuberculosis (TUB) band on LPA strip confirming identity as M. Tb or MTB Detected in CBNAAT, **N** if no TUB band on LPA strip or MTB Not Detected in CBNAAT

**‡ R** = Resistant Detected, **S** = Resistant not detected, I = Indeterminate, **NA** = no result, judged by no locus control band on LPA strip for *rpo-B* (RIF), or for *inh-A* or *kat-G* (INH) or for *gyr-A* or *gyr-B* for FLQ or *eis* for low level Km, or rrs for SLI. In case of CBNAAT, specify for NA, i.e. Error, Invalid, No Result

**§ Negative** = no growth, **Conta** = contaminated, **NTM** = Non-Tuberculosis Mycobacteria , **3+** = confluent growth, **2+** = >100 colonies, **1+** = 10–100 colonies; **Sc#** Scanty <10 .

**€ ICT** (Immuno-chromatographic Test ) **+Ve**=If test is positive, **-Ve**= if test is negative, **NA**= if test result is not available. M.tb confirmed by other biochemical (Niacin/ Nitrate reductase) / molecular methods are also to be indicated as +ve, -ve or NA as applicable.

## Annexure 15M: RNTCP PMDT Treatment Book

Patient's name: _____

Age: _____yrs Gender: ☐ Male ☐ Female ☐ Transgender

Address: _____

_____

Marital status: _____ Occupation: _____

Contact No: _____

Aadhar ID _____

Name, designation of treatment supporter:_____

_____

_____ Contact no:_____

State:_____ District: _____

TB Unit: _____ PHI:_____

Initial home visit: Date _____ By:_____

DR TB Centre:_____ District _____ State _____

| NIKSHAY ID | PMDT TB No | PMDT TB No | PMDT TB No |
|---|---|---|---|
| | | | |

| **Reason for Testing** | |
|---|---|
| ☐ New ☐ Previously Treated | |
| ☐ Presumptive TB ☐ Private Referral ☐ Presumptive NTM | |
| **Presumptive MDR TB** | ☐ At diagnosis<br>☐ Contact of MDR/RR TB<br>☐ Follow up Sm+ve at end IP<br>☐ Private referral |
| ☐ Presumptive H mono/poly | |
| **Presumptive XDR TB** | ☐ MDR/RR TB at diagnosis<br>☐ 4 months culture positive<br>☐ 3 monthly, for persistent culture positives (treatment month __)<br>☐ Culture reversion<br>☐ Failure of MDR/RR-TB regimen<br>☐ Recurrent case of second line treatment |

| Drug Susceptibility Test (DST) results at Diagnosis | | | | | | | | | | | | | | Spec. coll. Date: | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R | H (inhA/0.1) | H (katG/0.4) | S | E | Z | Km | Cm | Am | Lfx | Mfx (0.5) | Mfx (2) | FQ class | SLI class | SLI (eis) | Eto | PAS | Lzd | Cfz | Clr | Azi | Bdq | Dlm |
| | | | | | | | | | | | | | | | | | | | | | | |
| Name of the lab | | | | | | | | | | Lab.sr. No._____ | | | | | Result report date | | | | | | | |

*R: Resistant; S: Susceptible; C:* ***Contaminated****;* ***--*** *Not done/available*

**Contact Investigation**

| | |
|---|---|
| No of members screened | |
| No of presumptive TB cases identified | |
| No of presumptive TB cases evaluated | |
| No diagnosed with TB | |
| No of DR-TB diagnosed | |

**HIV Testing:**
Date:_____
Result:_____
PID number:_____
CPT start date:_____
ART start date:_____

**Blood Sugar Testing:**
Date:_____
RBS:_____
FBS:_____
ADT*
(*write date of starting)

| DR-TB Centre Committee meetings – dates and decisions | | |
|---|---|---|
| Date | Decision | Duration of indoor stay |
| | | |
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**TB Site:** ☐ Pulmonary ☐ Extra Pulmonary

If extra pulmonary, Specify _____

**Treatment regimen**

☐ Regimen for INH mono/poly resistant TB

☐ Regimen for MDR/RR TB

☐ Shorter MDR-TB Regimen

☐ Modified Regimen for MDR/RR-TB + FQ/SLI resistance

☐ Regimen for XDR TB

☐ Modified Regimen for mixed pattern resistance

☐ Regimen with New Drug for MDR-TB Regimen + FQ/SLI resistance

☐ Regimen with New Drug for XDR-TB

☐ Regimen with New Drug for failures of regimen for MDR-TB

☐ Regimen with New Drug for failures of regimen for XDR-TB

☐ Regimen with New Drug for mixed pattern resistance

**Initiation Date: _____ Registration Date: _____**

| Pre-treatment investigation | | | | | |
|---|---|---|---|---|---|
| **Test** | **Date** | **Result (units)** | **Test** | **Date** | **Result (units)** |
| ALT (SGPT) | | | Chest X-Ray findings<br>Cavities (Y/N) | | |
| AST (SGOT) | | | | | |
| Bilirubin- Direct<br>-Indirect | | | | | |
| Albumin | | | | | |
| WBC (TC/DC) | | | ECG (QTc & other findings) | | |
| Haemoglobin | | | Creatinine | | |
| ESR | | | Creatinine Clearance | | |
| Platelet count | | | Blood Urea | | |
| Lactic acid | | | Visual acuity | | |
| Amylase | | | Audiogram | | |
| Lipase | | | | | |
| CD4 Count | | | Psychiatric evaluation | | Yes/ No |
| Hepatitis markers | | | Surgical evaluation | | Yes/ No |
| TSH | | | Ophthalmic evaluation | | Yes/ No |
| Urine (R/M) | | | | | |
| UPT | | | | | |
| Potassium | | | | | |
| Magnesium | | | | | |
| Calcium | | | | | |

Initial Weight: _____kgs Height:_____cms

**Weight band:**

☐ <16 Kg ☐ 16-29 Kg ☐ 30-45 Kg ☐ 46-70 Kg ☐ >70 Kg

| Drug and Dosages | | |
|---|---|---|
| **Drugs** | **Dose during IP (mg)** | **Dose during CP (mg)** |
| H | | |
| R | | |
| E | | |
| Z | | |
| Km | | |
| Am | | |
| Cm | | |
| Lfx | | |
| Mfx | | |
| Cs | | |
| Eto | | |
| PAS | | |
| Lzd | | |
| Cfz | | |
| Amx Clv | | |
| Clr | | |
| BDQ | | |
| Dlm | | |
| | | |
| | | |

**Patient eligible and consented for New Drug*** ☐ **Yes** ☐ **No**
**If No, reason**_____

**Name & Signature of Treating Physician:**

_____

***Whenever available***

| Month of Treatment | Culture Results | | | Other Investigations | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Date | Lab No | Culture | S. Cr | LFT | ECG*-QTC | CBC/ Platelets | Electrolyte (K, Mg, Ca) | other |
| 1st week | | | | | | | | | |
| 2nd week | | | | | | | | | |
| 3rd week | | | | | | | | | |
| 1 | | | | | | | | | |
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** ECG to be done daily (first two weeks), weekly (for 3 months) then monthly for newer drug containing regimen under CAP.*

| Drug Susceptibility Testing (DST) Results | | | | | |
|---|---|---|---|---|---|
| **Lab name** | | | | | |
| **Lab. Sr. No** | | | | | |
| **Type of DST** (L J/LC/FL-LPA/SL-LPA/CBNAAT) | | | | | |
| **Specimen collection Date** | | | | | |
| **Result report date** | | | | | |
| follow up month | | | | | |
| DST results | | | | | |
| R | | | | | |
| H (inhA) | | | | | |
| H (katG) | | | | | |
| S | | | | | |
| E | | | | | |
| Z | | | | | |
| Km | | | | | |
| Cm | | | | | |
| Am | | | | | |
| Lfx | | | | | |
| Mfx (0.5) | | | | | |
| Mfx(2.0) | | | | | |
| FQ class | | | | | |
| SLID class | | | | | |
| SLI (eis) | | | | | |
| Eto* | | | | | |
| PAS* | | | | | |
| Lzd | | | | | |
| Cfz* | | | | | |
| Bdq* | | | | | |
| Dlm* | | | | | |

**Whenever available*

**Patient's Name:**_____

**Blood Sugar Testing:**

Date:_____

RBS:_____

FBS:_____

ADT*

(*write date of starting)

**Blood Sugar Testing:**

Date:_____

RBS:_____

FBS:_____

ADT*
(*write date of starting)

| **Thyroid Function Test** | | |
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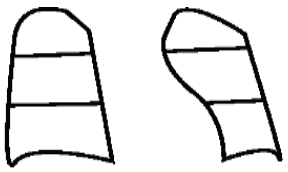
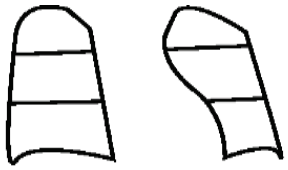
| Month | Zero | Six |
| Date | | |
| T3 | | |
| T4 | | |
| TSH | | |



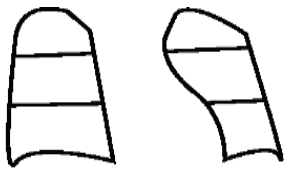
Date of X-ray

Findings_____



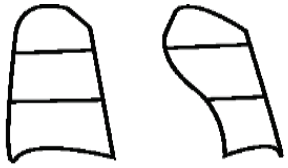
Date of X-ray

Findings_____



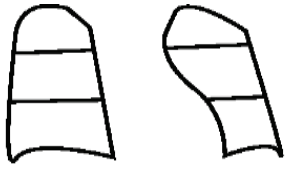
Date of X-ray

Findings_____



Date of X-ray

Findings_____



Date of X-ray

Findings_____

Date of starting intensive phase: _____

Date of starting continuation phase:_____

| Details of change in regimen drugs and dosages | | |
|---|---|---|
| Date | Changed regimen drugs and dosages | Reason for change |
| | | |
| | | |
| | | |
| | | |

## ADMINISTRATION OF DRUGS (one line per month)

| Month & Yr | Day | | | | | | | | | | | | | | | | | | | | | | | | | | | | | Wt in Kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | |
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Mark in the boxes: ✓= directly observed; (✓)= Unsupervised; ◯ = drugs not taken; X = initiation of new box; Recording of CP should start from fresh line.

| Month & Yr | Day | | | | | | | | | | | | | | | | | | | | | | | | | | | | | Wt in Kg |
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Mark in the boxes: ✓= directly observed; (✓) = Unsupervised; ◯ = drugs not taken; X = initiation of new box; Recording of CP should start from fresh line.

| Month & Yr | Day | | | | | | | | | | | | | | | | | | | | | | | | | | | | | Wt in Kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
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Mark in the boxes: ✓= directly observed; (✓)= Unsupervised; ◯ = drugs not taken; X = initiation of new box; Recording of CP should start from fresh line.

| Month & Yr | Day | | | | | | | | | | | | | | | | | | | | | | | | | | | | | Wt in Kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | |
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Mark in the boxes: ✓= directly observed; (✓) = Unsupervised; ( ) = drugs not taken; X = initiation of new box; Recording of CP should start from fresh line.

| Date of retrieval action | By whom | Who contacted | Reason for missed doses | Outcome of retrieval action |
|---|---|---|---|---|
| | | | | |
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| Date of adverse drug reaction | Details of symptoms | Action taken |
|---|---|---|
| | | |
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| | | |
| | | |

| Clinical Notes | |
|---|---|
| Date of visit:<br><br>Chief Complaints:<br><br><br><br><br><br>Clinical examination (major findings) :<br><br><br><br><br><br><br><br>Counselling notes:<br><br><br><br><br><br>Weight | Investigations<br><br><br><br><br><br>Treatment |

| Clinical Notes | |
|---|---|
| Date of visit:<br><br>Chief Complaints:<br><br><br><br><br><br>Clinical examination (major findings) :<br><br><br><br><br><br><br><br>Counselling notes:<br><br><br><br><br><br>Weight | Investigations<br><br><br><br><br><br>Treatment |

| Clinical Notes | |
|---|---|
| Date of visit:<br><br>Chief Complaints:<br><br><br><br><br><br>Clinical examination (major findings) :<br><br><br><br><br><br><br><br>Counselling notes:<br><br><br><br><br><br>Weight | Investigations<br><br><br><br><br><br>Treatment |

| Clinical Notes | |
|---|---|
| Date of visit:<br><br>Chief Complaints:<br><br><br><br><br><br>Clinical examination (major findings) :<br><br><br><br><br><br><br><br>Counselling notes:<br><br><br><br><br><br>Weight | Investigations<br><br><br><br><br><br>Treatment |

| Clinical Notes | |
|---|---|
| Date of visit:<br><br>Chief Complaints:<br><br><br><br><br><br>Clinical examination (major findings) :<br><br><br><br><br><br><br><br>Counselling notes:<br><br><br><br><br><br>Weight | Investigations<br><br><br><br><br><br>Treatment |

| Clinical Notes | |
|---|---|
| Date of visit:<br><br>Chief Complaints:<br><br><br><br><br><br>Clinical examination (major findings) :<br><br><br><br><br><br><br><br>Counselling notes:<br><br><br><br><br><br>Weight | Investigations<br><br><br><br><br><br>Treatment |

| Date of clinical follow up visits and observation | | |
|---|---|---|
| 1st visit:_____ | 6th visit:_____ | 11th visit:_____ |
| 2nd visit: _____ | 7th visit: _____ | 12th visit: _____ |
| 3rd visit:_____ | 8th visit:_____ | 13th visit:_____ |
| 4th visit:_____ | 9th visit:_____ | 14th visit:_____ |
| 5th visit: _____ | 10th visit: _____ | 15th visit: _____ |

| **Treatment outcome** ( Cured, Treatment completed, Died, Failure, Lost to follow up, Regimen Change, Not evaluated) | **Date** | **Remarks (**provide cause of death, reason for lost to follow up/Failure (culture non conversion, reversion, ADR, Additional drug resistance), latest TB no. in case of failure and put on treatment further) |
|---|---|---|
| | | |

| Post treatment follow up clinical & sputum (Result with date) | | | | | |
|---|---|---|---|---|---|
| **Follow up** | **Clinical** | **Smear** | **Culture** | **CXR** | **Impression** |
| **6 months of Rx** | | | | | |
| **12 months of Rx** | | | | | |
| **18 months of Rx** | | | | | |
| **24 months of RX** | | | | | |

| Do's and Don'ts for patient |
|---|
| **Infection Control** |
| **Do's**<br>• Cover your mouth with a tissue, handkerchief or upper sleeves of your clothing while coughing.<br>• Wash your hands with soap and water or an alcohol based hand wash.<br>• The room where the patient stays for a considerable time should be well ventilated and with proper sunlight.<br>• Any family member who develops cough should also follow similar cough etiquette. |
| **Don'ts**<br>• Do not cough and spit in the open.<br>• Do not close or obstruct your windows to ensure proper ventilation. |
| **Nutrition** |
| **Do's**<br>• Take complete meals inclusive of rice/roti, dal, vegetables, eggs, fish, meat (if available).<br>• At the start of the treatment, there may be some nausea and stomach upset. Kindly consult your doctor for the same. |
| **Don'ts**<br>• Do not stop or skip any meal. |
| **Side effects** |
| **Do's**<br>• If you have any side effects or discomfort on taking treatment, report to your doctor or treatment provider immediately.<br>• For any other illness developed during treatment, report to your doctor or treatment provider immediately. |
| **Don'ts**<br>• be complacent about side effects or discomfort.<br>• Do not try to take medications for side effects on your own. |
| Carry your treatment booklet whenever you visit any doctor.<br>Take your medicines regularly and complete the full course of treatment as prescribed by your doctor. |

## Annexure 15N: DR-TB counseling register

| S.No | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Nikshay ID | | | | | | |
| Name of the Patient | | | | | | |
| Age | | | | | | |
| Sex (M/F/TG) | | | | | | |
| Complete Address | | | | | | |
| Contact number | | | | | | |
| Date of Diagnosis | | | | | | |
| Type of test offered (CBNAAT/ LPA/ LJ/LC) | | | | | | |
| Date of Initiation on Treatment (DD/MM/YYYY) | | | | | | |
| Date of counseling at the time of PTE | | | | | | |
| Type of TB- RR/MDR/Pre- XDR/ XDR/DS TB | | | | | | |
| Is the patient aware of the treatment duration and follow up schedule? | | | | | | |
| Is the patient taking appropriate no. of medicines as per regimen prescribed? | | | | | | |
| No. of family members in house | | | | | | |
| No. of family members counseled during home visit | | | | | | |
| Additional Information from patient booklet | Key population type as per lab register | | | | | |
| | HIV Status | | | | | |
| | Comorbidity Status | | | | | |
| | H/O any side effect in the past | / | | | | |
| | Smoking/ Alcohol/ Tobacco | | | | | |
| | Status of nutrition | | | | | |
| | H/O Psychiatric illness | | | | | |
| | Coexisting depression, if any | | | | | |
| Major Observations at the time of counseling | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Remarks | | | | | | |
| Post treatment follow up sessions | 24 months | Session result | | | | | |
| | | Date (DD/MM/YYYY) | | | | | |
| | 18 months | Session result | | | | | |
| | | Date (DD/MM/YYYY) | | | | | |
| | 12 months | Session result | | | | | |
| | | Date (DD/MM/YYYY) | | | | | |
| | 6 months | Session result | | | | | |
| | | Date (DD/MM/YYYY) | | | | | |
| Treatment outcome with date | | | | | | | |
| Follow up counseling- Purpose | Session 4 | Session result | | | | | |
| | | Place (Facility/Home) | | | | | |
| | | Date (DD/MM/YYYY) | | | | | |
| | | Reason | | | | | |
| | Session 3 | Session result | | | | | |
| | | Place (Facility/Home) | | | | | |
| | | Date (DD/MM/YYYY) | | | | | |
| | | Reason | | | | | |
| | Session 2 | Session result | | | | | |
| | | Place (Facility/Home) | | | | | |
| | | Date (DD/MM/YYYY) | | | | | |
| | | Reason | | | | | |
| | Session 1 | Session result | | | | | |
| | | Place (Facility/Home) | | | | | |
| | | Date (DD/MM/YYYY) | | | | | |
| | | Reason | | | | | |

## Annexure 16: PMDT Monitoring Indicators

The source of information is Nikshay/eNikshay

| SN | Indicator | Numerator | Denominator |
|---|---|---|---|
| | | **Diagnosis** | |
| 1 | Proportion of notified TB patients tested for RR-TB<br>• New at diagnosis<br>• Previously treated at diagnosis<br>• Any follow up smear positive | Number of notified TB patients tested for RR-TB x 100<br>• New at diagnosis<br>• Previously treated at diagnosis<br>• Any follow up smear positive | Number of notified TB patients<br>• New<br>• Previously treated<br>• Any follow up smear positive |
| 2 | Proportion of TB detected with CBNAAT among Key population<br>• Paediatric age group<br>• People living with HIV<br>• Extra pulmonary | Number of TB patients diagnosed in specified group with CBNAAT Key population × 100<br>• Paediatric<br>• People living with HIV<br>• Extra pulmonary | Total number of TB patients of specific key population notified<br>• Paediatric patient<br>• People living with HIV<br>• Extra pulmonary |
| 3 | Proportion of RR-TB detected with CBNAAT among Key population<br>• Paediatric age group<br>• People living with HIV<br>• Extra pulmonary | Number of RR-TB patients diagnosed in specified group with CBNAAT Key population × 100<br>• Paediatric<br>• People living with HIV<br>• Extra pulmonary | Number of TB patients diagnosed in specified group with CBNAAT Key population<br>• Paediatric patient<br>• People living with HIV<br>• Extra pulmonary |
| 4 | Proportion of RS TB patient screened for Mono H resistance | Number of RS TB patients subjected to FL LPA × 100 | No. of RS TB confirmed patients diagnosed by CBNAAT |
| 5 | Proportion of Mono H resistant TB patient offered SL DST | Number of Mono H resistant TB patients offered SL DST × 100 | No. of Mono H resistant TB patients diagnosed during specified period |
| 6 | RR/MDR TB Notification Rate | Number of RR /MDRTB Patients Notified during specified Period | Projected Population in lac in a year |
| 7 | Mono H resistant TB Notification Rate | Number of Mono H resistant TB patients Notified during specified Period | Projected Population in lac in a year |

| | | | |
|---|---|---|---|
| 8 | Proportion of RR/MDR TB among New TB Patients | Number of RR/MDR TB Patients notified among New TB Patients during specified Period × 100 | Number of New TB Patients notified during specified Period |
| 9 | Proportion of RR/MDR TB among Previously treated TB Patients | Number of RR/MDR TB Patients notified among Previously treated TB Patients during specified Period × 100 | Number of Previously treated TB Patients notified during specified Period |
| 10 | Proportion of TB patient not responding to ( excluding failures) first line anti-TB treatment, tested for R resistance | Number of TB patient not responding to (excluding failures) first line anti-TB treatment, tested for Rif resistance | Total number TB patient not responding to (excluding failures) first line anti-TB treatment |
| 11 | Proportion of RR/MDR TB patients offered SL DST | Number of RR/MDR TB patients offered SL DST× 100 | No. of RR/MDR TB patients diagnosed during specified period |
| 12 | Proportion of RR/MDR TB diagnosed with additional drug resistance with SL- LPA<br>• FQ class resistance<br>• SLI class resistance<br>• Both FQ and SLI class resistance | No. of RR/MDR TB patients diagnosed with additional drug resistance with SL-LPA × 100<br>• FQ class resistance<br>• SLI class resistance<br>• Both FQ and SLI class resistance | No. of RR TB patients diagnosed for specified period |
| 13 | Proportion of any SL class resistant DR-TB patients subjected to LC –DST | Number of any SL class resistant DR-TB patients subjected to LC-DST× 100 | Number of any SL class resistant DR-TB patients diagnosed in specified period |
| **Process indicator** | | | |
| 1 | Average time taken for RR TB screening with CBNAAT among Presumptive DR-TB patients | Summation of days (difference between result date of TB diagnosis and result date of CBNAAT) | Total number of Presumptive DR patients screened for RR TB during the specified period |
| 2 | Average time taken for second line DST with SL- LPA among RR/MDR-TB patients | Summation of days (difference between result date of RR/MDR-TB diagnosis and test date for SL- LPA) | Total number of RR/MDR-TB patients subjected to SL – LPA during the specified period |

| | | | |
|---|---|---|---|
| 3 | Average time taken for second line DST with LC-DST for RR/MDR-TB + any class resistance patients | Summation of days (difference between result date for RR/ MDR TB + any class resistance with SL- LPA and result date of LC DST ) | Total number of RR/MDR TB + any class resistant patients subjected LC-DST during the specified period |
| 4 | Average time taken for initiation of RR/MDR treatment | Summation of days (difference between result date of RR/MDR TB diagnosis and date of treatment initiation) | Total number of RR/ MDRTB patients initiated on treatment during the specified period |
| 5 | Proportion of patients on RR/MDR-TB treatment who had reported adverse events | Number of RR/ MDR-TB patient reported any adverse events | Total number of RR/ MDR-TB patients on treatment during specified period |
| 6 | Proportion of RR/ DR-TB patients reported ADR affecting treatment continuation (partially or complete discontinuation of treatment) | Total number of DR-TB patients reported any ADR affecting treatment continuation. | Total number of patients on treatment during specified period |
| 7 | Proportion of DR TB patients on ICT enabled monitoring tool who interrupted treatment (missed 3 continuous doses >3 doses or total 10 doses) during month | Number of DR TB patients on ICT enabled monitoring tool missed doses (>3 doses) during month | Total number of patients on ICT enabled treatment monitoring tool during specified period |
| 8 | Extent of decentralised DR-TB treatment provision | Number of districts initiating DR-TB treatment for RR/ MDR TB at district or sub district level | Total number of districts in States/UTs |
| **Co-morbidity** | | | |
| 1 | Proportion of DR TB patients with known diabetes status | Number of DR TB patients (RR + Mono/ Poly) with known status for Diabetes | Number of DR TB patients (RR + Mono/ Poly) notified for specified period |
| 2 | Proportion of Diabetic patients among notified DR TB patients | Number of DR TB patients (RR + Mono/ Poly) diagnosed with Diabetes + known diabetic | Number of DR TB patients (RR + Mono/ Poly) screened for Diabetes during specified period |

| | | | |
|---|---|---|---|
| 3 | Proportion of DR TB patients with known HIV status | Number of DR TB patients (RR + Mono/ Poly) with Known HIV status | Number of DR TB patients (RR + Mono/ Poly) notified |
| 4 | Proportion DR TB patients with HIV infection | Number of DR TB patients (RR + Mono/ Poly) diagnosed with HIV + Known PLHIV | Number of DR TB patients (RR + Mono/ Poly) screened for HIV |
| 5 | Proportion of Paediatric Patients among DR TB Patients | Number of Paediatric DR TB Patients notified during specified Period × 100 | Total number of DR TB Patients notified during specified Period |
| 6 | Close contacts screening rate for DR-TB within one month of initiation of treatment | Number of pulmonary DR TB patients whose all close contacts were screened | Number of DR TB patients registered for treatment on month prior to specified period |
| 7 | DR TB rate among close contact screened | Number of DR TB patients diagnosed among screened close contacts | Number of close contacts screened |
| **Treatment** | | | |
| 1 | Proportion of RRMDR-TB patients initiated on treatment | Number of RR/MDR-TB patients initiated on treatment × 100 | Number of RR-TB patients diagnosed during specified period |
| 2 | Proportion of RR TB patients initiated on Shorter MDR-TB regimen | Number of RR-TB patients initiated on shorter treatment × 100 | Number of RR-TB patients diagnosed during specified period |
| 3 | Proportion of Mono/ poly initiated on mono/ poly treatment | Number of mono/ Poly DR TB patients initiated on Mono/ Poly DR TB treatment× 100 | Total number of mono/ poly DR TB patients diagnosed during specified period |
| 4 | Proportion of RR/MDR + any SLD resistant TB patient initiated on regimen containing newer drugs | No. of RR/MDR + any SLD resistance TB patient initiated on newer drugs containing regimen× 100 | No. of RR/MDR + any SLD resistant TB patients found eligible for newer drugs as per DST during specified period |
| 5 | Proportion of RR/MDR + any SLD resistant TB patient initiated on DST guided treatment | No. of RR/MDR + any SLD resistance TB patient initiated on DST guided treatment × 100 | No. of RR/MDR + any SLD resistant TB found during specified period |

| | | | |
|---|---|---|---|
| | ***Outcome Indicators*** | | |
| 1 | Proportion of RR/MDR-TB patients initiated on conventional regimen culture converted at 6 months | No. of RR/MDR-TB patients initiated on conventional regimen culture converted at end of 6 months | Total number RR/MDR-TB patients initiated on conventional regimen for selected period |
| 2 | Proportion of RR/MDR-TB patient on shorter regimen culture converted at 4 months | No. of RR/MDR-TB patient on shorter regimen culture converted at 4 months | Total number RR/MDR TB patients initiated on shorter RR/MDR-TB regimen for selected period |
| 3 | Proportion of mono/ poly- drug resistant pulmonary TB patients culture converted at 3 months | No. of mono- / poly- drug resistant TB patients culture converted at end of 3 months | Total number of mono- / poly- drug resistant TB patients initiated on treatment for specified period |
| 4 | Proportion of RR/MDR TB patients died at 6 months | No. of RR TB patients died at 6 months | Total number RR/MDR TB patients initiated on treatment for specified period |
| 5 | Proportion of RR/MDR TB patients lost to follow up at 6 months | No. of RR TB patients lost to follow up by at 6 months | Total number RR/MDR TB patients for specified period |
| 6 | Proportion of RR/MDR TB patients on shorter regimen patients declared (treatment outcome)<br>Cured<br>Treatment completed<br>Successfully treated<br>Died<br>Failure<br>Lost to follow up<br>Regimen changed<br>Not evaluated | No. of RR/MDR TB on shorter regimen patients declared specified (treatment outcome)<br>Cured<br>Treatment completed<br>Successfully treated<br>Died<br>Failure<br>Lost to follow up<br>Regimen changed<br>Not evaluated | Total No. of RR/MDR TB initiated on shorter regimen during specified period cohort (12 to 15 month prior) |

| | | | |
|---|---|---|---|
| 7 | Proportion of RR/ MDR TB (Conventional regimen) patients declared (treatment outcome)<br>Cured<br>Treatment completed<br>Successfully treated<br>Died<br>Failure<br>Lost to follow up<br>Regimen changed<br>Not evaluated | No. of RR/ MDR TB patients on conventional regimen declared specified (treatment outcome)<br>Cured<br>Treatment completed<br>Successfully treated<br>Died<br>Failure<br>Lost to follow up<br>Regimen changed<br>Not evaluated | Total No. of RR/MDR TB patients initiated on conventional regimen during specified period cohort (33 to 36 month prior) |
| 8 | Proportion of RR/ MDR TB patients on newer drug containing regimen declared (treatment outcome)<br>Cured<br>Treatment completed<br>Successfully treated<br>Died<br>Failure<br>Lost to follow up<br>Regimen changed<br>Not evaluated | No. of RR/ MDR TB patients on newer drug containing regimen declared specified (treatment outcome)<br>Cured<br>Treatment completed<br>Successfully treated<br>Died<br>Failure<br>Lost to follow up<br>Regimen changed<br>Not evaluated | Total No. of RR/ MDR TB initiated on newer drug containing regimen during specified period cohort (33 to 36 month prior) |

| | | | |
|---|---|---|---|
| 9 | Proportion of Mono H resistant TB patients declared (treatment outcome)<br>Cured<br>Treatment completed<br>Successfully treated<br>Died<br>Failure<br>Lost to follow up<br>Regimen changed<br>Not evaluated | No. of mono H resistant TB patient declared specified (treatment outcome)<br>Cured<br>Treatment completed<br>Successfully treated<br>Died<br>Failure<br>Lost to follow up<br>Regimen changed<br>Not evaluated | Total No. of DR TB initiated on regimen for mono H resistance during specified period cohort (12 to 15 month prior) |
| 10 | Recurrence free cure rate (up to 2 years) among successfully treated RR/ MDR TB with shorter regimen | Number of patients successfully treated with shorter regimen who do not develop TB within the 24 months of completion of treatment | Number of patients treated with shorter regimen whose treatment outcome was reported as successfully treated 24 months prior for selected cohort |
| **Private sector** | | | |
| 1 | Proportion of notified TB patients from private sector received DST within 1 month of treatment initiation. | Number of notified TB patients from private sector received DST within 1 month of treatment initiation. | Total number of TB patients notified from private sector. |
| 2 | Proportion of DR TB patient among notified private sector TB patients | Number of notified private sector DR TB patients | Total number of TB patients notified from private sector. |

## Annexure 17: Specifications of SLD monthly patient-wise boxes (PWB)

### Intensive Phase (1 month box)

**Box-A Type:**

- **Material – Punching Ring Flap model (1 side pin)**3 ply corrugated, Top paper 230 gsm White Colour, Ply 150 gsm & inner wall of 180 gsm Brown colour
- **Opening –** from Top with attached lid
- **Packing –** Box prepared by heavy stapler pins
- **Size** – 10.5" x 7.5" x 3.25" (Inch)
- **Colour** – Outer White & Inner Brown
- **Partitions in the box** – 3 ply White to White 2 Vertical & 12 Horizontal, Fitting in to vertical serially
- **Size of partition** – Horizontal 7.5" x 2.75"– Vertical 10.0" x 2.75"

Top Image: Box-A Type

Bottom Image: Box- B Type

**Box-B Type:**

a. **Material – Punching Ring Flap model (1 side pin)** 3 ply corrugated Top paper 230 gsm .White Colour, Ply 150 gsm & inner wall of 180 gsm & brown colour

**b. Opening –** from Top with attached lid

**c. Packing –** Box prepared by heavy stapler pins

**d. Size** – **11"x4"x4"** (Inch)

**e. Colour** – Outer White & Inner Brown

# Annexure 18: Prototype of label for SLD Patient Wise Boxes (PWB)

Conventional MDR TB Regimen

Conventional MDR TB Regimen: Type A (16-29 Kg) [2CRA1]

Weight Band : 16-29

| S.No. | Drug Name | UOM | Strength(No.) | Batch No. | DOE | No. Of Unit | Dosage Per Day |
|---|---|---|---|---|---|---|---|
| 1 | Cycloserine 250mg [PC24] | Cap | 250 MG | c102 | Dec/2017 | 30 | 1.0 |
| 2 | Ethambutol 400 mg [PC45] | Tab | 400 MG | 123 | Feb/2018 | 30 | 1.0 |
| 3 | Ethionamide 250 mg [PC20] | Tab | 250 MG | b-002 | Mar/2018 | 45 | 1.5 |
| 4 | Levofloxacin 250mg [PC28] | Tab | 250 MG | B-1 | Jul/2018 | 30 | 1.0 |
| 5 | Pyridoxine 100mg [PC26] | Tab | 100 MG | P-1 | Oct/2018 | 30 | 1.0 |

Lot No. : 1760000272 Total : 165

Date Of Expiry Of Box : Dec/2017

Store In Cool And Dark Place
Do Not Expose To Sunlight

SDS Assam
Assam

Conventional MDR TB Regimen

Conventional Mdr Tb Regimen:Type B(16-29Kg) [2CRB1]

Weight Band : 16-29

| S.No. | Drug Name | UOM | Strength(No.) | Batch No. | DOE | No. Of Unit | Dosage Per Day |
|---|---|---|---|---|---|---|---|
| 1 | Inj Kanamycin 1000mg [PC27] | Vial | 1000 MG | abc111 | Sep/2019 | 26 | 1.0 |
| 2 | Pyrazinamide 750mg [PC23] | Tab | 750 MG | abc111 | Apr/2018 | 30 | 1.0 |

Lot No. : 1760000273 Total : 56

Date Of Expiry Of Box : Apr/2018

Store In Cool And Dark Place
Do Not Expose To Sunlight

SDS Assam
Assam

Shorter MDR TB Regimen

Shorter Mdr Tb Regimen:Type A(46 -70kg) [2SRA3]

Weight Band : 46-70

| S.No. | Drug Name | UOM | Strength(No.) | Batch No. | DOE | No. Of Unit |
|---|---|---|---|---|---|---|
| 1 | Clofazimine 200mg [PC41] | Cap | 200 MG | RDN-01 | Oct/2017 | 3 |
| 2 | Ethambutol 800mg [PC10] | Tab | 800 MG | dfgrfg | May/2019 | 4 |
| 3 | Moxifloxacin 400mg [PC39] | Tab | 400 MG | f34fvsd | Oct/2017 | 6 |
| 4 | Pyrazinamide 750mg [PC23] | Tab | 750 MG | smoke01 | May/2018 | 6 |

Lot No. : 1760000429

Date Of Expiry Of Box : Oct/2017

Store In Cool And Dark Place
Do Not Expose To Sunlight

**Shorter MDR TB Regimen**

**Shorter Mdr Tb Regimen:Type B(29-45 Kg) [2SRB2]**

**Weight Band : 29-45**

| S.No. | Drug Name | UOM | Strength(No.) | Batch No. | DOE | No. Of Unit |
|---|---|---|---|---|---|---|
| 1 | Ethionamide 125mg [PC30] | Tab | 125 MG | Batch/01 | Oct/2017 | 12 |
| 2 | Inj Kanamycin 500mg [PC17] | Vial | 500 MG | aser123 | Jan/2018 | 2 |
| 3 | Isoniazid 300mg [PC11] | Tab | 300 MG | 454435 | Nov/2018 | 6 |

**Lot No. : 1760000432**

**Date Of Expiry Of Box : Oct/2017**

**Store In Cool And Dark Place**
**Do Not Expose To Sunlight**

**Regimen for H mono/poly DRTB**

**H Mono/Poly Drtb Regimen:Type A:(30-45 Kg) [2HRA2]**

**Weight Band : 29-45**

| S.No. | Drug Name | UOM | Strength(No.) | Batch No. | DOE | No. Of Unit |
|---|---|---|---|---|---|---|
| 1 | Ethambutol 800mg [PC10] | Tab | 800 MG | dfgrfg | May/2019 | 3 |
| 2 | Levofloxacin 500mg [PC29] | Tab | 500 MG | RNT002 | Mar/2020 | 4 |
| 3 | Pyrazinamide 150mg [PC55] | Tab | 150 MG | tst/2604 | May/2018 | 30 |
| 4 | Rifampicin 150mg [PC6] | Cap | 150 MG | fgxdgf | Apr/2018 | 8 |

**Lot No. : 1760000427**

**Date Of Expiry Of Box : Apr/2018**

**Store In Cool And Dark Place**
**Do Not Expose To Sunlight**

**Regimen for H mono/poly DRTB**

**H Mono/Poly Drtb Regimen:Type B:(30-45 Kg) [2HRB2]**

**Weight Band : 29-45**

| S.No. | Drug Name | UOM | Strength(No.) | Batch No. | DOE | No. Of Unit |
|---|---|---|---|---|---|---|
| 1 | Inj Kanamycin 500mg [PC17] | Vial | 500 MG | aser123 | Jan/2018 | 2 |

**Lot No. : 1760000428**

**Date Of Expiry Of Box : Jan/2018**

**Store In Cool And Dark Place**
**Do Not Expose To Sunlight**

## Annexure 19: Monthly Stock Statement for stocks at SDS Level

***(To be submitted to CTD each month by SDS)***

| Sr. No. | Nomenclature | UOM | Opening Balance | Receipts during the month | | Issues during the month | | | Balance Stock | DOM (One row for each drug) | DOE (One row for each drug) | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Receipt | Transfer In | Boxes | DRTB centre | Transfer Out | | | | |
| | | | (a) | (b) | ( c) | (d) | ( e) | (f) | (g)= (a+b+c-d-e-f) | | | |
| 1 | Kanamycin (Km) - 500 Mg | Vials | | | | | | | | | | |
| 2 | Kanamycin (Km) - 1000 Mg | Vials | | | | | | | | | | |
| 3 | Levofloxacin (Lfx)-250mg | Tabs | | | | | | | | | | |
| 4 | Levofloxacin (Lfx)-500mg | Tabs | | | | | | | | | | |
| 5 | Cycloserine (Cs) -250 Mg | Caps | | | | | | | | | | |
| 6 | Ethionamide (Eto) - 125 Mg | Tabs | | | | | | | | | | |
| 7 | Ethionamide (Eto) - 250 Mg | Tabs | | | | | | | | | | |
| 8 | Pyrazinamide (Z) - 500 Mg | Tabs | | | | | | | | | | |
| 9 | Pyrazinamide (Z) - 750 Mg | Tabs | | | | | | | | | | |
| 10 | Ethambutol(E) - 400 Mg | Tabs | | | | | | | | | | |
| 11 | Ethambutol(E) - 800 Mg | Tabs | | | | | | | | | | |
| 12 | Pyridoxine (X)-50mg | Tabs | | | | | | | | | | |
| 13 | Pyridoxine (X) - 100 Mg | Tabs | | | | | | | | | | |
| 14 | Na PAS - 4g | Sachets | | | | | | | | | | |
| 15 | Rifampicin (R)-150mg | Caps | | | | | | | | | | |
| 16 | Rifampicin (R)-450mg | Caps | | | | | | | | | | |
| 17 | Isoniazid (H)-100mg | Tabs | | | | | | | | | | |
| 18 | Isoniazid (H-300mg | Tabs | | | | | | | | | | |
| 19 | Linezolid (Lzd)-600mg | Tabs | | | | | | | | | | |
| 20 | Capreomycin (Cm)-750 Mg | Vials | | | | | | | | | | |
| 21 | Capreomycin (Cm)-1000 Mg | Vials | | | | | | | | | | |
| 22 | Moxifloxacin (Mfx)-400mg | Tabs | | | | | | | | | | |
| 23 | Clofazimine (Cfx)-100mg | Caps | | | | | | | | | | |
| 24 | Amoxyclav (Amx/Clv)1000mg | Tabs | | | | | | | | | | |
| 25 | Bedaquiline (Bdq)-100mg | Container | | | | | | | | | | |

| No. | Nomenclature | A/U | Opening Balance | Receipt during the month | Trf in | Trf Out | Qty issued | Closing Balance | D.O.E (One row for each box) |
|---|---|---|---|---|---|---|---|---|---|
| | | | (A) | (B) | (C ) | (D) | (E) | F = (A+B+C)-(D+E) | |
| | **Monthly Patient Wise Boxes** | | | | | | | | |
| 1 | Type-A ( <16 Kg Body Weight Patient ) | Drug Boxes | | | | | | | |
| 2 | Type-A ( 16-29 Kg Body Weight Patient ) | Drug Boxes | | | | | | | |
| 3 | Type-A ( 30- 45 Kg Body Weight Patient ) | Drug Boxes | | | | | | | |
| 4 | Type-A (46-70 Kg Body Weight Patient) | Drug Boxes | | | | | | | |
| 5 | Type-A (>70 Kg Body Weight Patient | Drug Boxes | | | | | | | |
| 6 | Type-B ( <16 Kg Body Weight Patient ) | Drug Boxes | | | | | | | |
| 7 | Type-B ( 16-29 Kg Body Weight Patient ) | Drug Boxes | | | | | | | |
| 8 | Type-B ( 30- 45 Kg Body Weight Patient) | Drug Boxes | | | | | | | |
| 9 | Type-B ( 46- 70 Kg Body Weight Patient) | Drug Boxes | | | | | | | |
| 10 | Type-B ( > 70 Kg Body Weight Patient ) | Drug Boxes | | | | | | | |

**(To be submitted for all standard regimen boxes)**

## Monthly Stock Report for stocking & indenting of PMDT at N/DDR-TB Centre
## (To be submitted to SDS/STO by N/DDR-TB Centre)

| Sr. No. | Nomenclature | UOM | Opening Balance (a) | Receipts during the month (b) | Issues during the month (c) | Balance Stock (d)= (a+b-c) | DOM (One row for each drug) | DOE (One row for each drug) | Remarks |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Kanamycin (Km) - 500 Mg | Vials | | | | | | | |
| 2 | Kanamycin (Km) - 1000 Mg | Vials | | | | | | | |
| 3 | Levofloxacin (Lfx)-250mg | Tabs | | | | | | | |
| 4 | Levofloxacin (Lfx)-500mg | Tabs | | | | | | | |
| 5 | Cycloserine (Cs) -250 Mg | Caps | | | | | | | |
| 6 | Ethionamide (Eto) - 125 Mg | Tabs | | | | | | | |
| 7 | Ethionamide (Eto) - 250 Mg | Tabs | | | | | | | |
| 8 | Pyrazinamide (Z) - 500 Mg | Tabs | | | | | | | |
| 9 | Pyrazinamide (Z) - 750 Mg | Tabs | | | | | | | |
| 10 | Ethambutol(E) - 400 Mg | Tabs | | | | | | | |
| 11 | Ethambutol(E) - 800 Mg | Tabs | | | | | | | |
| 12 | Pyridoxine (X)-50mg | Tabs | | | | | | | |
| 13 | Pyridoxine (X) - 100 Mg | Tabs | | | | | | | |
| 14 | Na PAS - 4g | Sachets | | | | | | | |
| 15 | Rifampicin (R)-150mg | Caps | | | | | | | |
| 16 | Rifampicin (R)-450mg | Caps | | | | | | | |
| 17 | Isoniazid (H)-100mg | Tabs | | | | | | | |
| 18 | Isoniazid (H-300mg | Tabs | | | | | | | |
| 19 | Linezolid (Lzd)-600mg | Tabs | | | | | | | |
| 20 | Capreomycin (Cm)-750 Mg | Vials | | | | | | | |
| 21 | Capreomycin (Cm)-1000 Mg | Vials | | | | | | | |
| 22 | Moxifloxacin (Mfx)-400mg | Tabs | | | | | | | |
| 23 | Clofazimine (Cfx)-100mg | Caps | | | | | | | |
| 24 | Amoxyclav (Amx/Clv)1000mg | Tabs | | | | | | | |
| 25 | Bedaquiline (Bdq)-100mg | Container | | | | | | | |

## Quarterly PMR for stocking & indenting of PMDT at DTC Level

**(To be submitted for all standard regimen boxes to CTD & STO/SDS by District) State:_____DTC: _____ Qtr-_____)**

| S.No | Item | UOM | Stock on first day of the Qtr | Stock received during the Qtr | Transferred In | Reconstitution | Consumption during the Qtr | Transferred Out | Stock on last day of the Qtr (c+d+e+f) – (g+h) | Quantity Requested for DTC (g/3 x 5) – f |
|---|---|---|---|---|---|---|---|---|---|---|
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | i | J |
| 1 | Type-A ( <16 Kg Body Weight Patient ) | Drug Boxes | | | | | | | | |
| 2 | Type-A ( 16-29 KgBody Weight Patient ) | Drug Boxes | | | | | | | | |
| 3 | Type-A ( 30- 45 Kg Body Weight Patient ) | Drug Boxes | | | | | | | | |
| 4 | Type-A (46-70 Kg Body Weight Patient) | Drug Boxes | | | | | | | | |
| 5 | Type-A (>70 Kg Body Weight Patient | Drug Boxes | | | | | | | | |
| 6 | Type-B ( <16 Kg Body Weight Patient ) | Drug Boxes | | | | | | | | |
| 7 | Type-B ( 16-29 KgBody Weight Patient ) | Drug Boxes | | | | | | | | |
| 8 | Type-B ( 30- 45 Kg Body Weight Patient) | Drug Boxes | | | | | | | | |
| 9 | Type-B ( 46- 70 Kg Body Weight Patient) | Drug Boxes | | | | | | | | |
| 10 | Type-B ( > 70 Kg Body Weight Patient ) | Drug Boxes | | | | | | | | |

## Stock Report for Stocks & Indenting of PMDT at DDS
## (To be submitted to DTO)

| Sr. No. | Nomenclature | UOM | Opening Balance | Receipts during the month | Issues during the month | Balance Stock | DOM (One row for each drug) | DOE (One row for each drug) | Remarks |
|---|---|---|---|---|---|---|---|---|---|
| | | | (a) | (b) | (c) | (d)= (a+b-c) | | | |
| 1 | Kanamycin (Km) - 500 Mg | Vials | | | | | | | |
| 2 | Kanamycin (Km) - 1000 Mg | Vials | | | | | | | |
| 3 | Levofloxacin (Lfx)-250mg | Tabs | | | | | | | |
| 4 | Levofloxacin (Lfx)-500mg | Tabs | | | | | | | |
| 5 | Cycloserine (Cs) -250 Mg | Caps | | | | | | | |
| 6 | Ethionamide (Eto) - 125 Mg | Tabs | | | | | | | |
| 7 | Ethionamide (Eto) - 250 Mg | Tabs | | | | | | | |
| 8 | Pyrazinamide (Z) - 500 Mg | Tabs | | | | | | | |
| 9 | Pyrazinamide (Z) - 750 Mg | Tabs | | | | | | | |
| 10 | Ethambutol(E) - 400 Mg | Tabs | | | | | | | |
| 11 | Ethambutol(E) - 800 Mg | Tabs | | | | | | | |
| 12 | Pyridoxine (X)-50mg | Tabs | | | | | | | |
| 13 | Pyridoxine (X) - 100 Mg | Tabs | | | | | | | |
| 14 | Na PAS - 4g | Sachets | | | | | | | |
| 15 | Rifampicin (R)-150mg | Caps | | | | | | | |
| 16 | Rifampicin (R)-450mg | Caps | | | | | | | |
| 17 | Isoniazid (H)-100mg | Tabs | | | | | | | |
| 18 | Isoniazid (H-300mg | Tabs | | | | | | | |
| 19 | Linezolid (Lzd)-600mg | Tabs | | | | | | | |
| 20 | Capreomycin (Cm)-750 Mg | Vials | | | | | | | |
| 21 | Capreomycin (Cm)-1000 Mg | Vials | | | | | | | |
| 22 | Moxifloxacin (Mfx)-400mg | Tabs | | | | | | | |
| 23 | Clofazimine (Cfx)-100mg | Caps | | | | | | | |
| 24 | Amoxyclav (Amx/Clv)1000mg | Tabs | | | | | | | |
| 25 | Bedaquiline (Bdq)-100mg | Container | | | | | | | |

## Quarterly PMR for stocking & indenting of PMDT at TU Level

**D.T.C._____TU: _____ Qtr-_____**

| S.No | Item | UOM | Stock on first day of the Qtr | Stock received during the Qtr | Consumption during the Qtr | Stock on last day of the Qtr (c+d) –e | Quantity Requested for TU (e/3 x 3) – f |
|---|---|---|---|---|---|---|---|
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) |
| 1 | Type-A ( <16 Kg Body Weight Patient ) | Drug Boxes | | | | | |
| 2 | Type-A ( 16-29 KgBody Weight Patient ) | Drug Boxes | | | | | |
| 3 | Type-A ( 30- 45 Kg Body Weight Patient ) | Drug Boxes | | | | | |
| 4 | Type-A (46-70 Kg Body Weight Patient) | Drug Boxes | | | | | |
| 5 | Type-A (>70 Kg Body Weight Patient | Drug Boxes | | | | | |
| 6 | Type-B ( <16 Kg Body Weight Patient ) | Drug Boxes | | | | | |
| 7 | Type-B ( 16-29 Kg Body Weight Patient ) | Drug Boxes | | | | | |
| 8 | Type-B ( 30- 45 Kg Body Weight Patient) | Drug Boxes | | | | | |
| 9 | Type-B ( 46- 70 Kg Body Weight Patient) | Drug Boxes | | | | | |
| 10 | Type-B ( > 70 Kg Body Weight Patient ) | Drug Boxes | | | | | |

**(To be submitted for all standard regimen boxes)**

## Monthly PMR for stocking & indenting of PMDT at PHI Level

**TU_____PHI: _____ Month-_____**

| S.No | Item | UOM | Stock on first day of the month | Stock received during the month | Consumption during the month | Stock on last day of the Month (c+d) –e | Quantity Requested for PHI (e x 3) – f |
|---|---|---|---|---|---|---|---|
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) |
| 1 | Type-A ( <16 Kg Body Weight Patient ) | Drug Boxes | | | | | |
| 2 | Type-A ( 16-29 KgBody Weight Patient ) | Drug Boxes | | | | | |
| 3 | Type-A ( 30- 45 Kg Body Weight Patient ) | Drug Boxes | | | | | |
| 4 | Type-A (46-70 Kg Body Weight Patient) | Drug Boxes | | | | | |
| 5 | Type-A (>70 Kg Body Weight Patient | Drug Boxes | | | | | |
| 6 | Type-B ( <16 Kg Body Weight Patient ) | Drug Boxes | | | | | |
| 7 | Type-B ( 16-29 Kg Body Weight Patient ) | Drug Boxes | | | | | |
| 8 | Type-B ( 30- 45 Kg Body Weight Patient) | Drug Boxes | | | | | |
| 9 | Type-B ( 46- 70 Kg Body Weight Patient) | Drug Boxes | | | | | |
| 10 | Type-B ( > 70 Kg Body Weight Patient ) | Drug Boxes | | | | | |

**(To be submitted for all standard regimen boxes**

## Annexure 20: Stock Register (SR)

Drug Item: Unit of Measurement (UOM) Patient-Wise Boxes/ blister: Folio No. :

| S.No. | Particulars of Receipts & Issues | | | | | | | Receipt (Qty.) | Issue (Qty.) | Balance (Qty.) | Date-Wise Expiry Details of Balance (Qty.) | | | | Signature of Store Keeper | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Date (dd/mm/yy) of Tran-saction (Receipt/ Issue) | Name of Party (GMSD/ SDS/ DTC/TU) | Receipt Voucher No. (For Receipts only) | Issue Voucher No.(For Issues only) | Date of Issue Voucher | Batch No. | Date of Expiry | | | | Expiry Date | Expiry Date | Expiry Date | Expiry Date | | |
| | | | | | | | | | | | | | | | | |
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## Annexure 21: Reconstitution Register (RR)

| S. NO. | DATE | DOT/ PHI (From which drugs have been trans-ferred) | TB REGIS-TER NO. | Category | INPUT (No. of Blisters) | | | | | OUTPUT (No. of Boxes) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | IP of PC-1 (24 Blister Combi-packs) | IP of PC-2 ( 36 Blister Combi-packs) | PP (12 Blisters in each PP) | CP of PC-1 (18 Multi-blister Combi-packs) | CP of PC-2 (22 Multi-blister Combi-packs) | Recon-stituted Drugs | Date of Expiry | Stock Register Folio No. |
| (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) |
| | | OPENING BALANCE | | | | | | | | | | |
| | | | | | | | | | | | | |
| | | | | | | | | | | | | |
| | | | | | | | | | | | | |
| | | | | | | | | | | | | |
| | | Total | | | | | | | | | | |

**KEY**: PC: Product Code; IP: Intensive Phase; CP: Continuation Phase

(1) Reconstituted IP & CP need to be put in poly bags with stickers on them, clearly mentioning that they comprise reconstituted IP or CP Pouches
(2) DOE of the reconstituted Patient Wise Box to be reported in Column (m) shall comprise the DOE of drugs used having the latest DOE
(3) Any loose drugs generated in the process may be used for patients put on non-DOTS treatment, or otherwise
(4) Reconstitution should be done under supervision of the DTO every quarter. DTO may instruct for reconstitution at shorter intervals, if required